Deployment, Convergence, and Upgrade

Get to VMware Cloud Foundation or VMware vSphere Foundation 9.1.x. Use **VCF Installer** and **VCF Operations** for automated deployment workflows and manually deploy the final components to complete your platform.

You can deploy a new VCF or vSphere Foundation platform on pre-installed ESX hosts by using VCF Installer.

You can also use your existing virtual infrastructure, in multiple scenarios, as building blocks and converge it to a VCF or vSphere Foundation platform.

You can upgrade your existing VCF 5.2.x or 9.0.x VCF environment to VCF 9.1.x.

You can only upgrade your VCF environment to a version that is released after the release date of your current VCF version. See VMware Interoperability Matrix.

In This Chapter

- Paths to Building 9.1.x Environments
- Components in VCF and vSphere Foundation
- Deploying a New VCF or vSphere Foundation Platform
- Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform
- Upgrading to VMware Cloud Foundation 9.1.x
- Upgrading to vSphere Foundation 9.1.x
- Upgrading vSphere 8 and Optionally vSAN and Aria Operations 8 to 9.1.x

VCF Upgrade Planner

You can now use the new VCF Upgrade Planning Tool to navigate you during the process of upgrading or converging your existing infrastructure to vSphere Foundation and VCF 9.1.x. Different tailored scenarios exist for starting from just vSphere or different other combinations that include NSX, Aria components and so on.

Getting to Know VCF

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes.

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

Paths to Building 9.1.x Environments

Multiple paths to get to VCF and vSphere Foundation 9.1 exist. Depending on your starting state, you follow one of the paths.

Depending on the starting state of your environment, you have three paths to VCF or vSphere Foundation environments:

- Deploy new VCF or vSphere Foundation environments.
- Use existing virtual infrastructure as building blocks to converge to VCF or vSphere Foundation environments.
- Upgrade your instances to VCF or vSphere Foundation 9.1.x.

The scope of the deploy, converge, and upgrade guidance for VCF is management domain-only as this is the minimum logical block to form the first VCF Instance in a VCF fleet. The version of the management domain also defines the VCF Instance version.

Deploying New VCF or vSphere Foundation 9.1.x

To deploy a new VCF fleet, VCF Instance, or vSphere Foundation without any existing components, see Deploying a New VCF or vSphere Foundation Platform.

Starting with version 9, the deployment process changes and you now install most of the components by using VCF Installer and VCF Operations workflows. Deployment of standalone components by deploying an OVA is typically not supported, apart from specific scenarios where you are explicitly instructed to do so.

- For information about the components installed by VCF Installer, see VCF Installer Overview.
- For information about the components you install after the VCF Installer workflow, see Completing the Deployment of Your Platform.

By following the deployment guidance, you can directly deploy VCF 9.1.1 without first deploying 9.1.0 and then patching to 9.1.1.

Converging to VCF 9.1.x

To converge your vSphere 8 and Aria 8 components to a VCF fleet or VCF Instance, you follow a specific convergence order. Depending on your existing components, proceed with one of the convergence scenarios:

- Converge a vCenter Instance, ESX Hosts, and Optionally vSAN and NSX
- Converge a vCenter Instance, ESX Hosts, VMware Aria Operations, and Optionally vSAN and NSX
- Converge a vCenter Instance, ESX Hosts, VMware Aria Operations, and Optionally vSAN and NSX
- Converge a vCenter Instance, ESX Hosts, VMware Aria Automation, and Optionally vSAN and NSX
- Converge a vCenter Instance, ESX Hosts, VMware Aria Operations, and VMware Aria Automation, and Optionally vSAN and NSX

For supported and unsupported configurations to converge to VCF, see Supported and Not Supported Configurations to Converge to VCF.

> **Note:**
>
> Convergence of vSphere Foundation 9.1 with deployed VCF management services to VCF 9.1 is not supported.

Converging to vSphere Foundation 9.1.x

To converge your vSphere 8 components to vSphere Foundation 9.1, you follow a specific convergence order. See Converge a vCenter Instance and ESX Hosts to vSphere Foundation.

Upgrading Existing VCF Instance or Fleet to 9.1.x

If you have an existing VCF 5.2.x or 9.0.x environment, see Upgrading to VMware Cloud Foundation 9.1.x.

Upgrading Existing vSphere Foundation to 9.1.x

If you have an existing vSphere Foundation 5.2.x or 9.0.x environment, see Upgrading to vSphere Foundation 9.1.x.

Upgrading vSphere 8 and Optionally Aria Operations 8 to 9.1.x

If you have an existing vSphere 8 environment that can also have Aria Operations 8 and you want to upgrade to 9.1 but you are not yet ready to deploy the entire VCF 9.1 stack, see Upgrading vSphere 8 and Optionally vSAN and Aria Operations 8 to 9.1.x.

Paths to Building Workload Domains With Version 9.1.x

After you create your management domain, you can proceed with the analogous, workload domain-specific guidance.

- To create a workload domain, see Create a New Workload Domain.
- To import existing infrastructure as a workload domain in a VCF Instance, see Import an Existing vCenter to Create a Workload Domain.
- To upgrade your workload domain components, see Upgrading VMware Cloud Foundation Workload Domains to 9.1.x.

Components in VCF and vSphere Foundation

The functionality of VCF and vSphere Foundation is implemented by a set of components. For VCF, these components are deployed in the management domain of the VCF Instances. For vSphere Foundation, you can deploy the components on a management vCenter.

Figure 392: Example Component Configuration for the First VCF Instance of a VCF Fleet

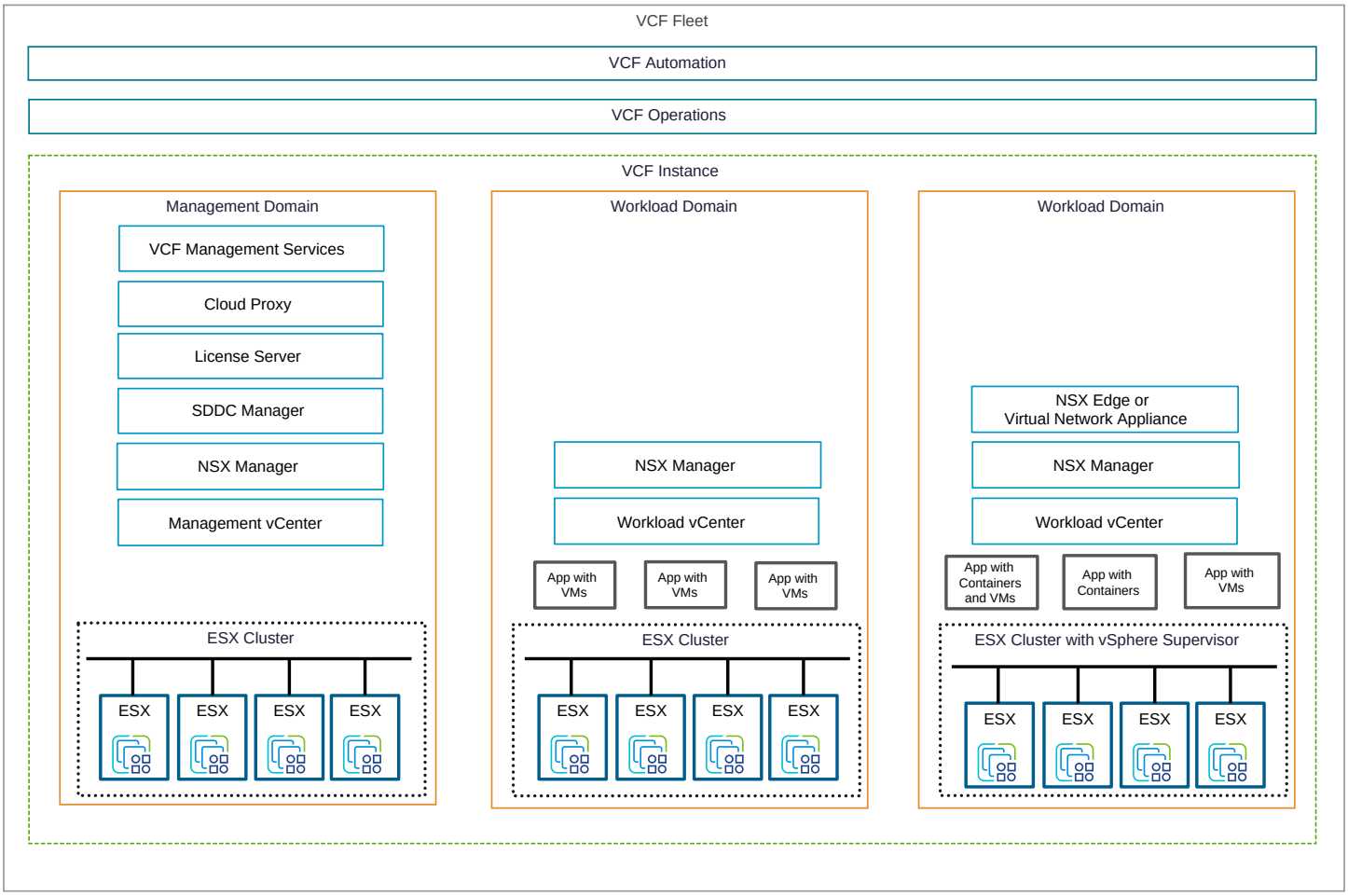
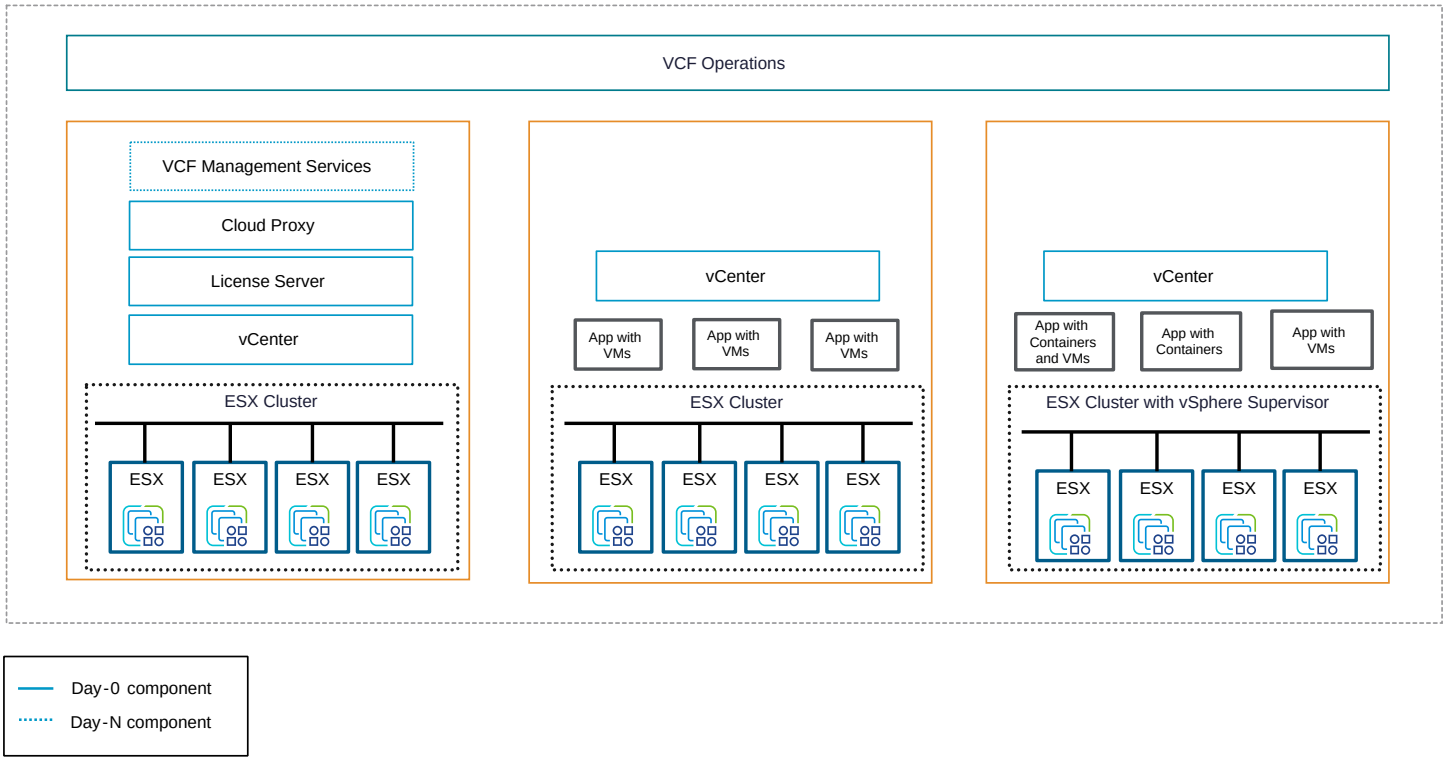


Figure 393: Example Component Configuration for vSphere Foundation



Component	Platform Type	Description
VCF Operations	• VCF • vSphere Foundation	See VCF Operations Overview.
Cloud proxy	• VCF • vSphere Foundation	A part of VCF Operations. Collects and monitors data locally in the platform

Component	Platform Type	Description
VCF Operations for networks	• VCF	Adds intelligent network analytics to VCF Operations.
VCF Operations HCX	• VCF	A part of VCF Operations. Supports workload migration between vCenter instances or between clusters.
VCF Automation	• VCF	See VCF Automation Overview.
SDDC Manager	• VCF	A part of VCF Operations. Provides support for the following functionality in VCF Operations: • Lifecycle management for ESX, vCenter, HCX, and NSX • Deployment of workload domains • Import of vCenter instances as workload domains • Configuration of vSAN stretched clusters
NSX Manager	• VCF	See NSX Overview.
NSX Edge	• VCF	Provides north-south traffic connectivity between the physical data center networks and the NSX SDN networks, and east-west traffic flow between virtualized workloads. It provides stateful services such as load balancing and DHCP.
Virtual Network Appliance	• VCF	A VM form factor that allows for NAT and load balancer stateful services in a distributed VPC.
vCenter	• VCF • vSphere Foundation	See vSphere Overview.
ESX	• VCF • vSphere Foundation	See vSphere Overview.
License server	• VCF • vSphere Foundation	A part of VCF Operations. Provides secure and integrated license management for your platform.

VCF Management Services

VCF 9.1 introduces VCF management services to provide a unified architecture for centralized lifecycle and operational functionalities. The VCF services runtime instance of the first VCF Instance hosts the fleet-level VCF management services components that perform global operations and the instance-level components for that VCF Instance. Every VCF Instance has a VCF services runtime instance that hosts the instance-level VCF management services that run local tasks. This foundation allows for a unified architecture, simplified lifecycle management, and streamlined backup and recovery of all hosted components.

The following components are part of VCF management services. Certain components are deployed as part of the initial deployment or upgrade to 9.1 and you deploy other components as a Day-N operation.

Table 956: VCF Management Services Components

Component	Platform Type	Deployment Scope for VCF	Description
Fleet lifecycle	• VCF • vSphere Foundation	• First VCF Instance	Orchestrates lifecycle operations across the fleet, including planning and sequencing upgrades across multiple instances.
Identity broker	• VCF	• First VCF Instance • Optionally, on other instances (Day-N)	Connects an identity provider (IdP) to the components of the VCF fleet for single sign-on.
Log management	• VCF • vSphere Foundation	• First VCF Instance (Day-N)	Adds intelligent logging and log analysis to VCF Operations.
Real-time metrics	• VCF	• Every VCF Instance (Day-N)	Collects and streams high-frequency performance data for monitoring.
Salt RaaS	• VCF	• First VCF Instance	Defines and manages desired state configurations and policies for VCF components.
Salt Master	• VCF	• Every VCF Instance	Enforces configuration states on the components in a VCF Instance.
SDDC lifecycle	• VCF • vSphere Foundation	• Every VCF Instance	Performs install, update, and, patch procedures for the VCF management components in a VCF Instance.
Software depot	• VCF • vSphere Foundation	• First VCF Instance • Optionally, on other instances (Day-N)	An internal store for install, upgrade, patch binaries, and OCI images. Can operate on the fleet level or optionally on an instance-level.
Telemetry	• VCF • vSphere Foundation	• Every VCF Instance	Sends telemetry data from the data center to Broadcom.
VCF services runtime	• VCF • vSphere Foundation	• Every VCF Instance	VCF services runtime is an implementation of the hosting for the containerized components that provides a consistent orchestration runtime.

For information about the architecture of these nodes, see Design.

Component Comparison Between VCF 5.2, VCF 9.0, and VCF 9.1

If you are converging or upgrading to VCF 9.1 from VCF 5.2.x or VCF 9.0.x, the following table maps the components of VCF 5.2.x to their equivalents in VCF 9.x.

VCF 5.2 Component	VCF 9.0 Component	VCF 9.1 Component
VMware Cloud Builder	VCF Installer	VCF Installer
SDDC Manager	SDDC Manager	SDDC Manager
VMware Aria Operations	VCF Operations	VCF Operations
VMware Aria Operations for logs	VCF Operations for logs	Log management
VMware Aria Operations for networks	VCF Operations for networks	VCF Operations for networks
VMware Aria Suite Lifecycle	VCF Operations fleet management	• Fleet lifecycle • SDDC lifecycle

VCF 5.2 Component	VCF 9.0 Component	VCF 9.1 Component
Workspace ONE Access	VCF Identity Broker	Identity broker
VMware HCX	VCF Operations HCX	VCF Operations HCX
VMware Aria Automation	VCF Automation	VCF Automation
VMware Aria Automation Orchestrator	VCF Operations orchestrator	VCF Operations orchestrator
VMware Cloud Director	N/A	VCF Automation
VMware vSphere with Tanzu/VMware IaaS Control Plane	vSphere Supervisor	vSphere Supervisor

Deploying a New VCF or vSphere Foundation Platform

The process of deploying VMware Cloud Foundation or VMware vSphere Foundation involves preparation, deployment, and post-deployment tasks.

Preparation:

- Preparing ESX Hosts for VCF or vSphere Foundation
- Deploy VCF Installer
- Downloading Binaries to VCF Installer

Deployment:

- Start a New VCF Fleet or a New VCF Instance Deployment by Using the VCF Installer Deployment Wizard
- Start a New vSphere Foundation Deployment by Using the VCF Installer Deployment Wizard
- Deploying VMware vSphere Foundation 9.1 Without VCF Management Services

Post-deployment:

- Deploying Deferred Components on Custom Networking
- Completing the Deployment of Your Platform

Table 957: VCF Installer Deployment Scope per Platform

Component Group	Component Node / Instance	VMware Cloud Foundation		vSphere Foundation
		Simple	High Availability	
VMware vCenter	vCenter	Yes	Yes	Yes
VCF Operations	operations (primary node)	Yes	Yes	Yes
	operations (replica node)	No	Yes	No
	operations (data node)	No	Yes	No
	cloud proxy	Yes	Yes	No
	license server	Yes	Yes	Yes
VMware NSX	NSX Manager (node 1)	Yes	Yes	No
	NSX Manager (node 2)	No	Yes	No
	NSX Manager (node 3)	No	Yes	No
VCF Automation*	automation (node 1)	Yes	Yes	No
	automation (node 2)	No	Yes	No
	automation (node 3)	No	Yes	No
SDDC Manager	SDDC Manager	Yes	Yes	No
VCF management services	VCF services runtime	Yes	Yes	Yes*
	Fleet lifecycle	Yes	Yes	Yes*
	SDDC lifecycle	Yes	Yes	Yes*

Component Group	Component Node / Instance	VMware Cloud Foundation		vSphere Foundation
		Simple	High Availability	
	Software depot	Yes	Yes	Yes*
	Identity broker	Yes	Yes	No
	Salt RaaS	Yes	Yes	No
	Salt master	Yes	Yes	No
	Telemetry	Yes	Yes	Yes

*You can skip the initial deployment and can deploy later as a Day-N operation.

VCF Installer can either deploy all components or use existing components to converge them to a VCF or VMware vSphere Foundation platform. For scenarios with existing components, see Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform.

Table 958: Post-VCF Installer Deployment Scope per Platform

Component Group	Component Node / Instance	VMware Cloud Foundation		vSphere Foundation
		Simple	High Availability	
VCF Operations	operations-networks (platform node)	Yes	Yes	No
	operations-networks (collector node 1)	Yes	Yes	No
	operations-networks (collector node 2)	No	Yes	No
	operations-orchestrator (primary node)	Yes	Yes	Yes
	operations-orchestrator (secondary node1)	No	Yes	Yes
	operations-orchestrator (secondary node2)	No	Yes	Yes
VCF management services	Log management	Yes	Yes	Yes
	Real-time metrics	Yes	Yes	No

For information about deploying components, not in scope for VCF Installer, see Completing the Deployment of Your Platform.

Preparing to Deploy a VCF or vSphere Foundation Platform

Before you can use the VCF Installer to deploy a new VMware Cloud Foundation or VMware vSphere Foundation platform, you must prepare your environment.

Before you start deploying a new platform, you must deploy and configure the necessary infrastructure. You use the planning section of VCF Installer to gather the information you require to deploy VCF or vSphere Foundation. Alternatively, you can use the *Planning and Preparation Workbook*. See Planning and Preparation. You can also refer to the VMware Cloud Foundation Design for specific requirements and recommendations for the design of each component.

Before you deploy a new platform, you must:

- Review the prerequisites page in VCF Installer.
  - Alternatively, review the prerequisite checklist in the *Planning and Preparation Workbook*. See Planning and Preparation.
- Image your ESX hosts. See Preparing ESX Hosts for VCF or vSphere Foundation.
- Download the VCF Installer appliance from the Broadcom Support portal.

- If the VCF Installer appliance does not have direct access to the Internet, to download VCF installation binaries from the Broadcom online depot (dl.broadcom.com), you download the VCF download tool from the Broadcom Support portal (support.broadcom.com). Then, you create an offline depot and download the VCF binaries to the offline depot or download and transfer the installation binaries manually to VCF Installer.

Preparing ESX Hosts for VCF or vSphere Foundation

Before you deploy a new VMware Cloud Foundation or VMware vSphere Foundation platform, you must prepare the ESX hosts.

Note:

The number of required hosts depends on the type of storage and the deployment model that you plan to use. You use the planning section of VCF Installer to determine the minimum number of ESX hosts required for your environment. Alternatively, you can use the *Planning and Preparation Workbook*. See Planning and Preparation.

Preparing the ESX hosts involves installing the correct version of ESX and performing some basic configuration tasks. For the supported ESX version, see the *VMware Cloud Foundation Release Notes*.

Note:

- VMware Cloud Foundation does not support stateless ESX hosts.
- VMware Cloud Foundation does not support heterogeneous cluster configurations during initial management domain deployment. During this phase, vCenter is not yet available to manage cluster composition, requiring all hosts in the initial cluster to share a uniform hardware profile.
- All VMware ESX hosts within the management domain must originate from a single hardware vendor. Cross-vendor hardware configurations in the management domain are not supported.

With VMware Cloud Foundation 9.0.1 and later, you can use non-certified ESA disks during vSAN ESA cluster configuration for proof-of-concept purposes only. For more information, see Broadcom knowledge base article 408300.

Custom ISO Image for ESX Hosts

If your environment requires a custom ISO file for ESX, you can create one using VCF PowerCLI or vSphere Lifecycle. See Creating a Custom ESX ISO Image Using VCF PowerCLI and Creating a Custom ESX ISO Image Using vSphere Lifecycle Manager.

You must create a custom ISO image for ESX in the following situations.

- The ESX version specified in the VMware Cloud Foundation Release Notes does not have an associated ISO file on the Broadcom Support Portal. This can be the case for ESX patch releases.
- You need an async patch version of ESX.
- You need a vendor-specific (OEM) ISO file.

Install ESX on Hosts Using an ISO Image

Install ESX on all hosts you plan to use in your VMware Cloud Foundation or VMware vSphere Foundation platform.

Download the ESX ISO from the Broadcom Support Portal. For the supported ESX versions, see the *VMware Cloud Foundation Release Notes*. If the required version of ESX does not have an ISO available on the Broadcom Support Portal, you can create one. See Creating a Custom ESX ISO Image Using VCF PowerCLI or Creating a Custom ESX ISO Image Using vSphere Lifecycle Manager.

1. Mount the ESX ISO on the host and restart the machine.
2. Set the BIOS or UEFI to boot from the mounted ISO.
3. On the welcome screen, press **Enter** to continue.
4. Accept the End User License Agreement by pressing **Enter**.
5. On the **Select a Disk to Install or Upgrade** screen, select the drive on which to install ESX on and press **Enter**.
6. Select the keyboard type for the host.
   You can change the keyboard type after installation in the direct console.
7. Enter the root password for the host.
8. In the **Confirm Install** screen, press **F11** to confirm the start of the installation.
9. On the **Installation Complete** screen, press **Enter** to reboot the host.
10. Set the first boot device to be the drive on which you installed ESX.
11. Repeat this procedure for all remaining hosts.

Configure the Network on the ESX Hosts

Use the ESX Direct Console User Interface (DCUI) for host network configuration and administrative access.

Perform the following tasks to configure the host network settings:

- Configure the network adapter (vmk0) and VLAN ID for the Management Network.
- Configure the IP address, subnet mask, gateway, DNS server, and FQDN for the ESX host.

Repeat this procedure for all hosts that you are adding to your VCF or VMware vSphere Foundation platform. If you completed the *Planning and Preparation Workbook*, refer to it for the required values.

Before the VCF deployment begins, network adapter (vmk0) must be connected to a single management NIC on a vSphere Standard Switch, for example, vmnic0. During the deployment process, VCF Installer migrates the ESX host to a vSphere Distributed Switch and adds the required uplinks.

1. Open the DCUI of the ESX host.
   a) Open a console window to the host.
   b) Press F2 to enter the DCUI.
   c) Log in using the `esx_root_user_password`.
2. Configure the network.
   a) Select **Configure Management Network** and press Enter.
   b) Select **VLAN (Optional)** and press Enter.
   c) Enter the VLAN ID for the ESX Management Network and press Enter.
   d) Select **IPv4 Configuration** and press Enter.
   e) Select **Set static IPv4 address and network configuration** and press the Space bar.
   f) Enter the IPv4 Address, Subnet Mask, Default gateway, and press Enter.
   g) If you plan to use dual-stack networking, configure an IPv6 address.
      1. Select **IPv6 Configuration** and press Enter.
      2. Select **Set static IPv6 address and network configuration** and press the Space bar.
      3. Enter the IPv6 Address, CIDR, Default gateway, and press Enter.

h) Select **DNS Configuration** and press Enter.
i) Select **Use the following DNS Server address and hostname** and press the Space bar.
j) Enter the Primary DNS Server, Alternate DNS Server and Hostname (FQDN) and press Enter.
k) Select **Custom DNS Suffixes** and press Enter.
l) Ensure that no suffixes are listed and press Enter.

3. Press Escape to exit and press Y to confirm the changes.
4. Repeat this procedure for all remaining hosts.

Configure NTP on the ESX Hosts

After you have configured the Virtual Machine Network port group, configure the NTP service to avoid time synchronization issues.

Repeat this procedure for all ESX hosts you are adding. If you completed the *Planning and Preparation Workbook*, refer to it for the required values.

1. In a web browser, log in to the ESX host using the VMware Host Client.
2. Configure and start the NTP service.
   a) In the navigation pane, click **Host** > **Configure** > **Time Configuration**.
   b) In the **Network Time Protocol** section, click **EDIT**.

      The **Edit Network Time Protocol** dialog box appears.
   c) Select the **Enable** check box.
   d) Enter the NTP Server FQDN or IP Address and click **OK**.
3. Optional: Verify that the service is running.
4. Repeat this procedure for all remaining hosts.

Regenerate the Self-Signed Certificate on the ESX Hosts

After you have configured the NTP, you must regenerate the self-signed certificate to ensure that the correct common name is defined in the certificate.

During the installation of ESX, the installer generates a self-signed certificate for each ESX host, but the process is performed prior to the ESX hostname being configured. This means that all ESX hosts have a common name in their self-signed certificate of `localhost.localdomain`. All communication between the VCF Installer and the ESX hosts is performed securely over HTTPS and as a result it validates the identity when making a connection by comparing the common name of the certificate against the FQDN provided within the VCF Installer deployment wizard.

To ensure that the connection attempts and validation do not fail, you must manually regenerate the self-signed certificate after the hostname has been configured.

> **Note:**
>
> If corporate policy requires that you use external CA-signed certificates instead of VMCA-signed certificates for ESX hosts, you can add external certificates to the hosts using the SDDC Manager API.

1. In a web browser, log in to the ESX host using the VMware Host Client.
2. In the **Actions** menu, click **Services** > **Enable Secure Shell (SSH)**.
3. Log in to the ESX host using an SSH client such as Putty.
4. Validate the ESX hostname and Fully Qualified Domain Name (FQDN).

   ```
   esxcli system hostname get
   ```

   If the FQDN is not configured correctly, you can edit it by running the `esxcli system hostname set --fqdn=`*`NEW_FQDN`* command.
5. Regenerate the self-signed certificate by executing the following command:

   ```
   /sbin/generate-certificates
   ```

6. Reboot the ESX host.
7. Log back in to the VMware Host Client and click **Services** > **Disable Secure Shell (SSH)** from the **Actions** menu.
8. Repeat this procedure for all remaining hosts.

Configure ESX Hosts with Signed Certificates

If corporate policy requires that you use external CA-signed certificates instead of VMCA-signed certificates for ESX hosts, you can manually add external certificates to the hosts.

External CA-signed certificate and key are available.

When you install ESX software on a server to create an ESX host, the host initially has an autogenerated certificate. By default, when the ESX host is added to a vSphere cluster during deployment of the management domain or other operations involving ESX hosts (for example, host commissioning, workload domain creation, and so on), the autogenerated certificate is replaced with a certificate that is signed by the VMware Certificate Authority (VMCA).

When you use external certificates during deployment, they are not replaced by VMCA-signed certificates. After you perform deployment with external certificates for ESX hosts, all future ESX hosts added to VCF must also use external certificates.

To deploy the management domain using external certificates, you must use a JSON specification file with the VCF Installer. You cannot use the VCF Installer deployment wizard. In the JSON specification file, use the `SecuritySpec` section in the `SddcSpec` to specify a value of `Custom` for `esxiCertsMode`. For example:

```
"securitySpec" : {
"esxiCertsMode" : "Custom",
"rootCaCerts" : [ {
"alias" : "Rainpole-CA",
"certChain" : [ "-----BEGIN CERTIFICATE-----
MICDczCCAlugAwIBAgIQI9xwbTkI9J5GhMffcP5CHDANBgkqhkiG9w0BAQsFADBM
MRIwEAYKCZImiZPyLGQBGRYCaW8xGDAWBgoJkiaJk/IsZAEZFghyYWlucG9sZTEc
MBoGA1UEAxMTcmFpbnBvbGUtZGMwMXJwbC1DQTAeFw0yMDAzMzAxNDQ2MTNaFw0y
NTAzMzAxNDU2MTNaMEwxEjAQBgoJkiaJk/IsZAEZFgJpbzEYMBYGCgmSJomT8ixk
ARkWCHJhaW5wb2xlMRwwGgYDVQQDExNyYWlucG9sZS1kYzAxcnBsLUNBMIIBIjAN
BgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAzpwkz7aPlQcfevcCelHc9DPswHkd
kjY96Vh3GvYlesaVEcy/q/BOvvh3KgLMLy8r7cy2cNPO3FANKOfqVdVx3ghfEUyL
g61W9BskAlwryzJRmjhOJJVqvB8CWjy+eCp7MejHGdEud6WdEvK8CaBcPngEg0KM
eLRNLGe8OCw8yY4GTrjU+H7PYQZtyD0kxxy5f47ueaDXat4ENRGcAuHEfCoMGfaR
bDue1OO4diHd900bCym5ggBNX0jhRudNULXPTayZl2ksImV0+QkaVeptQImXfCgb
kgnHQJ5CxK26up7fB5eAsmGLAsJLBnHuM7P9xvV09EvWjFCgLX/oBBDYTQIDAQAB
o1EwTzALBgNVHQ8EBAMCAYYwDwYDVR0TAQH/BAUwAwEB/zAdBgNVHQ4EFgQU7oOq
QBK8yg8mHnAfb+u6/GO0ZUcwEAYJKwYBBAGCNxUBBAMCAQAwDQYJKoZIhvcNAQEL
BQADggEBALYxZGj4vWjFDN1atOUsBx2jrmxbExgMAyRpNlSc2aj+7vzxHxUW5VbX
x9nc/BfkTiCK6c7Y9VYb+mgjb8z0kNv58sT4ar1yIl1n63VOCoyyLcaFB8HyEJpD
wUhz4RNPoSijZMpm+M5EuSLfWlhEJo7N8sLqHgvvk1dFpbK8fIHbPS5KJwJibbPe
w9UuNRdcxN9hFWKBC0SvfgX+1CJxVdvgfi65rSHPuWinJzrXXdH999DfpDESRzwH
0pqE3GtMCt1Nqalp2QJFdahbT+kxj7QWHTjUylSENDHjdln7a8WH8RGxvEy/97YZ
+crXmxvQ/bAgHk9vcRERbRjfyIs7v88=
-----END CERTIFICATE-----" ] } ] }
```

See the VCF Installer API Reference Guide for more information.

1. In a web browser, log in to the ESX host using the VMware Host Client.
2. In the **Actions** menu, click **Services** > **Enable ESX Shell**.
3. Log in to the ESX Shell for the first host, either directly from the DCUI or from an SSH client, as a user with administrator privileges.
4. In the directory `/etc/vmware/ssl`, rename the existing certificates using the following commands:

   ```
   mv rui.crt orig.rui.crt
   mv rui.key orig.rui.key
   ```

5. Copy the external certificate and key that you want to use to `/etc/vmware/ssl`.
6. Rename the external certificate and key to `rui.crt` and `rui.key`.
7. Restart the host management agents by running the following commands:

```
/etc/init.d/hostd restart
/etc/init.d/vpxa restart
```

8. In the VMware Host Client, select the **TSM-SSH** service for the ESX host and click **Stop**.
9. Repeat for all the ESX hosts that you are adding to VMware Cloud Foundation.

Creating a Custom ESX ISO Image Using VCF PowerCLI

You can use VCF PowerCLI to create a custom ISO for installing ESX.

- Verify that your VCF PowerCLI version is 9.1 or later.
- Verify that you downloaded the following `ZIP` files.
  - ESX patch for the ESX version specified in the VMware Cloud Foundation Release Notes or in the list of supported async patches in Broadcom knowledge base article 88287. You can download patches from the Broadcom Support Portal.
  - OEM add-on for ESX from the Broadcom Support Portal. If the ESX version specified in the BOM is not available, contact your vendor to determine which OEM add-on version to use.

1. Gather the required information for the software spec that is used to create the custom ISO.
   a) In VCF PowerCLI, use the `Get-DepotBaseImages` cmdlet to get the base image version from the zip file for the ESX patch that you downloaded from the patches portal.

      For example:

      ```
      Get-DepotBaseImages "c:\temp\VMware-ESX-9.1.x-xxxxxxx-depot.zip"
      ```

   b) Use the `Get-DepotAddons` cmdlet to get the add-on name and version from the zip file for the OEM add-on for ESX that you downloaded from the Broadcom Support Portal. (if applicable)

      For example:

      ```
      Get-DepotAddons "c:\temp\HPE-9.1.x.x.x-xxx-Synergy-Addon-depot.zip"
      ```

2. Create the software spec using the information you gathered in step 1.

   The software spec is a JSON file that contains information about the ESX version and vendor add-on (if applicable). For example:

```
{
"add_on": {
   "name": "HPE-Custom-Syn-AddOn",
   "version": "9.1.x.x.x.x-xx"
},
"base_image": {
   "version": "9.1.x-x.xx.xxxxxxxx"
},
"components": null,
"hardware_support": null,
"solutions": null
}
```

3. In VCF PowerCLI, use the `New-IsoImage` cmdlet to generate a custom ISO.

   For example:

   ```
   New-IsoImage -SoftwareSpec "c:\temp\HPE-70U1d-custom.JSON"  -Depots "c:\temp\VMware-ESX-9.0-17551050-depot.zip" , "c:\temp\HPE-9.1.x.xx.x.x.xx-xxxxxxx-Synergy-Addon-depot.zip" -Destination "c:\temp\HPE-9.1-custom.iso"
   ```

   Provide the path to the software spec you created in step 2.

   The depot(s) include the path to the zip files for the supported ESX version and vendor add-on.

The destination includes the path and file name for the custom ISO file.

For more information about the `New-IsoImage` cmdlet, see New-IsoImage.

Creating a Custom ESX ISO Image Using vSphere Lifecycle Manager

If you have access to a vSphere environment, you can use vSphere Lifecycle Manager to create and export a custom ISO.

- Verify that you imported the ESX patch and vendor add-ons to the vSphere Lifecycle Manager depot. See Import Updates to the vSphere Lifecycle Manager Depot.

1. Create a new ESX image.

   See Create an Image in Image Library.

2. Export the ESX image as an ISO file.

   See Export an Image from Image Library.

Use the exported custom ESX ISO image file to deploy your ESX hosts. See Install ESX on Hosts Using an ISO Image.

Mount NFS Datastore to an ESX Host

If you plan to deploy the VCF Installer appliance on an ESX host that will form the management domain and your storage type is NFS v3, you must manually mount the NFS datastore on that host. Follow the procedure that matches your deployment scenario.

You can use the following procedures when you plan to deploy VCF or vSphere Foundation by using the VCF Installer wizard or a JSON specification file. For JSON specification file examples, see the example SddcSpec.

1. No NFS VMkernel Interface

- Verify that the NFS server is reachable through the management VMkernel interface.
- Verify that no dedicated NFS VMkernel interface exists on the ESX host.

1. In the ESX CLI, mount the NFS datastore by running the following command. Replace the placeholder values in the command with values from your environment.

Placeholder Value	Description
NFS-Server-IP	IPv4 address of the NFS server
Datastore-Name	NFS datastore name as it appears in vSphere.
Share-Path	Export file path of the NFS share.

```
esxcli storage nfs add -H NFS-Server-IP -s Share-Path -v Datastore-Name
```

2. NFS and Management Traffic on the Same Distributed Virtual Switch

- Verify that you have a dedicated NFS VMkernel interface and both NFS and management traffic share the same distributed virtual switch.

1. On the ESX host, create a port group on `vSwitch0`.

   Note:

   The port group name must match the value of the `portgroupKey` field in the NFS network spec. If you do not set `portgroupKey`, use the default port group name `SDDC-DPortGroup-NFS`.

2. On the ESX host, create a VMkernel adapter for NFS traffic on the port group.
3. Assign an IP address to the VMkernel adapter from the NFS network IP range.

4. In the ESX CLI, mount the NFS datastore by running the following command. Replace the placeholder values in the command with values from your environment.

Placeholder Value	Description
NFS-Server-IP	IPv4 address of the NFS server
VMK-Adapter	VMkernel adapter. For example, `vmk1`.
Datastore-Name	NFS datastore name as it appears in vSphere.
Share-Path	Export file path of the NFS share.

   – If the `enableBindToVmknic` property in the `nfsDatastoreSpec` object is set to `true`, NFS traffic uses the VMkernel adapter and the NFS datastore binds to that adapter. To mount the NFS datastore with this binding, run the following command.

```
esxcli storage nfs add --host-vmknic=NFS-Server-IP:VMK-Adapter --volume-name=Datastore-Name --
share=Share-Path
```

   – If the `enableBindToVmknic` property in the `nfsDatastoreSpec` object is set to `false`, NFS traffic does not bind to a specific VMkernel adapter. To mount the NFS datastore without VMkernel adapter binding, run the following command.

```
esxcli storage nfs add -H NFS-Server-IP -s Share-Path -v Datastore-Name
```

CAUTION:

During the deployment workflows in VCF Installer, you must use the same port group name for the NFS vDS port group (*VMK-Adapter*) and define an NFS pool that includes the assigned IP address (*NFS-Server-IP*).

3. NFS and Management Traffic on Separate Distributed Virtual Switches

- Verify that the dedicated NFS VMkernel interface and management traffic run on separate distributed virtual switches.
- Verify that you have the `dvsSpec` configuration that defines the NFS network. The `dvsSpec` specifies the distributed virtual switch on which NFS traffic runs, separate from the switch used for Management traffic.

1. On the ESX host, create the NFS virtual switch according to the `dvsSpec` configuration that defines the NFS network.
2. Under the NFS virtual switch, create an NFS port group.

Note:

The port group name must match the value of the `portgroupKey` field in the NFS network spec. If you do not set `portgroupKey`, use the default port group name `SDDC-DPortGroup-NFS`.

3. On the NFS port group, create a VMkernel interface and assign it an IP address from the NFS network spec IP range.
4. In the ESX CLI, mount the NFS datastore by running the following command. Replace the placeholder values in the command with values from your environment.

Placeholder Value	Description
NFS-Server-IP	IPv4 address of the NFS server
VMK-Adapter	VMkernel adapter. For example, `vmk1`.
Datastore-Name	NFS datastore name as it appears in vSphere.
Share-Path	Export file path of the NFS share.

   – If the `enableBindToVmknic` property in the `nfsDatastoreSpec` object is set to `true`, NFS traffic uses the VMkernel adapter and the NFS datastore binds to that adapter. To mount the NFS datastore with this binding, run the following command.

```
esxcli storage nfs add --host-vmknic=NFS-Server-IP:VMK-Adapter --volume-name=Datastore-Name --
share=Share-Path
```

   – If the `enableBindToVmknic` property in the `nfsDatastoreSpec` object is set to `false`, NFS traffic does not bind to a specific VMkernel adapter. To mount the NFS datastore without VMkernel adapter binding, run the following command.

```
esxcli storage nfs add -H NFS-Server-IP -s Share-Path -v Datastore-Name
```

CAUTION:

During the deployment workflows in VCF Installer, you must use the same port group name for the NFS vDS port group (*VMK-Adapter*) and define an NFS pool that includes the assigned IP address (*NFS-Server-IP*).

Deploy VCF Installer

Deploy the VCF Installer appliance on a network that can access the ESX hosts and VM management network that you plan to use for the deployment. The appliance must be able to access all required external services, such as DNS and NTP.

- Verify that you downloaded the latest VCF Installer appliance (`VCF-SDDC-Manager-Appliance-<version.build_number>.ova`) from the Broadcom Support Portal.
- Verify that your environment meets the resource and network requirements for this process.

Resource Requirements	• 4 vCPUs • 16 GB RAM • 914 GB storage
Network Requirements	• Verify that the static IP address, DNS, and FQDN for the VCF Installer appliance are valid and can be resolved. • Verify that connectivity is in place from the VCF Installer appliance and the management VLAN used in the deployment. • Verify that connectivity is in place from the VCF management network to the VCF Installer appliance.

Where you deploy the VCF Installer appliance impacts how it can be used. You can deploy the appliance:

- Outside of the management infrastructure on any machine with connectivity to the management infrastructure. In this case, you can use the appliance to deploy multiple VCF or vSphere Foundation platforms. You can deploy the VCF Installer appliance on a vCenter instance or on a laptop running VMware Workstation or VMware Fusion.
- On one of the ESX hosts that will form the management domain. In this case, you can only use the appliance to deploy a single VCF platform. Then the appliance switches to SDDC Manager mode, can no longer be used in installer mode, and you use the VCF Installer FQDN for SDDC Manager.

This procedure deploys the VCF Installer appliance directly to an ESX host.

1. In a web browser, log in to the ESX host using the VMware Host Client.
2. In the navigation pane, select **Host**, and click **Actions** > **Deploy OVF Template**.
3. On the **Select an OVF template** page, enter a name for the virtual machine, browse to the `VCF-SDDC-Manager-Appliance-<version.build_number>.ova` file that you downloaded from the Broadcom Support Portal, and click **Next**.
4. On the **Review details** page, verify the template details and
5. On the **License agreements** page, select the **I accept all license agreements** check box and click **Next**.
6. On the **Select Storage** page, select **Thin Provision** disk format, select a datastore, and click **Next**.

VCF Installer deployment type	**Storage considerations**
Outside of the management infrastructure	Choose any valid datastore.
On one of the ESX hosts that will form the management domain	Choose a local disk. After the management domain storage is configured using the VCF Installer, the VCF Installer appliance is migrated to that storage with Storage vMotion and switched to SDDC Manager mode. The local disk remains empty and is not added to the management domain storage. In this case, the local disk must not be considered when planning for capacity.

7. On the **Select networks** page, select a source network, map it to a destination network, and click **Next**.
8. On the **Customize template** page, expand **Application**, enter the following values, and click **Next**.

Setting	Details
Root user password/Root user password confirm	**Note:** The root user password must be a minimum of 15 characters and include at least one uppercase, one lowercase, one digit, and one special character. Supported special characters: `@ ! # $ % ? ^` • Passwords must not contain a dictionary word or part of a dictionary word. • Passwords must not contain the user name or parts of the user name.
Local user password	You use the local user accounts (`admin@local` or `vcf`) to log in to VCF Installer. **Note:** The local user password must be a minimum of 15 characters and include at least one uppercase, one lowercase, one digit, and one special character. Supported special characters: `@ ! # $ % ? ^` • Passwords must not contain a dictionary word or part of a dictionary word. • Passwords must not contain the user name or parts of the user name.
Host Name	Enter the fully qualified domain name (FQDN) for the VCF Installer appliance. **Note:** If you are deploying VCF Installer on one of the ESX hosts that will form the management domain, the FQDN is used for SDDC Manager. After deployment the FQDN cannot be changed. If you are deploying VCF Installer with dual-stack networking, the FQDN must resolve to both the IPv4 and IPv6 addresses.
NTP Servers	Enter the NTP server(s). Use a comma if entering multiple NTP servers. NTP servers can be entered using FQDNs or IP addresses.

9. On the **Customize template** page, expand **Networking Configuration**, enter the following values, and click **Next**.

   To use dual-stack networking, you must deploy the VCF Installer appliance with configured IPv4 and IPv6 addresses.

   Dual-stack networking allows communication with components that support both network protocols, ensuring compatibility and continuous connectivity. While not all components support IPv6 networking, components that are IPv4 only can still communicate with the other components.

Setting	Details
Select IP Address Version	Select IPv4 only or IPv4 and IPv6.
Network 1 IPv4 Address	Enter the IPv4 address for the VCF Installer appliance.
Network 1 Subnet Mask	Enter the subnet mask for the VCF Installer appliance.
Network Default IPv4 Gateway	Enter the default IPv4 gateway for the VCF Installer appliance.
Network 1 IPv6 Address	Enter the IPv6 address for the VCF Installer appliance.

Setting	Details
Network 1 IPv6 Prefix Length	Enter the IPv6 prefix length for the VCF Installer appliance.
Network Default IPv6 Gateway	Enter the default IPv6 gateway for the VCF Installer appliance.
DNS Domain Name	Enter the DNS domain name. For example, `rainpole.io`.
Domain Search Path	Enter the domain search path(s). Use a comma if entering multiple search paths. For example, `rainpole.io, sfo.rainpole.io`.
Domain Name Servers	Enter the IP address of the primary and secondary DNS servers (comma separated). Do not specify more than two servers.

10. On the **Ready To Complete** page, review the virtual machine configuration.

    Note:

    Make sure your passwords meet the requirements specified above before clicking **Finish** or your deployment will not succeed.

- After the VCF Installer appliance is deployed, SSH in to the VM as the `vcf` user and the local user password provided in step 8.
- Ensure that you can ping the ESX hosts.
- Verify that the VCF Installer appliance has access to the required external services, such as DNS and NTP by performing forward and reverse DNS lookups for each host and the specified NTP servers.
- In a web browser, log in to the VCF Installer appliance at `https://installer_appliance_FQDN`. Accept the self-signed certificate and enter the `admin@local` username and the password that you provided when you deployed the appliance and then click **Log In**.

Downloading Binaries to VCF Installer

Before you can deploy VMware vSphere Foundation or VMware Cloud Foundation, you must download the required binaries to the VCF Installer appliance.

Table 959: VCF Installer Depot Connection Modes

You can configure the VCF Installer depot in the following connection modes.

Connection Mode	Description
Online Depot	If the VCF Installer appliance can connect to the Internet (either directly or through a proxy server), you can connect it to the Broadcom depot and download the binaries by using your VMware Cloud Foundation Business Services console account. For more information, see Connect VCF Installer to Broadcom or an Offline Depot and Download Binaries.
Offline Depot	If the VCF Installer appliance cannot connect to the Internet, you can connect it to an offline depot and use the VCF download tool to download the required binaries to the offline depot. For more information, see Connect VCF Installer to Broadcom or an Offline Depot and Download Binaries and Download Installation Binaries to an Offline Depot.
Manual Transfer	If the VCF Installer appliance cannot connect to an online or offline depot, you can use the VCF download tool to download binaries to a computer with access to the Internet and then upload them to the VCF Installer appliance. For more information, see Manually Transfer Binaries to VCF Installer.

Table 960: Required Installation Component Binaries for New Deployments

VMware Cloud Foundation	VMware vSphere Foundation
• Cloud proxy • Fleet lifecycle • Identity broker • License server • Migration service engine • Salt master • Salt RaaS • SDDC lifecycle • SDDC Manager • Software depot • Telemetry • VCF Automation • VCF Operations • VCF services runtime • VMware NSX • VMware vCenter	• Fleet lifecycle • License server • SDDC lifecycle • Software depot • Telemetry • VCF Operations • VCF services runtime • VMware vCenter

You do not need to download all of the binaries if you are reusing existing components. Even if you skip deployment of VCF Operations and VCF Automation in order to deploy them later with custom networking, you should still download the binaries, since they are required for the custom networking deployment.

Connect VCF Installer to Broadcom or an Offline Depot and Download Binaries

If VCF Installer can connect to the Internet, either directly or through a proxy server, you register it with Broadcom by using your VMware Cloud Foundation Business Services console account. If VCF Installer cannot connect to the Internet, you can connect it to an offline depot. After VCF Installer is connected, you can download the installation binaries from the configured source.

- If you configure an online depot, for information about the URLs that must be accessible to download bundles, see Broadcom knowledge base article 327186.
- If you configure an offline depot, verify that you deployed and configured an offline depot web server, and downloaded required component binaries to the offline depot. See Set Up an Offline Depot and Download Installation Binaries to an Offline Depot.
- Verify that your VCF Business Services console role has sufficient permissions. See Access Management.

You can only connect the VCF Installer appliance to one type of depot. If VCF Installer is connected to an online depot and you configure a connection to an offline depot, the online depot connection is deactivated and the configuration is deleted.

> **Note:**
>
> In environments where SSL/TLS interception is enabled, VCF Installer might not establish a trusted connection to the Broadcom authorization server. For more information on adding your proxy's CA certificate to the truststore of software depot, see Broadcom knowledge base article 442978.

1. Configure an online or offline depot.

Online Depot

1. In a web browser, log in to the VMware Cloud Foundation Installer appliance administration interface: `https://installer_appliance_FQDN`.
   Enter the `admin@local` user and password that you provided when you deployed the appliance and then click **Log In**.
2. Click **Depot Settings and Binary Management**.
3. Under **Online Depot**, click **Configure**.
4. Obtain and enter an activation code.
   a. Copy the software depot ID.

b. Log in to the VCF Business Services console.
c. Register the component with the correct tenant name.
   a. In the upper-left corner of the console, next to the tenant name, click the arrow to expand the section.
   b. From the drop-down menu, select the correct tenant name and siteID that map to your VCF or vSphere Foundation entitlement.
d. Navigate to **Software Depot Registrations** > **New Registration** and use your software depot ID to obtain an activation code for your depot.
e. Enter the activation code that you obtained from the VCF Business Services console.

5. If you must configure a proxy server, toggle the **Enable proxy server** setting to the on position.
   a. Select a **HTTP** or **HTTPS** protocol.
   b. Enter the proxy server FQDN or IP address and port number.
   c. If your proxy server requires authentication, toggle the **Authenticated** setting to the on position and enter the username and password.

      Note:

      VMware Cloud Foundation does not support proxy servers with Kerberos or NTLMv2 authentication.
   d. Click **Authenticate**.
   e. Review the certificate thumbprint and click **Accept**.

Offline Depot

1. If your offline depot web server is connected over HTTPS, you must import your TLS certificate to the certificate store of VCF Installer.
   a. SSH to VCF Installer.
   b. Log in with `vcf` user.
   c. Change to root with `su`.
   d. Download the certificate from your offline depot web server and copy the certificate to the VCF Installer appliance.

      ```
      openssl s_client -connect depot_server_FQDN:443 < /dev/null | sed -ne '/-
      BEGIN CERTIFICATE-/,/-END CERTIFICATE-/p' > /tmp/server.crt
      ```
   e. Import your certificate into the certificate store.

      ```
      keytool -import -trustcacerts -file /tmp/server.crt -keystore /etc/
      alternatives/jre/lib/security/cacerts -alias offline_depot_server -storepass
       changeit -noprompt
      ```
   f. Restart the VCF Installer services.

      ```
      echo 'y' | /opt/vmware/vcf/operationsmanager/scripts/cli/
      sddcmanager_restart_services.sh
      ```
2. Configure the offline depot.
   a. In a web browser, log in to the VMware Cloud Foundation Installer appliance administration interface: `https://installer_appliance_FQDN`.
      Enter the `admin@local` user and password that you provided when you deployed the appliance and then click **Log In**.
   b. Click **Depot Settings and Binary Management**.
   c. Under **Offline Depot**, click **Configure**.
   d. Enter configuration details for the offline depot.
      a. Enter the URL of the offline depot.
      b. If your proxy server requires authentication, toggle the **Authentication** setting to the on position and enter the username and password.

         Note:

         VMware Cloud Foundation does not support proxy servers with Kerberos or NTLMv2 authentication.
   e. Click **Configure**.

You are connected to a depot and can download installation binaries.

2. In the **Binary Management** section, select a product and version from the drop-down menus.

   The user interface lists all the required binaries for your deployment, including information about the file size and download status.

3. Select the binaries to download and click **Download**.

VCF Installer downloads and validates the binaries. The progress is displayed in the **Download Status** column.

After you deploy a VCF fleet, VCF Instance, or vSphere Foundation, your online or offline depot configuration and authentication is transferred to the software depot fleet-level instance. The binaries you downloaded in **Binary Management** are not transferred. If needed, you can reconfigure software depot. See Configure a Software Depot Connection Mode.

Download Installation Binaries to an Offline Depot

If you are connecting the VCF Installer to an offline depot, you must download the install binaries to the offline depot before you can use them to deploy VMware vSphere Foundation or VMware Cloud Foundation.

- Verify that you deployed and configured an offline depot web server. See Set Up an Offline Depot.
- The offline depot must have:
  - The latest version of the VCF Download Tool. You can download it from the Broadcom Support portal.
  - Internet connectivity (either directly or through a proxy) for downloading the bundles and other required files.
  - Configure TCP keepalive in your SSH client to prevent socket connection timeouts when using the VCF Download Tool for long-running operations.
- Connect the VCF Installer to the offline depot. See Connect VCF Installer to Broadcom or an Offline Depot and Download Binaries.
- Verify that your VCF Business Services console role has sufficient permissions. See Access Management.

You can use the VCF Download Tool to download install bundles to the offline depot.

> **Note:**
>
> The VCF Download Tool is the only supported method for downloading binaries for VCF 9.1 in environments without access to the Internet. Do not use third-party tools or other methods to download binaries.

In the steps below, where applicable, replace the following values.

Value	Description
`vcf-source-version`	Your current VCF version. For example, `9.1.1.0`.
`/path/to/binaries`	The download location for the binaries. For example, `/var/www`.
`/path/to/activation-code.txt`	Path to a text file that contains a single line with your activation code from the VCF Business Services Console.

If the computer hosting the offline depot can only access the Internet using a proxy server, use the following options when downloading binaries.

Option	Description
`--proxy-server, -s`	Provide the proxy server FQDN and port of the proxy server that the VCF Download Tool uses for Internet access. The required format is <FQDN:Port>. The proxy server must be reachable from the machine where the tool is running. For example: `--proxyServer proxy.example.com:3128`.

`--proxy-https`	Specifies whether the proxy server is configured to be used with HTTPS connections. To use this option, the proxy certificate must be added to the JRE default trust store of VCF Download Tool. The exact path to the `/cacerts` folder depends on the directory where the VCF Download Tool `tar` file is extracted.
	For example:
	• Unix
	`/opt/obtu/jre/lin64/bin/keytool -importcert -file ca-bundle.crt -keystore <tool root>/jre/lin64/lib/security/cacerts`
	• Windows
	`/opt/obtu/jre/lin64/bin/keytool -importcert -file ca-bundle.crt -keystore <tool root>/jre/win32/lib/security/cacerts`

`--proxy-user, -r`	If the proxy server enforces authentication, this parameter allows you to provide the necessary username required to authenticate to the proxy server. For example, `--proxy-user` *`username`*.
`--proxy-user-password-file`	Path to a text file *`path/to/password.txt`* containing the password for the provided `--proxy-user`. The file should contain the password on a single line. For example, `--proxyPasswordFile` *`path/to/password.txt`*.

Note:

VMware Cloud Foundation does not support proxy servers with Kerberos or NTLMv2 authentication.

Example that combines the options:

```
./vcf-download-tool binaries download --depot-download-activation-code-file /path/to/activation-code.txt --vcf-version vcf-source-version  --depot-store /path/to/binaries --proxy-server proxy.example.com:3128 --proxy-user username--proxy-user-password-file path/to/password.txt --proxy-https --automated-install --type INSTALL
```

1. Download the most recent version of the VCF Download Tool on a computer with Internet access.
   a) Log in to the Broadcom Support Portal and browse to **My Downloads** > **VMware Cloud Foundation**.
   b) Click the version of VMware Cloud Foundation to which you are upgrading.
   c) Click **Drivers & Tools**.
   d) Click the download icon for the VCF Download Tool.
   e) Extract `vcf-download-tool-<version.build>.tar.gz`.
   f) Navigate to the `vcf-download-tool-<version.build>/bin/` and confirm that you have execute permission on all folders.

      The directory includes both the Bundle Transfer Utility (`lcm-bundle-transfer-util`) and the VCF Download Tool (`vcf-download-tool`).
2. Obtain and enter a software depot activation code for VCF download tool.
   a) Generate and copy a software depot ID.

      ```
      vcf-download-tool configuration generate --software-depot-id
      ```

b) Log in to the VCF Business Services console.
c) In the upper-left corner of the console, next to the tenant name, click the arrow to expand the section.
d) From the drop-down menu, select the correct tenant name and siteID that map to your VCF or vSphere Foundation entitlement.
e) Navigate to **Software Depot Registrations** > **New Registration** and use your software depot ID (`software-depot-id`) to obtain an activation code (`depotDownloadActivationCode`) for your depot.
f) Enter the activation code that you obtained from the VCF Business Services console into a file, for example, `/path/to/activation-code.txt`.

3. On the computer hosting offline depot, list and download the installation binaries and metadata to the offline depot root directory.
   a) List the available binaries.

Product	Command
VMware Cloud Foundation	`./vcf-download-tool binaries list --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --type INSTALL --automated-install`
VMware vSphere Foundation	`./bin/vcf-download-tool binaries list --sku VCF --vcf-version vcf-source-version --sku VVF --depot-download-activation-code-file /path/to/activation-code.txt --type INSTALL --automated-install`

   b) Download the binaries and metadata.

Product	Command
VMware Cloud Foundation	`./vcf-download-tool binaries download --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --type INSTALL --automated-install --depot-store /path/to/binaries` The download command downloads both binaries and metadata. Make sure that the download destination (`/path/to/binaries`) has enough storage.
VMware vSphere Foundation	`./vcf-download-tool binaries download --sku VCF --vcf-version vcf-source-version --sku VVF --depot-download-activation-code-file /path/to/activation-code.txt --type INSTALL --automated-install --depot-store /path/to/binaries` The download command downloads both binaries and metadata. Make sure that the download destination (`/path/to/binaries`) has enough storage.

- Connect VCF Installer to Broadcom or an Offline Depot and Download Binaries.
- For information about additional binaries commands and options, see VCF Download Tool Binaries Commands and Options.

Manually Transfer Binaries to VCF Installer

If you cannot connect the VCF Installer appliance to an online or offline depot, use the VCF download tool to download binaries to a computer with access to the Internet and then upload them to the VCF Installer appliance.

- A Windows or Linux computer with Internet connectivity (either directly or through a proxy) for downloading the binaries and other required files.

- Configure TCP keepalive in your SSH client to prevent socket connection timeouts when using the VCF download tool for long-running operations.
- Obtain an activation code from the VCF Business Services Console.
- Create a text file on the VCF Installer appliance that contains the admin@local password.
- Verify that your VCF Business Services console role has sufficient permissions. See Access Management.

Note:

The VCF Download Tool is the only supported method for downloading binaries for VCF 9.1 in environments without access to the Internet. Do not use third-party tools or other methods to download binaries.

Using the VCF download tool to download binaries involves the following steps:

- Download the latest version of the VCF download tool from the Broadcom Support Portal.
- On a computer with access to the Internet, use the VCF download tool to download the binaries and metadata for VMware vSphere Foundation or VCF.
- Copy the binaries and metadata to the VCF Installer appliance.
- On the VCF Installer appliance, use the VCF download tool to upload the binaries and metadata to the internal repository.

In the steps below, where applicable, replace the following values.

Value	Description
vcf-source-version	Your current VCF version. For example, `9.1.1.0`.
`/path/to/binaries`	The download location for the binaries. For example, `/var/www`.
`/path/to/activation-code.txt`	Path to a text file that contains a single line with your activation code from the VCF Business Services Console.

If the computer with Internet access can only access the Internet using a proxy server, use the following options when downloading binaries.

Option	Description
`--proxy-server, -s`	Provide the proxy server FQDN and port of the proxy server that the VCF Download Tool uses for Internet access. The required format is <FQDN:Port>. The proxy server must be reachable from the machine where the tool is running. For example: `--proxyServer proxy.example.com:3128`.

`--proxy-https`	Specifies whether the proxy server is configured to be used with HTTPS connections. To use this option, the proxy certificate must be added to the JRE default trust store of VCF Download Tool. The exact path to the `/cacerts` folder depends on the directory where the VCF Download Tool `tar` file is extracted.
	For example:
	• Unix
	`/opt/obtu/jre/lin64/bin/keytool -importcert -file ca-bundle.crt -keystore <tool root>/jre/lin64/lib/security/cacerts`
	• Windows
	`/opt/obtu/jre/lin64/bin/keytool -importcert -file ca-bundle.crt -keystore <tool root>/jre/win32/lib/security/cacerts`

`--proxy-user, -r`	If the proxy server enforces authentication, this parameter allows you to provide the necessary username required to authenticate to the proxy server. For example, `--proxy-user` *`username`*.
`--proxy-user-password-file`	Path to a text file *`path/to/password.txt`* containing the password for the provided `--proxy-user`. The file should contain the password on a single line. For example, `--proxyPasswordFile` *`path/to/password.txt`*.

Note:

VMware Cloud Foundation does not support proxy servers with Kerberos or NTLMv2 authentication.

Example that combines the options:

```
./vcf-download-tool binaries download --depot-download-activation-code-file /path/to/activation-code.txt --vcf-version vcf-source-version  --depot-store /path/to/binaries --proxy-server proxy.example.com:3128 --proxy-user username--proxy-user-password-file path/to/password.txt --proxy-https --automated-install --type INSTALL
```

1. Download the most recent version of the VCF Download Tool on a computer with Internet access.
   a) Log in to the Broadcom Support Portal and browse to **My Downloads** > **VMware Cloud Foundation**.
   b) Click the version of VMware Cloud Foundation to which you are upgrading.
   c) Click **Drivers & Tools**.
   d) Click the download icon for the VCF Download Tool.
   e) Extract `vcf-download-tool-<version.build>.tar.gz`.
   f) Navigate to the `vcf-download-tool-<version.build>/bin/` and confirm that you have execute permission on all folders.

      The directory includes both the Bundle Transfer Utility (`lcm-bundle-transfer-util`) and the VCF Download Tool (`vcf-download-tool`).
2. Obtain and enter a software depot activation code for VCF download tool.
   a) Generate and copy a software depot ID.

      ```
      vcf-download-tool configuration generate --software-depot-id
      ```
   b) Log in to the VCF Business Services console.
   c) In the upper-left corner of the console, next to the tenant name, click the arrow to expand the section.
   d) From the drop-down menu, select the correct tenant name and siteID that map to your VCF or vSphere Foundation entitlement.
   e) Navigate to **Software Depot Registrations** > **New Registration** and use your software depot ID (`software-depot-id`) to obtain an activation code (`depotDownloadActivationCode`) for your depot.
   f) Enter the activation code that you obtained from the VCF Business Services console into a file, for example, `/path/to/activation-code.txt`.
3. List the available binaries.

Product	Command
VMware Cloud Foundation	`./vcf-download-tool binaries list --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --type INSTALL --automated-install`
VMware vSphere Foundation	`./bin/vcf-download-tool binaries list --sku VCF --vcf-version vcf-source-version --sku VVF --depot-download-activation-code-file /path/to/activation-code.txt --type INSTALL --automated-install`

4. Download the binaries and metadata to the computer with Internet access.

Product	Command
VMware Cloud Foundation	`./vcf-download-tool binaries download --sku VCF --vcf-version 9.1.1.0 --depot-download-activation-code-file /path/to/activation-code.txt --type INSTALL --automated-install --depot-store /path/to/binaries` The download command downloads both binaries and metadata. Make sure that the download destination (`/path/to/binaries`) has enough storage.
VMware vSphere Foundation	`./vcf-download-tool binaries download --sku VCF --vcf-version 9.1.1.0 --sku VVF --depot-download-activation-code-file /path/to/activation-code.txt --type INSTALL --automated-install --depot-store /path/to/binaries` The download command downloads both binaries and metadata. Make sure that the download destination (`/path/to/binaries`) has enough storage.

5. Download and extract `vcf-download-tool-<version.build>.tar.gz` to VCF Installer.

   Confirm that you have execute permission on all folders.

   ```
   $mkdir /nfs/vmware/vcf/nfs-mount/vcfdt
   $cd /nfs/vmware/vcf/nfs-mount/vcfdt
   $wget /download/path/to/vcf-download-tool-<version.build>.tar.gz
   $tar -xvf vcf-download-tool-<version.build>.tar.gz
   ```

6. Create a folder on VCF Installer and copy the downloaded binaries from the computer with Internet access.

   ```
   $mkdir /nfs/vmware/vcf/nfs-mount/depot
   $rsync -aP /path/to/binaries vcf@FQDN:/nfs/vmware/vcf/nfs-mount/depot
   ```

   For *FQDN*, enter the fully qualified domain name (FQDN) of the VCF Installer appliance.

7. Upload the downloaded binaries to the VCF Installer appliance.

   ```
   ./vcf-download-tool binaries upload --depot-store /nfs/vmware/vcf/nfs-mount/depot --sddc-manager-fqdn FQDN
    --sddc-manager-user username --sddc-manager-user-password-file /root/vcfdt/installer-password.txt
   ```

   For *FQDN*, enter the fully qualified domain name (FQDN) of the VCF Installer appliance.

   For *username*, enter your VCF Installer username, for example, `admin@local`.

   For */root/vcfdt/installer-password.txt*, enter the path to the VCF Installer password file.

After the binaries are successfully transferred to VCF Installer, you are ready to deploy.

Start a New VCF Fleet or a New VCF Instance Deployment by Using the VCF Installer Deployment Wizard

You can use the VCF Installer deployment wizard to plan, configure, and deploy a new VCF fleet. You can also deploy a new VCF Instance within an existing fleet.

- Verify that you deployed and configured ESX hosts. See Preparing ESX Hosts for VCF or vSphere Foundation.
- Verify that you deployed VCF Installer version 9.1.1 or later. See Deploy VCF Installer.
- Verify that you downloaded the required binaries for VCF. See Downloading Binaries to VCF Installer.
- Verify that you configured the required FQDNs and IP addresses for deployment. See VCF Components FQDNs and IP addresses.

- If you plan to use a load balancer in your VCF environment, verify it is deployed and configured.

A VCF fleet refers to an environment consisting of one or more VCF Instances and standalone vCenter instances that are managed by the same VCF Operations instance.

- When you deploy the first VCF Instance in a VCF fleet, you can deploy a new VCF Operations instance or connect to an existing one.
- When you deploy a new VCF Instance in an existing VCF fleet, you must connect to your existing VCF Operations instance.

After you choose a deployment path, you plan and gather all the information required for deployment. By specifying the scale of the deployment, network, and storage types, you define the required resources for your deployment.

After planning and preparation, you use the VCF Installer deployment workflows to specify deployment information specific for your environment such as networks, hosts, and other information. As a result, the VMware Cloud Foundation platform is automatically deployed and configured using the provided information.

This procedure uses the deployment wizard. You can also deploy by using a JSON specification file. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation for more information.

VCF Installer allows you to skip VCF Automation during the initial deployment and deploy it as a Day-N operation. See Deploying VCF Automation as a Day-N Operation.

1. In a web browser, log in to the VCF Installer appliance at `https://installer_appliance_FQDN`.
   Enter the `admin@local` user and password that you provided when you deployed the appliance and then click **Log In**.
2. Click **Deployment Wizard** > **VMware Cloud Foundation** .
3. Review the introductory information and click **Next**.
4. Select what you want to deploy.
   - Deploy a new VCF fleet
   - Deploy a new VCF Instance

   Some of the options in the following steps are different depending on which option you select.

   To deploy deferred components, see Deploying Deferred Components on Custom Networking.
5. Select the existing components.

Option	New VCF Fleet	VCF Instance in an existing VCF fleet
I have an existing VCF Operations 9.1 instance	Select this option to connect the VCF fleet to an existing VCF Operations instance. Otherwise a new VCF Operations instance is deployed. If you select a VCF Operations instance that is connected to a VCF Automation instance, that component is also added to this VCF fleet.	Option is not available. You must connect to your existing VCF Operations later in the deployment wizard.
I have an existing vCenter instance	Starting with VCF 9.1.1, it is mandatory that VCF Operations is always deployed in the management domain of the first VCF Instance in the VCF fleet. If you selected the existing VCF Operations option, this option is also always selected. If you have an existing vCenter instance, but no existing VCF Operations instance, this option is not applicable to deploying a new VCF fleet. Do not select this option. To use an existing vCenter, see Deploy Components by Using VCF Installer to Complete the Converging to VCF Process.	Not applicable to deploying a new VCF Instance. Do not select this option. To use an existing vCenter instance, see Deploy Components by Using VCF Installer to Complete the Converging to VCF Process.

Option	New VCF Fleet	VCF Instance in an existing VCF fleet
The vCenter instance is registered with NSX Manager	Not applicable to deploying a new VCF fleet. Do not select this option. To use an existing vCenter instance that is registered with NSX Manager, see Deploy Components by Using VCF Installer to Complete the Converging to VCF Process.	Not applicable to deploying a new VCF Instance. Do not select this option. To use an existing vCenter instance that is registered with NSX Manager, see Deploy Components by Using VCF Installer to Complete the Converging to VCF Process.
I have an existing VCF Automation instance or I will deploy later	Select the check box if you want to use VCF Operations to import your existing VCF Automation instance after the VCF Installer deployment workflows complete. You can also select this check box to deploy VCF Automation as Day-N operation, instead of using the VCF Installer deployment workflows. See Deploying VCF Automation as a Day-N Operation. To use an existing Aria Automation 8.x instance, after deploying the VCF fleet, you must import it to VCF Operations and upgrade it to VCF Automation 9.1. See Upgrading to VCF Automation 9.1.	Option is not available.

The VCF Installer validates the selected components to ensure that they are compatible.

6. Select a deployment model and the scale for your VCF Instance.

Deployment Model	You can choose the deployment model for new appliances that get deployed. **High Availability** is recommended for production environments. **Note:** The deployment model that you select does not impact existing appliances that you are using in your VCF fleet or VCF Instance. See the VMware Cloud Foundation Design documentation for more information. **Note:** New appliances that get deployed are automatically configured to comply with Federal Information Processing Standards (FIPS).
Size	You can choose the scale of the components in your VCF Instance. • Small – In VCF 9.1, this is the only available option for the simple deployment model. – In VCF 9.1.1 and later, this is the default option for the simple and HA deployment models. • Medium – In VCF 9.1, this is the default option for the HA deployment model. – In VCF 9.1.1 and later, this is an option for the HA deployment model. • Large – In VCF 9.1 and later, this is an option for the HA deployment model. For more information about deployment sizes, see VCF Fleet Sizing Models.

7. Select your network configuration.

   You can continue with the default values or click **Customize** to provide custom network settings.

Management network configuration	• VM management network: For core VCF components, such as vCenter, ESX, and NSX. You can share this network with the ESX management network or use a dedicated network. • VCF management network: For broader VCF management and automation components, such as VCF Operations and VCF Automation. You can share this network with the VM management network or use a dedicated network. You can use one of the networks to segment management traffic for security and compliance requirements. For more information, see Network Deployment Options for VCF Components.
IP Scheme	If Dual-stack is enabled for VCF Installer, you can enable the **Dual stack networking** toggle and use IPv4 and/or IPv6 Internet protocols for your management domain components. • vMotion network IP scheme (IPv4 or IPv6) • Storage network IP scheme (IPv4 or IPv6) Dual-stack networking allows communication with components that support both network protocols, ensuring compatibility and continuous connectivity. While not all components support IPv6 networking, components that are IPv4 only can still communicate with the other components. **Note:** The dual-stack networking option is not available, if VCF Installer does not have IPv4 and IPv6 configured. For more information, see Configure the Network on the ESX Hosts and Deploy VCF Installer.
Deploy the VCF Operations and VCF Automation to a specific vSphere distributed switch or NSX segment	If you are deploying a new VCF fleet, you can select this option to skip the deployment of VCF Operations, cloud proxy, license server, and VCF Automation. For more information about deploying these deferred components later, see Deploying Deferred Components on Custom Networking. **Note:** This option is not available when deploying a VCF Instance in an existing VCF fleet or when using an existing VCF Operations instance. When selecting this option, some of the pages in the VCF Installer wizard are not available.
VPC Network Configuration	**Full Stack VPC**: In version 9.1.1 and later, provides the complete VPC feature set, including all routing, switching, and gateways, to manage the virtual network environment. This option uses TEP-based overlay networking and supports both Distributed Connectivity and Centralized Connectivity models. Select this option to use the standard NSX VPC networking model.
	VLAN-Backed VPC: In version 9.1.1 and later, provides essential VPC services without TEPs, prioritizing the existing VLAN infrastructure. This option enables TEP-less VPC networking, where NSX provides only the overlay. If you select this option, the **VPC Gateway Connectivity** and **VNA Cluster Details** sections are not displayed, and Private CIDRs are not required for the Supervisor configuration.

VPC Gateway Connectivity	• **Centralized Connectivity**. Routes all north–south traffic through NSX Edge Nodes for centralized service delivery. If you select **Centralized Connectivity**, configure it after the workload domain is created to make it VPC-ready. See Configure Centralized Network Connectivity with Edge Clusters.
	• **Distributed Connectivity**. Allows for north–south traffic routing directly at the hypervisor level using a Transit Gateway with distributed external connectivity. If you select **Distributed Connectivity**, configure it in the **Prepare** section of VCF Installer, on the **NSX Manager** page. For Distributed Transit Gateways to provide stateful NAT and load-balancing services, you must later deploy a Virtual Network Appliance cluster. **Note:** VPC does not support IPv6. For more information about VPCs, see Managing Virtual Private Clouds in vCenter.

8. Choose a storage type for your management domain.

   Your selection determines the initial shared storage type for the default cluster in the management domain. This initial shared storage type is known as principal storage and the principal storage type for a cluster cannot be changed. However, you can add more clusters with different principal storage types to the management domain later.

Option	Details
vSAN	• Select vSAN ESA. – Allow auto claim of HCL incompatible disks • Select this option to allow the vSAN ESA automatic disk-claim process to claim storage devices that VCF Installer cannot verify against the vSAN Hardware Compatibility List (HCL). Use this option only for hardware that you have verified is HCL-certified. • Select vSAN OSA. – Select the number of host failures that are tolerated. – Activate deduplication and compression for this storage type. • vSAN Data-in-transit encryption (available for vSAN ESA and OSA) – vSAN data-in-transit encryption is a cluster-wide setting. All data and metadata between hosts in the same cluster and data and metadata between server and client clusters is encrypted. The rekey interval sets the time duration before vSAN automatically regenerates the host-to-host communication encryption keys. vSAN does not re-encrypt data at rest or reformat disk after the interval expires.
VMFS on Fibre Channel (FC)	The datastore must already be created and mounted on all ESX hosts.
NFS v3	The NFS Server must already be created and meet the NFS Storage Guidelines and Requirements found in the Guidelines and Requirements for NFS Storage with ESX. If VCF Installer is deployed on one of the ESX hosts that will form the management domain, the NFS datastore must be mounted on the host. See Mount NFS Datastore to an ESX Host. • Use the same network configuration as the ESX management network for the NFS storage network

Option	Details
	– Select this option to apply the ESX management network settings — gateway, VLAN, and MTU — to the NFS storage network.

9. Review the prerequisites for your deployment.

   Depending on your selections in the wizard, the prerequisites for the deployment are listed.

   You can click **PRE-FILL GENERATED FQDNs IN WIZARD** and specify the required FQDNs for the deployment. You can use the provided template or specify custom FQDNs. These values are later populated in the VCF Installer deployment workflows.

   You can download a JSON template that is based on your choices in the VCF Installer so far. You can populate it and deploy VCF by using a JSON specification file. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation for more information.

10. Enter the general configuration details for your deployment.

Option	Description
Version	Choose which VCF version to install.
VCF Instance Name	The VCF Instance Name is used to provide a unique name when creating the VMware Cloud Foundation Adapter within VCF Operations. A descriptive name provides for easy identification when you have multiple VCF Instances.
Management domain name	The name of the management domain that is created in SDDC Manager. This name is also used as a prefix for the default names of objects that are created or deployed by the VCF Installer. You can change these default names as you progress through the deployment wizard.
Customer Experience Improvement Program (CEIP)	Select an option to activate or deactivate CEIP across all deployed components. VMware's Customer Experience Improvement Program (CEIP) collects technical information from participating customers to provide better support. See Customer Experience Improvement Program for more information.
DNS servers	Enter the IP addresses of your DNS servers (comma separated). By default, this option is pre-populated with the value set of the VCF Installer appliance.
NTP servers	Enter one or more NTP servers (comma separated). NTP servers can be entered using FQDNs or IP addresses. By default, this option is pre-populated with the value set of the VCF Installer appliance.
DNS domain name	Enter the DNS domain name. For example, `rainpole.io`. By default, this option is pre-populated with the value set of the VCF Installer appliance.
Autogenerate passwords	Starting with VCF 9.1, select this option to let VCF Installer generate secure passwords for all deployed components. You can retrieve the passwords after the deployment begins. If you do not select this option, you must provide each password manually.

11. Enter the ESX host details.

    ESX hosts require passwords with predefined requirements. For more information, see ESX Passwords and Account Lockout.

12. Enter details about the networks.

    - ESX Management Network - VLAN separation for ESX Host management VMkernels
    - VM Management Network - distributed port group separation for management domain appliances

If you chose to share this network with the ESX management network, this option is not displayed.
- VCF Management Network - distributed port group separation for all management components
  If you chose to share this network with the VM management network, this option is not displayed.
- VCF Management Services IP Range
  - For deployments version 9.1.0.0 through 9.1.0.300, you specify the IP addresses as a contiguous IP range. Alternatively, you can specify non-contiguous IP addresses by using a JSON specification file. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation.
  - For deployments version 9.1.0.400 and later, you can specify the IP addresses as contiguous or non-contiguous IP range.
- VCF Automation IP Range
  - For deployments version 9.1.0.0 through 9.1.0.300, you specify a contiguous IP range for the VCF Automation nodes with 5 IP addresses.
  - For deployments version 9.1.0.400 and later, you specify 5 IP addresses as contiguous or non-contiguous IP range.
- vMotion Network - distributed port group separation for ESX Host vMotion VMkernel
- vSAN Network or NFS Network - distributed port group separation for ESX Host storage VMkernels
- NSX Host Overlay Network - dedicated VLAN separation for NSX tunnel endpoint VMkernels. With VCF 9.1.1 and later, this option is not required for VLAN backed VPCs.

Specify IP inclusion ranges for the VCF management, vMotion, vSAN, and NSX overlay networks of the management domain. IP addresses from the specified range are automatically assigned to ESX hosts, from vMotion, vSAN and NSX networks only.

- Ensure that the IP ranges include sufficient IP addresses for the deployment.
- The number of IP addresses must be at least equal to the number of ESX hosts deployed.
- All IPs within the range must be available for use or IP conflicts occur.
- IPs for each network must be part of the VLAN configured with the network's port group.

As an example, if you specify the range start value as 10.11.13.101 and end as 10.11.13.121, a total of 21 IP addresses would be available. Do not use special IP addresses, such as the network or broadcast address.

MTU determines the largest size of a data packet that can pass through for this traffic type.

13. Enter the details for your VCF management components.

I am deploying ...	Details
A new VCF fleet with a new VCF Operations instance	Enter all required configuration details for the VCF management components. For more information, see the VCF Management Configuration Details table below. If you specified the required FQDNs earlier, these are now populated. VCF Automation details section does not appear if you selected that you want to connect a VCF Automation instance later.
A new VCF fleet with an existing VCF Operations instance	• Connect to your existing VCF Operations instance. • Enter details for the new cloud proxy appliance. If a VCF Automation instance is connected to the VCF Operations instance, the appliance FQDN is displayed and you must enter the root password.
A new VCF Instance with an existing VCF Operations instance	Connect to the VCF Operations instance that manages the VCF fleet in which you want to deploy this VCF Instance. After connecting, enter details for the new cloud proxy appliance. If a VCF Automation instance is connected to the VCF Operations instance, the appliance FQDN is displayed and you must enter the root password.

Table 961: VCF Management Configuration Details

Each Fully Qualified Domain Name (FQDN) must point to a unique IP address.

Note:

If you are deploying a VCF fleet or a new VCF Instance with dual-stack networking, for VCF components that support the dual-stack model, the FQDNs must resolve to IPv4 and IPv6 addresses. See Dual Stack IP Addressing Model.

The FQDNs for VCF management services must point to unique IP addresses that are not assigned and the FQDNs must not be in the IP Ranges assigned to VCF management services nodes.

Component	Field	Description
VCF Operations	Primary node FQDN	Enter the FQDN of the VCF Operations primary node.
	Replica node FQDN	Enter the FQDN of the VCF Operations replica node.
	Data node FQDN	Enter the FQDN of the VCF Operations data node. This field appears only when deploying VCF in High Availability mode with a Medium or Large deployment size.
Load Balancer	Load balancer FQDN	(Optional) Enter the FQDN of the load balancer, that distributes the VCF Operations traffic. This field appears only when deploying VCF in High Availability mode.
Cloud Proxy	Cloud Proxy FQDN	Enter the FQDN of the cloud proxy that collects and monitors data locally in the VCF Instance.
License Server	License Server FQDN	Enter the FQDN of the license server. A license server is added to a VCF Operations instance to store the licenses you want to assign from this VCF Operations instance. You must add at least one license server to each VCF Operations instance that you use for license management. You can deploy license server as IPv4 only. In VCF 9.1.1 and later, after the deployment, you can change the networking configuration to IPv6. See Configure IPv6 on a License Server.
VCF Management Services	Fleet components FQDN	Enter the FQDN of the fleet-level components, such as fleet lifecycle, Salt RaaS, software depot, and so on. For more information, see Components in VCF and vSphere Foundation. **Note:** Do not use capital letters in the FQDN. Do not use a `.local` FQDN.
	Instance components FQDN	Enter the FQDN of the instance-level components, such as SDDC lifecycle, Salt master, real-time metrics, real-time metrics store, and telemetry. For more information, see Components in VCF and vSphere Foundation. **Note:** Do not use capital letters in the FQDN. Do not use a `.local` FQDN.
	Identity broker FQDN	Enter the FQDN of the identity broker component. **Note:** Do not use capital letters in the FQDN. Do not use a `.local` FQDN.
	VCF services runtime FQDN	Enter the FQDN for VCF services runtime. VCF services runtime is the common architectural runtime deployed to deliver the shared cluster of management services. The hostname from this FQDN is prefixed to the names of its node VMs and related objects. **Note:** Do not use capital letters in the FQDN. Do not use a `.local` FQDN.

Component	Field	Description
	Internal cluster CIDR IPv4	Select the VCF services runtime internal IPv4 network. • 198.18.0.0/15 • 240.0.0.0/15 • 250.0.0.0/15 Different VCF services runtime instances on the same network can work with the same internal network. For example, you can have the two instances of VCF services runtime for your VCF management services and VCF Automation using the same internal network.
	Internal cluster CIDR IPv6	Select the VCF services runtime internal IPv6 network. • fd00::/111 (default) • fd00::0/111 • fc00::0004:0/111 • fc00::4:0/111 • fc00::/111 • fc00::0/111 This option is only available if you have dual-stack networking enabled in your environment.
VCF Automation	VCF Automation FQDN	Enter the FQDN to assign to the VCF Automation instance. Later, you use it to access the VCF Automation user interface. **Note:** Do not use capital letters in the FQDN.
	VCF services runtime FQDN	Enter the FQDN for VCF services runtime to connect VCF Automation with the VCF fleet lifecycle component that manages the lifecycle of VCF Automation. The FQDN must be dedicated to the runtime of the VCF Automation cluster, not the VCF management services cluster. The hostname from this FQDN is prefixed to the names of its node VMs and related objects. **Note:** Do not use capital letters in the FQDN.
	Internal cluster CIDR IPv4	Select the VCF services runtime internal IPv4 network. • 198.18.0.0/15 • 240.0.0.0/15 • 250.0.0.0/15 Different VCF services runtime instances on the same network can work with the same internal network. For example, you can have the two instances of VCF services runtime for your VCF management services and VCF Automation using the same internal network.

Note:

If you selected the custom networking option for VCF Operations and VCF Automation, not all input fields are displayed.

Depending on your network options for the VM management network and the VCF management network, the network deployment location of VCF management services can vary. See Network Deployment Options for VCF Components.

14. Enter the vCenter details.
15. Enter storage details.

    Provide the required detail for the initial shared storage type for the default cluster in the management domain.

Option	Details
vSAN	• Enter a name for the vSAN datastore or use the pre-populated name.

Option	Details
VMFS on Fibre Channel (FC)	• Enter a name for the VMFS on FC datastore or use the pre-populated name.
NFS v3	• Enter a name for the NFS v3 datastore or use the pre-populated name. • Enter the path to the NFS share. • Enter the IPv4 address of the NFS server. • (optional) Choose to bind the NFS datastore to the VMkernel adapter specified when the datastore gets created. Indicates whether to bind the created NFS datastore to the Vmknic created based on NFS Network Spec. This is to prevent the unintentional flow of NFS traffic through any other VMkernel NIC, if such connectivity exists.

16. On the **vSphere Distributed Switch** page, configure the vSphere Distributed Switch details, configure the network traffic types, and click **Create Distributed Switch**.
    a) To use a preconfigured profile, select one of the profiles and click **Next**.

Option	Description
Default Configuration	Provides a unified fabric for all traffic types using a single vSphere Distributed Switch.
Storage Traffic Separation	This profile creates two vSphere Distributed Switches with separate physical NICs. One switch is dedicated for storage traffic, while the other is used for all other traffic.
NSX Traffic Separation	This profile creates two vSphere Distributed Switches with separate physical NICs. One switch is dedicated for NSX (Virtual Networking) traffic, while the other is used for all other traffic.
Custom Switch Configuration	Copy from a preconfigured profile or create a new distributed switch configuration. Multiple vSphere Distributed Switches can be configured. Each vSphere Distributed Switch can hold one or more network traffic configurations. Some network traffic types are mandatory. Switch configuration is incomplete until these mandatory traffic types are configured. VM Management, VCF Management, vMotion, vSAN, and NSX network types can be configured only once for a cluster. NSX traffic type can be configured in multiple vSphere Distributed Switches.

    b) To create a custom configuration for the distributed switch, click **Create custom configuration**.

Option	Description
Traffic types for the distributed switch	• ESX Management • VCF Management This option is not exposed for the default Network setting. • VM Management • vMotion • vSAN • NSX
Distributed Switch name	Enter a name for the distributed switch.
Uplinks / LAG configuration	VDS Uplinks represent the individual physical network adapters (vmnics) on an ESX host connecting to the physical switch. In this mode, the VDS manages load balancing and failover using software-based algorithms like Route Based on Physical NIC Load or Route Based on the Source of the Port ID. The physical switch ports do not need to be configured in a bond or channel; they operate as independent links.

Option	Description
	VDS LAG bundles multiple physical uplinks into a single logical channel using the Link Aggregation Control Protocol (LACP). This requires a corresponding LACP configuration on the physical ToR (Top-of-Rack) switches. The physical switch and VDS negotiate traffic hashing.
Number of uplinks	The number of uplinks dedicated to this distributed switch.
Map uplinks to physical NICs on host	Connects your logical virtual network to your physical network hardware. This mapping pairs a logical uplink (a software object on the distributed switch, with a physical adapter (hardware on the ESX host).

c) To configure the traffic types for the distributed switch, turn on the toggle for each traffic type and configure the settings.

Option	Description
Distributed port group name	Name of the distributed port group for this traffic type.
Load balancing	*Route based on the source of the port ID*: the distributed switch maps port IDs to specific uplinks based on the load balancing source of the NSX uplink profile.
	Route Based on Physical NIC Load (Default): the distributed switch monitors the saturation of the physical uplinks. If the primary uplinks exceed a set amount of utilization (75% default) for 30 seconds, the distributed switch moves some VM traffic to another active uplink.
	Route Based on Source MAC Hash: the distributed switch uses the VM's MAC address and hashes it to an uplink, which creates even distribution for VMs that might change port IDs but keep MAC addresses.
	Route Based on IP Hash: the distributed switch uses the source and destination IP address of every packet to decide which uplink to use. This allows a single VM to use multiple uplinks simultaneously. **Note:** You must configure LACP (Link Aggregation Control Protocol) on the physical Top-of-Rack switches.
	Use Explicit Failover Order: you define the order in which uplinks are used in case of failure.
Uplink	*Active*: This traffic type uses the uplink immediately during normal operation. In case of multiple active uplinks, traffic of this type is distributed across all active uplinks simultaneously.
	Standby: A standby uplink is connected, but no traffic is sent through it as long as at least one "Active" uplink is healthy. If all active uplinks fail, the distributed switch moves the standby uplink to active status.
	Unused: Traffic of this type never uses this uplink, regardless of the status of the remaining uplinks.

17. Enter the NSX Manager details.

    If you select **Distributed Connectivity**, enter the following details:

    - **VLAN ID**: A dedicated VLAN network is required for the distributed gateway's external connectivity.
    - **Gateway CIDR IPv4 Address**: The gateway CIDR IP address should be the same as the VLAN default gateway configured in the external router (TOR).

18. Enter the SDDC Manager details.

    Note:

    If VCF Installer is deployed on one of the management domain ESX hosts, details populate automatically.

19. Review the deployment information.

    To make changes, navigate back to the relevant page in the deployment wizard. You can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage.

20. Review the validation information.

    VCF Installer validates the information you provided in the deployment wizard and reports any errors or warnings.

Warnings	Review any warnings and either address them or click **Acknowledge** to proceed with deployment.
Errors	Review all failed validations and resolve the issue(s) using the remediation information provided in the error message. You can navigate back to the relevant pages in the deployment wizard to make updates and then re-run the validations. You can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage.

21. After addressing all validation errors and warnings, click **Deploy**.

    You can monitor the deployment progress. If there are errors, review the failed tasks in the Tasks panel, address the issue(s), and retry the deployment.

    CAUTION:

    After deployment completes, do not move the VCF management services or VCF Automation VMs to a different resource pool or folder. If you move the VMs to a new resource pool or folder, subsequent patching, deployment, or scale out operations fail.

- License your environment. See Registering VCF Operations and a License Server with the VCF Business Services Console.
- Click **Review Passwords** to review the passwords. To review passwords for past deployments, in the VCF Installer appliance, click the user name at the top-right corner of the window and select **Deployments**.

  Note:

  You can also retrieve some of the passwords using the SDDC Manager API and VCF Operations API.
- (Optional) Download the JSON specification file for the deployment.

  Note:

  The JSON specification file does not include passwords.
- Deploy the remaining VCF platform components that VCF Installer does not deploy automatically. See Completing the Deployment of Your Platform.
- To cover the platform after a failure, configure file-based backups for SDDC Manager and other management components. See Configure File-Based Backup Schedule for SDDC Manager.
- If you deployed VCF Operations in High Availability mode, to prevent VCF Operations cluster nodes from running on the same ESX host, apply a vSphere DRS anti-affinity rule.
- For platform administration tasks such as managing identity and access, certificates, passwords, component lifecycle, and backup and restore of your VCF platform, see the Fleet Management guide.
- For monitoring and troubleshooting tasks such as configuring alerts, analyzing logs, monitoring dashboards, and troubleshooting your VCF platform, see the Infrastructure Operations guide.

Start a New vSphere Foundation Deployment by Using the VCF Installer Deployment Wizard

You can use the VCF Installer deployment wizard to deploy a new VMware vSphere Foundation platform.

- Verify that you deployed and configured ESX hosts. See Preparing ESX Hosts for VCF or vSphere Foundation.
- Verify that you deployed VCF Installer version 9.1.1 or later. See Deploy VCF Installer.
- Verify that you downloaded the required binaries for VMware vSphere Foundation. See Downloading Binaries to VCF Installer.
- Verify that you configured the required FQDNs and IP addresses for deployment. See *vSphere Foundation Platform FQDNs and IP Addresses* in VCF Components FQDNs and IP addresses.

  Important:

  The VCF Installer deploys vSphere Foundation only with VCF management services. If you want to deploy vSphere Foundation without VCF management services, you can deploy manually. See Deploying VMware vSphere Foundation 9.1 Without VCF Management Services.

Plan and gather all the information that is required for deployment. By specifying the network and storage types, you define the required resources for your VMware vSphere Foundation deployment.

Use the VCF Installer deployment workflows to specify deployment information specific for your environment such as networks, hosts, and other information. As a result, the VMware vSphere Foundation software components are automatically deployed and configured using the information provided.

VCF Installer deploys vSphere Foundation in a simple mode with the following component sizes: medium vCenter, medium VCF Operations, and small VCF management services. To change the component sizes, instead of using this procedure to deploy vSphere Foundation by using the deployment wizard, you can use a `JSON` specification file. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation.

To create a VMware vSphere Foundation environment using existing vSphere infrastructure, see Converge a vCenter Instance and ESX Hosts to vSphere Foundation.

1. In a web browser, log in to the VCF Installer appliance at `https://installer_appliance_FQDN`.
   Enter the `admin@local` user and password you provided when you deployed the appliance and then click **Log In**.
2. Click **Deployment Wizard** > **VMware vSphere Foundation** .
3. Review the introductory information and click **Next**.
4. Select **Deploy new VMware vSphere Foundation**.
5. Do not select any existing components.
   To use an existing vCenter to create a VMware vSphere Foundation environment, see Converge a vCenter Instance and ESX Hosts to vSphere Foundation.

6. Select your network configuration.

   You can continue with the default values or provide custom network settings.

Management network configuration	• VM management network: For core VCF components, such as vCenter and ESX. You can share this network with the ESX management network or use a dedicated network. • VCF management network: For broader VCF management components, such as VCF Operations. You can share this network with the VM management network or use a dedicated network. You can use one of the networks to segment management traffic for security and compliance requirements. For more information, see Network Deployment Options for VCF Components.
IP Scheme	If Dual-stack is enabled for VCF Installer, you can enable the **Dual stack networking** toggle and use IPv4 and/or IPv6 Internet protocols for your management domain components. • vMotion network IP scheme (IPv4 or IPv6) • Storage network IP scheme (IPv4 or IPv6) Dual-stack networking allows communication with components that support both network protocols, ensuring compatibility and continuous connectivity. While not all components support IPv6 networking, components that are IPv4 only can still communicate with the other components. **Note:** The dual-stack networking option is not available, if VCF Installer does not have IPv4 and IPv6 configured. For more information, see Configure the Network on the ESX Hosts and Deploy VCF Installer.

7. Choose a storage type for your management domain.

   Your selection determines the initial shared storage type for the default cluster in the management domain. This initial shared storage type is known as principal storage and the principal storage type for a cluster cannot be changed. However, you can add additional clusters with different principal storage types to the management domain later.

Option	Details
vSAN	• Select vSAN ESA. – Allow auto claim of HCL incompatible disks • Select this option to allow the vSAN ESA automatic disk-claim process to claim storage devices that VCF Installer cannot verify against the vSAN Hardware Compatibility List (HCL). Use this option only for hardware that you have verified is HCL-certified. • Select vSAN OSA. – Select the number of host failures that are tolerated. – Activate deduplication and compression for this storage type. • vSAN Data-in-transit encryption (available for vSAN ESA and OSA) – vSAN data-in-transit encryption is a cluster-wide setting. All data and metadata between hosts in the same cluster and data and metadata between server and client clusters is encrypted. The rekey interval sets the time duration before vSAN automatically regenerates the host-to-host communication encryption keys. vSAN does not re-encrypt data at rest or reformat disk after the interval expires.

Option	Details
VMFS on Fibre Channel (FC)	The datastore must already be created and mounted on all ESX hosts.
NFS v3	The NFS Server must already be created and meet the NFS Storage Guidelines and Requirements. See Guidelines and Requirements for NFS Storage with ESX. If VCF Installer is deployed on one of the ESX hosts that will form the management domain, the NFS datastore must be mounted on the host. See Mount NFS Datastore to an ESX Host. • Use the same network configuration as the ESX management network for the NFS storage network – Select this option to apply the ESX management network settings — gateway, VLAN, and MTU — to the NFS storage network.

8. Review the prerequisites for your deployment.

   Depending on your selections in the wizard, the prerequisites for the deployment are listed.

   You can click **PRE-FILL GENERATED FQDNs IN WIZARD** and specify the required FQDNs for the deployment. You can use the provided template or specify custom FQDNs. These values are later populated in the VCF Installer deployment workflows.

   You can download a JSON template that is based on your choices in the VCF Installer so far. You can populate it and deploy VCF by using a JSON specification file. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation for more information.

9. Enter the general configuration details for your deployment.

Option	Description
Version	Choose which VMware vSphere Foundation version to install.
Customer Experience Improvement Program (CEIP)	Select an option to activate or deactivate CEIP across all deployed components. VMware's Customer Experience Improvement Program (CEIP) collects technical information from participating customers to provide better support. For more information, see Customer Experience Improvement Program.
DNS servers	Enter the IP addresses of your DNS servers (comma separated). By default, this option is pre-populated with the value set of the VCF Installer appliance.
NTP servers	Enter one or more NTP servers (comma separated). NTP servers can be entered using FQDNs or IP addresses. By default, this option is pre-populated with the value set of the VCF Installer appliance.
DNS domain name	Enter the DNS domain name. For example, `rainpole.io`. By default, this option is pre-populated with the value set of the VCF Installer appliance.
Autogenerate passwords	Starting with VCF 9.1, select this option to let VCF Installer generate secure passwords for all deployed components. You can retrieve the passwords after the deployment begins. If you do not select this option, you must provide each password manually.

10. Enter the ESX host details.

    ESX hosts require passwords with predefined requirements. For more information, see ESX Passwords and Account Lockout.

11. Enter details about the networks.
    - ESX Management Network - distributed port group separation for ESX Host management VMkernels
    - VM Management Network - distributed port group separation for management domain appliances
      If you chose to share this network with the ESX management network, this option is not displayed.
    - VCF Management Network - distributed port group separation for all management components
      If you chose to share this network with the VM management network, this option is not displayed.
    - VCF Management Services IP Range
      – For deployments version 9.1.0.0 through 9.1.0.300, you specify the IP addresses as a contiguous IP range. Alternatively, you can specify non-contiguous IP addresses by using a JSON specification file. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation.
      – For deployments version 9.1.0.400 and later, you can specify the IP addresses as contiguous or non-contiguous IP range.
    - vMotion Network - distributed port group separation for ESX Host vMotion VMkernel
    - vSAN Network or NFS Network - distributed port group separation for ESX Host storage VMkernels

    Specify IP inclusion ranges for the VCF management, vMotion, and vSAN networks of the management domain. IP addresses from the specified range are automatically assigned to ESX hosts, from vMotion, and vSAN networks only.
    - Ensure that the IP ranges include sufficient IP addresses for the deployment.
    - The number of IP addresses must be at least equal to the number of ESX hosts deployed.
    - All IPs within the range must be available for use or IP conflicts occur.
    - IPs for each network must be part of the VLAN configured with the network's port group.

    As an example, if you specify the range start value as 10.11.13.101 and end as 10.11.13.121, a total of 21 IP addresses would be available. Do not use special IP addresses, such as the network or broadcast address.

    MTU determines the largest size of a data packet that can pass through for this traffic type.
12. Enter the required configuration details for the VCF management components.

    If you specified the required FQDNs earlier, these are now populated.

Table 962: VCF Management Configuration Details

Each Fully Qualified Domain Name (FQDN) must point to a unique IP address.

> **Note:**
>
> For VCF components that support the dual-stack model, the FQDNs must resolve to IPv4 and IPv6 addresses.
>
> See Dual Stack IP Addressing Model.
>
> The FQDNs for VCF management services must point to unique IP addresses that are not assigned and the
>
> FQDNs must not be in the IP Ranges assigned to VCF management services nodes.

Component	Field	Description
VCF Operations	Primary node FQDN	Enter the FQDN of the VCF Operations primary node.
License Server	License Server FQDN	Enter the FQDN of the license server. A license server is added to a VCF Operations instance to store the licenses you want to assign from this VCF Operations instance. You must add at least one license server to each VCF Operations instance that you use for license management. You can deploy license server as IPv4 only. In vSphere Foundation 9.1.1 and later, after the deployment, you can change the networking configuration to IPv6. See Configure IPv6 on a License Server.
VCF Management Services	Fleet components FQDN	Enter the FQDN of the fleet-level components, such as fleet lifecycle, software depot, and so on. For more information, see Components in VCF and vSphere Foundation. **Note:** Do not use capital letters in the FQDN.

Component	Field	Description
	Instance components FQDN	Enter the FQDN of the instance-level components, such as SDDC lifecycle, and telemetry. For more information, see Components in VCF and vSphere Foundation. **Note:** Do not use capital letters in the FQDN.
	VCF services runtime FQDN	Enter the FQDN for VCF services runtime. VCF services runtime is the common architectural runtime deployed to deliver the shared cluster of management services. The hostname from this FQDN is prefixed to the names of its node VMs and related objects. **Note:** Do not use capital letters in the FQDN.
	Internal cluster CIDR IPv4	Select the VCF services runtime internal IPv4 network. • 198.18.0.0/15 • 240.0.0.0/15 • 250.0.0.0/15 Different VCF services runtime instances on the same network can work with the same internal network. For example, you can have the two instances of VCF services runtime for your VCF management services and VCF Automation using the same internal network.
	Internal cluster CIDR IPv6	Select the VCF services runtime internal IPv6 network. • fd00::/111 (default) • fd00::0/111 • fc00::0004:0/111 • fc00::4:0/111 • fc00::/111 • fc00::0/111 This option is only available if you have dual-stack networking enabled in your environment.

Depending on your network options for the VM management network and the VCF management network, the network deployment location of VCF management services can vary. See Network Deployment Options for VCF Components.

13. Enter the vCenter details.
14. Enter storage details.

    Provide the required detail for the initial shared storage type for the default cluster in the management domain.

Option	Details
vSAN	• Enter a name for the vSAN datastore or use the pre-populated name.
VMFS on Fibre Channel (FC)	• Enter a name for the VMFS on FC datastore or use the pre-populated name.
NFS v3	• Enter a name for the NFS v3 datastore or use the pre-populated name. • Enter the path to the NFS share. • Enter the IPv4 address of the NFS server. • (optional) Choose to bind the NFS datastore to the VMkernel adapter specified when the datastore gets created. Indicates whether to bind the created NFS datastore to the Vmknic created based on NFS Network Spec. This is to prevent unintentional flow of NFS traffic through any other VMkernel NIC, if such connectivity exists.

Option	Details
vSAN Data-in-transit encryption	• vSAN data-in-transit encryption is a cluster-wide setting. All data and metadata between hosts in the same cluster and data and metadata between server and client clusters is encrypted. The rekey interval sets the time duration before vSAN automatically regenerates the host-to-host communication encryption keys. vSAN does not re-encrypt data at rest or reformat disk after the interval expires.

15. On the **vSphere Distributed Switch** page, configure the vSphere Distributed Switch details, configure the network traffic types, and click **Create Distributed Switch**.
    a) To use a preconfigured profile, select one of the profiles and click **Next**.

Option	Description
Default Configuration	Provides a unified fabric for all traffic types using a single vSphere Distributed Switch.
Storage Traffic Separation	This profile creates two vSphere Distributed Switches with separate physical NICs. One switch is dedicated for storage traffic, while the other is used for all other traffic.
Custom Switch Configuration	Copy from a preconfigured profile or create a new distributed switch configuration. Multiple vSphere Distributed Switches can be configured. Each vSphere Distributed Switch can hold one or more network traffic configurations. Some network traffic types are mandatory. Switch configuration is incomplete until these mandatory traffic types are configured. VM Management, VCF Management, vMotion, and vSAN network types can be configured only once for a cluster.

    b) To create a custom configuration for the distributed switch, click **Create custom configuration**.

Option	Description
Traffic types for the distributed switch	• ESX Management • VCF Management • VM Management • vMotion • vSAN
Distributed Switch name	Enter a name for the distributed switch.
Uplinks / LAG configuration	VDS Uplinks represent the individual physical network adapters (vmnics) on an ESX host connecting to the physical switch. In this mode, the VDS manages load balancing and failover using software-based algorithms like Route Based on Physical NIC Load or Route Based on the Source of the Port ID. The physical switch ports do not need to be configured in a bond or channel; they operate as independent links.
	VDS LAG bundles multiple physical uplinks into a single logical channel using the Link Aggregation Control Protocol (LACP). This requires a corresponding LACP configuration on the physical ToR (Top-of-Rack) switches. The physical switch and VDS negotiate traffic hashing.
Number of uplinks	The number of uplinks dedicated to this distributed switch.
Map uplinks to physical NICs on host	Connects your logical virtual network to your physical network hardware. This mapping pairs a logical uplink (a software object on the distributed switch, with a physical adapter (hardware on the ESX host).

c) To configure the traffic types for the distributed switch, turn on the toggle for each traffic type and configure the settings.

Option	Description
Distributed port group name	Name of the distributed port group for this traffic type.
Load balancing	*Route Based on the Source of the Port ID*: the distributed switch maps port IDs to specific uplinks based on the virtual port where the traffic entered the switch.
	Route Based on Physical NIC Load (Default): the distributed switch monitors the saturation of the physical uplinks. If the primary uplinks exceed a set amount of utilization (75% default) for 30 seconds, the distributed switch moves some VM traffic to another active uplink.
	Route Based on Source MAC Hash: the distributed switch uses the VM's MAC address and hashes it to an uplink, which creates even distribution for VMs that might change port IDs but keep MAC addresses.
	Route Based on IP Hash: the distributed switch uses the source and destination IP address of every packet to decide which uplink to use. This allows a single VM to use multiple uplinks simultaneously. **Note:** You must configure LACP (Link Aggregation Control Protocol) on the physical Top-of-Rack switches.
	Use Explicit Failover Order: you define the order in which uplinks are used in case of failure.
Uplink	*Active*: This traffic type uses the uplink immediately during normal operation. In case of multiple active uplinks, traffic of this type is distributed across all active uplinks simultaneously.
	Standby: A standby uplink is connected, but no traffic is sent through it as long as at least one "Active" uplink is healthy. If all active uplinks fail, the distributed switch moves the standby uplink to active status.
	Unused: Traffic of this type never uses this uplink, regardless of the status of the remaining uplinks.

16. Review the deployment information.

    To make changes, navigate back to the relevant page in the deployment wizard. You can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage.

17. Review the validation information.

    The VCF Installer validates the information you provided in the deployment wizard and reports any errors or warnings.

Warnings	Review any warnings and either address them or click **Acknowledge** to proceed with deployment.
Errors	Review all failed validations and resolve the issue(s) using the remediation information provided in the error message. You can navigate back to the relevant pages in the deployment wizard to make updates and then re-run the validations. You can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage.

18. After addressing all validation errors and warnings, click **Deploy**.

    You can monitor the deployment progress. If there are errors, review the failed tasks in the Tasks panel, address the issue(s), and retry the deployment.

    CAUTION:

    After deployment completes, do not move the VCF management services VMs to a different resource pool or folder. If you move the VMs to a new resource pool or folder, subsequent patching, deployment, or scale out operations fail.

- License your environment. See Registering VCF Operations and a License Server with the VCF Business Services Console.
- Register your vCenter instance in VCF Operations. See Configuring a vCenter Account in VCF Operations.
- Click **Review Passwords** to review the passwords. To review passwords for past deployments, in the VCF Installer appliance, click the user name at the top-right corner of the window and select **Deployments**.

  Note:

  You can also retrieve some of the passwords using the VCF Operations API.
- (Optional) Download the JSON specification file for the deployment.

  Note:

  The JSON specification file does not include passwords.
- Deploy the remaining platform components that VCF Installer does not deploy automatically. See Completing the Deployment of Your Platform.
- For platform administration tasks such as managing identity and access, certificates, passwords, component lifecycle, and backup and restore of your vSphere Foundation platform, see the Fleet Management guide.
- For monitoring and troubleshooting tasks such as configuring alerts, analyzing logs, monitoring dashboards, and troubleshooting your vSphere Foundation platform, see the Infrastructure Operations guide.

Deploying VMware vSphere Foundation 9.1 Without VCF Management Services

Deploying VCF management services is not mandatory for VMware vSphere Foundation 9.1. You can build a fully operational 9.1 environment by deploying and configuring vSphere components manually without using VCF Installer.

Deploying and running vSphere Foundation 9.1 without VCF management services is a supported deployment model. The following functionalities are available without VCF management services.

- Compute, storage, networking, and administrative operations by using native vSphere management interfaces.
- Running, patching, and upgrading vSphere Foundation by using standard vSphere lifecycle mechanisms.

When using vSphere Foundation without VCF management services, the following functionalities are not available.

- Log management by using the log management component. See Components in VCF and vSphere Foundation.
- Binary management by using the software depot component. See Components in VCF and vSphere Foundation.
- Integrated lifecycle management of VCF Operations.

If you need VCF management services, later you can deploy VCF management services by using VCF Installer. After that you can deploy log management by using VCF Operations. See Deploy VCF Management Services and License Server for vSphere Foundation and Deploy Log Management.

Manual vSphere Foundation 9.1 Deployment

- Verify that you deployed and configured ESX hosts. See Installing and Setting Up ESX.

#	Component	9.1 Procedure
1	vCenter	Deploy vCenter Appliance
2	VCF Operations	1. Deploy VCF Operations 2. Configure a vCenter account in VCF Operations

#	Component	9.1 Procedure
3	License server	Deploy and configure the License Server

Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation

VCF Installer can deploy your platform by using a JSON file instead of the deployment wizard.

- Verify that you deployed VCF Installer version 9.1.1 or later. See Deploy VCF Installer.
- Download the required binaries for VMware Cloud Foundation or VMware vSphere Foundation. See Downloading Binaries to VCF Installer.
- Verify that you configured the required FQDNs and IP addresses for deployment. See VCF Components FQDNs and IP addresses.

You can download and complete the Planning and Preparation Workbook to help plan and gather all the information that is required for a VCF or VMware vSphere Foundation deployment. See Planning and Preparation. If you plan to use any existing components in your deployment, enter information about those components in the workbook.

A JSON specification file can be used to perform the following tasks.

- Deploying a new VCF fleet
- Deploying a new VCF Instance
- Deploying a new vSphere Foundation platform
- Deploying deferred components
- Converging existing vSphere infrastructure to a new VCF fleet
- Converging existing vSphere infrastructure to a new VCF Instance
- Converging existing vSphere infrastructure to a new vSphere Foundation platform
- Installing VCF Management Services and License Server for vSphere Foundation

If you are deploying multiple platforms, you might find it easier to start with an existing JSON specification file and modify it for the other deployments. See the VMware Cloud Foundation Installer API Reference Guide for information about the properties and parameters you can use to create a JSON specification file.

- For new deployments, see the following example JSON file with predefined values.
- For convergence, see the following example JSON file with predefined values.
- For deploying deferred components, see the examples in Deploying Deferred Components on Custom Networking.
- For installing VCF management services and license server for vSphere Foundation, see the following example JSON file with predefined values.

The following table provides guidance on certain properties that you specify, depending on the type of environment you want to create and what existing components you are using.

I want to...	`workflowType` property	VCF Operations `vcfOperationsSpec` `useExistingDeployment` property	vCenter `vcenterSpec` `useExistingDeployment` property	VCF management services `vspClusterSpec` ***	VCF Automation `vcfAutomationSpec` `useExistingDeployment` property	NSX Manager `nsxtSpec` `useExistingDeployment` property	License server `licenseServerSpec`	Identity broker `vidbSpec`
Deploy a new VCF fleet	VCF	false	false	true	true / false	n/a	true	true
Deploy a new VCF Instance	VCF_EXTEND	true	false	true	true	n/a	true *	true / false ****
Deploy deferred components	VCF_COMPLETE	true	false	false	true	false	true	n/a

I want to...	`workflowType` property	VCF Operations `vcfOperationsSpec` `useExistingDeployment` property	vCenter `vcenterSpec` `useExistingDeployment` property	VCF management services `vspClusterSpec` ***	VCF Automation `vcfAutomationSpec` `useExistingDeployment` property	NSX Manager `nsxtSpec` `useExistingDeployment` property	License server `licenseServerSpec`	Identity broker `vidbSpec`
Converge existing vSphere infrastructure to a new VCF fleet	VCF	true / false	true	true	true / false	true / false	true *	true *
Converge existing vSphere infrastructure to a new VCF instance	VCF_EXTEND	true	true	true	true	true / false	true *	true *
Deploy a new vSphere Foundation platform	VVF	false	false	true / false **	n/a	n/a	true	true
Converge existing vSphere infrastructure to a new vSphere Foundation platform	VVF	false	true	true / false **	n/a	n/a	true *	n/a
Install VCF Management Services and License Server for vSphere Foundation	VVF	true	true	true	false	false	true	n/a

* The component is required only when planning to converge an existing VCF Operations instance and the component is not already deployed and integrated with VCF Operations.

** You can deploy or converge to vSphere Foundation with or without VCF management services. For architectures without VCF management services, you must first apply a certain configuration to your VCF Installer appliance. The configuration step is provided in the procedure. You only perform that configuration for this specific deployment scenario.

*** You can specify the IP addresses and of VCF services runtime nodes by using a CIDR block, IP range, or list of IP addresses. VCF services runtime uses an internal CIDR block of 198.18.0.0/15. If this subnet is currently in use on your network, you must change the internal CIDR block for VCF management services to 240.0.0.0/15 or 250.0.0.0/15.

**** Identity broker is a mandatory component for the primary VCF Instance only. When deploying a subsequent VCF Instance, **`vidbSpec`** can be excluded from the JSON file.

CIDR Block

```
{
  "vspClusterSpec": {
    "ipv4Pool": {
      "cidr": "10.12.10.64/28",
      "excludedAddresses": [
        "10.12.10.90",
```

```
        "10.12.10.91"
      ]
    },
    "internalClusterCidrIpv4": "240.0.0.0/15"
  }
}
```

IP Range

```
{
  "vspClusterSpec": {
    "ipv4Pool": {
      "ipRange": {
        "startIpAddress": "10.12.10.86",
        "endIpAddress": "10.12.10.97"
      },
      "excludedAddresses": [
        "10.12.10.90",
        "10.12.10.91"
      ]
    },
    "internalClusterCidrIpv4": "240.0.0.0/15"
  }
}
```

List of IP Addresses

```
{
  "vspClusterSpec": {
    "ipv4Pool": {
      "addresses": [
        "10.12.10.86",
        "10.12.10.87",
        "10.12.10.88"
      ]
    },
    "internalClusterCidrIpv4": "240.0.0.0/15"
  }
}
```

1. If you want to deploy or converge to vSphere Foundation without VCF management services, reconfigure your VCF Installer to allow for that architecture.
   a) Open an SSH connection to the VCF Installer appliance and log in as root.
   b) Run the command to make the necessary configuration changes.

      ```
      echo 'explicit.management.components.deployment=true' >> /etc/vmware/vcf/domainmanager/
      application.properties
      ```

   c) Run the command to restart the affected service.

      ```
      systemctl restart domainmanager
      sleep 180
      curl localhost/domainmanager/about -> Expect json response and http 200
      ```

2. Refer to the *VMware Cloud Foundation Installer API Reference Guide* to create a JSON specification file using the information from your Planning and Preparation Workbook.
3. In a web browser, log in to the VCF Installer appliance at `https://installer_appliance_FQDN`.
   Enter the `admin@local` user and password that you provided when you deployed the appliance and then click **Log In**.

4. Click **Deploy Using JSON Spec** and upload the file.
5. Review the Summary and JSON Preview and click **Next**.
6. Review the validation information.

   The VCF Installer validates the information that you provided in the JSON specification and reports any errors or warnings.

Warnings	Review any warnings and either address them or click **Acknowledge** to proceed with deployment.
Errors	Review all failed validations and resolve the issues using the remediation information that is provided in the error message. You can modify and reupload the JSON specification file.

7. After addressing all validation errors and warnings, click **Deploy**.

   You can monitor the deployment progress. If there are errors, review the failed tasks in the Tasks panel, address the issue(s), and retry the deployment.

   CAUTION:

   After deployment completes, do not move the VCF management services or VCF Automation VMs to a different resource pool or folder. If you move the VMs to a new resource pool or folder, subsequent patching, deployment, or scale out operations fail.

- Download the JSON specification file.

  Note:

  The JSON specification file does not include passwords.
- License your environment. See Registering VCF Operations and a License Server with the VCF Business Services Console.
- Click **Review Passwords** to review the passwords. To review passwords for past deployments, in the VCF Installer appliance, click the user name at the top-right corner of the window and select **Deployments**.

  Note:

  You can also retrieve some of the passwords using the SDDC Manager API and VCF Operations API.
- To cover the platform after a failure, configure file-based backups for SDDC Manager and other management components. See Configure File-Based Backup Schedule for SDDC Manager.
- If you deployed VCF Operations in High Availability mode, to prevent VCF Operations cluster nodes from running on the same ESX host, apply a vSphere DRS anti-affinity rule.
- For platform administration tasks such as managing identity and access, certificates, passwords, component lifecycle, and backup and restore of your platform, see the Fleet Management guide.
- For monitoring and troubleshooting tasks such as configuring alerts, analyzing logs, monitoring dashboards, and troubleshooting your platform, see the Infrastructure Operations guide.

Troubleshooting VMware Cloud Foundation or vSphere Foundation Deployments

If you encounter errors during deployment, you can use the support bundle, log files, and the Supportability and Serviceability (SoS) Utility to help with troubleshooting.

Generate and Download a Support Bundle

If your current attempt to deploy a new VCF or VMware vSphere Foundation instance is not successful, you can generate and download a support bundle. The support bundle includes support logs from the components that you specified for the deployment.

1. In a web browser, log in to the VCF Installer appliance at `https://installer_appliance_FQDN`.
2. Enter the `admin@local` user and password that you provided when you deployed the appliance and then click **Log In**.
3. Click the user name at the top-right corner of the window and select **Support Bundle**.

4. Select the components to include in the bundle and click **GENERATE NEW SUPPORT BUNDLE**.
5. After the support bundle generation is completed, click **DOWNLOAD**.

Access Domain Manager Logs on the VCF Installer Appliance

If you experience validation or deployment failures, check the log files for the domain manager service on the VCF Installer appliance.

1. Log in to the VCF Installer appliance as **vcf** by using a Secure Shell (SSH) client or the vSphere Web Console.
2. Run the `su` command to switch to root account.
3. Access the domain manager service log at `/var/log/vmware/vcf/domainmanager/domainmanager.log`.

Recover Access to a Locked VCF Installer Appliance Account

If your local account is locked, you can recover access by restarting the `commonsvcs` service.

1. Log in to the VCF Installer appliance as `root` by using a Secure Shell (SSH) client or the vSphere Web Console.
2. Restart the `commonsvcs` service by running the `systemctl restart commonsvcs` command.
3. Wait approximately 5 minutes for the service to fully boot.

Retrieve VCF Management Services Passwords

After deploying the VCF management services components, you can retrieve the components' password.

1. Log in to SDDC Manager as `vcf` by using a Secure Shell (SSH) client.
2. Run the `su` command to switch to root account.
3. Turn on HTTP access to the domain manager's vault and restart the service.
   ```
   echo vcf.vault.http-access=true >> /etc/vmware/vcf/domainmanager/application.properties && systemctl
    restart domainmanager
   ```
4. Obtain the *`UUID`* of the deployment task.
   a. Log in to SDDC Manager UI with a user assigned the **Administrator** role.
   b. At the bottom of the page, expand the **Tasks** bar.
   c. Click the **Deploy VCF Management Components** task.
   d. Copy the *`UUID`* from the address bar of your Internet browser. For example, `9ccad9fd-a179-4538-822c-8bf6149ebc7d`.
5. Log back in to SDDC Manager as `vcf` by using a Secure Shell (SSH) client.
6. Use the *`UUID`* to copy the data for the deployment workflow into a temporary file.
   ```
   curl -X GET http://localhost/domainmanager/internal/vault/UUID | json_pp > /tmp/workflow.json
   ```
7. Obtain the `systemUserPassword` password from the `/tmp/workflow.json` file.
8. Delete the temporary file immediately after retrieving the password.
   ```
   shred -u /tmp/workflow.json
   ```
9. Turn off HTTP access to the domain manager's vault and restart the service.
   ```
   sed -i '/vcf.vault.http-access=true/d' /etc/vmware/vcf/domainmanager/application.properties && systemctl
    restart domainmanager
   ```

Supportability and Serviceability (SoS) Utility

You can run the SoS Utility on the VCF Installer and SDDC Manager appliances to perform health checks and collect logs for troubleshooting. After a successful deployment, run the SoS Utility from the SDDC Manager appliance. The SDDC Manager appliance supports a broader set of options than the VCF Installer appliance, including Fix-It-Up options, additional health checks, and additional log collection options. To see the options supported on a specific appliance, connect to it by SSH and run the `/opt/vmware/sddc-support/sos --help` command.

> **Note:**
>
> The SoS Utility is not a debug tool. It provides health check and log collection operations that you can use to facilitate troubleshooting.

To run the SoS Utility, SSH in to the appliance using the `vcf` account and then enter `su` to switch to the root user. Navigate to the `/opt/vmware/sddc-support` directory and type `./sos` followed by the options that are required for your desired operation.

```
./sos --option-1  --option-2...--option-n
```

In VCF Installer, the SoS Utility can collect support logs only for components from the latest deployment. The utility does not support logs from the following components: cloud proxy, VCF Operations for networks, telemetry, and license server. Bootstrap virtual machine logs for VCF management services and VCF Automation are collected only if a failure occurs.

Table 963: SoS Utility Help Options

Use the SoS Utility help to options to see information about the SoS tool itself.

Option	Description
`--help` or `-h`	Displays available options.
`--version` or `-v`	Displays the SoS Utility version.

Table 964: SoS Utility Generic Options

Option	Description
`--history`	Displays the last 20 SoS log locations.
`--force`	Allows SoS operations to run while workflows are in progress. Use this option with caution.
`--setup-json` *SETUP_JSON*	Uses a custom setup JSON file for log collection. See `setup.sample.json` for reference. On the VCF Installer appliance, the JSON file is located in the `/opt/vmware/sddc-support/setup.sample.json` directory. Supported components for this option: vCenter, ESX, NSX, VCF management services, VCF Automation, and VCF Operations. Sample command: `/opt/vmware/sddc-support/sos --setup-json setup.sample.json`
`--log-folder` *LOGFOLDER*	Specifies a name for the log directory.
`--log-dir` *LOGDIR*	Specifies the directory in which to store logs.
`--enable-stats`	Enables SoS execution statistics collection.
`--debug-mode`	Runs the SoS Utility in debug mode.
`--zip`	Creates a zipped TAR file of the output.
`--short`	Displays health check results for failures and warnings only.
`--skip-known-host-check`	Skips the SSL thumbprint check for hosts in the known hosts list.

The following options are available on the SDDC Manager appliance only.

Option	Description
`--domain-name` *DOMAINNAME*	Specifies the domain on which to run the SoS operation. Use `--domain-name ALL` for all domains. Default: `MANAGEMENT`. Add `--skip-cert-check` to include health checks for free or unassigned hosts.
`--clusternames` *CLUSTERNAMESß*	Specifies one or more cluster names for ESX or WCP log collection in a domain. Separate multiple cluster names with commas, for example, `--clusternames cluster1,cluster2`. Ignored when `--domain-name ALL` is used.
`--include-free-hosts`	Collects logs for free ESX hosts.

Table 965: SoS Utility VCF Summary Options

Option	Description	Appliance
`--get-vcf-summary`	Displays the VCF Summary.	• VCF Installer • SDDC Manager
`--get-vcf-tasks-summary`	Displays the VCF Tasks Summary.	• VCF Installer • SDDC Manager
`--get-vcf-services-summary`	Displays the VCF services details.	• SDDC Manager

SoS Utility Log Collection Options

Option	Description	Appliance
`--esx-logs`	Collects ESX logs from each host available in the deployment.	• VCF Installer • SDDC Manager
`--vc-logs`	Collects vCenter logs from each instance available in the deployment.	• VCF Installer • SDDC Manager
`--sddc-manager-logs`	Collects SDDC Manager logs.	• VCF Installer • SDDC Manager
`--nsx-logs`	Collects NSX logs.	• VCF Installer • SDDC Manager
`--vcf-mgmt-svcs-logs`	Collects VCF management services logs.	• VCF Installer • SDDC Manager
`--automation-logs`	Collects VCF Automation support logs.	• VCF Installer • SDDC Manager
`--operations-logs`	Collects VCF Operations support logs. Cloud proxy logs are not included. On the VCF Installer appliance, use this option directly. On the SDDC Manager appliance, use `--setup-json` to collect VCF Operations logs.	• VCF Installer • SDDC Manager
`--api-logs`	Collects output from APIs.	• VCF Installer • SDDC Manager
`--vm-screenshots`	Collects screenshots of all virtual machines.	• VCF Installer • SDDC Manager
`--no-clean-old-logs`	Retains existing log folders instead of removing them. By default, before writing the output to the directory, if present, the tool deletes the output files from the previous run. If you want to retain the older output files, specify this option.	• VCF Installer • SDDC Manager
`--no-health-check`	Skips the health check that runs as part of log collection.	• VCF Installer • SDDC Manager
`--collect-all-logs`	Collects logs for all components. On the SDDC Manager appliance, WCP and system debug logs are excluded. Use `--domain-name DOMAINNAME` for a specific domain, or `--domain-name ALL` for all domains. Default domain is `MANAGEMENT`.	• VCF Installer • SDDC Manager
`--vcf-installer-logs`	Collects VCF Installer appliance support logs.	• VCF Installer
`--wcp-logs`	Collects WCP cluster logs.	• SDDC Manager
`--hcx-manager-logs`	Collects HCX Manager logs.	• SDDC Manager
`--system-debug-logs`	Collects system logs for debugging abnormal scenarios such as memory leaks.	• SDDC Manager
`--test`	Verifies the integrity of collected logs.	• SDDC Manager

Table 966: SoS Utility Fix-It-Up options

The following options are available on the SDDC Manager appliance only.

Option	Description
`--enable-ssh-esxi`	Enables SSH on all ESX nodes in a domain. Use `--domain-name DOMAINNAME` to specify a domain, or `--domain-name ALL` for all domains. Default: `MANAGEMENT`.
`--disable-ssh-esxi`	Disables SSH on all ESX nodes in a domain. Use `--domain-name DOMAINNAME` to specify a domain or `--domain-name ALL` for all domains. Default: `MANAGEMENT`.
`--enable-ssh-vc`	Enables SSH on the vCenter Server instance in a domain. Use `--domain-name DOMAINNAME` to specify a domain or `--domain-name ALL` for all domains. Default: `MANAGEMENT`.
`--disable-ssh-vc`	Disables SSH on the vCenter Server instance in a domain. Use `--domain-name DOMAINNAME` to specify a domain or `--domain-name ALL` for all domains. Default: `MANAGEMENT`.
`--enable-lockdown-esxi`	Enables normal lockdown mode on all ESX nodes in a domain. Use `--domain-name DOMAINNAME` to specify a domain or `--domain-name ALL` for all domains. Combine with `--clusternames` to operate at the cluster level. Default: `MANAGEMENT`.
`--disable-lockdown-esxi`	Disables normal lockdown mode on all ESX nodes in a domain. Use `--domain-name DOMAINNAME` to specify a domain or `--domain-name ALL` for all domains. Combine with `--clusternames` to operate at the cluster level. Default: `MANAGEMENT`.
`--ondemand-service` *`ONDEMANDSERVICE`*	Executes commands on ESX hosts, vCenter instances, or SDDC Manager for a given domain. Requires an input `.yml` file. See `/opt/vmware/sddc-support/ondemand_command_sample.yml` for reference. Use this option with caution, as it can alter the state of the SDDC Manager environment and its components. Use `--domain-name` to specify the target domain. Default: `MANAGEMENT`.
`--refresh-ssh-keys`	Refreshes SSH keys.

SoS Utility Health Check Options

The SoS Utility can be used to perform health checks on various components, including connectivity, compute, and storage. Health checks do not include VCF management services, VCF Automation, and VCF Operations.

> **Note:**
>
> SoS Utility operations on the VCF Installer appliance are strictly intended for troubleshooting during initial deployment. Once deployment is complete, execute all SoS health checks and operations from the SDDC Manager appliance only. Running the SoS Utility from VCF Installer in a Day-N environment may lead to unexpected results, as VCF Installer is unaware of post-deployment configuration changes.

Option	Description	Appliance
`--health-check`	Performs all available health checks.	• VCF Installer • SDDC Manager
`--connectivity-health`	Performs a connectivity health check to inspect whether the different components of the system such as the ESX hosts, vCenter instances, NSX Manager nodes, and SDDC Manager can be pinged.	• VCF Installer • SDDC Manager
`--services-health`	Performs a services health check to confirm whether services are running.	• VCF Installer • SDDC Manager

Option	Description	Appliance
`--compute-health`	Performs a compute health check.	• VCF Installer • SDDC Manager
`--ntp-health`	Verifies that the time on components is synchronized with the NTP server. Also verifies that the hardware and software timestamps of ESX hosts are within 5 minutes of the SDDC Manager appliance.	• VCF Installer • SDDC Manager
`--general-health`	Verifies ESX entries across all sources, checks the Postgres DB operational status for hosts, checks ESX for error dumps, and gets NSX Manager and cluster status.	• VCF Installer • SDDC Manager
`--certificate-health`	Verifies that component certificates are valid and reports their expiration status. Certificates expiring in more than 30 days are shown as green. Certificates expiring in 1–30 days are shown as yellow.	• VCF Installer • SDDC Manager
`--get-host-ips`	Retrieves server information.	• VCF Installer • SDDC Manager
`--storage-health`	Performs a check on the vSAN disk health of the ESX hosts and vSphere clusters. Can be combined with `--run-vsan-checks`.	• SDDC Manager
`--run-vsan-checks`	Runs a VM creation test to verify the vSAN cluster health. Running the test creates a virtual machine on each host in the vSAN cluster. This option cannot be run on its own and must be combined with `--health-check` or `--services-health`.	• SDDC Manager
`--dns-health`	Performs a forward and reverse DNS health check.	• SDDC Manager
`--get-inventory-info`	Retrieves inventory details for SDDC Manager, such as vCenter, NSX, SDDC Manager, and ESX hosts. Optionally, add the flag `--domain-name ALL` to return details for all workload domains.	• SDDC Manager
`--password-health`	Checks password expiry status for components.	• SDDC Manager
`--version-health`	Checks the version of bill of materials (BOM) components. Compares the SDDC Manager inventory, the actual installed BOM component version, and the BOM component versions to detect any drift.	• SDDC Manager
`--json-output-dir` *`JSONDIR`*	Writes health check results to a JSON file in the specified *`JSONDIR`* directory.	• SDDC Manager

Deploying Deferred Components on Custom Networking

If you did not deploy VCF Operations and VCF Automation during the initial deployment of a VCF Instance, you can use the VCF Installer UI or the VCF Installer API to complete their deployment process to a custom network. Besides VCF Operations and VCF Automation, the workflow deploys cloud proxy and license server. You can use custom networking to separate user-facing networks from management networks to meet regulatory and security requirements.

By default, VCF Installer configures all VCF Operations and VCF Automation nodes to use the same distributed port group as the vCenter instance. By selecting the **Deploy VCF Operations and VCF Automation to a specific vSphere distributed switch or NSX segment** option you skip the deployment of these components when deploying a new VCF fleet.

Later, to deploy VCF Operations and VCF Automation on custom networking and complete the deployment of the management domain, you can create a separate network that is backed by a distributed port group or NSX segment and use this network to deploy the VCF components. After you complete the custom networking configuration, you select the **Deploy deferred components** option from the VCF Installer UI and complete the deployment process. Alternatively, you can complete the deployment by using the VCF Installer API.

For information about the network deployment locations, see Network Deployment Options for VCF Components.

You can deploy the VCF management components to the following custom networks:

- Deploy Deferred Components on the Shared Management Network

- Deploy Deferred Components on a Dedicated Management Network
- Deploy Deferred Components on NSX Overlay Segments

Deploy Deferred Components on the Shared Management Network

If you skipped deploying VCF Operations and VCF Automation during the initial deployment of a VCF Instance, you can use the VCF Installer user interface or the VCF Installer API to deploy these components on the same distributed port group as the VCF core components.

- Review Design Guide. See VCF Management Shared VLAN Network Model.
- Verify that you deployed VCF Installer version 9.1.1 or later. See Deploy VCF Installer.
- Download binaries for the VCF management components to VCF Installer. See Downloading Binaries to VCF Installer.
- Verify that the initial VCF fleet deployment is complete.
- If you plan to use a load balancer in your VCF environment, verify it is deployed and configured.
- Verify that you configured the required FQDNs and IP addresses for deployment. See VCF Components FQDNs and IP addresses.
- (API only) Verify that you have a machine with installed `curl` and `jq`, and network access to the VCF Installer appliance.

Follow this procedure to configure VCF Operations, VCF Automation, cloud proxy, and license server to use the same distributed port group as the VCF core components - usually the vCenter instance.

VCF Installer UI

1. In a web browser, log in to the VCF Installer appliance at `https://installer_appliance_FQDN`.

   Note:

   Enter the `admin@local` user and password you provided when you deployed the appliance and then click **Log In**.
2. Click **Deployment Wizard** > **VMware Cloud Foundation**.
3. Review the introductory information and click **Next**.
4. Select **Deploy deferred components**.
5. Enter the details of the existing SDDC Manager and click **Connect**.
6. Confirm the certificate thumbprints.

7. Select a deployment model, scale for your VCF Instance, and DNS and NTP servers.

Deployment model	You can choose the deployment model for new appliances that get deployed. **High Availability** is recommended for production environments. **Note:** The deployment model that you select does not impact existing appliances that you are using in your VCF fleet or VCF Instance. See the VMware Cloud Foundation Design documentation for more information. **Note:** New appliances that get deployed are automatically configured to comply with Federal Information Processing Standards (FIPS).
Size	You can choose the scale of the components in your VCF Instance. • Small – In VCF 9.1, this is the only available option for the simple deployment model. – In VCF 9.1.1 and later, this is the default option for the simple and HA deployment models. • Medium – In VCF 9.1, this is the default option for the HA deployment model. – In VCF 9.1.1 and later, this is an option for the HA deployment model. • Large – In VCF 9.1 and later, this is an option for the HA deployment model. For more information about deployment sizes, see VCF Fleet Sizing Models.
Customer Experience Improvement Program (CEIP)	Select an option to activate or deactivate CEIP across all deployed components. VMware's Customer Experience Improvement Program (CEIP) collects technical information from participating customers to provide better support. See Customer Experience Improvement Program for more information.
Autogenerate passwords	Starting with VCF 9.1, select this option to let VCF Installer generate secure passwords for all deployed components. You can retrieve the passwords after the deployment begins. If you do not select this option, you must provide each password manually.
Skip VCF Automation	In VCF 9.1.1 and later, select this option to skip the deployment of VCF Automation during the deferred components workflow. If you select this option, the UI does not prompt for VCF Automation FQDNs or IP addresses. You can deploy VCF Automation later as a Day-N operation. See Deploying VCF Automation as a Day-N Operation.

8. Provide networking details.
   a. Select the same distributed port group that the VCF core components use.

Note:

The VCF Management components network traffic will run on this port group.

b. Enter the network gateway address.
c. Enter IP details for VCF Automation.
   If you chose to deploy VCF Automation later as a Day-N operation, this option is not displayed.
   - For deployments version 9.1.0.0 through 9.1.0.300, you specify a contiguous IP range for the VCF Automation nodes with 5 IP addresses.
   - For deployments version 9.1.0.400 and later, you specify 6 IP addresses as contiguous or non-contiguous IP range.

   Note:

   The sixth additional IP address is requested by the UI but is not consumed by the deployment.

9. Enter the details for the VCF management components.

Table 967: VCF Management Configuration Details

Each Fully Qualified Domain Name (FQDN) must point to a unique IP address.

Note:

For VCF components that support the dual-stack model, the FQDNs must resolve to IPv4 and IPv6 addresses. See Dual Stack IP Addressing Model.

Component	Field	Description
VCF Operations	Primary node FQDN	Enter the FQDN of the VCF Operations primary node.
	Replica node FQDN	Enter the FQDN of the VCF Operations replica node.
	Data node FQDN	Enter the FQDN of the VCF Operations data node. This field appears only when deploying VCF in High Availability mode with a Medium or Large deployment size.
Load Balancer	Load balancer FQDN	(Optional) Enter the FQDN of the load balancer that distributes the VCF Operations traffic. This field appears only when deploying VCF in High Availability mode.
Cloud Proxy	Cloud Proxy FQDN	Enter the FQDN of the cloud proxy. Cloud proxy collects and monitors data locally in a VCF Instance.
License Server	License Server FQDN	Enter the FQDN of the license server. A license server is added to a VCF Operations instance to store the licenses you want to assign from this VCF Operations instance. You must add at least one license server to each VCF Operations instance that you use for license management. You can deploy license server as IPv4 only. In VCF 9.1.1 and later, after the deployment, you can change the networking configuration to IPv6. See Configure IPv6 on a License Server.
VCF Automation *	VCF Automation FQDN	Enter the FQDN to assign to the VCF Automation instance. Later, you use it to access the VCF Automation user interface. **Note:** Do not use capital letters in the FQDN.

Component	Field	Description
	VCF services runtime FQDN	Enter the FQDN for VCF services runtime to connect VCF Automation with the VCF fleet lifecycle component that manages the lifecycle of VCF Automation. The FQDN must be dedicated to the runtime of the VCF Automation cluster, not the VCF management services cluster. The hostname from this FQDN is prefixed to the names of its node VMs and related objects. **Note:** Do not use capital letters in the FQDN.
	Internal cluster CIDR IPv4	Select the VCF services runtime internal IPv4 network. • 198.18.0.0/15 • 240.0.0.0/15 • 250.0.0.0/15 Different VCF services runtime instances on the same network can work with the same internal network. For example, you can have the two instances of VCF services runtime for your VCF management services and VCF Automation using the same internal network.

* If you chose to deploy VCF Automation later as a Day-N operation, these options are not displayed.

10. Review the deployment information.

    Note:

    To make changes, navigate back to the relevant page in the deployment wizard. You can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage.

11. Validate and deploy the VCF management components.

VCF Installer API

1. Authenticate to the VCF Installer API.
   a. On the machine with access to VCF Installer, set the environment variables. Replace all placeholder values with values specific to your environment.

      ```
      vcfInstallerFqdn='vcf-installer.rainpole.io'
      vcfInstallerUser='admin@local'
      vcfInstallerPass='<password>'
      ```

   b. Authenticate and capture the access token.

      ```
      vcfInstallerToken=$(curl -k -X POST \
        https://$vcfInstallerFqdn/v1/tokens \
        -H 'Content-Type: application/json' \
        -d '{"username": "'"$vcfInstallerUser"'", "password":
       "'"$vcfInstallerPass"'"}' \
        | jq -r '.accessToken')
      ```

   c. (Optional) Verify that the token is not empty.

      ```
      echo $vcfInstallerToken
      ```

2. Create a JSON file with information about the VCF management components to use for deployment and save the file. For example, as `deferred-components.json`. Replace all placeholder values with values specific to your environment.

   ```
   {
     "sddcId": "MGMT",
     "workflowType": "VCF_COMPLETE",
     "vcenterSpec": {
       "vcenterHostname": "vcenter-1.vrack.rainpole.io",
       "sslThumbprint": "<valid-thumbprint>",
       "adminUserSsoPassword": "<password>",
       "adminUserSsoUsername": "administrator@vsphere.local",
       "useExistingDeployment": true
     },
   ```

```
  "sddcManagerSpec": {
    "useExistingDeployment": true,
    "sslThumbprint": "<valid-thumbprint>",
    "rootPassword": "<password>",
    "hostname": "sddc-manager.vrack.rainpole.io",
    "sshPassword": "<password>",
    "localUserPassword": "<password>"
  },
  "ceipEnabled": true,
  "vcfInstanceName": "sddcId-1001",
  "vcfOperationsSpec": {
      "applianceSize": "medium",
      "useExistingDeployment": false,
      "adminUserPassword": "<password>",
      "nodes": [
          {
              "hostname": "flt-ops01a.rainpole.io",
              "rootUserPassword": "<password>",
              "type": "master"
          },
          {
              "hostname": "flt-ops01b.rainpole.io",
              "rootUserPassword": "<password>",
              "type": "replica"
          },
          {
              "hostname": "flt-ops01c.rainpole.io",
              "rootUserPassword": "<password>",
              "type": "data"
          }
      ]
  },
  "vcfOperationsCollectorSpec": {
      "hostname": "vcfopscp-1.vrack.rainpole.io",
      "rootUserPassword": "<password>",
      "applianceSize": "small",
      "useExistingDeployment": false
  },
  "licenseServerSpec": {
    "hostname": "ls.vrack.rainpole.io"
  },
  "vcfAutomationSpec": {
      "ipPool": [
          "10.11.10.51",
          "10.11.10.52",
          "10.11.10.53",
          "10.11.10.54",
          "10.11.10.55"
      ],
      "nodePrefix": "flt-auto",
      "hostname": "flt-auto01.rainpole.io",
      "platformFqdn": "vcfa-platform1.vrack.rainpole.io",
      "internalClusterCidr": "198.18.0.0/15",
      "adminUserPassword": "<password>"
  }
}
```

3. Validate your JSON file.

```
curl -k -X POST \
  https://$vcfInstallerFqdn/v1/sddcs/validations \
  -H "Authorization: Bearer $vcfInstallerToken" \
  -H 'Content-Type: application/json' \
  -d @deferred-components.json
```

4. Check the response of the validation.

```
curl -k -s -X GET \
  https://$vcfInstallerFqdn/v1/sddcs/validations/latest \
  -H "Authorization: Bearer $vcfInstallerToken" \
  -H 'Content-Type: application/json' \
  | jq '{executionStatus: .executionStatus, resultStatus: .resultStatus, failed:
 [.validationChecks[] | select(.resultStatus == "FAILED")]}'
```

5. Deploy the VCF management components.

```
curl -k -X POST \
  https://$vcfInstallerFqdn/v1/sddcs \
  -H "Authorization: Bearer $vcfInstallerToken" \
  -H 'Content-Type: application/json' \
  -d @deferred-components.json
```

6. (Optional) Monitor the deployment.

```
curl -k -X GET \
  https://$vcfInstallerFqdn/v1/sddcs \
  -H "Authorization: Bearer $vcfInstallerToken" \
  -H 'Content-Type: application/json'
```

7. When the deployment completes, open the VCF Operations UI to apply licensing and perform other post-deployment configuration.

Deploy Deferred Components on a Dedicated Management Network

If you skipped deploying VCF Operations and VCF Automation during the initial deployment of a VCF Instance, you can use the VCF Installer user interface or the VCF Installer API to deploy these components on a dedicated distributed port group.

- Review Design Guide. See VCF Management Dedicated VLAN Network Model.
- Verify that you deployed VCF Installer version 9.1.1 or later. See Deploy VCF Installer.
- Download binaries for the VCF management components to VCF Installer. See Downloading Binaries to VCF Installer.
- Verify that the initial VCF fleet deployment is complete.
- Create a distributed port group with a dedicated VLAN on the management vCenter vSphere Distributed Switch.
- If you plan to use a load balancer in your VCF environment, verify it is deployed and configured.
- Verify that you configured the required FQDNs and IP addresses for deployment. See VCF Components FQDNs and IP addresses.
- (API only) Verify that you have a machine with installed `curl` and `jq`, and network access to the VCF Installer appliance.

Follow this procedure to configure VCF Operations, VCF Automation, and license server to use a different distributed port group than the VCF core components that use the same distributed port group as the vCenter instance. The cloud proxy appliances are deployed to the VM management network.

VCF Installer UI

1. In a web browser, log in to the VCF Installer appliance at `https://installer_appliance_FQDN`.

   Note:

   Enter the `admin@local` user and password you provided when you deployed the appliance and then click **Log In**.
2. Click **Deployment Wizard** > **VMware Cloud Foundation**.
3. Review the introductory information and click **Next**.
4. Select **Deploy deferred components**.
5. Enter the details of the existing SDDC Manager and click **Connect**.

6. Confirm the certificate thumbprints.
7. Select a deployment model, scale for your VCF Instance, and DNS and NTP servers.

Deployment model	You can choose the deployment model for new appliances that get deployed. **High Availability** is recommended for production environments. **Note:** The deployment model that you select does not impact existing appliances that you are using in your VCF fleet or VCF Instance. See the VMware Cloud Foundation Design documentation for more information. **Note:** New appliances that get deployed are automatically configured to comply with Federal Information Processing Standards (FIPS).
Size	You can choose the scale of the components in your VCF Instance. • Small – In VCF 9.1, this is the only available option for the simple deployment model. – In VCF 9.1.1 and later, this is the default option for the simple and HA deployment models. • Medium – In VCF 9.1, this is the default option for the HA deployment model. – In VCF 9.1.1 and later, this is an option for the HA deployment model. • Large – In VCF 9.1 and later, this is an option for the HA deployment model. For more information about deployment sizes, see VCF Fleet Sizing Models.
Customer Experience Improvement Program (CEIP)	Select an option to activate or deactivate CEIP across all deployed components. VMware's Customer Experience Improvement Program (CEIP) collects technical information from participating customers to provide better support. See Customer Experience Improvement Program for more information.
Autogenerate passwords	Starting with VCF 9.1, select this option to let VCF Installer generate secure passwords for all deployed components. You can retrieve the passwords after the deployment begins. If you do not select this option, you must provide each password manually.
Skip VCF Automation	In VCF 9.1.1 and later, select this option to skip the deployment of VCF Automation during the deferred components workflow. If you select this option, the UI does not prompt for VCF Automation FQDNs or IP addresses. You can deploy VCF Automation later as a Day-N operation. See Deploying VCF Automation as a Day-N Operation.

8. Select a network.
   a. Select a dedicated distributed port group.

Note:

The VCF Management components network traffic will run on this port group.

b. Enter the network gateway address.
c. Enter IP details for VCF Automation.
   If you chose to deploy VCF Automation later as a Day-N operation, this option is not displayed.
   - For deployments version 9.1.0.0 through 9.1.0.300, you specify a contiguous IP range for the VCF Automation nodes with 5 IP addresses.
   - For deployments version 9.1.0.400 and later, you specify 6 IP addresses as contiguous or non-contiguous IP range.

   Note:

   The sixth additional IP address is requested by the UI but is not consumed by the deployment.

9. Enter the details for the VCF management components.

Table 968: VCF Management Configuration Details

Each Fully Qualified Domain Name (FQDN) must point to a unique IP address.

Note:

For VCF components that support the dual-stack model, the FQDNs must resolve to IPv4 and IPv6 addresses. See Dual Stack IP Addressing Model

Component	Field	Description
VCF Operations	Primary node FQDN	Enter the FQDN of the VCF Operations primary node.
	Replica node FQDN	Enter the FQDN of the VCF Operations replica node.
	Data node FQDN	Enter the FQDN of the VCF Operations data node. This field appears only when deploying VCF in High Availability mode with a Medium or Large deployment size.
Load Balancer	Load balancer FQDN	(Optional) Enter the FQDN of the load balancer that distributes the VCF Operations traffic. This field appears only when deploying VCF in High Availability mode.
Cloud Proxy	Cloud Proxy FQDN	Enter the FQDN of the cloud proxy. Cloud proxy collects and monitors data locally in a VCF Instance.
License Server	License Server FQDN	Enter the FQDN of the license server. A license server is added to a VCF Operations instance to store the licenses you want to assign from this VCF Operations instance. You must add at least one license server to each VCF Operations instance that you use for license management. You can deploy license server as IPv4 only. In VCF 9.1.1 and later, after the deployment, you can change the networking configuration to IPv6. See Configure IPv6 on a License Server.
VCF Automation *	VCF Automation FQDN	Enter the FQDN to assign to the VCF Automation instance. Later, you use it to access the VCF Automation user interface. **Note:** Do not use capital letters in the FQDN.

Component	Field	Description
	VCF services runtime FQDN	Enter the FQDN for VCF services runtime to connect VCF Automation with the VCF fleet lifecycle component that manages the lifecycle of VCF Automation. The FQDN must be dedicated to the runtime of the VCF Automation cluster, not the VCF management services cluster. The hostname from this FQDN is prefixed to the names of its node VMs and related objects. **Note:** Do not use capital letters in the FQDN.
	Internal cluster CIDR IPv4	Select the VCF services runtime internal IPv4 network. • 198.18.0.0/15 • 240.0.0.0/15 • 250.0.0.0/15 Different VCF services runtime instances on the same network can work with the same internal network. For example, you can have the two instances of VCF services runtime for your VCF management services and VCF Automation using the same internal network.

* If you chose to deploy VCF Automation later as a Day-N operation, these options are not displayed.

10. Review the deployment information.

    Note:

    To make changes, navigate back to the relevant page in the deployment wizard. You can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage.

11. Validate and deploy the VCF management components.

VCF Installer API

1. Authenticate to the VCF Installer API.
   a. On the machine with access to VCF Installer, set the environment variables. Replace all placeholder values with values specific to your environment.

      ```
      vcfInstallerFqdn='vcf-installer.rainpole.io'
      vcfInstallerUser='admin@local'
      vcfInstallerPass='<password>'
      ```

   b. Authenticate and capture the access token.

      ```
      vcfInstallerToken=$(curl -k -X POST \
        https://$vcfInstallerFqdn/v1/tokens \
        -H 'Content-Type: application/json' \
        -d '{"username": "'"$vcfInstallerUser"'", "password":
       "'"$vcfInstallerPass"'"}' \
        | jq -r '.accessToken')
      ```

   c. (Optional) Verify that the token is not empty.

      ```
      echo $vcfInstallerToken
      ```

2. Create a JSON file with information about the VCF management components to use for deployment and save the file. For example, as `deferred-components.json`.

   Replace all placeholder values with values specific to your environment. For example, replace *<port-group-name>* with the name of the dedicated distributed port group.

   ```
   {
     "sddcId": "MGMT",
     "workflowType": "VCF_COMPLETE",
     "vcenterSpec": {
       "vcenterHostname": "vcenter-1.vrack.rainpole.io",
       "sslThumbprint": "<valid-thumbprint>",
       "adminUserSsoPassword": "<password>",
       "adminUserSsoUsername": "administrator@vsphere.local",
       "useExistingDeployment": true
   ```

```
},
"sddcManagerSpec": {
  "useExistingDeployment": true,
  "sslThumbprint": "<valid-thumbprint>",
  "rootPassword": "<password>",
  "hostname": "sddc-manager.vrack.rainpole.io",
  "sshPassword": "<password>",
  "localUserPassword": "<password>"
},
"ceipEnabled": true,
"vcfInstanceName": "sddcId-1001",
"vcfOperationsSpec": {
    "applianceSize": "medium",
    "useExistingDeployment": false,
    "adminUserPassword": "<password>",
    "nodes": [
        {
            "hostname": "flt-ops01a.rainpole.io",
            "rootUserPassword": "<password>",
            "type": "master"
        },
        {
            "hostname": "flt-ops01b.rainpole.io",
            "rootUserPassword": "<password>",
            "type": "replica"
        },
        {
            "hostname": "flt-ops01c.rainpole.io",
            "rootUserPassword": "<password>",
            "type": "data"
        }
    ]
},
"vcfOperationsCollectorSpec": {
    "hostname": "vcfopscp-1.vrack.rainpole.io",
    "rootUserPassword": "<password>",
    "applianceSize": "small",
    "useExistingDeployment": false
},
"licenseServerSpec": {
  "hostname": "ls.vrack.rainpole.io"
},
"vcfAutomationSpec": {
    "ipPool": [
        "10.11.10.51",
        "10.11.10.52",
        "10.11.10.53",
        "10.11.10.54",
        "10.11.10.55"
    ],
    "nodePrefix": "flt-auto",
    "hostname": "flt-auto01.rainpole.io",
    "platformFqdn": "vcfa-platform1.vrack.rainpole.io",
    "internalClusterCidr": "198.18.0.0/15",
    "adminUserPassword": "<password>"
},
"vcfManagementComponentsInfrastructureSpec": {
```

```
            "xRegionNetwork": {
            "networkName": "<port-group-name>",
            "subnetMask": "255.255.255.0",
            "gateway": "10.11.99.1"
          }
        }
      }
```

3. Validate your JSON file.

```
curl -k -X POST \
  https://$vcfInstallerFqdn/v1/sddcs/validations \
  -H "Authorization: Bearer $vcfInstallerToken" \
  -H 'Content-Type: application/json' \
  -d @deferred-components.json
```

4. Check the response of the validation.

```
curl -k -s -X GET \
  https://$vcfInstallerFqdn/v1/sddcs/validations/latest \
  -H "Authorization: Bearer $vcfInstallerToken" \
  -H 'Content-Type: application/json' \
  | jq '{executionStatus: .executionStatus, resultStatus: .resultStatus, failed:
 [.validationChecks[] | select(.resultStatus == "FAILED")]}'
```

5. Deploy the VCF management components.

```
curl -k -X POST \
  https://$vcfInstallerFqdn/v1/sddcs \
  -H "Authorization: Bearer $vcfInstallerToken" \
  -H 'Content-Type: application/json' \
  -d @deferred-components.json
```

6. (Optional) Monitor the deployment.

```
curl -k -X GET \
  https://$vcfInstallerFqdn/v1/sddcs \
  -H "Authorization: Bearer $vcfInstallerToken" \
  -H 'Content-Type: application/json'
```

7. When the deployment completes, open the VCF Operations UI to apply licensing and perform other post-deployment configuration.

Deploy Deferred Components on NSX Overlay Segments

If you skipped deploying VCF Operations and VCF Automation during the initial deployment of a VCF Instance, you can use the VCF Installer user interface or the VCF Installer API to deploy these components to a specific NSX segment. The cloud proxy appliances are deployed to the VM management network.

- Review Design Guide. See VCF Management Dedicated VLAN and NSX Overlay Segment Network Model.
- Verify that you deployed VCF Installer version 9.1.1 or later. See Deploy VCF Installer.
- Download binaries for the VCF management components to VCF Installer. See Downloading Binaries to VCF Installer.
- Verify that the initial VCF fleet deployment is complete.
- Verify that you configured the required FQDNs and IP addresses for deployment. See VCF Components FQDNs and IP addresses.
- (API only) Verify that you have a machine with installed `curl` and `jq`, and network access to the VCF Installer appliance.
- Create an NSX Edge cluster. See Create an NSX Edge Cluster and Add Edge Nodes.
- If you plan to import the local NSX Manager into an NSX Global Manager to stretch a network across regions, do not deploy a centralized transit gateway. The centralized transit gateway contains objects that cannot be imported into NSX Global Manager. See Broadcom knowledge base article 439149.
- Add an NSX tier-0 gateway. See Add an NSX Tier-0 Gateway.
- Create an NSX tier-1 gateway. See Add an NSX Tier-1 Gateway.
  - Set the HA mode to **Active Standby**.

  - Linked to the tier-0 gateway.
  - Select the Edge cluster you created.
  - Under **Route Advertisement**, select **All Connected Segments and Service Ports**.
- Create an NSX segment on the management domain overlay transport zone. See Create an NSX Segment.
  - For the **Connected Gateway**, select the tier-1 gateway.
  - For the **Transport Zone**, select the overlay transport zone.
- If you plan to use a load balancer in your VCF environment, verify it is deployed and configured.

Follow this procedure to configure VCF Operations, VCF Automation, and license server to use an NSX overlay segment. The cloud proxy appliances are deployed to the VM management network.

Note:

VCF services runtime uses an internal CIDR block of 198.18.0.0/15. If this subnet is currently in use on your network, you must change the internal CIDR block for VCF management services to 240.0.0.0/15 or 250.0.0.0/15. You can specify the internal network to be used during the deployment by using a JSON specification file for VCF and vSphere Foundation. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation. Alternatively, you can specify the internal network by using the SDDC Manager API for VCF management services or by using the fleet lifecycle API for VCF Automation.

VCF Installer UI

1. In a web browser, log in to the VCF Installer appliance at `https://installer_appliance_FQDN`.

   Note:

   Enter the `admin@local` user and password you provided when you deployed the appliance and then click **Log In**.
2. Click **Deployment Wizard** > **VMware Cloud Foundation**.
3. Review the introductory information and click **Next**.
4. Select **Deploy deferred components**.
5. Enter the details of the existing SDDC Manager and click **Connect**.
6. Confirm the certificate thumbprints.

7. Select a deployment model, scale for your VCF Instance, and DNS and NTP servers.

Deployment model	You can choose the deployment model for new appliances that get deployed. **High Availability** is recommended for production environments. **Note:** The deployment model that you select does not impact existing appliances that you are using in your VCF fleet or VCF Instance. See the VMware Cloud Foundation Design documentation for more information. **Note:** New appliances that get deployed are automatically configured to comply with Federal Information Processing Standards (FIPS).
Size	You can choose the scale of the components in your VCF Instance. • Small – In VCF 9.1, this is the only available option for the simple deployment model. – In VCF 9.1.1 and later, this is the default option for the simple and HA deployment models. • Medium – In VCF 9.1, this is the default option for the HA deployment model. – In VCF 9.1.1 and later, this is an option for the HA deployment model. • Large – In VCF 9.1 and later, this is an option for the HA deployment model. For more information about deployment sizes, see VCF Fleet Sizing Models.
Customer Experience Improvement Program (CEIP)	Select an option to activate or deactivate CEIP across all deployed components. VMware's Customer Experience Improvement Program (CEIP) collects technical information from participating customers to provide better support. See Customer Experience Improvement Program for more information.
Autogenerate passwords	Starting with VCF 9.1, select this option to let VCF Installer generate secure passwords for all deployed components. You can retrieve the passwords after the deployment begins. If you do not select this option, you must provide each password manually.
Skip VCF Automation	In VCF 9.1.1 and later, select this option to skip the deployment of VCF Automation during the deferred components workflow. If you select this option, the UI does not prompt for VCF Automation FQDNs or IP addresses. You can deploy VCF Automation later as a Day-N operation. See Deploying VCF Automation as a Day-N Operation.

8. Select a network.
   a. Select an NSX overlay segment that is backed by a distributed port group.

Note:

The VCF Management components network traffic will run on this port group.

b. Enter the network gateway address.
c. Enter IP details for VCF Automation.
   If you chose to deploy VCF Automation later as a Day-N operation, this option is not displayed.
   - For deployments version 9.1.0.0 through 9.1.0.300, you specify a contiguous IP range for the VCF Automation nodes with 5 IP addresses.
   - For deployments version 9.1.0.400 and later, you specify 6 IP addresses as contiguous or non-contiguous IP range.

   Note:

   The sixth additional IP address is requested by the UI but is not consumed by the deployment.

9. Enter the details for the VCF management components.

Table 969: VCF Management Configuration Details

Each Fully Qualified Domain Name (FQDN) must point to a unique IP address.

Note:

For VCF components that support the dual-stack model, the FQDNs must resolve to IPv4 and IPv6 addresses. See Dual Stack IP Addressing Model

Component	Field	Description
VCF Operations	Primary node FQDN	Enter the FQDN of the VCF Operations primary node.
	Replica node FQDN	Enter the FQDN of the VCF Operations replica node.
	Data node FQDN	Enter the FQDN of the VCF Operations data node. This field appears only when deploying VCF in High Availability mode with a Medium or Large deployment size.
Load Balancer	Load balancer FQDN	(Optional) Enter the FQDN of the load balancer that distributes the VCF Operations traffic. This field appears only when deploying VCF in High Availability mode.
Cloud Proxy	Cloud Proxy FQDN	Enter the FQDN of the cloud proxy. Cloud proxy collects and monitors data locally in a VCF Instance.
License Server	License Server FQDN	Enter the FQDN of the license server. A license server is added to a VCF Operations instance to store the licenses you want to assign from this VCF Operations instance. You must add at least one license server to each VCF Operations instance that you use for license management. You can deploy license server as IPv4 only. In VCF 9.1.1 and later, after the deployment, you can change the networking configuration to IPv6. See Configure IPv6 on a License Server.
VCF Automation *	VCF Automation FQDN	Enter the FQDN to assign to the VCF Automation instance. Later, you use it to access the VCF Automation user interface. **Note:** Do not use capital letters in the FQDN.

Component	Field	Description
	VCF services runtime FQDN	Enter the FQDN for VCF services runtime to connect VCF Automation with the VCF fleet lifecycle component that manages the lifecycle of VCF Automation. The FQDN must be dedicated to the runtime of the VCF Automation cluster, not the VCF management services cluster. The hostname from this FQDN is prefixed to the names of its node VMs and related objects. **Note:** Do not use capital letters in the FQDN.
	Internal cluster CIDR IPv4	Select the VCF services runtime internal IPv4 network. • 198.18.0.0/15 • 240.0.0.0/15 • 250.0.0.0/15 Different VCF services runtime instances on the same network can work with the same internal network. For example, you can have the two instances of VCF services runtime for your VCF management services and VCF Automation using the same internal network.

* If you chose to deploy VCF Automation later as a Day-N operation, these options are not displayed.

10. Review the deployment information.

    Note:

    To make changes, navigate back to the relevant page in the deployment wizard. You can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage

11. Validate and deploy the VCF management components.

VCF Installer API

1. Authenticate to the VCF Installer API.
   a. On the machine with access to VCF Installer, set the environment variables. Replace all placeholder values with values specific to your environment.

      ```
      vcfInstallerFqdn='vcf-installer.rainpole.io'
      vcfInstallerUser='admin@local'
      vcfInstallerPass='<password>'
      ```

   b. Authenticate and capture the access token.

      ```
      vcfInstallerToken=$(curl -k -X POST \
        https://$vcfInstallerFqdn/v1/tokens \
        -H 'Content-Type: application/json' \
        -d '{"username": "'"$vcfInstallerUser"'", "password":
       "'"$vcfInstallerPass"'"}' \
        | jq -r '.accessToken')
      ```

   c. (Optional) Verify that the token is not empty.

      ```
      echo $vcfInstallerToken
      ```

2. Create a JSON file with information about the VCF management components to use for deployment and save the file. For example, as `deferred-components.json`.
   Replace all placeholder values with values specific to your environment. For example, replace *<overlay-segment-name>* with the name of the overlay segment.

   ```
   {
     "sddcId": "MGMT",
     "workflowType": "VCF_COMPLETE",
     "vcenterSpec": {
       "vcenterHostname": "vcenter-1.vrack.rainpole.io",
       "sslThumbprint": "<valid-thumbprint>",
       "adminUserSsoPassword": "<password>",
       "adminUserSsoUsername": "administrator@vsphere.local",
       "useExistingDeployment": true
   ```

```
},
"sddcManagerSpec": {
  "useExistingDeployment": true,
  "sslThumbprint": "<valid-thumbprint>",
  "rootPassword": "<password>",
  "hostname": "sddc-manager.vrack.rainpole.io",
  "sshPassword": "<password>",
  "localUserPassword": "<password>"
},
"ceipEnabled": true,
"vcfInstanceName": "sddcId-1001",
"vcfOperationsSpec": {
    "applianceSize": "medium",
    "useExistingDeployment": false,
    "adminUserPassword": "<password>",
    "nodes": [
        {
            "hostname": "flt-ops01a.rainpole.io",
            "rootUserPassword": "<password>",
            "type": "master"
        },
        {
            "hostname": "flt-ops01b.rainpole.io",
            "rootUserPassword": "<password>",
            "type": "replica"
        },
        {
            "hostname": "flt-ops01c.rainpole.io",
            "rootUserPassword": "<password>",
            "type": "data"
        }
    ]
},
"vcfOperationsCollectorSpec": {
    "hostname": "vcfopscp-1.vrack.rainpole.io",
    "rootUserPassword": "<password>",
    "applianceSize": "small",
    "useExistingDeployment": false
},
"licenseServerSpec": {
  "hostname": "ls.vrack.rainpole.io"
},
"vcfAutomationSpec": {
    "ipPool": [
        "10.11.10.51",
        "10.11.10.52",
        "10.11.10.53",
        "10.11.10.54",
        "10.11.10.55"
    ],
    "nodePrefix": "flt-auto",
    "hostname": "flt-auto01.rainpole.io",
    "platformFqdn": "vcfa-platform1.vrack.rainpole.io",
    "internalClusterCidr": "198.18.0.0/15",
    "adminUserPassword": "<password>"
},
"vcfManagementComponentsInfrastructureSpec": {
```

```
            "xRegionNetwork": {
            "networkName": "<overlay-segment-name>",
            "subnetMask": "255.255.255.0",
            "gateway": "192.168.11.1"
          }
        }
      }
```

3. Validate your JSON file.
```
curl -k -X POST \
  https://$vcfInstallerFqdn/v1/sddcs/validations \
  -H "Authorization: Bearer $vcfInstallerToken" \
  -H 'Content-Type: application/json' \
  -d @deferred-components.json
```
4. Check the response of the validation.
```
curl -k -s -X GET \
  https://$vcfInstallerFqdn/v1/sddcs/validations/latest \
  -H "Authorization: Bearer $vcfInstallerToken" \
  -H 'Content-Type: application/json' \
  | jq '{executionStatus: .executionStatus, resultStatus: .resultStatus, failed:
 [.validationChecks[] | select(.resultStatus == "FAILED")]}'
```
5. Deploy the VCF management components.
```
curl -k -X POST \
  https://$vcfInstallerFqdn/v1/sddcs \
  -H "Authorization: Bearer $vcfInstallerToken" \
  -H 'Content-Type: application/json' \
  -d @deferred-components.json
```
6. (Optional) Monitor the deployment.
```
curl -k -X GET \
  https://$vcfInstallerFqdn/v1/sddcs \
  -H "Authorization: Bearer $vcfInstallerToken" \
  -H 'Content-Type: application/json'
```
7. When the deployment completes, open the VCF Operations UI to apply licensing and perform other post-deployment configuration.

Completing the Deployment of Your Platform

After the automated VCF Installer deployment of components for your VMware Cloud Foundation or vSphere Foundation platform, you must deploy the remaining components to complete the platform. You deploy VCF Operations orchestrator manually for both VCF and vSphere Foundation. For VCF, you deploy the remaining components by using automation that is provided by VCF Operations.

Note:

If deployment of a component fails and the component appears listed in the UI, there is no option to delete the listing. To remove the component from the UI, see Broadcom knowledge base article 441333.

If you did not deploy VCF Automation by using the VCF Installer workflows, see Deploying VCF Automation as a Day-N Operation.

Table 970: Post-VCF Installer Deployment Scope per Platform

Component Group	Component Node / Instance	VMware Cloud Foundation		vSphere Foundation
		Simple	High Availability	
VCF Operations	operations-networks (platform node)	Yes	Yes	No

Component Group	Component Node / Instance	VMware Cloud Foundation		vSphere Foundation
		Simple	High Availability	
	operations-networks (collector node 1)	Yes	Yes	No
	operations-networks (collector node 2)	No	Yes	No
	operations-orchestrator (primary node)	Yes	Yes	Yes
	operations-orchestrator (secondary node1)	No	Yes	Yes
	operations-orchestrator (secondary node2)	No	Yes	Yes
VCF management services	Log management	Yes	Yes	Yes
	Real-time metrics	Yes	Yes	No

Important:

After deployment of any VCF services runtime components, note the following guidelines:

- For automatic backups during management component patching and upgrading, configure SFTP after the management services installation completes. See Configure an External SFTP Server for VCF Management Services and VCF Automation Backups.
- VCF management services uses the VCF services runtime template to deploy VMs during scale and remediation operations. Do not remove the VCF services runtime template.
- Do not rename or delete datastores, network objects, folders, or VMs associated with VCF services runtime directly in vCenter. These modifications are not supported and cause synchronization failures.

Deploy Log Management

Log management collects, analyzes, and visualizes logs across your VMware Cloud Foundation platform. The component is now part of VCF management services and you deploy it from VCF Operations. You also use this procedure for scenarios where a redeployment is required for VCF 5.2.x to 9.1 upgrade.

Table 971: Resource Requirements for Log Management Deployments

Size	vCPUs per Node	Memory per Node	Storage per Node	Number of Nodes
Small	8	16 GB	575 GB	1-19
Medium	24	48 GB	575 GB	3-19
Large	32	64 GB	575 GB	3-19

Note:

Log management is hosted by VCF management services. The resources in the table are in addition to the footprint required by VCF management services. Depending on your choice for sizing and number of log management nodes, VCF management services may scale out to accommodate the log management deployment.

Procedure

1. Log in to the VCF Operations user interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Navigate to **Build** > **Lifecycle** > **VCF Management**.

3. On the Components tab, select **Add component** > **Log management** and configure the parameters for the deployment.

Option	Description
Version	Select the version for the instance.
Import configuration from an earlier Operations for logs instance	Select to migrate a VMware Aria Operations for Logs 8.x configuration. You also enter the FQDN, admin username, and password for the instance that you migrate from.
Log management FQDN	Enter the new Fully Qualified Domain Name to assign to that instance. The FQDN must be unique and should not point to any IP addresses already assigned to VCF services runtime nodes. If you have a deployment, the FQDN for the new deployment must be different than the existing one. **Note:** Do not use capital letters in the FQDN.
Size	Select a deployment size. The deployment size determines the compute and storage resources allocated to each node. For maximum limits and sizing specifics, see VMware Configuration Maximums.
Number of Replicas	Specify the number of replicas for the current deployment. The number of replicas is restricted based on the deployment size.

4. Click **Next** to proceed to the Summary page for final review before deployment.
5. Click **Finish** to start the deployment.
6. If you imported a configuration from an earlier version, after the upgrade completes, review the details of the procedure and download in the detailed report.
   a. Click the **Tasks** tab and click the log management deployment/upgrade task to see the details.
   b. Click the **Upgrade report** icon to download the report and review the details.

After Completing the Task

If you are importing a configuration from an earlier Operations for logs instance, to fully transition to the new 9.1 component, you perform the necessary post-deployment configuration steps.

Post-Upgrade Action	Description	Documentation
1. Configure agents	• If you are importing from 9.0 to 9.1 and have log collection configured in VCF Operations 9.0 UI, the agent and log sources reconfiguration is automatic. • If you are importing from 8.18 to 9.1 and have log collection configured in the VMware Aria Operations for Logs UI, you must manually reconfigure the agents and log sources. In these scenarios, legacy appliances continue collecting data until you manually reconfigure log sources and agents to send logs to the 9.1 component.	Configuring Agents
2. Configure log sources	Reconfigure required log sources to point to your new instance if they are not automatically transitioned.	Configuring Log Sources for Log Management

Post-Upgrade Action	Description	Documentation
3. Configure log forwarding	Reconfigure log forwarding based on your import scenario: • If any log forwarder is sending logs to the existing VMware Aria Operations for Logs 8.18 or VCF Operations for logs 9.0 cluster, you must reconfigure the log forwarder to send logs to the new 9.1 log management component FQDN. • When importing the configuration from VMware Aria Operations for Logs 8.18 or VCF Operations for logs 9.0, forwarders are copied over but are not active. You must manually activate them.	Configuring Log Forwarding
4. Migrate custom content	Custom content (dashboards, queries, alerts) is not transferred automatically. Create a Content Pack for your custom content, use the conversion tool to generate a Management Pack, and install it into VCF Operations.	Converting Content Packs to Management Packs
5. Transfer historical log data	You can access historical log data in the following ways: • You can use the data transfer utility. • You can import archived data and can query it in the 9.1 log management component UI. • For 90 days after the upgrade, you can query logs from the old cluster by using the 9.1 log management component UI.	Log Data Transfer

After you complete the post-deployment configuration, you can shut down and remove the legacy appliances.

Deploy VCF Operations for Networks

VCF Operations for networks provides a comprehensive overview of your VMware Cloud Foundation platform network health, issues, and operations.

Initially, you deploy a single platform node and a single collector node. After deployment, you can scale up or down the node size, see Scaling VCF Management Components. Also, you can scale out the deployment by adding platform and collector nodes, see Add Nodes in VCF Operations and VCF Operations for networks.

You must allocate a static IP for every VCF Operations for networks node you deploy.

You size your VCF Operations for networks deployment based on the size of your VCF fleet. See VCF Operations for Networks System Recommendations and Requirements.

After deployment of VCF Operations for networks, the platform and collector VM names contain their respective IP addresses. The VM names are not specified during deployment. Do not change the names of the VMs manually, they are managed by VCF Operations.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Build**.
3. In the left pane, click **Lifecycle**.
4. Click **VCF Management** and click the **Components** tab.
5. From the **Add component** drop-down menu, select **VCF Operations for networks**.
   The **Add VCF Operations for Networks** wizard opens.
6. Deploy VCF Operations for networks.

a. On the **Add options** page, configure the settings and click **Next**.

Setting	Description
Version	Select the specific version of the VCF Operations for networks component.
Import configuration	Click to import a VMware Aria Operations for Networks 6.x configuration. If left unchecked, VCF Operations for networks is installed as a new deployment.
Size	Select a deployment size. The deployment size determines the compute and storage resources allocated to each node.

b. On the **Parameters** page, configure the settings and click **Next**.

Setting	Description
Password	Type a custom password or click **Regenerate** to create a secure password. This password is used for the **console-user**, **support**, and **admin@local** accounts.
Platform Node IPv4 Address	Enter the static IPv4 address for the platform node.
Platform Node IPv6 Address	This option is available if you select Dual-stack. Enter the static IPv6 address for the platform node.
Collector Node IP Address	Enter the static IP address for the collector node.
Collector Node IPv6 Address	This option is available if you select Dual-stack. Enter the static IPv6 address for the collector node.
Dual-stack	This option turns on IPv4 and IPv6 networking for the VCF Operations for networks nodes.

c. On the **Summary** page, review the configuration and click **Finish**.

Deploy Real-Time Metrics

Real-time metrics allows you to collect the real-time, granular data from your core VCF infrastructure, such as vCenter, ESX, NSX, and vSAN. As part of the deployment the real-time metrics store component is also deployed.

The real-time metrics component collects performance data directly from vCenter, ESX, NSX, and vSAN. This outbound connection originates from the real-time metrics component that is operating within VCF management services and goes to both the vCenter instance and the ESX hosts. To facilitate access to the ESX hosts, verify that network connectivity is established through port 443 over HTTPS.

Perform the following steps to deploy the real-time metrics component.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. (Optional) If integrating VCF instances with shared NSX components, verify that **System Managed Credential** is enabled to ensure that real-time metrics data collection succeeds.
   a) Navigate to **Operate** > **Administration** > **Integrations**.
   b) On the **Accounts** tab, expand the VMware Cloud Foundation section.
   c) Expand your VCF instance and click the domain.
   d) On the domain page, click the **NSX** tab.
   e) If **System Managed Credential** is unchecked, click it on.
3. Navigate to **Build** > **Lifecycle** > **VCF Management** > **Add Components**.
4. On the Components tab, select **Real-Time Metrics**.
   a) Provide the information required for deployment.
      - Version: Select the version of the Real-time metrics component.

- VCF Instance: All VCF Instances managed by SDDC Manager instances that run version 9.1 are selected for installation by default. To customize the selection, click **Change Selection** and deselect any VCF Instances where you do not want real-time metrics installed.

b) Click **Install** to start the real-time metrics component deployment.

The deployment configures the adapter and invokes data collection from core components such as vCenter, ESX, NSX, and vSAN. When the deployment completes, the real-time metrics appear on the observability workbench for the selected inventory object.

Note:

If real-time metrics is not enabled, some charts that are displayed in dashboards and consume aggregated metrics from real-time metrics have empty values.

When you add VCF Instances to your fleet, repeat the procedure to install the component on the new instances.

Deploy VCF Operations Orchestrator

Before you can access the VCF Operations orchestrator content in your VCF environment, deploy the VCF Operations orchestrator appliance.

- Verify that the host on which you are deploying the VCF Operations orchestrator appliance meets the minimum hardware requirements.
  - 4 CPUs
  - 12 GB memory
  - 200 GB storage

  Note:

  Do not reduce the default memory size, because the VCF Operations orchestrator server requires at least 8 GB of free memory.
- Download the `.ova` file for the appliance from the Broadcom Support portal.
  - If you plan to create a cluster, download and deploy three standalone instances.

1. Log in to the vSphere Client as an **administrator**.
2. Select an inventory object that is a valid parent object of a virtual machine, such as a data center, folder, cluster, resource pool, or host.
3. Select **Actions** > **Deploy OVF Template**.
4. Enter the file path or the URL to the `.ova` file and click **Next**.
5. Enter a name and location for the VCF Operations orchestrator appliance, and click **Next**.
6. Select a host, cluster, resource pool, or vApp as a destination on which you want the appliance to run, and click **Next**.
7. Review the deployment details, and click **Next**.
8. Accept the terms in the license agreement and click **Next**.
9. Select the storage format that you want to use for the VCF Operations orchestrator appliance.

Format	Description
Thick Provisioned Lazy Zeroed	Creates a virtual disk in a default thick format. The space required for the virtual disk is allocated when the virtual disk is created. If any data remains on the physical device, it is not erased during creation, but is zeroed out on demand later on first write from the virtual machine.
Thick Provisioned Eager Zeroed	Supports clustering features such as Fault Tolerance. The space required for the virtual disk is allocated when the virtual disk is created. If any data remains on the physical device, it is zeroed

Format	Description
	out when the virtual disk is created. It might take much longer to create disks in this format than to create disks in other formats.
Thin Provisioned Format	Saves hard disk space. For the thin disk, you provision as much datastore space as the disk requires based on the value that you select for the disk size. The thin disk starts small and, at first, uses only as much datastore space as the disk needs for its initial operations.

10. Click **Next**.
11. Configure the network settings and enter the **root** password.

    When configuring the network settings of the VCF Operations orchestrator appliance, you must use the IPv4 protocol. For both DHCP and Static network configurations, you must add a fully qualified domain name (FQDN) for your VCF Operations orchestrator appliance.

    If the host name displayed in the shell of the deployed VCF Operations orchestrator appliance is `photon-machine`, the preceding network configuration requirements are not met.
12. Optional: Configure additional network settings for the VCF Operations orchestrator appliance.

    Note:
    When configuring a Kubernetes network, the values of the internal cluster CIDR and internal service CIDR must allow for at least 1024 hosts. Because of this requirement, the network mask value must be 22 or less. Network mask values higher than 22 are invalid. The Kubernetes network properties have the following default values:

Kubernetes network property	Default value	Property description
Kubernetes internal cluster CIDR	10.244.0.0/22	The CIDR used for pods running inside the Kubernetes cluster.
Kubernetes internal service CIDR	10.244.4.0/22	The CIDR used for Kubernetes services inside the Kubernetes cluster.

    Note:

    You can also change the Kubernetes CIDR network properties after deployment.
13. Click **Next**.
14. Review the **Ready to complete** page and click **Finish**.

The VCF Operations orchestrator appliance is successfully deployed.

- Log in to the VCF Operations orchestrator appliance command line as **root** and confirm that you can perform a forward or reverse DNS lookup.
  - To perform a forward DNS lookup, run the `nslookup` *`your_orchestrator_FQDN`* command. The command must return the VCF Operations orchestrator appliance IP address.
  - To perform a reverse DNS lookup, run the `nslookup` *`your_orchestrator_IP`* command. The command must return the VCF Operations orchestrator appliance FQDN.
- To verify all pods are running successfully, run the `kubectl get pods -A` command. The VCF Operations orchestrator UI might be unresponsive for a few minutes, until you configure an authentication provider for your VCF Operations orchestrator instance.
- Configure authentication for your VCF Operations orchestrator integration.
  - Configure VCF Operations Orchestrator for VCF Automation All Apps Organizations
  - Configure VCF Operations Orchestrator for VCF Automation VM Apps Organizations
  - Configure VCF Operations Orchestrator for vSphere
  - Configure VCF Operations Orchestrator for vSphere by Using VCF SSO
  - Configure VCF Operations Orchestrator for VCF Operations

- To set up a cluster, connect the three standalone VCF Operations orchestrator instances that you deployed. See Configure a VCF Operations Orchestrator Cluster.

Configure VCF Operations Orchestrator for VCF Automation All Apps Organizations

As a provider administrator, you integrate a VCF Automation for All Apps organization with VCF Operations orchestrator so that the organization administrator and project users can use VCF Operations orchestrator workflows in VCF Automation.

In a VCF Automation for All Apps organization, a single external VCF Operations orchestrator instance is deployed, configured, and added as an integration by the provider administrator on a per-request basis. Other VCF Operations orchestrator instances cannot be added. If the organization already has a VCF Operations orchestrator integration, you must first delete the existing integration before you can add a different one.

The external instance you configure is restricted and registered to that VCF Automation All Apps organization only. The registered VCF Operations orchestrator instance cannot be registered to other organizations even if the initial organization is deleted.

Step 1. Configure Authentication for a VCF Operations orchestrator Integration in VCF Automation for All Apps

When you configure an authentication provider for VCF Operations orchestrator, that VCF Operations orchestrator instance is restricted to the respective VCF Automation organization.

1. Log in to the VCF Operations orchestrator appliance as `root`.
2. Configure the authentication provider.
   ```
   vracli vro authentication set -p tm -u PROVIDER_ADMINISTRATOR_ACCOUNT -hn https://FQDN_VCF_AUTOMATION --
   tenant ALL_APPS_ORGANIZATION_NAME
   ```
3. Redeploy the VCF Operations orchestrator pods.
   ```
   /opt/scripts/deploy.sh
   ```

The VCF Operations orchestrator instance is configured and associated with the All Apps organization. The configuration process can take up to several minutes to complete.

> **Note:**
>
> You can unregister the current authentication provider by running the `vracli vro authentication unregister` command from the command-line interface.

Step 2. Add a VCF Operations orchestrator Integration in VCF Automation for All Apps

You integrate VCF Operations orchestrator with your All Apps organization to use it for extensibility automation tasks.

Only one external VCF Operations orchestrator instance can be integrated with the organization.

1. Log in to the VCF Automation Provider Management Portal with system administrator credentials.
2. Navigate to your target organization and select **Launch Organization Portal**.
3. In the organization UI, navigate to **Administer** > **Connections** > **Integrations** and select **Add Integration**.
4. Select **Orchestrator**.
5. Enter a name and description for the VCF Operations orchestrator integration.
6. Under **VCF Operations Orchestrator URL**, enter the fully qualified domain name (FQDN) of your external VCF Operations orchestrator instance.
   For example, `https://my_Orchestrator_FQDN.com:443`.

   > **Note:**
   >
   > If you have a clustered VCF Operations orchestrator environment, enter the URL of the load balancer.
7. To validate the integration, click **Validate**.
8. When prompted to confirm the certificate, review the certificate and click **Accept**.
9. Click **Add**.
   The VCF Operations orchestrator integration is saved.
10. To verify that the integration is configured and that the workflows are added, navigate to the **Orchestrate** tab.

The VCF Operations orchestrator instance is ready and you can now enable extensibility with VCF Operations orchestrator for the organization. To access VCF Operations orchestrator, select the **Orchestrate** tab in the VCF Automation organization UI.

After you configure VCF Operations orchestrator for the organization, VCF Operations orchestrator starts to perform periodic data collection for the integration after about 10 minutes.

Step 3. Enable VCF Operations orchestrator for the VCF Automation All Apps Organization

To enable extensibility through VCF Operations orchestrator for the VCF Automation organization, you must publish the Orchestrator rights bundle to that organization.

1. Log in to the VCF Automation Provider Management Portal with system administrator credentials.
2. Navigate to **Administration** > **Access Control** > **Rights Bundles**.
3. Select the **Orchestrator Rights Bundle**.
4. On the **Orchestrator Rights Bundle** page, click **Publish**.
5. Select the target organization and click **Save**.
6. Verify that the organization is added by navigating to **Administration** > **Access Control** > **Rights Bundles** and checking the link in the **Published To** list of the **Orchestrator Rights Bundle** tab.

You have registered VCF Operations orchestrator for use in your organization. Next, the organization administrator can manage extensibility with VCF Operations orchestrator-related authorizations for users in that organization.

Using the VCF Operations orchestrator Plug-Ins

VCF Operations orchestrator provides a standard set of preinstalled plug-ins, including plug-ins for vCenter and VCF Automation for All Apps, to allow you to orchestrate tasks in the different environments that the plug-ins expose. For more information, go to Using the VCF Operations orchestrator Plug-ins.

> **Note:**
>
> If your VCF Operations orchestrator integration authenticates with VCF Automation, when using the vCenter plug-in, you must use shared session and not session per user. For more information, go to Configure the Connection to a vCenter Instance.

Configure VCF Operations Orchestrator for VCF Automation VM Apps Organizations

As an organization administrator, you integrate a VCF Automation for VM Apps organization with VCF Operations orchestrator so that you and your project users can use VCF Operations orchestrator workflows in VCF Automation.

In VCF Automation for VM Apps organizations, you can register multiple VCF Operations orchestrator instances with the organization. Both embedded and external VCF Operations orchestrator instances can be registered with the same organization.

> **Important:**
>
> If you upgrade an existing VMware Aria Automation 8.x deployment to VCF Automation 9, your organizations are automatically mapped to VCF Automation for VM Apps organizations.
>
> - If you are upgrading from VMware Aria Automation 8.x, any existing Automation Orchestrator integrations and their content are preserved as-is in the VCF Automation for the VM Apps organization.
> - Any external Automation Orchestrator instances must be upgraded after first upgrading VMware Aria Automation 8.x to VCF Automation. The integrations are then available in the corresponding target VM apps organizations.

Before you begin

- You are the organization administrator of a VM Apps organization in VCF Automation. For more information on VCF Automation user roles, see Predefined VCF Automation Roles and Their Rights.

- Users have been imported into your organization. For more information on importing users, see Import Provider Users to Your VCF Automation.

Step 1. Configure Authentication for a VCF Operations orchestrator Integration in VCF Automation for VM Apps

You configure an authentication provider for VCF Operations orchestrator so that you can integrate it with a VM Apps organization in VCF Automation.

1. Log in to the VCF Operations orchestrator appliance as *root*.
2. Configure the authentication provider.
   ```
   vracli vro authentication set -p tm -u ORGANIZATION_ADMINISTRATOR_ACCOUNT -hn https://FQDN_VCF_AUTOMATION
    --tenant VM_APPS_ORGANIZATION_NAME
   ```
3. Redeploy VCF Operations orchestrator pods.
   ```
   /opt/scripts/deploy.sh
   ```

The VCF Operations orchestrator instance is configured and associated with the VM Apps organization in VCF Automation.

> **Note:**
>
> You can unregister the current authentication provider by running the `vracli vro authentication unregister` command from the command-line interface.

Step 2. Add a VCF Operations orchestrator Integration in VCF Automation for VM Apps

You integrate the VCF Operations orchestrator with your VM apps organization to use it for extensibility automation tasks.

1. Log in to the VM Apps organization as an organization administrator.
2. Select **Infrastructure** > **Integrations** and click **Add Integration**.
3. Select **Orchestration**.
4. Enter a name for the VCF Operations orchestrator integration.
5. (Optional) Enter a description for the VCF Operations orchestrator integration.
6. Select your integration type.
   - To integrate an external VCF Operations orchestrator, select **User-deployed VCF Operations Orchestrator**.
   - To integrate an embedded VCF Operations orchestrator, select **Embedded VCF Operations Orchestrator**.

     > **Note:**
     >
     > The embedded VCF Operations orchestrator instance is the default VCF Operations orchestrator instance that is part of your VCF Automation deployment.
7. Click **Add**.
   The VCF Automation integration is saved. The **Integrations** page opens.
8. On the **Manage** page, verify the resource consumption by the embedded VCF Operations orchestrator.
   If there is an issue with the resource consumption, the instance appears as shut down.
9. Start data collection so that VCF Operations orchestrator workflows are synchronized with VCF Automation for your organization.
10. (Optional) Add capability tags.
    You use capability tags to distinguish between multiple VCF Operations orchestrator instances and to explicitly direct calls from the VCF Automation to a specific integration. For more information on capability tags, see Using capability tags in VCF Automation.
11. To verify that the integration is configured and that the workflows are collected, select **Orchestrator** > **Library** > **Workflows**.

VCF Operations orchestrator is ready to be used for extensibility automation tasks in the VM Apps organization. To access VCF Operations orchestrator, select the **Orchestrator** tab in the VCF Automation for VM Apps organization. If you have multiple VCF Operations orchestrator integrations, you use the drop-down to select the instance you want to work in.

Step 3. Define Authorization Restrictions

1. Log in to the VM Apps organization as an organization administrator.
2. Select **Infrastructure** > **Access Control** > **Users**.
3. Select a user to whom you want to assign permissions to design and run workflows and actions for automation tasks in VCF Operations orchestrator.
4. Click **Edit**.
5. Under **Assigned roles**, select the *Orchestrator Workflow Designer* role.
6. Click **Save**.

You assigned a VCF Operations orchestrator role to a user in your organization. This user can now create VCF Operations orchestrator content for automation tasks in your organization.

Configure the VCF Operations orchestrator plug-in for VCF Automation for VM Apps

The VCF Operations orchestrator plug-in for VCF Automation for VM Apps is used to integrate VCF Operations orchestrator with VCF Automation for VM Apps.

When upgrading from VMware Aria Automation 8.18.1, the plug-in works as expected without additional user action required. However, if you are using a new deployment, you must manually configure the plug-in.

- For single-tenant environments, configure the plug-in by following the instructions in Broadcom knowledge base article 401216.
- For multi-tenant environments, configure the plug-in by following this procedure:
  a. Configure vIDM as the identity provider for VCF Automation Provider Management by following the instructions in Broadcom knowledge base article 401216.
  b. Configure the organization hostname.

     Note:

     Verify that the hostnames include the tenant identifier. For each tenant that is not the default tenant, DNS records must be configured. This is required as the behavior remains consistent with VMware Aria Automation 8.x where the tenant identifier is included in the hostname for tenants as a subdomain. In VCF Automation 9.x, the tenant should be exposed in the hostname.

```
# Example of multiple Soft Tenants

# CSP API will work
second-tenant.vcfa.com/csp/gateway/am/api/login?access_token

# CSP API will work
third-tenant.vcfa.com/csp/gateway/am/api/login?access_token

# This will attempt to log into the default tenant. The defaultLoginOrg should be configured in TM as
 described in step 2 of the above section: Set Up Default Login Tenant
```

Configure VCF Operations Orchestrator for vSphere

You can integrate VCF Operations orchestrator with vSphere by using vSphere as the authentication provider for core infrastructure automation tasks.

Before You Begin

- Verify that you have installed and configured VCF Operations orchestrator. For information on deploying VCF Operations orchestrator, go to Deploying VCF Operations orchestrator. For clustered deployments, verify that you have deployed three VCF Operations orchestrator instances.
- For VCF Operations orchestrator clusters, deploy a load balancer server.
- You have installed and configured vSphere.

Step 1. Create an Administrative Group for VCF Operations orchestrator

Users assigned to this group have administrative rights in the VCF Operations orchestrator instance.

1. Log in to the vSphere Client as an administrator.
2. Navigate to **Menu** > **Administration**.
3. Under **Single Sign On**, select **Users and Groups**.
4. Click the **Groups** tab.
5. To create a group, click **Add**.
6. Configure the group.
   a. Enter a group name.
   b. (Optional) Enter a description.
   c. Select the group domain. For local vCenter groups, the group domain is usually `vsphere.local`.
7. To add users to the group, search for existing users in the **Add Members** field, and add them to the group.
8. To create the group, click **Add**.

Step 2. Configure Authentication for a VCF Operations orchestrator Integration in vSphere

1. Log in to the VCF Operations orchestrator appliance as `root`.
2. Configure the authentication provider.

   Note:

   When running the `vracli vro authentication set` command, you are prompted for the password for `administrator@vsphere.local`.

   ```
   vracli vro authentication set -p vsphere \
   --hostname https://FQDN_SPHERE \
   --username administrator@vsphere.local \
   --tenant vsphere.local \
   --admin-group Administrators \
   --admin-group-domain vsphere.local
   ```
3. Verify the configuration of the authentication provider and check the property values.
   ```
   vracli vro authentication
   ```
4. Redeploy the VCF Operations orchestrator pods.
   ```
   /opt/scripts/deploy.sh
   ```

You have configured VCF Operations orchestrator with vSphere as the authentication provider.

Note:

You can unregister the current authentication provider by running the `vracli vro authentication unregister` command from the command-line interface.

Step 3. Configure VCF Operations orchestrator plug-ins

To manage the inventory of a vSphere instance, you must register vCenter in VCF Operations orchestrator using the vCenter plug-in. In addition, the VCF Operations orchestrator plug-in for the vSphere client allows you to directly select VCF Operations orchestrator workflows to run, to schedule, and to answer user interaction directly from the vSphere Client.

1. Register vCenter using the vCenter plug-in. See Configuring the vCenter Server Plug-In in VCF Operations Orchestrator.
2. Configure the VCF Operations orchestrator plug-in for the vSphere client. See Configure the VCF Operations Orchestrator Plug-in for vSphere Web Client.
3. Add custom actions on the vCenter inventory objects. See Create a context action.

Configure VCF Operations Orchestrator for vSphere by Using VCF SSO

VCF Operations orchestrator can be integrated with vSphere by using VCF SSO as the authentication provider for core infrastructure automation tasks. This scenario covers integrating VCF Operations orchestrator by using VCF SSO as the authentication provider.

Before You Begin

- You have installed and configured VCF Operations orchestrator. For information on deploying VCF Operations orchestrator, go to Deploying VCF Operations orchestrator.
- You have installed and configured VCF Operations.
- You have installed and configured a vSphere instance and applied VCF SSO through VCF Operations.

Step 1. Create an OAuth client in VCF Operations

This client is used to configure VCF SSO as authentication provider for the VCF Operations orchestrator instance.

1. Log in to VCF Operations as a VCF Operations administrator.
2. Navigate to **Fleet Management** > **Identity & Access** > **VCF Other Components** and click **Add**.
3. Enter a name for the client.
4. Select an identity broker.
5. Leave the redirect URL and post-log out redirect URLs blank.
6. Click **Generate OIDC Client**.
7. Click **Save**.

Now you have an OIDC client to use for your VCF Operations orchestrator configuration.

Step 2. Configure VCF SSO as Authentication Provider for VCF Operations orchestrator

You use above OIDC client to configure VCF Operations orchestrator with enough privileges in VCF SSO to request and renew VCF SSO OAuth tokens.

1. Log in to the VCF Operations orchestrator appliance as `root`.
2. Configure the authentication provider.

```
vracli vro authentication set -p vidb \
--hostname https://FQDN_IDENTITY_BROKER/acs/t/CUSTOMER \
--client-id 'CLIENT_ID_OF_ABOVE_OIDC_CLIENT' \
--client-secret 'CLIENT_SECRET_OF_ABOVE_OIDC_CLIENT' \
--admin-group O11N_ADMIN_GROUP@DOMAIN
```

3. Verify the configuration of the authentication provider and check the property values.

```
vracli vro authentication
```

4. Redeploy the VCF Operations orchestrator pods.

```
/opt/scripts/deploy.sh
```

The VCF Operations orchestrator instance is configured with the same VCF SSO as the vSphere instance you want to integrate to.

> **Note:**
>
> You can unregister the current authentication provider by running the `vracli vro authentication unregister` command from the command-line interface.

Step 3. Configure VCF Operations orchestrator plug-ins

To manage the inventory of a vSphere instance, you must register vCenter in VCF Operations orchestrator using the vCenter plug-in. In addition, the VCF Operations orchestrator plug-in for the vSphere client allows you to directly select VCF Operations orchestrator workflows to run, to schedule, and to answer user interaction directly from the vSphere Client.

1. Register vCenter using the vCenter plug-in. See Configuring the vCenter Server Plug-In in VCF Operations Orchestrator.
2. Configure the VCF Operations orchestrator plug-in for the vSphere client. See Configure the VMware Cloud Foundation Operations orchestrator Plug-in for vSphere Web Client.
3. Add custom actions on the vCenter inventory objects. See Create a context action.

Configure VCF Operations Orchestrator for VCF Operations

You can integrate VCF Operations orchestrator with VCF Operations by installing the relevant management pack.

Before You Begin

- Verify that your VCF Operations orchestrator instance is authenticated with vSphere or VCF SSO.

Integrating VCF Operations orchestrator by Using a Management Pack

To integrate VCF Operations orchestrator, you must install the relevant management pack in VCF Operations.

Management packs are hosted on the VCF Solutions Catalog (formerly VMware Marketplace) and can either be downloaded as PAK files or applied directly from **Marketplace** tab in VCF Operations.

For more information on adding and managing integrations in VCF Operations, go to Adding and Managing Management Packs. For more information on using the VCF Operations orchestrator management pack, see Configuring Automation Orchestrator.

Configure a VCF Operations Orchestrator Cluster

You can configure your new VCF Operations orchestrator deployment to run in high availability by deploying three nodes and connecting them as a cluster.

- Download and deploy three standalone VCF Operations orchestrator instances.

  Note:

  A clustered VCF Operations orchestrator environment must consist of three nodes.
- Configure a load balancer server to direct the traffic to the VCF Operations orchestrator nodes.

A VCF Operations orchestrator cluster consists of three VCF Operations orchestrator instances.

To create a VCF Operations orchestrator cluster, you must select one VCF Operations orchestrator instance to be the primary node of the cluster. After configuring the primary node, you join the secondary nodes to it.

The VCF Operations orchestrator cluster you create is pre-configured with automatic failover.

Note:
Failure of the automatic failover can lead to loss of data.

1. Configure the primary node.
   a) Log in to the VCF Operations orchestrator primary node over SSH as **root**.
   b) To configure the cluster load balancer server, run the `vracli load-balancer set` *`load_balancer_FQDN`* command.
   c) Configure the authentication provider by running the `vracli vro authentication set` command.

      Optionally, you can add the FQDN address of the load balancer server by adding the `--fqdn` parameter to the `vracli authentication set` command.

2. Join secondary nodes to primary node.
   a) Log in to the VCF Operations orchestrator secondary node over SSH as **root**.
   b) To join the secondary node to the primary node, run the `vracli cluster join` *`primary_node_hostname_or_IP`* command.
   c) Enter the root password of the primary node.
   d) Repeat the procedure for the other secondary node.
3. Optional: If your primary node uses a custom certificate, you must either set the certificate in the appliance or generate a new certificate. For more information, go to Generate a custom TLS certificate for VCF Operations orchestrator.

   Note:
   The file containing the certificate chain must be PEM-encoded.
4. Finish the cluster deployment.
   a) Log in to the VCF Operations orchestrator primary node over SSH as **root**.
   b) To confirm that all nodes are in a ready state, run the `kubectl -n prelude get nodes` command.
   c) Run the `/opt/scripts/deploy.sh` script and wait for the deployment to finish.

You have created a VCF Operations orchestrator cluster. After you create the cluster, you can access your VCF Operations orchestrator environment only from the FQDN address of your load balancer server.

Note:

You cannot update the configuration of a cluster node if it has a different root password from the password that is used by the load balancer. To edit the configuration of this node, remove it from the load balancer, edit the configuration, and add the node back to the load balancer.

To monitor the state of the VCF Operations orchestrator cluster, log in to the VCF Operations orchestrator client and navigate to the **System** tab of the dashboard.

Note:

If you are unable to access VCF Operations orchestrator, you can also monitor the states of your cluster nodes by running the `kubectl get pods -n prelude` command from the VCF Operations orchestrator appliance command line.

To check the health status of your load balancer, navigate to `https://<load_balancer_FQDN>/vco/api/healthstatus`. The health check port is 8008.

Deploy VCF Operations HCX

VCF Operations HCX is a workload mobility platform that simplifies workload migration across data centers. VCF Operations HCX is an optional component that you can deploy on a VCF Instance domain.

- Verify that the VCF Operations HCX deployment binaries are downloaded to software depot in the VCF Instance where you want to deploy. See Download Binaries to Software Depot.
- Verify that the Fully Qualified Domain Name (FQDN) for VCF Operations HCX is resolvable over DNS. The IP address to which the FQDN resolves must not be already used.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Navigate to **Build** > **Lifecycle** and select the VCF Instance and domain.
3. On the Component Versions tab, click **Manage Components**.
4. On the VCF Operations HCX card, click **Install** to start the installation wizard.
   a) On General, select the VCF Operations HCX version to install. Only versions that are compatible with the domain components are listed.
   b) On Deployment Settings, provide the admin and root user passwords for VCF Operations HCX.
      - Password length must be greater than 12 characters.

- Password cannot be a palindrome.
- Password must have at least 5 different characters.
- Password must have 1 lower case, 1 upper case, 1 number, 1 special character.

c) Validations check the VCF Operations HCX version compatibility, the FQDN, and the root and admin passwords.
d) On the **Summary** page, review the VCF Operations HCX deployment configuration.
e) Review the next steps and click **Finish**.

When the installation wizard completes, the deployment begins.

5. To monitor the progress of the deployment, navigate to **Build** > **Tasks**, select the VCF Instance, and click the VCF Operations HCX deployment task.

   On the subtask list that appears, verify the status of each subtask. If any binary download validations fail, you can use binary management to reinstall the binaries.

VCF Operations HCX is deployed in the same cluster as the target vCenter instance and is automatically configured with the target NSX and vCenter instances.

After a successful installation, go to the **Software Versions** tab of the domain. In the **Domain Management Component**s table, verify that the current version and target version of VCF Operations HCX match.

Import Existing VCF Operations HCX to a VCF Domain

Instead of deploying a new VCF Operations HCX you can import an existing instance for VCF Operations HCX version 9.1 or later. The import version must also be the same or earlier than the version of SDDC Manager.

- Verify that you have the FQDN and admin user password of your existing HCX appliance.
- Verify that you have the IPv4 address and SSL certificate thumbprint, so that you can confirm automatically retrieved information when HCX Manager is imported into the VCF Domain.
- Verify that the IP address and Fully Qualified Domain Name (FQDN) for VCF Operations HCX are resolvable over DNS.
- Verify that the existing VCF Operations HCX is located either in the management vCenter instance or in the domain where you plan to import it.
- Verify that you have a valid certificate with Common Names (CN) and Subject Alternative Names (SAN).

Perform the following steps to import VCF Operations HCX to an existing management or workload domain.

Note:

If the VCF Operations HCX admin user password is not compliant with the password policy, you must change the password in VCF Operations HCX .

After the import operation completes, any custom backup server that is configured in the VCF Operations HCX instance is overwritten by the backup server settings that are defined in SDDC Manager.

1. Log in to the VCF Operations interface at `https://<vcf_operations_fqdn>` as a user assigned the Administrator role.
2. Navigate to **Build** > **Lifecycle** and select the VCF Instance and domain.
3. On the **Component Versions** tab, click **Manage Components**.
4. On the **VCF Operations HCX** card, click **Install**.
   a) On the **General** page, select **Import an already configured instance**.
   b) On the **Import Settings** page, provide the details of your existing VCF Operations HCX :
      - Enter the FQDN for VCF Operations HCX .
      - Confirm the displayed IP address and SSL thumbprint.
      - Enter the admin user password for VCF Operations HCX
   c) On the **Validations** page, verify that there are no validation errors.

      Validations include password checks, compatibility checks, and checks for vCenter and NSX configurations, and reachability.

d) On the **Summary** page, review the VCF Operations HCX import configuration.
e) Review the next step and click **Finish**.

When the installation wizard completes, the import procedure begins. The procedure for import includes:

- Creation and update of necessary service accounts for HCX to vCenter, SDDC Manager to HCX and HCX to NSX communications.
- If VCF Operations HCX has a valid self-signed certificate, the certificate is replaced with a new one that is issued by the Management vCenter VMCA.

  Note:

  If VCF Operations HCX is assigned a CA signed certificate, VCF Operations HCX does not replace the certificate.

After a successful import, the VCF Operations HCX services are restarted.

5. Optional: To monitor the progress of the import procedure, navigate to **Build** > **Tasks**, select the VCF instance, and click the VCF Operations HCX import task.

   On the subtask list that appears, verify the status of each subtask.

After a successful import, go to the **Software Versions** tab of the domain. In the **Domain Management Components** table, verify that the current version and target version of VCF Operations HCX match.

Configuring Advanced Architectures for VCF Operations

VCF Operations supports architectures that are not automated by VCF Installer. You must perform a manual configuration to achieve the required states.

Configure VCF Operations Cluster for Continuous Availability

VCF Installer supports the deployment of VCF Operations in either simple mode or high availability mode during the initial VCF deployment. After the initial deployment, the VCF Operations cluster can be converted to a continuous availability architecture.

Identify three separate fault domains for the installation of the VCF Operations primary node and data nodes, the VCF Operations primary replica node and data nodes, and the witness node, respectively. The fault domains can be located in either a different workload domain, a separate availability zone, or a different region. In a continuous availability architecture, there must be at least one primary node and one data node within the primary fault domain. Additionally, the replica fault domain should have the same number of nodes as the primary fault domain, and there should be one witness node in the witness fault domain. The VCF Operations nodes in the primary fault domain, the VCF Operations nodes in the replica fault domain, and the witness node in the witness fault domain must be able to communicate over the routable network, despite being in separate fault domains. Consult the VCF Operations sizing guidelines for the sizing and network latency requirements between the three distinct fault domains.

1. Deploy VCF Operations using VCF Installer in either Simple mode or High Availability mode within the fault domain designated for the primary node.
2. Deploy the witness node in the fault domain designated for it, by following these steps:
   a) Download the VCF Operations ova file from the Broadcom support portal to your local computer. Alternatively, you can deploy the OVA directly from vCenter.
   b) If you have downloaded the OVA to your local computer, upload the VCF Operations ova file to the vCenter, in the fault domain designated for the witness node.
   c) Deploy the VCF Operations OVF template and select **Witness** as the configuration for deployment.
   d) Note down the IP address and the hostname or the FQDN of the witness node.
   e) Log in to the VCF Operations administrator interface in the primary fault domain. The administrator interface can be accessed from `https://primary-node-name-or-ip-address/admin`.
   f) In the VCF Operations administrator interface, click the **+** icon in the **Nodes in the VCF Operations Cluster** section, to add the witness node. The **Add new nodes to cluster** wizard opens.
   g) Add the newly deployed witness node to the VCF Operations cluster. Enter the **Node Name**, **Node Address** inputs. In the **Current Cluster Role** option, select **Witness Node** from the drop-down list. Use the IP address

and hostname or the FQDN of the witness node, as recorded in the previous steps. Click **SAVE** and then after the thumbprint of the node is displayed, click **NEXT**.

h) A dialog box with the post-expansion certificate management message is displayed. Select the checkbox to accept the message, and click **NEXT**.

i) In the **Primary Password** step, enter the password of the primary node to add nodes to the cluster. Click **FINISH**. The cluster initialization begins.

j) After the cluster initialization completes, click **FINISH ADDING NEW NODE(S)** in the Cluster Status card. Click OK in the confirmation dialog box which appears.

3. If VCF Operations was deployed in a High Availability architecture during the initial setup, follow the steps below:
   a) Deactivate High Availability in the VCF Operations administrator console `https://primary-node-name-or-ip-address/admin`.
   b) The primary replica cluster node is converted into a standard data node within the cluster, which can then be removed if an extra data node is not needed. The remaining nodes in the cluster maintain their current roles.
   c) Ensure that the VCF Operations cluster has at-least one primary node and one data node in the primary fault domain. Consult the VCF Operations sizing guidelines for the maximum number of data nodes that can be added in the primary fault domain. Note that the number of VCF Operations nodes in the primary fault domain and the replica fault domain must be equal.
4. The replica fault domain must contain the same number of data nodes as the primary fault domain to ensure data redundancy and consistency across the two domains. Deploy the same number of VCF Operations nodes in the designated replica fault domain as in the primary fault domain, following the steps below:
   a) If the designated replica fault domain is managed by the VCF Operations in the primary fault domain, then use **Add Node** capability in Lifecycle, within VCF Operations to add the same number of VCF Operations nodes in the replica fault domain as in the primary fault domain. You can add replica nodes and data nodes using Lifecycle. Witness nodes can only be added from vCenter.
   b) If the designated replica fault domain is NOT managed by the VCF Operations in the primary fault domain, then continue with the the steps to deploy the witness node or data nodes.
   a) Download the VCF Operations ova file from the Broadcom support portal.
   b) Upload the VCF Operations ova file to the vCenter in the fault domain designated for the replica fault domain.
   c) Use the VCF Operations OVF template to deploy an equal number of VCF Operations nodes as in the primary fault domain, ensuring that the selected size matches that of the nodes deployed in the primary fault domain.
   d) Note down the IP address and the hostname of the VCF Operations nodes deployed in the replica fault domain.
   e) Log in to the VCF Operations administrator console in the replica fault domain. The administrator interface can be accessed from `https://replica-node-name-or-ip-address/admin`.
   f) In the VCF Operations administrator interface, click the **+** icon in the **Nodes in the VCF Operations Cluster** section, to add the node. The **Add new nodes to cluster** wizard opens.
   g) Add the newly deployed data node to the VCF Operations cluster. Enter the **Node Name**, **Node Address**. In the **Current Cluster Role** option, select **Data Node** from the drop-down list. Use the IP address and hostname or the

FQDN of the replica data node that you deployed from the OVF template, as recorded in the previous steps. Click **SAVE** and then after the thumbprint of the node is displayed, click **NEXT**.

h) A dialog box with the post-expansion certificate management message is displayed. Select the checkbox to accept the message, and click **NEXT**.

i) In the **Primary Password** step, enter the password of the data node to add nodes to the cluster. Click **FINISH**. The cluster initialization begins.

5. Click **Activate CA** in the VCF Operations Admin console.
6. The Continuous Availability wizard opens. The witness node exists outside the fault domains. The primary node is already assigned to Fault Domain. Perform the following steps:
   a) To pair with the VCF Operations nodes in the primary fault domain, move the VCF Operations data nodes deployed in the replica fault domain to Fault Domain.
   b) Ensure an even distribution of VCF Operations nodes between the primary and replica fault domains to enable Continuous Availability. The number of nodes in both fault domains must be equal.
7. After successfully activating CA, click **Build** in the top menu, and then **Lifecycle** in the left navigation bar. Within VCF Operations click **trigger inventory sync** on the Operations component.

The Lifecycle page for the VCF Operations component now displays Continuous Availability as the configuration for the VCF Operations cluster.

VCF Operations Analytics Node Data Storage and Fault Domain Behavior

VCF Operations uses a robust architecture to ensure data availability and fault tolerance across analytics nodes. Data is stored in pairs of analytics nodes distributed across two fault domains, ensuring continuous availability and synchronization. Each object is stored in two nodes (primary and replica) across fault domains, providing redundancy and resilience in case of failures.

Table 972: Key Concepts of Data Storage and Fault Domain Behavior

Key Concept	Details
Data Storage and Synchronization	When an object is discovered, VCF Operations determines the primary node to store the data and automatically replicates it to its paired node in the other fault domain.
	Synchronization between the primary and replica nodes ensures data consistency across fault domains
Fault Domain Resilience	Each fault domain contains half of the analytics nodes, and each node has a paired replica in the other fault domain.
	The cluster continues to function without data loss as long as one node from each pair is available.
Continuous Availability	The cluster can tolerate the loss of one fault domain or individual nodes without data loss, but the system will operate in a degraded state until the issue is resolved.
Latency and Network Requirements	Latency between fault domains must be <10 ms, with peaks up to 20 ms during 20-second intervals. High latency can impact synchronization but does not stop the process.
Witness Node Role	The witness node ensures cluster health and determines which fault domain remains online during communication failures.

Table 973: Key Scenario and Behavior

Scenario	Behavior
Data Storage	Data is stored in a primary node and replicated to its paired node in the other fault domain.
Loss of One Fault Domain	The cluster operates in a degraded state but continues without data loss. The fault domain must be repaired or replaced to restore full functionality.
Loss of Two Nodes in One Fault Domain	The cluster continues to operate as long as one node from each pair is available.
Loss of Connectivity Between Fault Domains	One fault domain will go offline automatically. After connectivity is restored, the admin must manually bring the fault domain back online to synchronize data.
Witness Node Unavailability	If both fault domains are healthy, the witness node's unavailability has no impact. If communication issues occur, the witness determines which fault domain remains online.
Node Replacement	Use the "Replace nodes of cluster" functionality in the admin UI. Sync time depends on object count, historical data, network bandwidth, and cluster load.
Latency Exceeds 20 ms	Synchronization continues but might be delayed. Latency must return to <10 ms for optimal performance.
Primary Node Failover	If the primary node becomes unavailable, the primary replica node is promoted. The original primary node becomes the replica when it returns online.
Entire Fault Domain Replacement	If a fault domain is unrecoverable, it can be replaced with newly deployed nodes. The admin UI allows replacing the primary replica node.
Load Balancer Support	Load balancers are supported for continuous availability.
Node Reboot or Offline	Nodes automatically resume operation after reboot or being brought offline/online. No additional steps are required.

Configure a VCF Operations Cluster for High Availability

VCF Installer supports the manual deployment of VCF Operations in simple mode for VMware vSphere Foundation. After the initial deployment, the VCF Operations cluster can be converted to a high availability architecture. To activate high availability (HA) for a VCF Operations cluster, specify one of the data nodes to become a replica of the primary node.

- Create and configure the primary node.
- Create and configure a data node with a static IP address.
- Note the fully qualified domain name (FQDN) or IP address of the primary node.

> **Note:**
> If the cluster is running, activating HA restarts the cluster.

You can add HA to the VCF Operations cluster at installation time or after VCF Operations is up and running. Adding HA at installation is less intrusive because the cluster has not yet started.

1. In a Web browser, navigate to the VCF Operations primary node administration interface.
   `https://primary-node-fqdn-or-ip-address/admin`

2. Enter the VCF Operations administrator user name of `admin`.
3. Enter the VCF Operations administrator password and click **Log In**.
4. To add a new data node, click **Add new Nodes**.
   a) In the **Add new nodes to cluster** wizard, click the **Add Nodes** icon.
   b) For **Node Name**, enter a generic name to identify the node.
   c) For **Node Address**, enter the IP address of the node you want to add.
   d) For **Current Cluster Role**, select **Data**, to add a data node to be used as the primary replica node.
   e) Click **Save**, review and close the thumbprint information, and then click **Next**.
   f) For **Primary Node Password**, enter the administrator password used during log in, and then click **Finish**.

   If the cluster was online, the administration interface displays that the cluster initialization is in progress as the new data node gets added to the VCF Operations cluster. You must wait for the cluster to come **Online** again before activating HA.
5. Under High Availability, click **Activate**.
6. In the **Activate High Availability** wizard, select a data node to serve as the replica for the primary node, and click **OK**.
7. To activate HA, you must restart your cluster. Review the **Confirm Cluster Restart** message and click **Yes**.

   If the cluster was online, the administration interface displays progress as VCF Operations configures, synchronizes, and rebalances the cluster for HA.
8. If the primary node and replica node go offline, and the primary remains offline for any reason while the replica goes online, the replica node does not take over the primary role, take the entire cluster offline, including data nodes and log in to the replica node command-line console as a root.
9. Open `$ALIVE_BASE/persistence/persistence.properties` in a text editor.
10. Locate and set the following properties:

    ```
    db.role=PRIMARY
    db.driver=/data/vcops/xdb/vcops.bootstrap
    ```
11. Save and close *`persistence.properties`*.
12. In the administration interface, bring the replica node online, and verify that it becomes the primary node and bring the remaining cluster nodes online.

High Availability is activated on your VCF Operations cluster.

After creating a primary replica node, you have the following options.

- New, unstarted clusters:
  – Create and add data nodes.
  – Click **Start** VCF Operations to start the cluster, and log in to finish configuring the product.
    The cluster might take from 10 to 30 minutes to start, depending on the size of your cluster and nodes. Do not make changes or perform any actions on cluster nodes while the cluster is starting.
- Established, running clusters:
  – Create and add data nodes.
- To deactivate high availability, click **Deactivate HA** and follow the steps in the admin UI. The deactivation timeline depends on the cluster size and the data retention period. Please wait for the deactivation to complete.

Converging Existing Virtual Infrastructure to a VCF or a vSphere Foundation Platform

When you converge your existing virtual infrastructure by using VCF Installer, the existing components become part of the management domain of a new VCF Instance in a new or an existing VCF fleet. The process for converging includes complying with the general requirements and scenario-specific prerequisites, manual upgrades of your existing components, and deployments and configurations performed by VCF Installer.

The following configurations, requirements, and converge scenarios are applicable for your management domain only. You import and upgrade workload domains in VCF Operations, after you deployed the management domain.

The versions of the converged VCF components determine which VCF features are available in your environment and the component with the earliest version defines the effective VCF version for the entire instance. VCF features that are introduced in later versions are available only after all VCF components reach the target version. To access the full feature set of VCF 9.1.x, upgrade all components to version 9.1.x.

VCF Upgrade Planner

You can now use the new VCF Upgrade Planning Tool to navigate you during the process of converging your existing infrastructure to vSphere Foundation and VCF 9.1.x. Different tailored scenarios exist for starting from just vSphere or different other combinations that include NSX, Aria components, and so on.

Upgrading Converged Components

VCF enforces strict upgrade path validation. You can only upgrade your VCF components to a target version that is released after the release date of your current version. The target version can be a major, minor, or a maintenance release. In general, you can upgrade directly to a patch release only if you can also upgrade to its parent maintenance release. For example:

- Supported: VCF 5.2.0 (25 JUN 2024) to VCF 9.1.0 (11 MAY 2026).
- Not supported: VCF 5.2.4 (27 MAY 2026) to VCF 9.1.0 (11 MAY 2026).
- Supported: VCF 5.2.4 (27 MAY 2026) to VCF 9.1.1 (03 SEP 2026).

This is valid for all components in a VCF source BoM. For example:

- Supported upgrade to VCF 9.1.0 (11 MAY 2026): vCenter 8.0 Update 3a (18 JUN 2024) and ESX 8.0 Update 3 (25 JUN 2024), which are part of VCF 5.2.0 BoM.
- Not supported upgrade to VCF 9.1.0 (11 MAY 2026): vCenter 8.0 Update 3j (27 MAY 2026) and ESX 8.0 Update 3j (27 MAY 2026), which are part of VCF 5.2.4 BoM.
- Supported upgrade to VCF 9.1.1 (03 SEP 2026): vCenter 8.0 Update 3j (27 MAY 2026) and ESX 8.0 Update 3j (27 MAY 2026), which are part of VCF 5.2.4 BoM.

VCF components might have further limitations, see VMware Interoperability Matrix.

Supported and Not Supported Configurations to Converge to VCF

Before you proceed with a supported scenario, verify that your existing configuration is supported.

vCenter and NSX Considerations

- For vCenter instances with existing NSX registration, the convergence procedure brings your vCenter environment with an existing NSX deployment under SDDC Manager as the management domain. SDDC Manager imports NSX Manager credentials, registers them in inventory, and explicitly trusts node certificates. vSphere clusters not prepared for NSX remain unconfigured for NSX.
- For vCenter instances without existing NSX registration, SDDC Manager deploys NSX Manager VMs, configures the NSX cluster, and registers vCenter as a Compute Manager in NSX. All eligible vSphere clusters in the domain are prepared for NSX.

Convergence of vCenter instances with existing NSX registrations and without Enhanced Linked Mode (ELM) is supported. Alternatively, you can deploy a new NSX 9.1.x instance during the convergence operation by using VCF Installer. The version of the deployed NSX is the latest compatible NSX version for the related vCenter instance. To change the NSX version, see Broadcom knowledge base article 429205.

- When deploying a new NSX Manager, you can choose between an HA deployment model with a cluster of 3 NSX Manager nodes or a simple deployment model with 1 NSX Manager.
- You can choose to configure overlay traffic on a management VMkernel or on a VLAN-backed segment.

Note:

After the convergence, you can only upgrade your VCF environment to a version that is released after the release date of the earliest version of the existing VCF components, that you selected for convergence. If the NSX version is released after a specific VCF minor or maintenance release, you cannot upgrade the management domain to that specific VCF minor or maintenance version. See VMware Interoperability Matrix.

If you are converging an NSX Federation configuration, only the Local Manager is converted. To upgrade such an environment, see Upgrading NSX Global Manager Nodes in a Federated Environment.

When you converge your existing infrastructure into a management domain, the associated NSX Manager can be shared with workload domains in the VCF Instance.

Note:

During convergence, VCF Installer treats the network on which the existing vCenter instance resides as the management network. All VCF management components, both newly deployed by the VCF Installer and existing, including SDDC Manager, NSX Manager, VCF Operations, and VCF Automation, must reside on the same subnet as the existing vCenter instance. For information about network deployment options, see Network Deployment Options for VCF Components.

Supported and Not Supported Configurations

The following table lists supported and not supported configurations for converging your existing infrastructure to a management domain.

For storage and compute, maintain an additional 20% free capacity beyond the deployment resource requirements. If the cluster does not meet this guideline, VCF Installer displays a warning, but does not block the deployment.

Category	Supported	Not Supported
Storage	• Enough free space for the full deployment of VMware Cloud Foundation. See VMware Configuration Maximums. • Datastore shared across all hosts in the cluster, accessible from all hosts in the cluster, and writable by all hosts in the cluster • Minimum number of ESX hosts for new deployments of VCF Operations, VCF Automation, and NSX Manager – Simple deployments • Minimum 3 ESX hosts with vSAN Storage or minimum 2 ESX hosts with external storage – High availability deployments • minimum 4 ESX hosts • Any supported vSphere storage type. If the default cluster includes multiple datastore types, VCF determines the primary datastore using the following priority: 1. vSAN 2. NFS v3 3. VMFS 4. NFS 4.1 5. iSCSI 6. vVols* **Note:** If you converge a cluster that runs vCenter in an environment with multiple mounted datastores of the same type, the datastore on which the vCenter runs is designated as the principal storage for the cluster. If you are using a `JSON` specification file to converge, you can use the `existingDatastoreName` property to override the priority order and use a specific datastore. • In VCF 9.1, vSAN Stretched Clusters (ESA, OSA, or ESA storage clusters) with at least 3 ESX hosts in each availability zone and a witness host deployed in a vSAN witness zone that is different from the availability zones. • In VCF 9.1.1 and later, vSAN Stretched Clusters (ESA, OSA, or ESA storage clusters) with at least 2 ESX hosts in each availability zone and a witness host deployed in a vSAN witness zone that is different from the availability zones. • Two-node vSAN clusters for remote offices/branch offices (ROBO) with two ESX hosts and one witness host for VCF simple deployments **Note:** For two-node vSAN clusters, the admission control policy should be configured to >= 50% for both memory and CPU. • vSAN OSA clusters, where deduplication and compression are either both activated or both deactivated • ~~vSAN OSA clusters with activated compression~~ only	
Network	• vSphere Distributed Switch (VDS) 8.0 or later **Note:**	• Cisco virtual switches • vCenter instances without a vSphere Distributed

Category	Supported	Not Supported
Compute	• vCenter instances without ELM and with existing NSX registrations • vCenter instances without ELM and with existing NSX Local Managers that are part of an NSX Federation configuration • Virtual machine of vCenter instance hosted on one of the clusters that is managed by the same instance **Note:** You can move the virtual machine to a supported location by using vSphere vMotion. See Migrating Virtual Machines Between vCenter Systems. • Clusters using vSphere Configuration Profiles • Clusters using vSphere Lifecycle Manager images **Note:** You can convert clusters that use baselines to use images before upgrading to ESX 9.1.x or later. See Convert a Cluster or a Host That Uses Baselines Into a Cluster or a Host That Uses vSphere Lifecycle Manager Images. • Clusters using fully automated vSphere DRS • Standalone ESX hosts and single ESX host clusters, if your vCenter instance has an additional cluster that meets the cluster requirements **Note:** You can manually perform the lifecycle management of the standalone ESX hosts and single ESX host clusters by using the vSphere Client. • Clusters with vSphere Supervisor enabled • Clusters with vSphere Supervisor enabled and Avi Load Balancer • vCenter instance with multiple clusters – All clusters managed by the instance are imported together into the management domain. The cluster hosting the virtual machine of the vCenter instance is automatically designated as the primary cluster of the management domain.	• vCenter instances with ELM **Note:** If your vCenter instance does not have existing NSX registrations, you can manually upgrade your vCenter instances to version 9.1.x or later and then break ELM for all vCenter instances. See Deactivate Enhanced Link Mode for Standalone vCenter Instances Using the cmsso-util break-elm Utililty. • vCenter instances with existing NSX registrations and with ELM • vCenter instance VM hosted on a cluster that is managed by a different vCenter instance • Clusters using baselines for lifecycle management • Clusters using manual or partially automated vSphere DRS **Note:** You can set the automation level to fully automated. See Edit Cluster Settings. • Clusters that use vCenter High Availability (VCHA) • vCenter instances that are already connected to an SDDC Manager • Partial import of clusters – All clusters managed by a vCenter instance are imported together.

* vSphere Virtual Volumes (vVols) framework is deprecated in version 9 and using it is not recommended.

Important:

Some functionalities are not available for converged clusters:

- Adding hosts to converged clusters without the use of the vSphere Client.
- Password Management of imported ESX hosts via the Password Management Console of VCF Operations. To create the missing ESX root credential entries in SDDC Manager to enable password management for these hosts, you can use the workaround in the Broadcom knowledge base article 388859.

Supported Components and Scenarios to Converge to VCF

Evaluate the components in your existing environment and determine which supported convergence scenario to follow.

Supported Components

The following table lists pre-version 9 components and each table row provides information whether the component can be converged to VCF 9.1.x.

Table 974: Supported Components to Converge by Using VCF Installer

Pre-Version 9 Component	Converge Existing Component
Aria Operations	Yes
Aria Automation	Yes
NSX	Yes *
vCenter	Yes
ESX	Yes
vSAN	Yes

Note:

Your pre-version 9 components can be managed by VMware Aria Suite Lifecycle. VMware Aria Suite Lifecycle is not part of VCF 9.1.x. Post-convergence, the newly introduced Fleet lifecycle and SDDC lifecycle components, part of VCF management services, perform that functionality.

* NSX 4.2.1 or later deployments that are registered with vCenter instances that do not use Enhanced Linked Mode are converged to VCF with their current version. After the convergence completes, you can upgrade NSX to version 9.1.x.

Table 975: Supported Components to Manually Upgrade Post-Convergence

Pre-Version 9 Component	Post-Convergence Procedure
Aria Operations for Logs	Deploy Log Management
Aria Operations for Networks	Upgrading to VCF Operations for Networks 9.1
Aria Automation Orchestrator	Upgrade to VCF Operations Orchestrator 9.1

Supported Scenarios

During the convergence process, you use VCF Installer to specify deployment information for your environment such as your existing vSphere infrastructure, VMware Aria infrastructure, and other information in the deployment wizard. Any missing VMware Cloud Foundation software components are automatically deployed and configured using the provided information. Existing vSphere infrastructure is used to create the management domain of your VCF Instance.

- Converge a vCenter Instance, ESX Hosts, and Optionally vSAN and NSX
- Converge a vCenter Instance, ESX Hosts, VMware Aria Operations, and Optionally vSAN and NSX
- Converge a vCenter Instance, ESX Hosts, VMware Aria Operations, and Optionally vSAN and NSX
- Converge a vCenter Instance, ESX Hosts, VMware Aria Automation, and Optionally vSAN and NSX
- Converge a vCenter Instance, ESX Hosts, VMware Aria Operations, and VMware Aria Automation, and Optionally vSAN and NSX

After you select a convergence path, depending on the version of your standalone components, you might need to upgrade some of them before you can proceed with converging the components to a supported vSphere Foundation or VCF deployment. For either path, first verify that you meet the minimum version requirements and supported configurations.

Converge a vCenter Instance, ESX Hosts, and Optionally vSAN and NSX

To converge your existing vCenter instance and ESX hosts to a management domain, ensure that you comply with the prerequisites, follow the procedure, and perform the next steps. This scenario supports existing environments with and without NSX.

- Verify that your existing configuration is supported. See Supported and Not Supported Configurations to Converge to VCF.
- Verify that your environment satisfies the minimum component versions.

Item	VCF 9.1.x Requirements
Minimum component versions	• VMware vCenter 8.0 Update 3a • VMware ESX 8.0 Update 3 • (Optional) vSAN 8.0 Update 3 • (Optional) VMware NSX 4.2.1

To ensure a successful convergence process, you must follow this high-level convergence procedure in the provided order. Each step includes a dedicated and detailed procedure, that you must complete, before proceeding with the next one.

1. Deploy the VCF Installer appliance.

   See Deploy VCF Installer.

2. Download the binaries for the components that VCF Installer deploys.

   See Downloading Binaries to VCF Installer.

3. Ensure that your vCenter instance runs a supported version for your convergence scenario.

   - If your vCenter instance is without an existing NSX registration, upgrade your vCenter instance to version 9.1.x. See Upgrading the vCenter Appliance.
   - If your vCenter instance is version 8.0 Update 3a or later and has an existing NSX registration with version 4.2.1 or later, proceed without upgrading.

4. Converge your existing environment to a VCF platform by using VCF Installer.

   VCF Installer validates and uses your existing components, deploys required new components, and configures the platform. See Deploy Components by Using VCF Installer to Complete the Converging to VCF Process.

5. If VCF Installer deploys a new NSX instance, in VCF Operations, synchronize the inventory of the management domain.

   See Remediate VCF Domain Configuration Drift.

To ensure your VCF environment is fully operational, after completing the required steps in the main convergence procedure, perform the following additional tasks.

- License your environment. See Registering VCF Operations and a License Server with the VCF Business Services Console.
- Deploy the remaining components. See Completing the Deployment of Your Platform.
- Import your existing infrastructure as a workload domain. See Import an Existing vCenter to Create a Workload Domain.
- Upgrade your workload domains. See Upgrading VMware Cloud Foundation Workload Domains to 9.1.x.
- Integrate your workload domains in VCF Automation. See Add Workload Domains to Your Cloud Organization Structure.

Converge a vCenter Instance, ESX Hosts, VMware Aria Operations, and Optionally vSAN and NSX

To converge your existing vCenter instance, ESX hosts, VMware Aria Suite Lifecycle, and VMware Aria Operations, ensure that you comply with the prerequisites, follow the procedure, and perform the next steps. This scenario supports existing environments with and without NSX.

- Verify that your vCenter instance is registered in VMware Aria Operations. See Configure a vCenter Server Cloud Account in VMware Aria Operations.

- Verify that your existing configuration is supported. See Supported and Not Supported Configurations to Converge to VCF.
- Verify that your environment satisfies the minimum component versions.

Item	VCF 9.1.0.x Requirements	VCF 9.1.1.x Requirements
Minimum component versions	• VMware Aria Operations 8.18 • VMware vCenter 8.0 Update 3a • VMware ESX 8.0 Update 3 • (Optional) vSAN 8.0 Update 3 • (Optional) VMware NSX 4.2.1	• VMware Aria Operations 8.18.6 • VMware vCenter 8.0 Update 3a • VMware ESX 8.0 Update 3 • (Optional) vSAN 8.0 Update 3 • (Optional) VMware NSX 4.2.1

To ensure a successful convergence process, you must follow this high-level convergence procedure in the provided step order. Each step includes a dedicated and detailed procedure, that you must complete, before proceeding with the next one.

Note:

A VCF fleet is managed by a single VCF Operations instance. Only the instance you converge during this procedure to the VCF fleet performs fleet, lifecycle, certificate, and license management of your VCF environment.

If your existing environment has additional VMware Aria Operations instances, the instances operate in a monitoring-only capacity only. You can manually upgrade an additional instance to VCF Operations 9.x or if you no longer need the instance, you can power off and delete the virtual machine from the vCenter inventory.

This scenario also supports environments that include Aria Operations for Logs, Aria Operations for Networks, and Aria Automation Orchestrator.

1. Deploy the VCF Installer appliance.

   See Deploy VCF Installer.

2. Download the binaries for the components that VCF Installer deploys.

   See Downloading Binaries to VCF Installer.

3. Upgrade your VMware Aria Operations to VCF Operations version 9.1.x.

   See Upgrading to VCF Operations 9.1

4. Ensure that your vCenter instance runs a supported version for your convergence scenario.
   - If your vCenter instance is without an existing NSX registration, upgrade your vCenter instance to version 9.1.x. See Upgrading the vCenter Appliance.
   - If your vCenter instance is version 8.0 Update 3a or later and has an existing NSX registration with version 4.2.1 or later, proceed without upgrading.
5. Converge your existing environment to a VCF platform by using VCF Installer.

   VCF Installer validates and uses your existing components, deploys required new components, and configures the platform. See Deploy Components by Using VCF Installer to Complete the Converging to VCF Process.

6. If VCF Installer deploys a new NSX instance, in VCF Operations, synchronize the inventory of the management domain.

   See Remediate VCF Domain Configuration Drift.

To ensure your VCF environment is fully operational, after completing the required steps in the main convergence procedure, perform the following additional tasks.

- License your environment. See Registering VCF Operations and a License Server with the VCF Business Services Console.
- Deploy the remaining components. See Completing the Deployment of Your Platform.
- If you have VMware Aria components that are not subject to upgrade to version 9.1.x, where supported, manually upgrade the components and import configurations from the old appliances to the new ones. See Upgrading to VCF

Operations for Networks 9.1 and Upgrade to VCF Operations Orchestrator 9.1. Otherwise, perform a fresh component deployment. See Deploy Log Management.

- If your environment has additional cloud proxy components that are deployed in other data centers, you can add the components in VCF Operations. See Add a Cloud Proxy.
- Import your existing infrastructure as a workload domain. See Import an Existing vCenter to Create a Workload Domain.
- Upgrade your workload domains. See Upgrading VMware Cloud Foundation Workload Domains to 9.1.x.
- Integrate your workload domains in VCF Automation. See Add Workload Domains to Your Cloud Organization Structure.

If you have an existing VMware Aria Suite Lifecycle, the instance is not upgraded to version 9.1.x. After VCF is deployed, the Fleet lifecycle and SDDC lifecycle components, part of VCF management services, perform that functionality.

- Remove your VMware Aria components from the VMware Aria Suite Lifecycle 8.x environment without deleting the associated VM from vCenter. See Delete a Product from an Environment.
  - If your VMware Aria Suite Lifecycle appliance no longer manages any deployments, you can delete the appliance.
  - If your VMware Aria Suite Lifecycle appliance manages VMware Identity Manager, which is used as an authentication source for a VCF Automation instance, do not delete the component. You must also preserve the appliance to perform all Day-N operations for VMware Identity Manager 3.3.x.

Converge a vCenter Instance, ESX Hosts, VMware Aria Automation, and Optionally vSAN and NSX

To converge your existing vCenter instance, ESX hosts, VMware Aria Suite Lifecycle, and VMware Aria Automation, ensure that you comply with the prerequisites, follow the procedure, and perform the next steps. This scenario supports existing environments with and without NSX.

- Verify that your existing configuration is supported. See Supported and Not Supported Configurations to Converge to VCF.
- Verify that your environment satisfies the minimum component versions.

Item	VCF 9.1.x Requirements
Minimum component versions	• VMware Aria Automation 8.18.1 • VMware vCenter 8.0 Update 3a • VMware ESX 8.0 Update 3 • (Optional) vSAN 8.0 Update 3 • (Optional) VMware NSX 4.2.1

To ensure a successful convergence process, you must follow this high-level convergence procedure in the provided step order. Each step includes a dedicated and detailed procedure, that you must complete, before proceeding with the next one.

Note:

If your environment has multiple VMware Aria Automation instances, only the first instance can be managed by VCF Operations in your VCF management domain. Any subsequent VMware Aria Automation instances must be managed by other dedicated VCF Operations instances.

This scenario also supports environments that include Aria Operations for Logs, Aria Operations for Networks, and Aria Automation Orchestrator.

1. Deploy the VCF Installer appliance.

   See Deploy VCF Installer.

2. Download the binaries for the components that VCF Installer deploys.

   See Downloading Binaries to VCF Installer.

3. Ensure that your vCenter instance runs a supported version for your convergence scenario.
   - If your vCenter instance is without an existing NSX registration, upgrade your vCenter instance to version 9.1.x. See Upgrading the vCenter Appliance.
   - If your vCenter instance is version 8.0 Update 3a or later and has an existing NSX registration with version 4.2.1 or later, proceed without upgrading.
4. Converge your existing environment to a VCF platform by using VCF Installer.

   VCF Installer validates and uses your existing components, deploys required new components, and configures the platform. See Deploy Components by Using VCF Installer to Complete the Converging to VCF Process.
5. If VCF Installer deploys a new NSX instance, in VCF Operations, synchronize the inventory of the management domain.

   See Remediate VCF Domain Configuration Drift.
6. Import your VMware Aria Automation in VCF Operations and upgrade to VCF Automation version 9.1.x.

   See Upgrading to VCF Automation 9.1 and Perform the Upgrade to VCF Automation 9.1.

To ensure your VCF environment is fully operational, after completing the required steps in the main convergence procedure, perform the following additional tasks.

- License your environment. See Registering VCF Operations and a License Server with the VCF Business Services Console.
- Deploy the remaining components. See Completing the Deployment of Your Platform.
- If you have VMware Aria components that are not subject to upgrade to version 9.1.x, where supported, manually upgrade the components and import configurations from the old appliances to the new ones. See Upgrading to VCF Operations for Networks 9.1 and Upgrade to VCF Operations Orchestrator 9.1. Otherwise, perform a fresh component deployment. See Deploy Log Management.
- Import your existing infrastructure as a workload domain. See Import an Existing vCenter to Create a Workload Domain.
- Upgrade your workload domains. See Upgrading VMware Cloud Foundation Workload Domains to 9.1.x.
- Integrate your workload domains in VCF Automation. See Add Workload Domains to Your Cloud Organization Structure.

If you have an existing VMware Aria Suite Lifecycle, the instance is not upgraded to version 9.1.x. After VCF is deployed, the Fleet lifecycle and SDDC lifecycle components, part of VCF management services, perform that functionality.

- Remove your VMware Aria components from the VMware Aria Suite Lifecycle 8.x environment without deleting the associated VM from vCenter. See Delete a Product from an Environment.
  - If your VMware Aria Suite Lifecycle appliance no longer manages any deployments, you can delete the appliance.
  - If your VMware Aria Suite Lifecycle appliance manages VMware Identity Manager, which is used an authentication source for a VCF Automation instance, do not delete the component. You must also preserve the appliance to perform all Day-N operations for VMware Identity Manager 3.3.x.

Converge a vCenter Instance, ESX Hosts, VMware Aria Operations, and VMware Aria Automation, and Optionally vSAN and NSX

To successfully converge your existing vCenter instance, ESX hosts, VMware Aria Suite Lifecycle, VMware Aria Operations, and VMware Aria Automation, you must ensure that you comply with the prerequisites, follow the procedure, and perform the next steps. This scenario supports existing environments with and without NSX.

- Verify that your vCenter instance is registered in VMware Aria Operations. See Configure a vCenter Server Cloud Account in VMware Aria Operations.

- Verify that your existing configuration is supported. See Supported and Not Supported Configurations to Converge to VCF.
- Verify that your environment satisfies the minimum component versions.

Item	VCF 9.1.0.x Requirements	VCF 9.1.1.x Requirements
Minimum component versions	• VMware Aria Operations 8.18 • VMware Aria Automation 8.18.1 • VMware vCenter 8.0 Update 3a • VMware ESX 8.0 Update 3 • (Optional) vSAN 8.0 Update 3 • (Optional) VMware NSX 4.2.1	• VMware Aria Operations 8.18.6 • VMware Aria Automation 8.18.1 • VMware vCenter 8.0 Update 3a • VMware ESX 8.0 Update 3 • (Optional) vSAN 8.0 Update 3 • (Optional) VMware NSX 4.2.1

To ensure a successful convergence process, you must follow this high-level convergence procedure in the provided step order. Each step includes a dedicated and detailed procedure, that you must complete, before proceeding with the next one.

Note:

A VCF fleet is managed by a single VCF Operations instance. Only the instance you converge during this procedure to the VCF fleet performs fleet, lifecycle, certificate, and license management of your VCF environment.

If your existing environment has additional VMware Aria Operations instances, the instances operate in a monitoring-only capacity only. You can manually upgrade an additional instance to VCF Operations 9.x or if you no longer need the instance, you can power off and delete the virtual machine from the vCenter inventory.

This scenario also supports environments that include Aria Operations for Logs, Aria Operations for Networks, and Aria Automation Orchestrator.

1. Deploy the VCF Installer appliance.

   See Deploy VCF Installer.

2. Download the binaries for the components that VCF Installer deploys.

   See Downloading Binaries to VCF Installer.

3. Upgrade your VMware Aria Operations to VCF Operations version 9.1.x.

   See Upgrading to VCF Operations 9.1.

4. Ensure that your vCenter instance runs a supported version for your convergence scenario.
   - If your vCenter instance is without an existing NSX registration, upgrade your vCenter instance to version 9.1.x. See Upgrading the vCenter Appliance.
   - If your vCenter instance is version 8.0 Update 3a or later and has an existing NSX registration with version 4.2.1 or later, proceed without upgrading.

5. Converge your existing environment to a VCF platform by using VCF Installer.

   VCF Installer validates and uses your existing components, deploys required new components, and configures the platform. See Deploy Components by Using VCF Installer to Complete the Converging to VCF Process.

6. If VCF Installer deployed an NSX instance, in VCF Operations, synchronize the inventory of the management domain.

   See Remediate VCF Domain Configuration Drift.

7. Import your VMware Aria Automation in VCF Operations and upgrade to VCF Automation version 9.1.x.

   See Upgrading to VCF Automation 9.1 and Perform the Upgrade to VCF Automation 9.1.

To ensure your VCF environment is fully operational, after completing the required steps in the main convergence procedure, perform the following additional tasks.

- License your environment. See Registering VCF Operations and a License Server with the VCF Business Services Console.

- Deploy the remaining components. See Completing the Deployment of Your Platform.
- If you have VMware Aria components that are not subject to upgrade to version 9.1.x, where supported, manually upgrade the components and import configurations from the old appliances to the new ones. See Upgrading to VCF Operations for Networks 9.1 and Upgrade to VCF Operations Orchestrator 9.1. Otherwise, perform a fresh component deployment. See Deploy Log Management.
- If your environment has additional cloud proxy components that are deployed in other data centers, you can add the components in VCF Operations. See Add a Cloud Proxy.
- Import your existing infrastructure as a workload domain. See Import an Existing vCenter to Create a Workload Domain.
- Upgrade your workload domains. See Upgrading VMware Cloud Foundation Workload Domains to 9.1.x.
- Integrate your workload domains in VCF Automation. See Add Workload Domains to Your Cloud Organization Structure.

If you have an existing VMware Aria Suite Lifecycle, the instance is not upgraded to version 9.1.x. After VCF is deployed, the Fleet lifecycle and SDDC lifecycle components, part of VCF management services, perform that functionality.

- Remove your VMware Aria components from the VMware Aria Suite Lifecycle 8.x environment without deleting the associated VM from vCenter. See Delete a Product from an Environment.
  - If your VMware Aria Suite Lifecycle appliance no longer manages any deployments, you can delete the appliance.
  - If your VMware Aria Suite Lifecycle appliance manages VMware Identity Manager, which is used an authentication source for a VCF Automation instance, do not delete the component. You must also preserve the appliance to perform all Day-N operations for VMware Identity Manager 3.3.x.

Deploy Components by Using VCF Installer to Complete the Converging to VCF Process

By using VCF Installer, you use your existing vSphere and Aria infrastructure to converge it to a new VCF fleet or a new VCF Instance in an existing VCF fleet. During the process, VCF Installer deploys all required components that you do not have in your existing environment and uses your vCenter instance and ESX hosts to create the management domain for the new instance.

- Download the required binaries for VCF. See Downloading Binaries to VCF Installer.
- Verify that you deployed VCF Installer version 9.1.1 or later. See Deploy VCF Installer.
- Verify that you configured the required FQDNs and IP addresses for deployment. See VCF Components FQDNs and IP addresses.
- Verify that you meet the requirements and have completed the pre-requisite tasks based on your convergence scenario. See Supported Components and Scenarios to Converge to VCF.
- If you plan to use a load balancer in your VCF environment, verify it is deployed and configured.

A VCF fleet refers to an environment consisting of one or more VCF Instances and standalone vCenter instances that are managed by the same VCF Operations instance. If you have a VCF fleet, you can converge your infrastructure to a new VCF Instance within the fleet by connecting to your existing VCF Operations instance or you can deploy a new VCF Operations instance to create a new VCF fleet. If you have an existing VCF Operations instance, you can also use it to create a new VCF fleet.

During the procedure, you plan and gather all the information required for deployment. By specifying the scale of the deployment and network types, you define the required resources for your deployment. You use the VCF Installer deployment workflows to specify deployment information specific for your existing environment and other information. As a result, the VMware Cloud Foundation platform is automatically deployed and configured using the information provided.

You can use the VCF Installer deployment wizard or a custom `JSON` specification file. For more information, see Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation.

VCF Installer allows you to skip VCF Automation during the initial deployment and deploy it as a Day-N operation. See Deploying VCF Automation as a Day-N Operation.

1. In a web browser, log in to the VCF Installer appliance at `https://installer_appliance_FQDN`.

   Enter the `admin@local` user and password that you provided when you deployed the appliance and click **Log In**.

2. Under **Deploy**, select **Deployment Wizard** > **VMware Cloud Foundation** .
3. Review the introductory information and click **Next**.
4. Select **Deploy a new VCF fleet** or **Deploy a VCF Instance in an existing VCF fleet**.
5. On the **Existing Components** page, select your existing infrastructure components.

   When you connect to your existing components, VCF Installer validates them to ensure compatibility.

   When you select an existing VMware vCenter infrastructure, the **Storage**, **Hosts**, and **Distributed Switch** pages of the wizard are not available.

   You can also select the **I have an existing VCF Automation instance or I will deploy later** check box to deploy VCF Automation as Day-N operation, instead of using the VCF Installer deployment workflows. See Deploying VCF Automation as a Day-N Operation.

   Starting with VCF 9.1.1, it is mandatory that VCF Operations is always deployed in the management domain of the first VCF Instance in the VCF fleet. If you selected the **I have an existing VCF Operations 9.1 instance** option, the **I have an existing VMware vCenter** option is also always selected.
6. Select a deployment model and the scale for your VCF Instance.

Deployment model	You can choose the deployment model for new appliances that get deployed. **High Availability** is recommended for production environments. **Note:** The deployment model that you select does not impact existing appliances that you are using in your VCF fleet or VCF Instance. See the VMware Cloud Foundation Design documentation for more information. **Note:** New appliances that get deployed are automatically configured to comply with Federal Information Processing Standards (FIPS).
Size	You can choose the scale of the components in your VCF Instance. • Small – In VCF 9.1, this is the only available option for the simple deployment model. – In VCF 9.1.1 and later, this is the default option for the simple and HA deployment models. • Medium – In VCF 9.1, this is the default option for the HA deployment model. – In VCF 9.1.1 and later, this is an option for the HA deployment model. • Large – In VCF 9.1 and later, this is an option for the HA deployment model. For more information about deployment sizes, see VCF Fleet Sizing Models.

7. Select your network configuration.

   Note:

   VCF Installer supports only vCenter instances with IPv4 network configuration and deploys the remaining components with IPv4 only. If your vCenter instance uses dual-stack (IPv4 and IPv6) networking, perform your deployment by using a JSON spec, where all remaining components get deployed in the same configuration.

To use dual-stack networking, the VCF Installer virtual machine and VCF Installer must have IPv4 and IPv6 configured.

Management network configuration	• VM management network: For core VCF components, such as vCenter, ESX, and NSX. You can share this network with the ESX management network or use a dedicated network. • VCF management network: For broader VCF management and automation components, such as VCF Operations and VCF Automation. You can share this network with the VM management network or use a dedicated network. You can use one of the networks to segment management traffic for security and compliance requirements. For more information, see Network Deployment Options for VCF Components.

If you are deploying a new VCF fleet, you can select **Deploy VCF Operations and VCF Automation to a specific vSphere distributed switch or NSX segment**. When you select this option, VCF Installer skips the deployment of VCF Operations, cloud proxy, license server, and VCF Automation. For more information about deploying these deferred components later, see Deploying Deferred Components on Custom Networking.

Note:

This option is not available when deploying a VCF Instance in an existing VCF fleet or when using an existing VCF Operations instance.

When selecting this option, some of the pages in the VCF Installer wizard are not available.

8. Review the prerequisites for your deployment.

   Depending on your selections in the wizard, the prerequisites for the deployment are listed.

   You can click **PRE-FILL GENERATED FQDNs IN WIZARD** and specify the required FQDNs for the deployment. You can use the provided template or specify custom FQDNs. These values are later populated in the VCF Installer deployment workflows.

   You can download a JSON template that is based on your choices in the VCF Installer so far. You can populate it and deploy VCF by using a JSON specification file. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation for more information.

9. On the **General information** page, enter the configuration details for your deployment.

   Additional information:

Option	Description
Version	Choose which VCF version to install.
VCF Instance Name	The VCF Instance Name is used to provide a unique name when creating the VMware Cloud Foundation Adapter within VCF Operations. A descriptive name provides for easy identification when you have multiple VCF Instances.
Management domain name	The name of the management domain that is created in SDDC Manager. This name is also used as a prefix for the default names of objects that are created or deployed by the VCF Installer. You can change these default names as you progress through the deployment wizard.
Customer Experience Improvement Program (CEIP)	Select an option to activate or deactivate CEIP across all deployed components. VMware's Customer Experience Improvement Program (CEIP) collects technical information from participating customers to provide better support. See Customer Experience Improvement Program for more information.

10. On the **VCF Management** page, enter the details for your VCF management components.

I am deploying ...	Details
A new VCF fleet with a new VCF Operations instance	Enter all required configuration details for the VCF management components. For more information, see the VCF Management Configuration Details table below. If you specified the required FQDNs earlier, these are now populated. VCF Automation details section does not appear if you selected that you want to connect a VCF Automation instance later.
A new VCF fleet with an existing VCF Operations instance	• Connect to your existing VCF Operations instance. • Enter details for the new cloud proxy appliance. If a VCF Automation instance is connected to the VCF Operations instance, the appliance FQDN is displayed and you must enter the root password.
A new VCF Instance with an existing VCF Operations instance	Connect to the VCF Operations instance that manages the VCF fleet in which you want to deploy this VCF Instance. After connecting, enter details for the new cloud proxy appliance. If a VCF Automation instance is connected to the VCF Operations instance, the appliance FQDN is displayed and you must enter the root password.

Note:

This page does not appear if you selected the custom networking option for VCF Operations and VCF Automation.

Table 976: VCF Management Configuration Details

Each Fully Qualified Domain Name (FQDN) must point to a unique IP address.

Note:

If you are deploying a VCF fleet or a new VCF Instance with dual-stack networking, for VCF components that support the dual-stack model, the FQDNs must resolve to IPv4 and IPv6 addresses. See Dual Stack IP Addressing Model.

The FQDNs for VCF management services must point to unique IP addresses that are not assigned and the FQDNs must not be in the IP Ranges assigned to VCF management services nodes.

Component	Field	Description
VCF Operations	Primary node FQDN	Enter the FQDN of the VCF Operations primary node.
	Replica node FQDN	Enter the FQDN of the VCF Operations replica node.
	Data node FQDN	Enter the FQDN of the VCF Operations data node. This field appears only when deploying VCF in High Availability mode with a Medium or Large deployment size.
Load Balancer	Load balancer FQDN	(Optional) Enter the FQDN of the load balancer, that distributes the VCF Operations traffic. This field appears only when deploying VCF in High Availability mode.
Cloud Proxy	Cloud Proxy FQDN	Enter the FQDN of the cloud proxy. Cloud proxy collects and monitors data locally in a VCF Instance.
License Server	License Server FQDN	Enter the FQDN of the license server. A license server is added to a VCF Operations instance to store the licenses you want to assign from this VCF Operations instance. You must add at least one license server to each VCF Operations instance that you use for license management. You can deploy license server as IPv4 only. In VCF 9.1.1 and later, after the deployment, you can change the networking configuration to IPv6. See Configure IPv6 on a License Server.

Component	Field	Description
VCF Management Services	Fleet components FQDN	Enter the FQDN of the fleet-level components, such as fleet lifecycle, Salt RaaS, software depot, and so on. For more information, see Components in VCF and vSphere Foundation. **Note:** Do not use capital letters in the FQDN.
	Instance components FQDN	Enter the FQDN of the instance-level components, such as SDDC lifecycle, Salt master, real-time metrics, real-time metrics store, and telemetry. For more information, see Components in VCF and vSphere Foundation. **Note:** Do not use capital letters in the FQDN.
	Identity broker FQDN	Enter the FQDN of the identity broker component. **Note:** Do not use capital letters in the FQDN.
	VCF services runtime FQDN	Enter the FQDN for VCF services runtime. VCF services runtime is the common architectural runtime deployed to deliver the shared cluster of management services. The hostname from this FQDN is prefixed to the names of its node VMs and related objects. **Note:** Do not use capital letters in the FQDN.
	Internal cluster CIDR IPv4	Select the VCF services runtime internal IPv4 network. • 198.18.0.0/15 • 240.0.0.0/15 • 250.0.0.0/15 Different VCF services runtime instances on the same network can work with the same internal network. For example, you can have the two instances of VCF services runtime for your VCF management services and VCF Automation using the same internal network.
VCF Automation	VCF Automation FQDN	Enter the FQDN to assign to the VCF Automation instance. Later, you use it to access the VCF Automation user interface. **Note:** Do not use capital letters in the FQDN.
	VCF services runtime FQDN	Enter the FQDN for VCF services runtime to connect VCF Automation with the VCF fleet lifecycle component that manages the lifecycle of VCF Automation. The FQDN must be dedicated to the runtime of the VCF Automation cluster, not the VCF management services cluster. The hostname from this FQDN is prefixed to the names of its node VMs and related objects. **Note:** Do not use capital letters in the FQDN.
	Internal cluster CIDR IPv4	Select the VCF services runtime internal IPv4 network. • 198.18.0.0/15 • 240.0.0.0/15 • 250.0.0.0/15 Different VCF services runtime instances on the same network can work with the same internal network. For example, you can have the two instances of VCF services runtime for your VCF management services and VCF Automation using the same internal network.

Depending on your network options for the VM management network and the VCF management network, the network deployment location of VCF management services can vary. See Network Deployment Options for VCF Components.

11. On the **Existing vCenter** page, enter the required details of your existing vCenter instance.

    To set up the single sign-on with identity broker, select a vCenter instance where identity broker or Identity Manager is used as the identity provider.

12. Enter details about the networks.

    - VCF Management Services IP Range
      - For deployments version 9.1.0.0 through 9.1.0.300, you specify the IP addresses as a contiguous IP range. Alternatively, you can specify non-contiguous IP addresses by using a JSON specification file. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation.
      - For deployments version 9.1.0.400 and later, you can specify the IP addresses as contiguous or non-contiguous IP range.
    - VCF Automation IP Range

      This option is not displayed if you selected the **I have an existing VCF Automation instance** check box.
      - For deployments version 9.1.0.0 through 9.1.0.300, you specify a contiguous IP range for the VCF Automation nodes with 5 IP addresses.
      - For deployments version 9.1.0.400 and later, you specify 5 IP addresses as contiguous or non-contiguous IP range.

13. On the **NSX Manager** page, enter the required details for NSX Manager appliance.

    If the existing NSX Manager version is below 9.0, you cannot activate NSX Edge cluster synchronization. After the convergence operation completes, you can upgrade NSX Manager to 9.1 or later and use the **Enable Edge cluster sync and import NSX Edge node VMs** option in the VCF Operations user interface. See Remediate VCF Domain Configuration Drift.

14. On the **SDDC Manager** page, enter the required details.

15. On the **Review** page, review the deployment information.

    To make changes, navigate back to the relevant page in the deployment wizard. You can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage.

16. On the **Validate & Deploy** page, review the validation information.

    The VCF Installer validates the information that you provided in the deployment wizard and reports any errors or warnings.

Warnings	Review any warning messages. To proceed with your deployment, address the warnings or click **Acknowledge.**
Errors	Use the Review all failed validations and resolve the issues by using the provided remediation information. You can navigate back to the relevant pages in the deployment wizard to make updates and then re-run the validations. Alternatively, you can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage.

17. After addressing all validation errors and warnings, click **Deploy**.

    You can monitor the deployment progress. If there are errors, review the failed tasks in the Tasks panel, address the issue(s), and retry the deployment.

    CAUTION:

    After deployment completes, do not move the VCF management services or VCF Automation VMs to a different resource pool or folder. If you move the VMs to a new resource pool or folder, subsequent patching, deployment, or scale out operations fail.

- License your environment. See Registering VCF Operations and a License Server with the VCF Business Services Console.

- Click **Review Passwords** to review the passwords. To review passwords for past deployments, in the VCF Installer appliance, click the user name at the top-right corner of the window and select **Deployments**.

  Note:

  You can also retrieve some of the passwords using the SDDC Manager API and VCF Operations API.
- (Optional) Download the JSON specification file for the deployment.

  Note:

  The JSON specification file does not include passwords.
- To cover the platform after a failure, configure file-based backups for SDDC Manager and other management components. See Configure File-Based Backup Schedule for SDDC Manager.
- For platform administration tasks such as managing identity and access, certificates, passwords, component lifecycle, and backup and restore of your VCF platform, see the Fleet Management guide.
- For monitoring and troubleshooting tasks such as configuring alerts, analyzing logs, monitoring dashboards, and troubleshooting your VCF platform, see the Infrastructure Operations guide.

Day-N lifecycle management and topology expansion are divided as follows.

- Lifecycle management for the converged vCenter, NSX Manager, and ESX hosts is orchestrated by VCF Operations.
- Topology expansion is performed directly in the VCF components.
  - Add ESX hosts in vCenter.
  - Add new clusters by using the SDDC Manager workflow or directly in vCenter.
  - Add NSX Edge nodes or Edge clusters in vCenter or NSX Manager.

Converge a vCenter Instance and ESX Hosts to vSphere Foundation

To successfully converge your existing vCenter instance and ESX hosts, you must ensure that you comply with the prerequisites, follow the procedure, and perform the next steps.

- Download the required binaries for VCF. See Downloading Binaries to VCF Installer.
- Verify that you deployed VCF Installer version 9.1.1 or later. See Deploy VCF Installer.
- Verify that you configured the required FQDNs and IP addresses for deployment. See VCF Components FQDNs and IP addresses.
- Verify that you satisfy the minimum component versions.

Item	vSphere Foundation 9.1 Requirements
Minimum component versions	• VMware vCenter 9.1 • VMware ESX 8.0 Update 3

Supported vSphere configurations are also supported for convergence to vSphere Foundation.

During the procedure, you plan and gather all the information that is required for deployment. By specifying the network types, you define the required resources for your deployment. You use VCF Installer to specify deployment information for your existing vCenter instance and ESX hosts, and to deploy VCF Operations, VCF management services, and the license server component to complete the converging to VMware vSphere Foundation platform. You can also manually deploy VCF Operations, see Deploy VCF Operations as Part of a 9.1 Upgrade.

VCF Installer deploys vSphere Foundation in a simple mode with the following component sizes: medium VCF Operations, small VCF management services, and the existing vCenter instance is converged as is. To change the component sizes, converge to vSphere Foundation by using a `JSON` specification file. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation.

To converge to vSphere Foundation 9.1 without deploying VCF management services, you must manually upgrade your existing components. See Upgrading vSphere 8 and Optionally vSAN and Aria Operations 8 to 9.1.x.

1. Upgrade your existing vCenter instance to version 9.1.

   See Upgrading the vCenter Appliance.

2. Deploy the VCF Installer appliance.

   See Deploy VCF Installer.

3. Download bundles for the VCF Installer appliance.

   See Downloading Binaries to VCF Installer.

4. Use VCF Installer to deploy your vSphere Foundation platform.
   a) In a web browser, log in to the VCF Installer appliance at `https://installer_appliance_FQDN`.

      Enter the `admin@local` user and password that you provided when you deployed the appliance and click **Log In**.
   b) Under **Deploy**, select **Deployment Wizard** > **VMware vSphere Foundation**.
   c) Review the introductory information and click **Next**.
   d) Select **Deploy new VMware vSphere Foundation**.
   e) On the **Existing Components** page, select **VMware vCenter**.

      When you select an existing VMware vCenter infrastructure, the **Storage**, **Hosts**, **Networks**, and **Distributed Switch** pages of the wizard are not available.
   f) Select your network configuration.

      You can proceed with default network configuration or click **Customize**.

Management network configuration	• VCF management network: For broader VCF management components, such as VCF Operations. You can share this network with the VM management network or use a dedicated network. You can use one of the networks to segment management traffic for security and compliance requirements. For more information, see Network Deployment Options for VCF Components.

Note:

VCF Installer supports only vCenter instances with IPv4 network configuration and deploys the remaining components with IPv4 only. If your vCenter instance uses dual-stack (IPv4 and IPv6) networking, perform your deployment by using a JSON spec, where all remaining components get deployed in the same configuration. To use dual-stack networking, the VCF Installer virtual machine and VCF Installer must have IPv4 and IPv6 configured.

   g) Review the prerequisites for your deployment.

      Depending on your selections in the wizard, the prerequisites for the deployment are listed.

      You can click **PRE-FILL GENERATED FQDNs IN WIZARD** and specify the required FQDNs for the deployment. You can use the provided template or specify custom FQDNs. These values are later populated in the VCF Installer deployment workflows.

      You can download a JSON template that is based on your choices in the VCF Installer so far. You can populate it and deploy VCF by using a JSON specification file. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation for more information.
   h) On the **General information** page, enter the configuration details for your deployment.

      Additional information:

Option	Description
Version	Choose which VMware vSphere Foundation version to install.
Customer Experience Improvement Program (CEIP)	Select an option to activate or deactivate CEIP across all deployed components. VMware's Customer Experience Improvement Program (CEIP) collects technical information from participating customers to provide better support. For more information, see Customer Experience Improvement Program.

i) On the **VCF Management** page, enter the details for your VCF management components.

If you specified the required FQDNs earlier, the same are now populated.

Table 977: VCF Management Configuration Details

Each Fully Qualified Domain Name (FQDN) must point to a unique IP address.

Note:

For VCF components that support the dual-stack model, the FQDNs must resolve to IPv4 and IPv6 addresses. See Dual Stack IP Addressing Model.

The FQDNs for VCF management services must point to unique IP addresses that are not assigned and the

FQDNs must not be in the IP Ranges assigned to VCF management services nodes.

Component	Field	Description
VCF Operations	Primary node FQDN	Enter the FQDN of the VCF Operations primary node.
License server	License Server FQDN	Enter the FQDN of the license server. A license server is added to a VCF Operations instance to store the licenses you want to assign from this VCF Operations instance. You must add at least one license server to each VCF Operations instance that you use for license management. You can deploy license server as IPv4 only. In vSphere Foundation 9.1.1 and later, after the deployment, you can change the networking configuration to IPv6. See Configure IPv6 on a License Server.
VCF management services	Fleet components FQDN	Enter the FQDN of the fleet-level components, such as fleet lifecycle, Salt RaaS, software depot, and so on. For more information, see Components in VCF and vSphere Foundation. **Note:** Do not use capital letters in the FQDN.
	Instance components FQDN	Enter the FQDN of the instance-level components, such as SDDC lifecycle, Salt master, real-time metrics, real-time metrics store, and telemetry. For more information, see Components in VCF and vSphere Foundation. **Note:** Do not use capital letters in the FQDN.
	VCF services runtime FQDN	Enter the FQDN for VCF services runtime. VCF services runtime is the common architectural runtime deployed to deliver the shared cluster of management services. The hostname from this FQDN is prefixed to the names of its node VMs and related objects. **Note:** Do not use capital letters in the FQDN.
	Internal cluster CIDR IPv4	Select the VCF services runtime internal IPv4 network. • 198.18.0.0/15 • 240.0.0.0/15 • 250.0.0.0/15 Different VCF services runtime instances on the same network can work with the same internal network. For example, you can have the two instances of VCF services runtime for your VCF management services and VCF Automation using the same internal network.

Component	Field	Description
	Internal cluster CIDR IPv6	Select the VCF services runtime internal IPv6 network. • fd00::/111 (default) • fd00::0/111 • fc00::0004:0/111 • fc00::4:0/111 • fc00::/111 • fc00::0/111 This option is only available if you have dual-stack networking enabled in your environment.

Depending on your network options for the VM management network and the VCF management network, the network deployment location of VCF management services can vary. See Network Deployment Options for VCF Components.

j) On the **Existing vCenter** page, enter the required details of your existing vCenter instance.

VCF Installer validates your existing vCenter instance to ensure compatibility.

Note:

An NSX Manager cannot be connected to the vCenter.

To set up the single sign-on with identity broker, select a vCenter instance where identity broker or Identity Manager is used as the identity provider.

k) On the **Networks** page, enter the IP range for VCF management services.

Provide a minimum of 12 IP addresses for the VCF services runtime nodes. The IP addresses must be located on the VCF management network.

- In VCF Installer version 9.1.0.0 through 9.1.0.300, you specify the IP addresses as a contiguous IP range. Alternatively, you can specify non-contiguous IP addresses by using a JSON specification file. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation.
- In VCF Installer version 9.1.0.400 and later, you can specify the IP addresses as contiguous or non-contiguous IP range.

l) On the **Review** page, review the deployment information.

To make changes, navigate back to the relevant page in the deployment wizard. You can also download the `JSON` specification file, make the necessary changes, and then upload the modified `JSON` specification file from the VCF Installer homepage.

m) On the **Validate & Deploy** page, review the validation information.

The VCF Installer validates the information that you provided in the deployment wizard and reports any errors or warnings.

Warnings	Review any warning messages. To proceed with your deployment, address the warnings or click **Acknowledge.**
Errors	Use the Review all failed validations and resolve the issues by using the provided remediation information. You can navigate back to the relevant pages in the deployment wizard to make updates and then re-run the validations. Alternatively, you can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage.

n) After addressing all validation errors and warnings, click **Deploy**.

You can monitor the deployment progress. If there are errors, review the failed tasks in the Tasks panel, address the issue(s), and retry the deployment.

CAUTION:

After deployment completes, do not move the VCF management services VMs to a different resource pool or folder. If you move the VMs to a new resource pool or folder, subsequent patching, deployment, or scale out operations fail.

- License your environment. See Registering VCF Operations and a License Server with the VCF Business Services Console.
- Register your vCenter instance in VCF Operations. See Configuring a vCenter Account in VCF Operations.
- Click **Review Passwords** to review the passwords. To review passwords for past deployments, in the VCF Installer appliance, click the user name at the top-right corner of the window and select **Deployments**.

  Note:

  You can also retrieve some of the passwords using the VCF Operations API.
- (Optional) Download the JSON specification file for the deployment.

  Note:

  The JSON specification file does not include passwords.
- For platform administration tasks such as managing identity and access, certificates, passwords, component lifecycle, and backup and restore of your vSphere Foundation platform, see the Fleet Management guide.
- For monitoring and troubleshooting tasks such as configuring alerts, analyzing logs, monitoring dashboards, and troubleshooting your vSphere Foundation platform, see the Infrastructure Operations guide.

Day-N lifecycle management and topology expansion are divided as follows.

- Lifecycle management for the converged vCenter and ESX hosts is orchestrated by VCF Operations.
- Topology expansion is performed directly in the components.
  - Add ESX hosts in vCenter.
  - Add new clusters in vCenter.

Upgrading to VMware Cloud Foundation 9.1.x

As a mandatory part of the upgrade to version 9.1.x, you must upgrade your components that run on the fleet level and the management domain to version 9.1.x. Before you upgrade, it is important to understand how VCF 9 is different from earlier versions. Upgrading your workload domains is not a mandatory part of the upgrade and can be performed later as a Day-N procedure.

By following the upgrade guidance, you can directly upgrade to VCF 9.1.1 without first upgrading to 9.1.0 and then patching to 9.1.1.

Forward-Only Upgrade Requirement

VCF enforces strict upgrade path validation. You can only upgrade your VCF components to a target version that is released after the release date of your current version. The target version can be a major, minor, or a maintenance release. In general, you can upgrade directly to a patch release only if you can also upgrade to its parent maintenance release. For example:

- Supported: VCF 5.2.0 (25 JUN 2024) to VCF 9.1.0 (11 MAY 2026).
- Not supported: VCF 5.2.4 (27 MAY 2026) to VCF 9.1.0 (11 MAY 2026).
- Supported: VCF 5.2.4 (27 MAY 2026) to VCF 9.1.1 (03 SEP 2026).

This is valid for all components in a VCF source BoM. For example:

- Supported upgrade to VCF 9.1.0 (11 MAY 2026): vCenter 8.0 Update 3a (18 JUN 2024) and ESX 8.0 Update 3 (25 JUN 2024), which are part of VCF 5.2.0 BoM.
- Not supported upgrade to VCF 9.1.0 (11 MAY 2026): vCenter 8.0 Update 3j (27 MAY 2026) and ESX 8.0 Update 3j (27 MAY 2026), which are part of VCF 5.2.4 BoM.
- Supported upgrade to VCF 9.1.1 (03 SEP 2026): vCenter 8.0 Update 3j (27 MAY 2026) and ESX 8.0 Update 3j (27 MAY 2026), which are part of VCF 5.2.4 BoM.

VCF components might have further limitations, see VMware Interoperability Matrix.

VCF Components

Before version 9, VCF always included the following core components:

- SDDC Manager
- NSX Manager
- vCenter
- ESX

Starting with VCF 9, the management components are part of the VCF fleet.

- VCF Operations (previously Aria Operations)
- Cloud proxy (previously Aria Operations Cloud Proxy)
- VCF Automation (previously Aria Automation, can be deployed as a Day-N operation after the upgrade)
- VCF management services (introduced in 9.1)
- License server (introduced in 9.1)

You begin the upgrade process of your SDDC with the VCF Instance that hosts the VCF Operations instance dedicated to managing your VCF fleet.

> **Note:**
>
> The SDDC Manager UI is being deprecated and will be removed in a future release. After your upgrade to VCF 9.1 completes, use VCF Operations to perform lifecycle management activities.

High-Level Upgrade Order For a VCF 9 Fleet

1. Upgrade the management components and the management domain of the VCF Instance where the management components are deployed by strictly following the upgrade order.
2. Upgrade the management domains of the remaining VCF instances in the fleet by upgrading any instance-level components that are available for upgrade.

Upgrading the workload domains for VCF instances is optional and can be performed at any time after the management domain of the respective instance is already upgraded. For information on how to upgrade your workload domains, see Upgrading VMware Cloud Foundation Workload Domains to 9.1.x.

VCF Upgrade Planner

You can now use the new VCF Upgrade Planning Tool to navigate you during the process of upgrading or converging your existing infrastructure to VCF 9.1. Different tailored scenarios exist for starting from different VCF versions.

Upgrading a VCF Instance from version 9.0.x to version 9.1.x

VCF 9.1 transitions the lifecycle management of VCF Operations, VCF Operations for logs, VCF Operations for networks, VCF Automation, and VCF Identity Broker to the new fleet lifecycle and SDDC lifecycle components. The transition occurs during the VCF Operations upgrade to 9.1.x. The upgrade process validates credentials, migrates critical data, including component inventory, certificates, and service accounts, and automatically decommissions the VCF Operations fleet management appliance.

The following table lists VCF 9.0.x and other compatible components in the order that you must upgrade each component and a link to a dedicated upgrade procedure. The upgrade also includes installation of new 9.1 components.

#	VCF 9.0 Component	Upgrade to version 9.1.x
-	VCF Operations fleet management *	No
0	VCF Identity Broker ***	Redeploy VCF Identity Broker 9.0.x to a Supported Network and Datastore for Upgrade to 9.1
1	VCF Operations & cloud proxy	Upgrade to VCF Operations 9.1

#	VCF 9.0 Component	Upgrade to version 9.1.x
2	vSphere Replication	Converge or Upgrade to Protection and Recovery 9.1
3	VMware Live Site Recovery **	
4	vSAN Data Protection **	
5	Avi Load Balancer **	Upgrade to Avi Load Balancer 32.1.1
6	SDDC Manager	Upgrade to SDDC Manager 9.1
7	(New components in 9.1) VCF management services and license server	1. Deploy VCF Management Services and License Server 2. Transfer licenses
8	VCF Identity Broker ***	Upgrade to Identity Broker 9.1
9	VCF Automation	Upgrade to VCF Automation 9.1
10	VCF Operations orchestrator	Upgrade to VCF Operations Orchestrator 9.1
11	VCF Operations for networks	Upgrade to VCF Operations for Networks 9.1
12	VCF Operations for logs	Deploy Log Management 9.1
13	VCF Operations HCX	Upgrade to VCF Operations HCX 9.1
14	Data Services Manager	Upgrade to Data Services Manager 9.1.x
15	NSX Global Manager	Upgrading NSX Global Manager Nodes in a Federated Environment
16	NSX Manager	Upgrade vCenter and NSX Manager to 9.1
17	vCenter	
18	vSphere Kubernetes Service	Upgrade vSphere Kubernetes Service to 3.6.0 or later
19	Supervisor	Upgrade a Supervisor to 9.1
20	vSAN witness host	Upgrade ESX to 9.1
21	ESX	
22	NSX Edge & NSX finalize	Upgrade the NSX Edge Cluster and Finalize NSX Upgrade
23	vSphere Distributed Switch	Upgrade a vSphere Distributed Switch to a Later Version
24	Data Services Manager databases	Upgrading Databases in VCF Automation
25	VMware Tools	Upgrade VMware Tools to 13.1
26	Virtual machine compatibility	Upgrade Virtual Machines
27	vSAN On-Disk Format versions	Upgrade vSAN On-Disk Format Versions
28	vSAN File Service	Upgrade vSAN File Service

* In 9.1, the newly introduced Fleet lifecycle and SDDC lifecycle components, part of VCF management services, perform the VCF Operations fleet management appliance functionality and the appliance is decommissioned during the upgrade. After you upgrade VCF Operations to 9.1.x, you cannot manage certificates and passwords for VCF Automation, VCF Identity Broker, VCF Operations for logs, and VCF Operations for networks until you upgrade these components to version 9.1, as these components are managed by VCF Operations fleet management.

** Component is not part of the VCF SKU.

*** VCF Identity Broker 9.0.x in appliance mode has a dedicated upgrade to identity broker 9.1.x. If VCF Identity Broker 9.0.x in appliance mode is deployed on an NSX overlay segment not on the VCF management network, you must deploy a new instance before the upgrade. VCF Identity Broker in embedded mode is automatically upgraded when the vCenter instance is upgraded.

Upgrading a VCF Instance from Version 5.2.x to Version 9.1.x

You can perform a sequential or skip-level upgrade to VCF 9.1.x from VCF 5.2.x. If your platform is at a version earlier than 5.2, you must upgrade the management domain and all workload domains to VCF 5.2.x before you can upgrade to VCF 9.1.x.

The process depends on which, if any, VMware Aria Suite components you have deployed as part of VCF 5.2.x.

The following table lists VCF 5.2.x and other compatible with VCF 5.2.x components in the order that you must upgrade each component and a link to a dedicated upgrade procedure. The upgrade also includes installation of new 9.1 components.

#	VCF 5.2 Component	Upgrade to version 9.1.x
-	Aria Suite Lifecycle *	No
1	Aria Operations & Cloud Proxy	Upgrade to VCF Operations 9.1
2	vSphere Replication	Converge or Upgrade to Protection and Recovery 9.1
3	VMware Live Site Recovery **	
4	vSAN Data Protection **	
5	Avi Load Balancer **	Upgrade to Avi Load Balancer 32.1.1
6	SDDC Manager	Upgrade to SDDC Manager 9.1
7	(New components in 9.1) VCF management services and license server	Deploy VCF Management Services and License Server as Part of VCF Upgrade to 9.1
8	VMware Identity Manager ***	
9	Aria Automation	Upgrade to VCF Automation 9.1
10	Aria Automation Orchestrator	Upgrade to VCF Operations Orchestrator 9.1
11	Aria Operations for Networks	Upgrade to VCF Operations for Networks 9.1
12	Aria Operations for Logs ****	Deploy Log Management 9.1
13	VMware HCX *****	Upgrade to VCF Operations HCX 9.1
14	NSX Global Manager	Upgrading NSX Global Manager Nodes in a Federated Environment
15	NSX Manager	Upgrade vCenter and NSX Manager to 9.1
16	vCenter	
17	vSphere Kubernetes Service	Upgrade vSphere Kubernetes Service to 3.6.0 or later
18	Supervisor	Upgrade a Supervisor to 9.1
19	vSAN Witness Host	Upgrade ESX to 9.1
20	ESX	
21	NSX Edge & NSX finalize	Upgrade the NSX Edge Cluster and Finalize NSX Upgrade
22	vSphere Distributed Switch	Upgrade a vSphere Distributed Switch to a Later Version
23	VMware Tools	Upgrade VMware Tools to 13.1
24	Virtual machine compatibility	Upgrade Virtual Machines
25	vSAN On-Disk Format versions	Upgrade vSAN On-Disk Format Versions
26	vSAN File Service	Upgrade vSAN File Service

* VMware Aria Suite Lifecycle has no upgrade to version 9.1. In 9.1, the newly introduced Fleet lifecycle and SDDC lifecycle components, part of VCF management services, perform that functionality. After you upgrade VMware Aria Automation 8.x to VCF Automation 9.1.x, VMware Aria Suite Lifecycle 8.x stops managing VCF Automation 9.1.x, but continues to manage VMware Identity Manager. Preserve the VMware Aria Suite Lifecycle 8.x instance to perform all Day-N operations for VMware Identity Manager 3.3.x and do not delete VMware Identity Manager as it is still an authentication source for VCF Automation 9.1.x.

** Component is not part of the VCF SKU.

*** No upgrade path for VMware Identity Manager. Identity broker 9.1.x is deployed as part of the deployment of the VCF management services during the upgrade.

**** No upgrade path for VMware Aria Operations for Logs. You install the newly introduced in 9.1, log management component as a Day-N operation.

***** After you upgrade your VMware HCX appliance to 9.1.x, upgrade all paired appliances to VCF Operations HCX 9.0 or later. Appliances with version 9.0 and later are not compatible with earlier versions and pairing breaks.

Upgrading vSphere 8 and Aria 8 Components to 9.1.x

As a first phase to upgrade your vSphere 8 and Aria 8 virtual infrastructure to 9.1.x, follow the order of upgrade and deployment procedures in the dedicated scenario. This scenario takes you to a stable environment that can later be extended to a full VCF environment. See Upgrading vSphere 8 and Optionally vSAN and Aria Operations 8 to 9.1.x

Upgrading vSphere Foundation to version 9.1.x

To upgrade your vSphere Foundation 5.2 and 9.0.x deployments to 9.1.x, follow the order of upgrade and deployment procedures. See Upgrading to vSphere Foundation 9.1.x.

Redeploy VCF Identity Broker 9.0.x to a Supported Network and Datastore for Upgrade to 9.1

Before you upgrade VCF Operations to 9.1, you must perform this procedure if your VCF Identity Broker 9.0.x is deployed on an NSX overlay segment or on a network or datastore that are different than the ones where you plan to deploy the VCF management services.

> **Important:**
>
> If VCF Operations is already on version 9.1.x, do not perform this procedure. To remediate the upgrade to identity broker 9.1, follow the guidance in Broadcom knowledge base article 444734.

- Redeployment of VCF Identity Broker involves the back-up of the current configuration and importing the configuration in the new cluster. This requires the newly deployed cluster to have the same build number as the existing cluster (the same maintenance and patch version). Download and stage the necessary VCF Identity Broker binaries to your software depot.

With VCF 9.1, VCF Identity Broker transitions to being part of VCF management services. To successfully upgrade to identity broker 9.1, VCF Identity Broker 9.0.x and VCF management services must be deployed on the same network and on the same datastore. For available VCF management services networking models, see VCF Management Services Models.

You perform this procedure if:

- VCF Identity Broker in appliance mode is deployed on an NSX overlay segment. You cannot deploy VCF management services on an NSX overlay segment.
- VCF Identity Broker in appliance mode is deployed on a different network than VCF management services.
- VCF Identity Broker in appliance mode is deployed on a different datastore than VCF management services.

You deploy a new VCF Identity Broker and reuse the existing certificate and FQDN. You reconfigure your DNS to repoint the new IP address of the cluster to the existing FQDN. You then import the original configuration in the new VCF Identity Broker to restore full functionality. Configured permissions are not affected. This process is a service-disruptive change and requires a maintenance window.

1. Back up VCF Identity Broker. See On-Demand Backup of a VCF Identity Broker Appliance Cluster.
2. Power off the VCF Identity Broker virtual machines by using the vSphere Client.
3. Delete the VCF Identity Broker component in VCF Operations. See Delete VCF Identity Broker.

   The first initiation of the delete workflow fails. You must retry the workflow with the `Skip task` property set to *`True`*.

4. Allow for a VCF Identity Broker redeployment in VCF Operations. See Enable VCF Identity Broker Redeployment.
5. Reconfigure your DNS to repoint the VCF Identity Broker FQDN to the new IP address that you use during deployment of the new VCF Identity Broker instance. This step is only required if you are changing the deployment network of the new cluster.
6. Deploy a new VCF Identity Broker in appliance mode by using VCF Operations. See Deploying VCF Identity Broker.
   - Reuse the same FQDN and certificate (stored in the VCF Operations fleet management locker) from the old VCF Identity Broker cluster. Ensure that the certificate you use during deployment of the new cluster does not include an IP address in the Subject Alternative Name (SAN). Provide only the VCF Identity Broker FQDN in the Subject Alternative Name of the certificate.
   - Deploy the new cluster on the VCF management network or on a custom VLAN-backed network, where you plan to deploy VCF management services.
   - Deploy the new cluster on the datastore where you plan to deploy VCF management services.
   - Deploy the new cluster to the same maintenance and patch version as the old cluster.
7. Restore the VCF Identity Broker configuration from the backup by using VCF Operations. See Restore a VCF Identity Broker Appliance Cluster.

After you validate that VCF Identity Broker is operational, proceed with the VCF Operations upgrade. See Upgrading to VCF Operations 9.1.

Upgrading to VCF Operations 9.1

You begin a VCF fleet upgrade to 9.1 by upgrading the VCF Operations instance in the first VCF Instance.

Upgrading from 5.2.x

If your environment has VMware Aria Operations 8.x, you upgrade your Aria Operations instance by uploading the VCF Operations 9.1 PAK file directly in the Aria Operations administration interface. For instances that are managed by VMware Aria Suite Lifecycle 8.x, after you upgrade to VCF Operations 9.1, VMware Aria Suite Lifecycle 8.x can continue to manage VMware Identity Manager. Do not delete VMware Identity Manager as it can still be an authentication source for a VCF Automation instance. Preserve the VMware Aria Suite Lifecycle 8.x instance to perform all Day-N operations for VMware Identity Manager 3.3.x.

After you complete the upgrade of VMware Aria Operations 8.x to VCF Operations 9.1, you must remove VCF Operations from the VMware Aria Suite Lifecycle 8.x environment without deleting the associated VM from vCenter. See Delete a Product from an Environment.

If your environment does not have VMware Aria Operations 8.x, you must deploy a new VCF Operations instance on the management domain vCenter that hosts the core components. See Deploy VCF Operations as Part of a 9.1 Upgrade.

Upgrading from 9.0.x

Before you begin the overall upgrade to 9.1.x, ensure your existing 9.0.x VCF Automation is deployed in the management domain and not in a workload domain. If your environment has VCF Operations 9.0.x, you upgrade your VCF Operations instance by uploading the VCF Operations 9.1 PAK file directly in the VCF Operations administration interface.

The upgrade process manages the transition of lifecycle orchestration from VCF Operations fleet management to VCF management services. During upgrade, VCF Operations connects to the VCF Operations fleet management appliance and collects inventory data, such as component inventory, service registry, user imported certificates, certificate signing requests, content management endpoints, marketplace configuration, and service accounts. After the data is transferred, the VCF Operations fleet management appliance is automatically decommissioned and shut down. The data is later transferred to the new for VCF 9.1 fleet lifecycle and SDDC lifecycle components. Before VCF management services are deployed and the data is transferred, you cannot perform certificate, password, and lifecycle operations by using VCF Operations.

Certificates and Password Management for Management Components During an Upgrade from 9.0.x

When you are upgrading from 9.0.x, after you upgrade VCF Operations to 9.1, you cannot manage certificates and passwords for your management components from the VCF Operations UI. VCF Automation, VCF Identity Broker,

VCF Operations for logs, and VCF Operations for networks in version 9.0.x are managed by VCF Operations fleet management, that is powered off after the VCF Operations upgrade to 9.1. Management from the VCF Operations UI is not possible until you upgrade these components to version 9.1.

Internal certificates are renewed when you upgrade VCF Operations. Automatic root-CA certificate renewal occurs when the cloud proxy component is upgraded. After each upgrade, the cluster receives a new root-CA certificate with a five-year validity period.

Note:

Automatic certificate renewal does not affect custom certificates.

Certificate Subject Alternative Names

VCF 9.1 transitions lifecycle management of components to the new fleet lifecycle and SDDC lifecycle components and enforces stricter certificate validation for management services. During the VCF Operations upgrade, the process validates credentials, migrates component inventory and service accounts, and migrates certificates. Certificates with incomplete Subject Alternative Names (SANs) are no longer supported. See Subject Alternative Name Requirements.

Before you begin the upgrade, verify the certificate Subject Alternative Names for the following components:

- VCF Operations
- VCF Automation
- VCF Operations for networks

Upgrading Cloud Proxies

A cloud proxy in the VCF Instance is required for the deployment of VCF Management services and the integration between VCF Operations, SDDC Manager, and VCF Management services.

To upgrade the primary VCF Instance that hosts VCF Operations, the instance must have a cloud proxy. If a cloud proxy is not present, upgrade VCF Operations, deploy a new cloud proxy, and then complete the VCF Instance upgrade. See Deploy Cloud Proxy as Part of a VCF 9.1 Upgrade.

If your environment has cloud proxies, they are upgraded automatically during the VCF Operations upgrade. Cloud proxies do not collect any data during this period. Data collection resumes after the upgrade completes. The upgrade process runs concurrently for up to 10 cloud proxy instances. If you have more than 10 cloud proxy instances, the upgrade process runs sequentially and the next cloud proxy upgrade starts after any one of the 10 running upgrades completes or fails. In case the automatic upgrade fails, upgrade your cloud proxy manually by using the CLI. See Using the Cloud Proxy Command-Line Interface.

Unsupported Upgrade Scenario for Aria Operations Cloud Proxy 8.18.x with Log Forwarding

If your Aria Operations Cloud Proxy 8.18.x has all four of the following settings, it is not supported after upgrade to version 9.1:

- The Aria Operations Cloud Proxy is on version 8.18.x.
- The cloud proxy is not deployed in the first VCF Instance, where VCF Operations is deployed.
- The cloud proxy has an active integration with VMware Aria Operations for Logs.
- The cloud proxy has log collection configured during initial deployment with a one-time key.

If even one of the conditions is not met, the Cloud Proxy configuration is supported for upgrade to 9.1.

If your Aria Operations Cloud Proxy 8.18.x configuration meets all of these conditions, the Aria Operations Cloud Proxy is upgraded, but it is not supported in VCF 9.1. An unsupported cloud proxy is shut down after the upgrade. You can view the status of the cloud proxy in the VCF Operations UI.

You must deploy a new cloud proxy in the first VCF Instance, where VCF Operations is deployed. You deploy the new cloud proxy instance after you complete the upgrade of VCF Operations. After you deploy the new cloud proxy, you must configure the cloud proxy to collect data and metrics from SDDC Manager. See Deploy Cloud Proxy as Part of a VCF 9.1 Upgrade.

You can shut down and delete the unsupported for upgrade Aria Operations Cloud Proxy 8.18.x before you upgrade VCF Operations.

Create Snapshots of the VCF Operations Nodes Before You Upgrade

Create a snapshot of each node before you upgrade the VCF Operations cluster. After the upgrade completes, delete the snapshots to avoid performance degradation.

For more information about snapshots, see the *vSphere Virtual Machine Administration* documentation.

1. Log in to the VCF Operations Administrator interface at `https://<primary-node-FQDN-or-IP-address>/admin`.
2. On the **System status** page, in the **Cluster status** section, click **Take Cluster Offline**.
3. In the **Take Cluster Offline** dialog box, enter a reason and click **OK**.

   When you take the cluster offline, VCF Operations services are stopped, but the VCF Operations virtual machines are not shut down.
4. When all nodes are offline, log in to the vSphere Client where VCF Operations is deployed.
5. Right-click a VCF Operations node and select **Snapshots** > **Take Snapshot**.
   a) Enter the name for the snapshot.
   b) Deselect the **Include virtual machine's memory** check box.
   c) Deselect the **Quiesce guest file system(requires VM tools)** check box.
   d) Click **Create**.
6. Repeat the previous step for each node in the cluster.

After you create snapshots of the VCF Operations cluster, bring back the cluster to `Online` status before you can proceed with the upgrade.

Back up Customized Content in VCF Operations Before an Upgrade

When you upgrade VCF Operations, the upgrade process overwrites out-of-the-box content and resets it to the latest default versions. Out-of-the-box content includes alert definitions, symptom definitions, recommendations, dashboards, views, reports, and widgets. If you modified any out-of-the-box content directly, those customizations are lost during the upgrade. This behavior is by design: the upgrade does not preserve manual edits made to the original content.

To preserve your customizations through an upgrade, follow the best practice of cloning out-of-the-box content before you customize it. The upgrade does not overwrite user-created cloned objects, so customizations applied to clones persist after the upgrade.

If you must customize any of the existing **out-of-the-box-content** that is provided with VCF Operations, clone the content and then customize the cloned content.

After you have cloned and customized your content, create a backup before you begin the upgrade. The backup ensures that you can recover your customizations if they are lost during or after the upgrade.

1. Log in to VCF Operations.
2. In the left pane, navigate to **Infrastructure Operations**.
3. Clone **out-of-the-box-content** and customize the cloned content.

VCF Operations 9.0.x	1. Clone default policy definitions. a. In the left pane, navigate to **Infrastructure Operations** > **Configurations**. b. Click the **Policy Definition** tile. c. Click **Add**. d. On the **Create New Policy** page, enter a name and a description. e. From the **Inherit from** drop-down menu, select the policy you want to clone. f. Click **Create Policy**. 2. Clone default alert definitions. a. In the left pane, navigate to **Infrastructure Operations** > **Configurations**.

b. Click the **Alert Definitions** tile.
c. Click the ellipsis icon next to the default alert definition you want to clone and, from the drop-down menu, select **Clone**.
d. On the **Create Alert Definition** page, configure the settings and click **Create**.

3. Clone default symptom definitions.
   a. In the left pane, navigate to **Infrastructure Operations** > **Configurations**.
   b. Click the **Symptom Definitions** tile.
   c. Click the ellipsis icon next to the default symptom definition you want to clone and, from the drop-down menu, select **Clone**.
   d. On the **Clone Symptom Definition** page, configure the settings and click **Save**.
   For additional information on symptom definitions, see Symptom Definitions in VCF Operations.
4. Clone default recommendations.
   a. In the left pane, navigate to **Infrastructure Operations** > **Configurations**.
   b. Click the **Recommendations** tile.
   c. Click the ellipsis icon next to the default recommendation you want to clone and, from the drop-down menu, select **Clone**.
   d. On the **Create New Recommendation** page, configure the settings and click **Create**.
5. Clone default dashboards.
   a. In the left pane, navigate to **Infrastructure Operations** > **Dashboards and Reports**.
   b. Click **Dashboards**.
   c. Click the **Manage** tab.
   d. Click the ellipsis icon next to the default dashboard you want to clone and, from the drop-down menu, select **Clone**.
   e. In the **Clone Dashboard** dialog box, enter a name for the cloned dashboard and click **OK**.
6. Clone default views.
   a. In the left pane, navigate to **Infrastructure Operations** > **Dashboards and Reports**.
   b. Click **Views**.
   c. Click the **Manage** tab.
   d. Click the ellipsis icon next to the default view you want to clone and, from the drop-down menu, select **Clone**.
   e. On the **Clone Vlew** page, configure the settings and click **Create**.
7. Clone default report templates.
   a. In the left pane, navigate to **Infrastructure Operations** > **Dashboards and Reports**.
   b. Click **Reports**.
   c. Click the **Report Templates** tab.
   d. Click the ellipsis icon next to the default report template you want to clone and, from the drop-down menu, select **Clone**.
   e. On the **Clone Report Template** page, configure the settings and click **Create**.

Aria Operations 8.18.x

1. Clone default policy definitions.
   a. In the left pane, navigate to **Operations** > **Configurations**.
   b. Click the **Policy Definition** tile.
   c. Click **Add**.
   d. On the **Create New Policy** page, enter a name and a description.
   e. From the **Inherit from** drop-down menu, select the policy you want to clone.
   f. Click **Create Policy**.
2. Clone default alert definitions.
   a. In the left pane, navigate to **Operations** > **Configurations**.
   b. Click the **Alert Definitions** tile.
   c. Click the ellipsis icon next to the default alert definition you want to clone and, from the drop-down menu, select **Clone**.
   d. On the **Create Alert Definition** page, configure the settings and click **Create**.
3. Clone default symptom definitions.
   a. In the left pane, navigate to **Operations** > **Configurations**.
   b. Click the **Symptom Definitions** tile.
   c. Click the ellipsis icon next to the default symptom definition you want to clone and, from the drop-down menu, select **Clone**.
   d. On the **Clone Symptom Definition** page, configure the settings and click **Save**.

4. Clone default recommendations.
   a. In the left pane, navigate to **Operations** > **Configurations**.
   b. Click the **Recommendations** tile.
   c. Click the ellipsis icon next to the default recommendation you want to clone and, from the drop-down menu, select **Clone**.
   d. On the **Create New Recommendation** page, configure the settings and click **Create**.
5. Clone default dashboards.
   a. In the left pane, navigate to **Operations** > **Dashboards**.
   b. In the left pane, click **Manage**.
   c. Click the ellipsis icon next to the default dashboard you want to clone and, from the drop-down menu, select **Clone**.
   d. In the **Clone Dashboard** dialog box, enter a name for the cloned dashboard and click **OK**.
6. Clone default views.
   a. In the left pane, navigate to **Operations** > **Views**.
   b. In the left pane, click **Manage**.
   c. Click the ellipsis icon next to the default view you want to clone and, from the drop-down menu, select **Clone**.
   d. On the **Clone VIew** page, configure the settings and click **Create**.
7. Clone default report templates.
   a. In the left pane, navigate to **Operations** > **Reports**.
   b. In the left pane, click **Manage**.
   c. Click the ellipsis icon next to the default report template you want to clone and, from the drop-down menu, select **Clone**.
   d. On the **Clone Report Template** page, configure the settings and click **Create**.

Note:

Ensure that the original OOTB version is deactivated in the active policy to avoid duplicate alerts.

4. Create a backup of the customized content.

   See Creating a Backup and Configure Image-Based Backup of VCF Operations and VCF Operations for Networks.

   For upgrading from VCF 5.2, see Creating a Backup.

The customized content is backed up and you can proceed with the upgrade. Post the upgrade, review the upgraded **out-of-the-box-content** in VCF Operations, and compare them with the preserved customized content to determine whether to keep your previously modified versions, or to use the upgraded versions.

Perform the Upgrade to VCF Operations 9.1

You upgrade VCF Operations 9.0.x and Aria Operations 8.18.x to version 9.1 by using a PAK file in the VCF Operations administration interface. For maintenance upgrades, the PAK upgrade method is not recommended and the user interface suggests you use alternatives. For upgrades from versions 8.x and 9.0.x, PAK upgrade is the only method.

- The upgrades to VCF Operations 9.1.0.x or VCF Operations 9.1.1.x have specific source version requirements:
  – If you are upgrading to VCF Operations 9.1.0.x, verify that your Aria Operations version is 8.18 or later.
  – If you are upgrading to VCF Operations 9.1.1.x, verify that your Aria Operations version is 8.18.6 or later.

  See Broadcom Interoperability Matrix.
- If you upgrade from VCF 9.0.x, verify that all certificates for VCF Operations for networks 9.0.x meet the SAN attribute requirements. The VCF Operations for networks certificates must have FQDNs and IPs for all platform and collector nodes in the certificate SAN attributes. To replace non-compliant certificates, see Replace a Certificate with a CA-Signed Certificate.
- Verify that the Subject Alternative Name (SAN) field of the VCF Operations certificate contains the FQDNs of all VCF Operations nodes and the FQDN of the Load Balancer / VIP. To replace non-compliant certificates:
  – For VCF Operations, see Replace a Certificate with a Configured CA-Signed Certificate.
  – For Aria Operations, see Replace VMware Aria Automation Certificate.

- The vCenter field in all vCenter integrations must not contain a URL or an IP address, or an FQDN that includes slashes or sub-paths.
- Ensure that you have the VCF Operations fleet management appliance root user password to facilitate the inventory transfer.
- If your environment has additional management components and your VCF Operations has a configured proxy server with SSL termination, you must remove the proxy server configuration before you upgrade VCF Operations.
- If you have VCF Automation 9.0.x in your environment, ensure it is deployed in the management domain and not in a workload domain.

The process first upgrades the VCF Operations nodes in the cluster and then automatically upgrades the cloud proxy for the VCF Operations instance.

If you are upgrading from VCF Operations 9.0.x, the inventory of the VCF Operations fleet management appliance is transferred to the VCF Operations instance. When the transfer completes, the fleet management appliance is decommissioned. VCF Operations temporarily keeps the inventory until you deploy the VCF management services that substitute the fleet management appliance functionalities.

1. Download the VCF Operations 9.1 upgrade PAK file from the Broadcom Support Portal.
   a) Log in to the Broadcom Support Portal.
   b) Click **My Downloads**.
   c) Search for `VMware Cloud Foundation`, select **VMware Cloud Foundation 9**, and click **9.1.1.0**.
   d) Read both terms and conditions to activate the agreement check box and accept the terms.
   e) For **VCF Operations**, click **View group** and download the VCF Operations upgrade PAK file.
2. Log in to the primary VCF Operations node administrator interface at `https://<primary-node-FQDN>/admin` with the `admin` user.
3. In the left pane, click **Software Update** and, in the main pane, click **Install a Software Update**.
4. In the **Recommended Actions for the .PAK File Upload** dialog box, accept the terms and conditions, and click **Next**.

   The **Add Software Update** wizard opens.
5. On the **Select Software Update** page, browse to the upgrade PAK file, click **Upload**, and when the upload completes click **Next**.
6. On the **End User License Agreement** page, click **Next**.
7. On the **Update information** page, and click **Next**.
8. On the **Install Software Update** page, click **Install**.

   After you start the upgrade procedures, the VCF Operations administrator interface restarts, and you are logged out.
9. Reload the VCF Operations administrator interface, log in again, and click **Software Update** to track progress.
10. If you are upgrading from VCF Operations 9.0.x, migrate the inventory of the respective VCF Operations fleet management appliance 9.0.x to the upgraded VCF Operations 9.1 instance.
    a) On the Software Update page, wait for the **Fleet Management v9.0 Node data Migration to VCF Operations** dialog box to appear.
    b) Enter the VCF Operations fleet management appliance root user password, select the **I confirm that I have reviewed the reference documentation and verified that the system configuration is correct** confirmation box, and click **OK**.
11. After all validations complete, the upgrade of the VCF Operations nodes starts.

    The VCF Operations administrator interface restarts.

    a) Reload the VCF Operations administrator interface and log in again.

       The **System Status** page shows that the cluster status is `offline` and then `Going Online` when the installation is in progress.

b) Wait for all VCF Operations nodes and the cluster to show an `online` status.

If a cluster fails and the status changes to `offline` during the installation process of a PAK file update, then some nodes become unavailable. Access the administrator interface and manually take the cluster `online` and click **Finish Installation** to continue the installation process.

12. Click **Software Update** to verify that the update of the VCF Operations cluster is complete.

A confirmation message that the update was completed successfully appears in the main pane.

Note:

After the VCF Operations cluster upgrade completes, the respective cloud proxy for the VCF Operations instance is also upgraded. During the cloud proxy upgrade process, the component reboots and is expected to be temporarily offline. For more information on the cloud proxy status, see Monitoring the Health of Cloud Proxies.

- The VCF Operations instance is now upgraded.
- The VCF Operations fleet management appliance 9.0.x is decommissioned and powered off. You can manually delete the appliance.

Multiple snapshots can degrade performance, so delete your pre-upgrade snapshots after the upgrade completes.

After the upgrade of VCF Operations from version 9.0.x, and before you deploy and register a license server, you see a `Licenses could not be synchronized with vCenter instances` error. The error disappears after you deploy and register the license server, and the licenses are transferred from the VCF Operations instance to the license server.

After the upgrade of VCF Operations from version 9.0.x, and before you deploy the VCF management services and the corresponding components, VCF Operations does not show the certificates for the identity broker and the log management components.

After you complete the upgrade of VMware Aria Operations 8.x to VCF Operations 9.1, you must remove VCF Operations from the VMware Aria Suite Lifecycle 8.x environment without deleting the associated VM from vCenter. See Delete a Product from an Environment.

Important:

To continue the upgrade, you deploy a new cloud proxy if:

- Your environment does not have a cloud proxy
- The existing cloud proxy is now showing as unsupported in the VCF Operations UI.

See Deploy Cloud Proxy as Part of a VCF 9.1 Upgrade.

After VCF Operations and the cloud proxy are successfully upgraded and in a supported state, proceed with the SDDC Manager upgrade. See Upgrade to SDDC Manager 9.1.

Deploy VCF Operations as Part of a 9.1 Upgrade

VCF Operations is a required component in VCF 9 and later. If you do not have an existing deployment of VMware Aria Operations 8.18.x or VCF Operations 9.0.x, you must deploy it before you can proceed with the VCF 9.1 upgrade. You deploy the VCF Operations nodes from the vSphere Client by using an OVA file.

After you deploy the VCF Operations nodes, you connect your instance to SDDC Manager to activate the fleet management capabilities in VCF Operations.

1. Download the VCF Operations 9.1 Install OVA file from the Broadcom Support Portal.
   a) Log in to the Broadcom Support Portal.
   b) Click **My Downloads**.
   c) Search for `VMware Cloud Foundation`, select **VMware Cloud Foundation 9**, and click **9.1.1.0**.
   d) Read both terms and conditions to activate the agreement check box and accept the terms.
   e) For **VCF Operations**, click **View group** and download the VCF Operations Install OVA file.
2. Log in to vCenter using the vSphere Client.
3. In the **Hosts and clusters** inventory, navigate to the target cluster.
4. Right-click the cluster and, from the drop-down menu, select **Deploy OVF Template**.

   The **Deploy OVF Template** wizard opens.
5. Deploy VCF Operations.
   a) On the **Select an OVF template** page, select **Local file**, click **Upload Files**, navigate to the VCF Operations `.ova` file, and click **Open**.
   b) On the **Select a name and folder** page, enter a unique name for the VCF Operations VM, select the location for the VM, and click **Next**.
   c) On the **Select a compute resource** page, select the target cluster and click **Next**.
   d) On the **Review details** page, click **Next**.
   e) On the **License agreement** page, accept the license agreement and click **Next**.
   f) On the **Configuration** page, select the VCF Operations deployment size and click **Next**.

      The deployment size does not affect the disk size for the VM.
   g) On the **Select storage** page, select the virtual disk format, select the VM storage policy, select the target datastore, and click **Next**.
   h) On the **Select network** page, select the target network and click **Next**.

      You can deploy VCF Operations on the same distributed port group as the vCenter instance, or to a specific vSphere distributed switch or NSX segment. See Network Deployment Options for VCF Components.
   i) On the **Customize template** page, configure the settings and click **Finish**.
   j) On the **Ready to complete** page, click **Next**.

   To create a multi-node cluster, repeat the steps to deploy each node.

   After the deployment of the additional nodes, you can convert the cluster to continuous availability architecture. See Configure VCF Operations Cluster for Continuous Availability.

   After the deployment of the additional nodes, you can convert the cluster to high availability architecture. See Configure a VCF Operations Cluster for High Availability.
6. Power on the VCF Operations VM.
   a) In vSphere Client, navigate to the VCF Operations VM.
   b) Right-click the VM and select **Power** > **Power On**.
   c) Wait for the VM to power on.
7. Perform the initial configuration of VCF Operations.
   a) In a Web browser, navigate to `https://vcf_operations_primary_IP`.
   b) On the **Get started** page, click **New Installation**.

      The **VCF Operations Initial Setup** wizard opens.
   c) On the **Getting Started** page, click **Next**.
   d) On the **Set Administrator Credentials** page, enter the password for the **admin** account and click **Next**.
   e) On the **Choose Certificate** page, use the default certificate or upload a custom certificate and click **Next**.
      - To use the included certificate, click **Use the default certificates**, review the certificate information and click **Next**.
      - To install a custom certificate, click **Install a certificate**.
        a. Click **Browse** to locate the certificate file.

b. Click **Open** to load the file in the Certificate text box.
c. Click **Next**

f) On the **Deployment settings** page, configure the settings and click **Next**.

Setting	Description
Cluster primary node name	Enter the cluster primary node name.
NTP	To have VCF Operations manage its own time synchronization, leave the shared IP address blank. All nodes synchronize with the primary and replica nodes.

g) On the **Configure Availability** page, leave the **Availability Mode** toggle unselected and click **Next**.
h) On the **Nodes** page, click **Next**.
i) On the **Ready to Complete** page, click **Finish**.

The VCF Operations administrative interface opens.

j) To start the cluster, click **Start VCF Operations**.
k) In the **Confirm First Application Startup** dialog box, click **Yes**.

Note:

Depending on the size of the cluster and nodes, deployment might take 10 to 30 minutes to start. Do not make any changes or perform any actions on cluster nodes while the cluster is starting.

VCF Operations is deployed in evaluation mode. Before adding other components, assign the VCF entitlement type. See Registering VCF Operations and a License Server with the VCF Business Services Console.

1. To integrate VMware Cloud Foundation with VCF Operations, add a VCF Instance integration. See Configuring a Standalone VCF Account in VCF Operations.
2. A cloud proxy for the VCF Instance is required for the deployment of VCF management services. See Deploy Cloud Proxy as Part of a VCF 9.1 Upgrade.

Deploy Cloud Proxy as Part of a VCF 9.1 Upgrade

You need cloud proxies to collect data from physical data centers. For high availability or load balancing, you deploy two or more cloud proxies to create a cloud proxy group. A cloud proxy collects data from the end-point environment and uploads it to VCF Operations. Cloud proxies can support multiple vCenter instances. For more information on cloud proxies, see the Cloud Proxy FAQ.

- Verify that you have an IP address and a DNS configuration for the cloud proxy.
- Verify that you have proper permissions to deploy OVF templates in vSphere. For more information, see Deploy and Export OVF and OVA Templates.
- Allow outgoing HTTPS traffic for cloud proxy over port 443.
- Allow incoming traffic to cloud proxy over ports 443, 4505, and 4506 for Telegraf based application monitoring.

  Note:

  For Telegraf based application monitoring in VCF 9.1 and later versions.

  Port 8443 will reach end of support after a few releases. It is recommended that you use port 443 so that the end point VM/physical server has access to port 443 of an individual cloud proxy or cloud proxies that are a part of a cloud proxy group.

  - As a new user, enable port 443 instead of port 8443 so that the end point VM/physical server has access to port 443 instead of port 8443 of an individual cloud proxy or cloud proxies that are a part of a cloud proxy group.
  - For existing users who upgrade to VCF 9.1, it is recommended that you reconfigure your open source and product managed Telegraf workloads to use port 443 instead of ports 8443 within the next few releases.
- Allow incoming traffic to cloud proxy over port 443 for push model adapters or Suite-API on cloud proxy.
- Verify that the user has the required cloud proxy roles to access configuration templates. For more information, see the Access Control: Roles Tab.

- Add a vCenter cloud account and provide an account with the following read and write privileges:
  - vCenter IP address or FQDN.
  - Permissions required to install a cloud proxy on the vCenter Server.

  For more information on privileges, see Privileges Required for Configuring a vCenter Account.
- Cloud proxies must have a proper DNS resolution to the VCF Operations nodes when using short/long FQDN names.

  Note:

  Using a firewall to restrict traffic by IP is not recommended since IPs can change without notice. Restricting traffic must be performed via FQDNs only.

  Refer to the cloud proxy sizing requirements before configuring a cloud proxy. For more information, see Collecting Data with Cloud Proxy in VCF Operations.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Cloud Proxies**, and then click **Add**.
2. Click **Broadcom Support Portal** to navigate to the Broadcom Support Portal, log in using your credentials and download the VMware Cloud Foundation Operations collector (Cloud proxy) OVA file. Deploy the VMware Cloud Foundation Operations collector (Cloud proxy) OVA file in a unified cloud proxy configuration.

Cloud Proxy Type	VM Scale Requirements	CPU Requirements	Memory Requirements
Unified Small Cloud Proxy	Up to 16,000 VMs	4 vCPUs	16GB
Unified Standard Cloud Proxy	Between 16,000 to 80,000 VMs	8 vCPUs	48 GB

   VCF Operations supports both log collection and log assist capabilities. To configure log collection, deploy the Cloud Proxy OVA using the Unified Cloud Proxy configuration.

   You can deploy a classic cloud proxy in VCF Operations, to collect and monitor data from your remote data centers.

   CAUTION:

   Do not deploy the OVA using the Classic Cloud Proxy configuration unless explicitly instructed by technical support.

3. Navigate to your vSphere Client, select the name of your vCenter cluster, and select **Deploy OVF Template** from the **Actions** menu.
4. Insert the OVA link and then click **Next**.
   - Paste the cloud proxy OVA link in the **URL** field.
   - Click the **Local File** option, browse, and select the downloaded OVA file.
5. Follow the prompts to install the OVA on your vCenter.

   For the most current information about sizing and scaling, see Broadcom knowledge base article 324340.
6. When prompted to enter the Unique Registration Key in the **Customize template** screen, return to the Install Cloud Proxy page in VCF Operations.
7. Optional: (Optional) Activate **Data Persistence** to store data in the cloud proxy in case of connectivity issues. For more information, see Activating Data Persistence in Cloud Proxy.
8. Click the **Copy Key** icon.

   The unique registration key expires 24 hours after generation. To avoid using an expired key, click **Regenerate Key** before proceeding. The unique registration key is used by the cloud proxy to authenticate to VCF Operations.

   Note:

   Upon clicking Regenerate Key, the unique registration key is refreshed and a new key is generated if you reload the cloud proxy page. A new unique registration key is generated if you activate data persistence, or log forwarding, or both.

9. Return to vSphere and paste the key in the **Unique Registration Key** text box to install the VCF Operations Cloud Proxy.
10. Optional: (Optional) Enter an IP in CIDR format for **Docker Subnet**.

    The Docker Subnet value in CIDR format must be /27 or larger (for example, /27, /26, /25, /24 etc). If a custom value is not provided, the Docker's default value gets assigned automatically (for example, /16 subnets starting from 172.17.0.0). For more information on docker subnet allocation, see Automatic subnet allocation.

    Note:

    The Docker Subnet creates a network subnet for the Docker containers running inside this cloud proxy, to prevent any network overlap. For information related to connectivity issues between cloud proxy and connected nodes due to network overlap, see Broadcom knowledge base article 392302.
11. Optional: (Optional) Select **Prefer IPv6** to use IPv6 for internal communications.
12. Optional: (Optional) Set up a proxy server in the **Customize template** screen.
    a) Enter details in the **Network Proxy IP Address** and **Network Proxy Password** properties.

       Note:

       If you use a network proxy for log forwarding, port 9543 must be open.
    b) To activate SSL, select the **Use SSL connection to proxy** check box.
    c) If you are using SSL, you can verify the certificate of the proxy server. Public certificate authorities are used to verify the proxy server certificate. To activate this, select the **Verify proxy's SSL cert** check box in the **Verify SSL cert** property.
    d) You can specify the IP/FQDN URL that is used to access the system when a load balancer is used.
    e) If you have a custom certificate authority, paste the root certificate authority in the **Custom CA** property to verify the certificate of the proxy server. Include the following lines when you copy the root certificate authority:

       ```
       "-----BEGIN CERTIFICATE-----"

       "-----END CERTIFICATE-----"
       ```
13. (Optional) Starting with version 9.1.1, you can set up Outbound Network Proxy Settings to route Broadcom Portal traffic through a dedicated proxy. Expand **Outbound Network Proxy Settings** and enter the proxy IP address, port, credentials, SSL, and custom certificate authority options. For more information, see Configuring Outbound Network Proxy Settings on an Upgraded Cloud Proxy.
14. Click **Finish**.

    The deployment takes a few minutes to finish.
15. Locate the cloud proxy you installed, select the VCF Operations Cloud Proxy, and click **Power on**.

    Note:

    Power on the cloud proxy within 24 hours of registration. After 24 hours, the Unique Registration Key expires. If the key expires, delete the proxy and deploy a new one.
16. Return to the Cloud Proxy page in VCF Operations to view the status of the cloud proxy you installed. For more information, see Monitoring the Health of Cloud Proxies.
17. After you deploy and configure the cloud proxy, assign the new cloud proxy to a new or existing VMware Cloud Foundation (VCF) instance and its associated workload domains. For the existing domains of the same VCF Instance, update the integrations for all vCenter instances to use the new Cloud proxy/ group. For more information, see Configuring a Standalone VCF Account in VCF Operations.

    Important:

    VCF 8.18.x did not include unified cloud proxies. When you upgrade your VCF instance from 8.18.x to 9.1, you must assign any newly deployed cloud proxy to your VCF instance. After installation, navigate to **Operate** > **Administration** > **Integrations** and edit the VCF instance settings and replace the default cloud proxy with the new cloud proxy.

18. To view the accounts that are using this connection, click the Cloud Proxy.
19. Optional: To delete a cloud proxy, click the vertical ellipsis and then click **Delete**. For more information, see Deleting Cloud Proxies .

Upgrade to SDDC Manager 9.1

After you complete the upgrade of VCF Operations, you continue by upgrading SDDC Manager in the first VCF Instance.

- If you have an AVI Load Balancer in your environment, verify that the version is compatible with the target version for SDDC Manager. See the Product Interoperability Matrix.
- Download the SDDC Manager upgrade binary for your target release.
  - If you are upgrading from SDDC Manager 5.2.x, see Download VMware Cloud Foundation Upgrade Bundles.
  - If you are upgrading from SDDC Manager 9.0.x, see Download VMware Cloud Foundation 9.0 Upgrade Bundles Using an Online or Offline Depot.
  - If your SDDC Manager does not have access to the Internet, you can download the upgrade binary by using the VCF Download Tool to an offline depot or to the SDDC Manager. See Download Binaries to an Offline Depot by Using the VCF Download Tool or Download VMware Cloud Foundation Upgrade Bundles Without an Online or Offline Depot. You can also download the required binaries to deploy VCF management services and license server, which are required in the subsequent procedure. See Deploy VCF Management Services and License Server as Part of VCF Upgrade to 9.1.
- Ensure that you have a recent successful backup of SDDC Manager using an external SFTP server.
- Ensure that you have taken a snapshot of the SDDC Manager appliance.
- Ensure that you have recent successful backups of the components that are managed by SDDC Manager.
- Perform upgrade prechecks in SDDC Manager and resolve any issues.

Upgrade from 9.0.x to 9.1	1. Log in to VCF Operations. 2. From the navigation bar at the top, click **Build**. 3. In the left pane, click **Lifecycle**. 4. Expand **VCF Instances**, select the respective VCF Instance that hosts the SDDC Manager to upgrade. 5. On the **SDDC Manager Updates** tab, click the **Available updates**, and click **Update Now**. 6. Click **Finish**.
Upgrade from 5.2.x to 9.1	1. Log in to SDDC Manager at `https://<sddc_manager_fqdn>` with a user assigned the **ADMIN** role. 2. On the **Workload Domains** page, click the management domain and then click the **Updates** tab. 3. In the **Available Updates** section, click **Plan Upgrade**. 4. On the **Plan Upgrade** page, select the target major version, and click **Next**. 5. On the **Change Summary** page, select the specific version and click **Confirm**. 6. On the **Plan Upgrade** page, click **Done**. 7. Click **Update Now** or **Schedule Update** next to the VMware Cloud Foundation upgrade bundle. 8. If you selected **Schedule Update**, select the date and time for the bundle to be applied and click **Schedule**. 9. Click **Finish**.

After upgrading SDDC Manager to version 9.1, log in to your SDDC Manager instance, navigate to **Administration** > **Depot Settings**, and reconfigure an online depot to use an activation code from the VCF Business Services console for authentication, instead of a download token. See Configure a Software Depot Connection Mode.

Deploy VCF Management Services and License Server as Part of VCF Upgrade to 9.1

After your VCF Operations and SDDC Manager instances are at version 9.1, to continue with the upgrade, you deploy VCF management services. You can perform this procedure by using the VCF Operations UI or the SDDC Manager API that provides more options to customize the configuration for the deployment.

Before you can deploy the VCF management services and license server components, ensure that the following prerequisites are met.

- Verify that all required ports are open. See VMware Ports and Protocols.

- Verify that you configured the required FQDNs and IP addresses for deployment. See VCF Components FQDNs and IP addresses.
- Verify that VCF Operations and SDDC Manager are at version 9.1.x.
- Download the required install binaries in the software depot for the components that you install during the upgrade of the first VCF instance that hosts VCF Operations.

  Within a VCF fleet, as the required fleet level components are already deployed, during upgrades of additional VCF instances, only VCF services runtime, Salt master, SDDC lifecycle, and telemetry are deployed.

  For environments, where a license server already exists, this workflow does not deploy a second license server.

Table 978: Required Install Binaries for Fleet Upgrade

Component	Binary Required for First Instance	Binary Required for Additional Instances
VCF services runtime	Yes	Yes
SDDC lifecycle	Yes	Yes
Salt master	Yes	Yes
Telemetry	Yes	Yes
Software depot	Yes	No*
Salt RaaS	Yes	No
Fleet lifecycle	Yes	No
Identity broker	Yes	No*
License server	Yes	No*

* As part of the upgrade process, software depot, identity broker, and license server are deployed only in the first VCF instance. You can later deploy them in any of the additional VCF instances per the requirements of your environment. See Adding or Removing VCF Components Post Deployment.

If your SDDC Manager does not have access to the Internet, you can download the binaries for the following components by using the VCF Download Tool to an offline depot or to SDDC Manager. See Download Binaries to an Offline Depot by Using the VCF Download Toolor Download VMware Cloud Foundation Upgrade Bundles Without an Online or Offline Depot.

- The VCF management services FQDNs are checked for validity, the FQDN validation relies on RFC1034 and RFC1123, and only top level domains of the internet are validated. The `.local` domain is not supported as it does not fall into the list of top-level domains in the Domain Name System (DNS) of the Internet.
- Obtain administrative credentials for the VCF Operations instance to register the components with.
- If your environment does not have a cloud proxy for the VCF instance that hosts VCF Operations, you must deploy and configure a new cloud proxy instance. See Deploy Cloud Proxy as Part of a VCF 9.1 Upgrade.
- If your environment has additional management components and your VCF Operations has a configured proxy server with SSL termination, you must remove the proxy server configuration before you deploy VCF management services.

Within a VCF fleet, as the required fleet level components are already deployed, during upgrades of additional VCF instances, only VCF services runtime, Salt master, SDDC lifecycle, and telemetry are deployed.

> **CAUTION:**
>
> After deployment, do not move the VCF management services VMs to a different resource pool or VM folder. If you move the VMs to a new resource pool or VM folder, subsequent patching, deployment, or scale out operations fail.
>
> Storage DRS (SDRS) is disabled by default during VCF management services deployment starting with VCF 9.1 EP2 (see VCF Services Runtime 9.1.0.0200 Release Notes). Do not manually use storage vMotion to migrate the VCF management services virtual machines to a different datastore. Storage vMotion of the VCF management services nodes is not supported and can cause subsequent patching, deployment, or scale-out operations to fail.

You can perform this procedure by using the VCF Operations UI or the SDDC Manager API. When using the SDDC Manager API, you use the `VcfManagementComponentsSpec` JSON file to configure the VCF management services

and license server deployment parameters. See Deploy VCF Management Components. Each method has specific requirements and advantages. For additional information, see the following table.

Deployment Configuration	Description
Network placement	• For target versions 9.1.0.x, with the UI, you deploy VCF management services and license server on the VCF management network. If your environment does not have a dedicated VCF management network, you can deploy the components to the VM management network. See Network Deployment Options for VCF Components. • For target versions 9.1.1 and later, with the UI, you can select the network where VCF management services and license server are deployed to. • With the API, you can deploy VCF management services and license server on the VCF management network or on a VLAN-backed network by using the `xRegionNetwork` parameter in the `VcfManagementComponentsSpec` JSON file. This option allows you to avoid IP exhaustion on the VCF management network.
Dual-stack networking	For dual-stack networking and VCF components that support the dual-stack model, the FQDNs must resolve to IPv4 and IPv6 addresses. See Dual Stack IP Addressing Model. • With the UI, you can provide FQDNs that resolve to and IPv4 and IPv6 addresses. • With the API, the necessary IPv6 parameters in the `VcfManagementComponentsSpec` JSON file are included but commented out. To configure dual-stack networking for VCF management services, you must uncomment and provide values for the respective IPv6 parameters.
VCF Identity Broker	If you have VCF Identity Broker 9.0.x in appliance mode, identity broker 9.1 deployment is skipped and you can later upgrade, see Upgrade to Identity Broker 9.1. If you have embedded VCF Identity Broker 9.0.x, it is upgraded to 9.1 during the vCenter instance upgrade but an identity broker 9.1 instance is also deployed as part of this procedure. You can later transition your 9.1 embedded instance to the highly available 9.1 instance. See Migration of Identity Broker Embedded to Identity Broker Instance. • With the API, the sample `VcfManagementComponentsSpec` JSON file contains specification for deploying the identity broker component. If you have an existing VCF Identity Broker in appliance mode, you must remove the `vidbSpec` specification or the deployment validation fails. • If you have an existing VCF Identity Broker for the primary VCF instance, when deploying a subsequent VCF Instance, `vidbSpec` can be excluded from the JSON file. With VCF 9.1, VCF Identity Broker transitions from being hosted by a dedicated VCF services runtime to being hosted by the shared VCF services runtime, which hosts all VCF management services. To successfully upgrade to the 9.1 identity broker component, both VCF management services and VCF Identity Broker must reside on the same network. • With the API, you can specify a custom network where you deploy VCF management services by using the `xRegionNetwork` parameter. You can deploy VCF management services on the same VLAN-backed network as VCF Identity Broker.
IP addresses for VCF services runtime	A minimum of 12 IP addresses are required for deployment (/28 CIDR block). A recommended number of 30 IP addresses in total can be added. The additional IP addresses are reserved for new VCF management services components or scaling out existing components (/27 CIDR block). • With the UI, for target versions 9.1.0.0 through 9.1.0.300, you allocate a /28 or a /27 CIDR block. • With the UI, for target versions 9.1.0.400 and later: — You can now exclude IP addresses from the provided CIDR block (comma-separated list). Excluded IP addresses cannot be used for deploying additional VCF components. — You can now provide a comma-separated list of IP addresses. The IP addresses can be contiguous or non-contiguous. • With the API, you can use CIDR, an IP range, or a list of IP addresses (can be non-contiguous). If your VCF management network does not have sufficient IP addresses, you can use the API to deploy to a custom VLAN-backed network.

Deployment Configuration	Description
VCF services runtime internal network	VCF services runtime uses an internal CIDR block of 198.18.0.0/15. If this subnet is in use on your network, you must change the internal CIDR block for VCF management services to 240.0.0.0/15 or 250.0.0.0/15. Different VCF services runtime instances on the same network can work with the same internal network. For example, you can have the two instances of VCF services runtime for your VCF management services and VCF Automation using the same internal network. • With the VCF 9.1.0.x UI, you cannot change the default internal network. With VCF 9.1.1 and later, you can change the default internal network. Select a CIDR block that does not overlap with any of your infrastructure networks, including management, VM, and IP pool networks. An overlap causes routing conflicts and breaks connectivity to or from the VCF services runtime cluster. • With the API, you can specify the internal CIDR block by using the `internalClusterCidrIpv4` or `internalClusterCidrIpv6` JSON keys.
FQDNs for VCF management services	You must provide three FQDNs for the VCF services runtime, the fleet component, and the instance component. In the context of deploying VCF management services, the required FQDN for the fleet component includes the fleet lifecycle, Salt RaaS, and software depot components. The FQDN for the instance component includes the SDDC lifecycle, Salt master, real-time metrics, real-time metrics store, and telemetry components. These FQDNs must resolve to unique and unused IP addresses outside of the IP range provided for VCF services runtime, but still on the network that hosts the VCF services runtime component.
FQDNs for identity broker and license server	You must provide one FQDN for identity broker and one FQDN for license server. These FQDNs must resolve to unique IP addresses outside of the IP range provided for the VCF services runtime, but still on the network that hosts the VCF services runtime component. You can deploy license server as IPv4 only. In VCF 9.1.1 and later, after the new deployment, you can change the networking configuration to IPv6. If you upgrade a license server from version 9.1.0 to version 9.1.1, you cannot configure IPv6 on that license server instance. See Configure IPv6 on a License Server.

UI Procedure

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Build**.
3. In the left pane, click **Lifecycle**.
4. Expand **VCF Instances** and navigate to the VCF Instance that hosts VCF Operations.
5. Click the **SDDC Manager Upgrades** tab, click the **Install Management Services** tab, and, in the **Next steps** section, click **Install**.
6. In the **Components Installation** dialog box, enter the VCF Operations load balancer FQDN or VCF Operations primary node FQDN, administrator username and password, and click **Connect**.
7. In the **Components Installation** dialog box, configure the settings and click **Install**.

The FQDNs for VCF management services must point to unique, unassigned IP addresses and the FQDNs must not be in the IP range assigned to VCF management services.

Setting	Description
Deployment network	This option is only available in VCF 9.1.1 and later. Select the network where VCF management services and license server are deployed. If you leave this option blank, VCF management services and license server are deployed by default to the VM management network. You can place VCF management services on the ESX management network, on the VM Management Network, or on a custom NSX network. See Network Deployment Options for VCF Components. **Note:** The UI displays all available networks (including portgroups and segments) that are available in the management vCenter. If this vCenter manages more than one data center, all available networks for all managed data centers are displayed.
Gateway CIDR	This option is only available if you select the VCF management services deployment network. Enter the Gateway IP and netmask in CIDR notation.
VCF services runtime IP pool	• For target versions 9.1.0.0 through 9.1.0.300 – Allocate a minimum of 12 IP addresses expressed as a /28 CIDR block. – Allocate a recommended 30 IP addresses expressed as a /27 CIDR block. These IP addresses are reserved for new VCF management services components or scaling out existing components. • For target versions 9.1.0.400 and later, you can also: – Exclude IP addresses from the provided CIDR block (comma-separated list). Excluded IP addresses cannot be later used for deploying additional VCF components. – Provide a comma-separated list of IP addresses. The IP addresses can be contiguous or non-contiguous.
VCF services runtime FQDN	FQDN of the VCF services runtime component. This FQDN must resolve to a unique IP outside of the VCF services runtime IP range. The hostname from this FQDN is prefixed to the names of its node VMs and related objects. **Note:** Do not use capital letters in the VCF services runtime FQDN. Do not use a `.local` FQDN.

Setting	Description
VCF services runtime size	Select the size for the VCF services runtime component. The VCF services runtime sizing determines the sizing of all remaining VCF management services components. You do not specify the size of separate components. For information on the number and size of nodes for your VCF management services and license server, see VCF Fleet Sizing Models. For information on CPU, RAM, and storage requirements, see the *Management Domain Sizing* tab of the *Planning and Preparation Workbook*. See Planning and Preparation.
Internal cluster CIDR IPv4	This option is only available in VCF 9.1.1 and later. Select the VCF services runtime internal IPv4 network. Available blocks are: • 198.18.0.0/15 • 240.0.0.0/15 • 250.0.0.0/15 Select a CIDR block that does not overlap with any of your infrastructure networks, including management, VM, and IP pool networks. An overlap causes routing conflicts and breaks connectivity to or from the VCF services runtime cluster.
Instance component FQDN	FQDN of the instance-level components: SDDC lifecycle, Salt master, real-time metrics, real-time metrics store, and telemetry. This FQDN must resolve to a unique IP outside of the VCF services runtime IP range. **Note:** Do not use capital letters in the instance component FQDN. Do not use a `.local` FQDN.
VCF services runtime password	Password for the local administrative accounts `vmware-system-user` and `admin@vsp.local`, used when deploying VCF services runtime and integrating with other VCF components.
Fleet component FQDN	FQDN of the fleet-level components: fleet lifecycle, Salt RaaS, software depot. This FQDN must resolve to a unique IP outside of the VCF services runtime IP range. **Note:** Do not use capital letters in the fleet component FQDN. Do not use a `.local` FQDN.
Identity broker FQDN	FQDN of the Identity Broker component. This FQDN must resolve to a unique IP outside of the VCF services runtime IP range. **Note:** Do not use capital letters in the identity broker component FQDN. Do not use a `.local` FQDN. This option is available if your environment does not have a VCF Identity Broker already deployed.

Setting	Description
License server FQDN	FQDN of the license server. This FQDN must resolve to a unique IP outside of the VCF services runtime IP range. **Note:** License server can be deployed in IPv4 only.

8. Wait for the process to complete.
   If you encounter issues during the deployment, see Troubleshooting VMware Cloud Foundation or vSphere Foundation Deployments.

API Procedure

1. Retrieve the SSL thumbprint of the VCF Operations certificate.
   a. From a computer with network access to the VCF Operations primary node, run the command.
      ```
      echo | openssl s_client -connect <vcfoperations-primary.rainpole.io>:443 2>/dev/
      null | openssl x509 -noout -fingerprint -sha256
      ```
      The command returns the SHA-256 fingerprint of the certificate in the format `Fingerprint=2E:3D:88:49:3D:A7:C9:69:96:B4:85:29:6A:E6:E2:D0:C5:52:84:B5:4`
   b. Save the SSL thumbprint to later use in the `VcfManagementComponentsSpec` JSON specification file.
2. Log in to VCF Operations.
3. From the navigation bar at the top, click **Build**.
4. In the left pane, navigate to **Developer Center** > **APIs and SDKs**.
5. On the **APIs and SDKs** page, in the SDDC Manager API card, click **API Explorer**.
6. Select the first VCF Instance.
7. In the **API Categories** section, expand **VCF Management Components**.
8. Modify the sample `VcfManagementComponentsSpec` JSON specification file with values for your environment, expand **/v1/vcf-management-components/validations**, paste the content of the modified JSON in the body of the `POST` request, and click **Execute**.

   Note:

   This JSON file serves as an example and illustrates multiple available options for configuring your VCF management services. The provided JSON deploys highly available VCF management services with a `Medium` size by using an IP range on a custom VLAN-backed network. License server and identity broker are also deployed on the same custom VLAN-backed network. If you have an existing VCF Identity Broker for the primary VCF instance or an existing VCF Identity Broker in appliance mode, you must remove the `vidbSpec` specification. Some properties in the JSON are commented out - they can be used according to the needs of your environment.

```
{
  "vcfOperationsSpec": {
    "useExistingDeployment": true,
    "adminUserPassword": "VCF-operations-admin-password",
    "nodes": [
      {
        "hostname": "vcfoperations-master.rainpole.io",
        "type": "master",
        "sslThumbprint":
"B8:08:7B:92:4D:8D:6D:5E:58:13:C3:B3:9F:91:11:78:DC:7C:0B:36:70:2A:29:4F:AE:CE:44:14:62:04:D
      }
    ]
  },
  "vspClusterSpec": {
    "platformFqdn": "platform-vsp-cluster.rainpole.io",
    //FQDN of the VCF services runtime component. This FQDN must resolve to a
unique IP outside of the VCF services runtime IP range. The hostname from this
FQDN is prefixed to the names of its node VMs and related objects.
    "instanceFqdn": "instance-vsp-cluster.rainpole.io",
```

```
    //FQDN of the instance-level components: SDDC lifecycle, Salt master, real-
time metrics, real-time metrics store, and telemetry. This FQDN must resolve to a
 unique IP outside of the VCF services runtime IP range.
    "fleetFqdn": "fleet-vsp-cluster.rainpole.io",
    //FQDN of the fleet-level components: fleet lifecycle, Salt RaaS, software
 depot. This FQDN must resolve to a unique IP outside of the VCF services runtime
 IP range.
    "internalClusterCidrIpv4": "198.18.0.0/15",
    //"internalClusterCidrIpv6": "fd00::/111",
    "size": "medium",
    "systemUserPassword": "vmware-system-user-admin@vsp.local-password",
    "ipv4Pool": {
      "ipRange": {
        "startIpAddress": "10.10.20.100",
        "endIpAddress": "10.10.20.120"
      }
      // "cidr": "10.10.20.0/24",
      // "addresses": [
      //   "10.10.20.100",
      //   "10.10.20.101",
      //   "10.10.20.102"
      // ],
      // "excludedAddresses": [
      //   "10.10.20.105"
      // ]
    }
    // "ipv6Pool": {
    //   "ipRange": {
    //     "startIpAddress": "fd10::100",
    //     "endIpAddress": "fd10::120"
    //   },
    //   "cidr": "fd10::/64",
    //   "addresses": [
    //     "fd10::100",
    //     "fd10::101",
    //     "fd10::102"
    //   ],
    //   "excludedAddresses": [
    //     "fd10::105"
    //   ]
    // }
  },
  "licenseServerSpec": {
    "hostname": "license-server.rainpole.io"
  },
   "vidbSpec": {
     "hostname": "vidb.rainpole.io"
  },
  "vcfManagementComponentsInfrastructureSpec": {
    "xRegionNetwork": {
//Specifies the custom VLAN backed network where VCF management services and
license server are deployed.
      "networkName": "vlan-backed-custom-network",
      "subnetMask": "255.255.255.0",
      "gateway": "10.10.20.1",
      //"ipv6Gateway": "fd10::1",
      //"ipv6Prefix": 64
```

```
        }
      }
    }
```

9. Retrieve the validation ID from the body of the response.
10. Expand **/v1/vcf-management-components/validations/{validationID}**, paste the validation ID in the `GET` request body, and click **Execute**.
11. Review the results and remediate any errors.
12. To deploy VCF management services, identity broker, and license server, expand **/v1/vcf-management-components**, paste the content of the validated JSON specification file in the `POST` request body, and click **Execute**.

After the deployment completes, license server and VCF management services are available for your fleet.

To retrieve the VCF management services components system-managed service account password, see Retrieve VCF Management Services Passwords.

- License your environment.
  - If you upgrade from VCF 5.2.x, see Registering VCF Operations and a License Server with the VCF Business Services Console.
  - If you upgrade from VCF 9.0.x, see Transfer Licenses After You Upgrade VCF Operations from Version 9.0.x.
- After you deploy the VCF management services components, configure your fleet-level software depot before you can proceed with the upgrade of the remaining components. See Binary Management for VMware Cloud Foundation.
- To perform lifecycle management operations on VCF Automation, VCF Identity Broker, VCF Operations for networks, and VCF Operations for logs, you must upgrade these components to version 9.1.

Transfer Licenses After You Upgrade VCF Operations from Version 9.0.x

Starting with version 9.1, the license server component is mandatory to license a VCF Instance or vSphere Foundation. After the deployment of the license server as part of an upgrade from 9.0.x to 9.1, your licenses are transferred from the VCF Operations instance to the license server.

After the deployment of the license server, you might see duplicate licenses in VCF Operations while the licenses are transferred. The duplicate licenses disappear when the transfer completes. During that period, avoid assigning licenses to assets as license assignments might fail.

If your VCF Operations instance is in connected mode, the license server is automatically registered and licenses are automatically transferred to the license server.

If your VCF Operations instance is in disconnected mode, you must register the license server with the VCF Business Services console. Licenses are then automatically transferred from VCF Operations to the license server. You must generate a new license file from the VCF Business Services console, and import it in VCF Operations.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Manage** and in the left navigation pane, click **Licensing** > **Licenses & Registration**.
3. In the **Register & License VCF Operations** pane, click **Continue**.

   The **Register and License VCF Operations** workflow appears.

   The pre-registration steps of the procedure are complete because a license server is deployed during the upgrade of the VCF Operations to version 9.1, and the connection mode remains unchanged.

4. Download the registration file.
   a) In the **Registration** section, in the **Download Registration File** card, click **Download**.
   b) Save the registration file.
5. Use a computer with Internet connection and log in to the VCF Business Services console with your Broadcom Support Portal credentials.
   a) In the VCF Operations instance, in the **Import Registration File to the VCF Business Services console** card, click **Start**.

      A new tab opens in your browser, and you are redirected to the VCF Business Services console at https://vcf.broadcom.com/.
   b) Enter your Broadcom Support Portal credentials.
   c) Select the Site ID to which you want to register this VCF Operations instance.

      If you have only one site, it is selected by default.
6. Import the registration file in the VCF Business Services console.
   a) Navigate to **Licensing** > **VCF Operations Registrations**.
   b) In the **License Your Environment** card, click **Upload Registration File**.
   c) In the **VCF Operations Registrations** page, click **New Registration**.

      The **Register & License VCF Operations** workflow appears.
   d) In the **Registration** section, in the **Import Registration File** card, click **Start**.

      A dialog box appears.
   e) In the **Import Registration File** dialog box, click **Browse**, and select and import the license file.

      Note:

      If you upload a registration file that is already used to register another VCF Operations instance, the operation fails with an error.
7. In the VCF Business Services console, enter a display name for your VCF Operations instance.
   a) In the Registration section, in the **Name VCF Operations** card, click **Start**.
   b) Enter a unique name for your VCF Operations instance, and click **Save**.

      By default the FQDN of VCF Operations is prepopulated as a display name when you start the registration process in the VCF Business Services console. The FQDN is only saved if you do not enter another display name. If you change the FQDN with another display name, the connectivity between VCF Operations and the VCF Business Services console is not impacted. You can change the display name at any time after the registration.
8. Complete the verification process by exchanging the verification and confirmation files between the VCF Business Services console and the VCF Operations instance.
   a) In the Registration section, in the **Download Verification File** card, click **Start**, and save the verification file.
   b) In your VCF Operations instance, in the **Import Verification File** card, click **Import**, and upload the verification file.

      After you import the verification file, the confirmation file is ready for download.
   c) In the **Download Confirmation File** card, click **Download** to download the confirmation file.
   d) To go back to the VCF Business services console, in the **Upload the Confirmation File to the VCF Business Services console** card, click **Start**.
   e) To verify that you uploaded the verification file in VCF Operations, in the Upload the Verification file to VCF Operations card, click **Mark as Complete**.
   f) In the **Import Confirmation File** card, click **Start**, and upload the confirmation file that you downloaded from VCF Operations.

      After you complete the registration of the license server, the licenses are automatically transferred from the VCF Operations instance to the license server. The **Add Licenses** section is automatically populated.
9. In the Download license file card, click **Start**, and download the license file.

   You must import the license file in the VCF Operations instance to complete the registration process.

10. In the VCF Operations instance, import the license file that you downloaded from the VCF Business Services console.
    a) In the **Import License File** card, click **Import**, import the licenses file that you downloaded from the VCF Business Services console, and click **Complete**.
    b) In the **License File Imported** dialog box, review the licenses imported to VCF Operations, and click **Close**.

The transfer of licenses is in progress. The licenses are automatically reassigned to the assets that they are assigned to before the upgrade. You might see duplicate licenses while the licenses are transferred. The duplicate licenses disappear when the transfer is complete. During that period, avoid assigning licenses to assets as license assignments might fail.

Upgrading to VCF Automation 9.1

You use VCF Operations to manage the lifecycle of VCF Automation. Depending on your environment, you can import an existing VMware Aria Automation 8.x instance and upgrade it to VCF Automation 9.1, or directly upgrade VCF Automation from 9.0.x to 9.1.

The upgrade uses a blue-green deployment to provision a new VCF Automation instance while the existing instance remains active during the upgrade. The upgrade deploys a dedicated VCF services runtime instance that hosts the VCF Automation 9.1 nodes. During the upgrade, the system migrates data and configurations to the new environment before switching network traffic to the 9.1 instance. After the upgrade completes, the source Automation nodes are shut down automatically.

Networking Details

To allow for reduced risk and downtime during the upgrade process, you must provide additional IP addresses and an FQDN for the VCF Automation 9.1 instance. You reuse the existing FQDN that you use to access the Automation UI. You deploy the new VCF Automation 9.1 nodes on the same network as your existing Automation nodes. The network requirements differ based on your Automation source and target version.

- For target versions 9.1.0.0 through 9.1.0.300, you must provide a unique network CIDR block that contains 8 sequential IP addresses.
  - The required block is a /29 subnet mask representing 8 sequential IP addresses.
  - All 8 IP addresses must reside on the same network where your current Automation nodes are deployed.
    - 3 of the IP addresses are assigned to the VCF Automation 9.1 nodes
    - 2 of the IP addresses are a buffer for automatic redeployment of failed nodes or for rolling updates in the future.
    - 3 of the IP addresses are not used.
- For target versions 9.1.0.400 and later, you can now also provide 5 unique IP addresses that reside on the same network where your current Automation nodes are deployed. The IP addresses can be contiguous or non-contiguous.
  - 3 of the IP addresses are assigned to the VCF Automation 9.1 nodes
  - 2 of the IP addresses are a buffer for automatic redeployment of failed nodes or for rolling updates in the future.
- 1 New FQDN with a unique corresponding IP for the VCF services runtime instance that hosts the 9.1 VCF Automation nodes. The corresponding IP must be outside of the CIDR that you already dedicated. This FQDN is used only internally between VCF components and is not user-facing.
- The FQDN from the existing Automation instance is automatically transferred as the VCF Automation FQDN of the 9.1 instance.
- If your existing Automation instance uses an external load balancer, no changes in its configuration are required.
- If you upgrade from Aria Automation 8.18.x, the IP addresses of the existing nodes are automatically assigned as VIPs for the 9.1 instance.
- If you upgrade from VCF Automation 9.0.x, the IP addresses of the existing nodes are freed after the upgrade.

Account Details for the New 9.1 Instance

The upgrade wizard prompts for a single password that is assigned for both the **vmware-system-user** and **admin@vsp.local** accounts for the newly deployed VCF Automation 9.1 instance.

Import an Existing VMware Aria Automation Instance in VCF Operations

If your environment has a VMware Aria Automation instance, to upgrade to VCF Automation 9.1, you first import VMware Aria Automation into VCF Operations and then use VCF Operations to upgrade VCF Automation.

Before starting the import and upgrade:

- Verify that your VMware Aria Automation 8.18.1 and later instance resides in the management domain of the VCF Instance where you want to import it to. To move VMware Aria Automation to a different vCenter, you must reconfigure the network. See Change IP addresses of VMware Aria Automation node or cluster.
- Because the upgrade introduces architectural changes that could affect existing cloud account and integration configurations, verify that your environment does not include any unsupported endpoints by reviewing the precheck requirements and remediation steps in Broadcom knowledge base article 89563.
- Verify that you have configured a software depot. See Configure a Software Depot Connection Mode.
- Verify that you have downloaded the upgrade binaries for VCF Automation. See Download Binaries to Software Depot.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Build**.
3. In the left pane, click **Lifecycle**.
4. Click **VCF Management** and click the **Components** tab.
5. From the **Add Component** drop-down menu, select **VCF Automation**.

   The **Add VCF Automation** wizard opens.
6. On the **Deployment Type** page, select **Import vRealize Automation 8.x or VCF Automation 9.0.x**, configure the settings, and click **Next**.

Setting	Description
VCF Instance	Select the VCF Instance to host the imported VCF Automation instance.
Primary node FQDN or IP address	FQDN or IP address of the primary node of the Aria Automation instance that you import.
Primary node SSH user password	The password for the SSH user of the primary node of the Aria Automation instance.

7. In the **Certificate Security Alert** dialog box, click **Trust Certificate**.
8. On the **Summary** page, click **Finish**.
9. To synchronize the VCF Operations inventory and continue with the upgrade, on the **Components** tab, click **Sync**.

Infrastructure data, such as DNS and NTP settings, is not available in the fleet lifecycle component before VMware Aria Automation is upgraded to VCF Automation 9.1.

Perform the Upgrade to VCF Automation 9.1

You use VCF Operations to upgrade VCF Automation. Before you proceed, ensure that you are familiar with the networking and account details for the new VCF Automation 9.1 instance. See Upgrading to VCF Automation 9.1.

- If you are upgrading from VCF 5.2.x, verify that you imported and integrated the VMware Aria Automation instance version 8.18.1 and later to VCF Operations. See Import an Existing VMware Aria Automation Instance in VCF Operations.
- If your VMware Aria Automation 8.18 has a vSphere or an NSX-T endpoint with a certificate signed by the same CA as the Aria Automation certificate or by a well-known external CA, Aria Automation automatically trusts the connection because the CA is already present in the truststore. During the upgrade to VCF Automation 9.1, the truststore is not migrated and these endpoints become untrusted, causing TLS failures. Before you upgrade, follow the guidance in

Broadcom knowledge base article 425489. If the issue is not resolved, follow the guidance in Broadcom knowledge base article 447667.

- VCF Automation 9.x enforces FIPS 140-2 compliance for all external integration endpoints. The pre-upgrade check validates that every integration communicates over a secure, certificate-backed channel. The upgrade might be blocked for environments with any of the following configurations:
  - An IPAM integration endpoint configured with skip certificate check flag set to `ON`.
  - A VMware Aria Operations integration endpoint.
  - An LDAP directory integration configured using plain LDAP.

    See Broadcom knowledge base article 447667.
- Verify that you have an FQDN for the VCF services runtime that hosts VCF Automation in version 9.1. The FQDN must resolve to an IP address that is not part of the IP addresses dedicated for the VCF services runtime to host the VCF Automation nodes.
- For target versions 9.1.0.0 through 9.1.0.300, verify that you have a dedicated CIDR for use by the VCF services runtime instance that hosts the VCF Automation nodes. For example, a /29 subnet mask as VCF Automation requires 5 IP addresses.
- For target versions 9.1.0.400 and later, verify that you have dedicated 5 unique IP addresses for use by the VCF services runtime instance that hosts the VCF Automation nodes. The IP addresses can be contiguous or non-contiguous.
- Verify that you have configured a software depot. See Configure a Software Depot Connection Mode.
- Verify that you have the necessary install and upgrade binaries. See Download Binaries to Software Depot.
  - For 9.0.x to 9.1.x upgrade, verify that you have the 9.1.x upgrade binary for VCF Automation and the 9.1.x install binary for the migration service engine component.
  - For 8.18.x to 9.1.x upgrade, verify that you have the 9.1.x install binary for VCF Automation and the 9.1.x install binary for the migration service engine component.
- Verify that the Subject Alternative Name (SAN) field of the VCF Automation certificate contains the FQDNs and IP addresses of all VCF Automation nodes and the FQDN and IP address of the Load Balancer / VIP.
  - To replace non-compliant certificates for VCF Automation, see Replace a Certificate with a Configured CA-Signed Certificate.
  - To replace non-compliant certificates for Aria Automation, see Replace VMware Aria Automation Certificates.
- Verify that in VCF Operations, the cloud accounts for the integration of VCF Automation are working properly by explicitly clicking the **Validate Connection** button.
- If you have a Data Services Manager instance integrated with VCF Automation, before you upgrade VCF Automation to 9.1.x, see Upgrading your Data Services Manager in a VCF Environment.
- If your 8.18.x or 9.x Automation instance is not deployed on the primary datastore of your VCF Instance, you can upgrade to VCF Automation 9.1.1.0200 and later if your fleet lifecycle, SDDC lifecycle, and SDDC Manager components are on version 9.1.1.0200 or later.
- If your Aria Automation 8.18.x instance is a single-node deployment of extra-large size, scale it out to a 3-node cluster before you start the upgrade.

Upgrading to VCF Automation starts with creating an upgrade plan to ensure the correct binary is mapped to the upgrade. After you create the plan, you start the upgrade process.

During the upgrade from Aria Automation 8.18 to VCF Automation 9.1.x, VCF Automation retains and reuses the certificate of the Aria Automation 8.18 instance.

If your existing Automation instance uses an external load balancer, no changes in its configuration are required.

> **Note:**
>
> After you upgrade VMware Aria Automation 8.x to VCF Automation 9.1, VMware Aria Suite Lifecycle 8.x stops managing VCF Automation 9.1. Preserve the VMware Aria Suite Lifecycle 8.x instance to perform all Day-N operations for VMware Identity Manager 3.3.x. Do not delete VMware Identity Manager as it is still an authentication source for VCF Automation 9.1.

VCF services runtime uses an internal CIDR block of 198.18.0.0/15. If this subnet is in use on your network, you must change the internal CIDR block for VCF management services to 240.0.0.0/15 or 250.0.0.0/15. You can specify the internal network to be used during the deployment by using a JSON specification file for VCF and vSphere Foundation.

See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation. Alternatively, you can specify the internal network by using the fleet lifecycle API.

Different VCF services runtime instances on the same network can work with the same internal network. For example, you can have the two instances of VCF services runtime for your VCF management services and VCF Automation using the same internal network.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Build**.
3. In the left pane, click **Lifecycle**.
4. Click **VCF Management** and click the **Upgrade** tab.
5. Next to the **VCF Automation** instance, click **Configure**.

   The **Configure VCF Automation Upgrade** wizard opens.
6. On the **Parameters** page, configure the settings and click **Next**.

   Note:

   VCF Automation is hosted by a dedicated VCF services runtime instance. The IP addresses and FQDN for the VCF services runtime instance for VCF Automation must be unique. You cannot reuse the CIDR block and FQDN from the VCF services runtime instance for VCF management services.

Setting	Description
• (In versions 9.1.0.0 through 9.1.0.300) VCF services runtime nodes CIDR • (In versions 9.1.0.400 and later) VCF services runtime nodes IP pool	Enter the pool of IP addresses that the dedicated VCF services runtime instance can assign to VCF Automation nodes it hosts. 3 of the addresses are consumed right away for the deployment of the nodes and 2 are reserved as a buffer for future lifecycle management operations. • In versions 9.1.0.0 through 9.1.0.300, enter a valid CIDR. • In versions 9.1.0.400 and later, enter a CIDR or 5 comma-separated individual IP addresses. • In versions 9.1.0.400 and later, you can exclude individual IP addresses from the CIDR. This option is only available if you select CIDR.
VCF services runtime FQDN	Enter the FQDN for VCF services runtime dedicated to host the VCF Automation cluster. **Note:** Do not use capital letters in the FQDN.
VCF Automation and VCF services runtime password	Enter the password for the VCF Automation **vmware-system-user** and **admin@vsp.local** accounts. **Note:** The password must be a minimum of 15 characters and include at least one uppercase, one lowercase, one digit, and one special character. Supported special characters: `@ ! # $ % ? ^` Passwords must not contain a dictionary word or part of a dictionary word. Passwords must not contain the user name or parts of the user name.

7. On the **Summary** page, review the details and click **Finish**.
8. On the **Upgrade** tab, next to the **VCF Automation** instance, click **Run Prechecks**.

   Upgrade prechecks start. To view the progress of the workflow, click **Precheck details**.

After the process completes, if any failures occured, remediate them and run the prechecks again.

9. In the **Upgrade** tab, next to the **VCF Automation** instance, click **Upgrade**.

   Upgrade starts. To view the progress of the workflow, click **Tasks**.

Your VCF Automation instance is now version 9.1.x. The nodes that were part of the old Automation instance are shut down and, if no longer needed, you can delete them.

Migrate VCF Automation Single Sign-On Configuration from VMware Identity Manager to VCF Identity Broker

VCF Automation 9.1 uses identity broker and VCF SSO to manage authentication, replacing the legacy VMware Identity Manager (vIDM) model in Aria Automation. Complete the procedures in this section to migrate authentication for both the Provider Management and the Organization Portal after upgrading to VCF Automation 9.1.

Aria Automation uses VMware Identity Manager to manage authentication. When you upgrade Aria Automation to VCF Automation 9.1, the VMware Identity Manager configuration is preserved for backward compatibility. VCF Automation 9.1 introduces a different authentication model that uses VCF SSO and the identity broker component. Complete the migration to replace the legacy vIDM configuration with the VCF Automation 9.1 recommended authentication model.

VCF Automation 9.1 introduces two distinct portals, each with a separate authentication model:

- The VCF Automation Provider Management authenticates users by using identity provider and VCF SSO. The Provider Management UI must be configured after upgrading to VCF Automation 9.1.
- VCF Automation Provider Management uses a separate identity provider configuration from the VCF Automation organization UI. After upgrade, configure an identity provider in VCF Automation Provider Management. Including VCF Automation Provider Management in the fleet-wide VCF SSO is highly recommended.
- The VCF Automation organization UI authenticates end users through a direct connection (such as OIDC, LDAP, or SAML) to an external identity provider or identity broker. Before the migration, the Organization for VM Apps, created from the existing Aria Automation configuration during the upgrade, uses the legacy VMware Identity Manager OIDC connection for authentication.

After you complete all procedures in this section, the legacy Aria Suite Lifecycle and VMware Identity Manager deployments are no longer required for the operation of VCF Automation.

Configure Authentication for VCF Automation Provider Management

Configure authentication for the VCF Automation Provider Management by verifying that the identity broker is set as the identity provider, then add the users and groups that require Provider Management access.

- Aria Automation is upgraded to VCF Automation 9.1, and the existing configuration is converted to an Organization for VM Apps.
- VCF Single Sign-On is configured. See Configuring VCF Single Sign-On.
- If you are migrating groups or users with administrative rights, VMware Identity Manager and identity broker must be configured to use the same identity provider.

The Provider Management UI is the VCF Automation 9.1 interface through which provider administrators manage organizations, identity providers, and tenants. The Provider Management UI does not exist in Aria Automation, there is no configuration for identity providers and access control after upgrade to VCF Automation 9.1. Authentication for VCF Automation Provider Management must be configured before provider administrators can access it.

Follow the guidance in the table to configure the authentication for VCF Automation Provider Management.

Step	Description
1. Configure the VCF Automation integration with identity broker and VCF Single Sign-On.	Configure VCF Single Sign-On for VCF Operations and VCF Automation Provider
2. Import users to the System organization.	Import Provider Users to Your VCF Automation

Step	Description
3. Import groups to the System organization.	Import a Provider Group to Your VCF Automation

Configure Authentication for the VCF Automation Organization Portal

After you upgrade Aria Automation to VCF Automation 9.1, the Organization for VM Apps continues to use the legacy VMware Identity Manager OIDC connection for authentication. Complete the procedures in this section to replace VMware Identity Manager-based authentication with a direct identity provider connection in the VCF Automation Organization Portal.

- Aria Automation is upgraded to VCF Automation 9.1, and the existing configuration is converted to an Organization for VM Apps.
- VMware Identity Manager and identity broker must be configured to use the same identity provider.

The Organization Portal is the VCF Automation interface through which users access and manage their assigned organizations. When Aria Automation is upgraded to VCF Automation 9.1, the Organization for VM Apps retains the VMware Identity Manager authentication configuration from the source environment. This configuration uses an OIDC connection to VMware Identity Manager as the authentication source.

VCF Automation 9.1 supports direct integration between the Organization Portal and an external identity provider. Migrating from VMware Identity Manager-based authentication to direct identity provider integration allows the Organization Portal to authenticate users through a standard LDAP or SAML connection that is managed by the identity broker component.

Although VCF Automation 9.1 continues to support VMware Identity Manager-based authentication for the Organization for VM Apps after an upgrade, direct identity provider integration is the recommended authentication model for VCF Automation 9.1 and later.

Configure the First User Account for the Organization for VM Apps

The first user account is a local account that provides access to the Organization for VM Apps during the transition from VMware Identity Manager-based authentication to direct identity provider integration. Create this account before you configure the identity broker component as the identity provider for VCF Automation.

The first user account is a local application account with full administrative access to the Organization for VM Apps in VCF Automation. This account does not depend on an external identity provider and retains access to the Organization for VM Apps when you disconnect VMware Identity Manager authentication and configure the direct identity provider integration.

You must assign all four administrator and owner roles to the first user account. If you do not assign all four roles, you lose administrative access to the organization during the identity provider migration.

1. Log in to the VCF Automation Provider Management UI using `System` as the organization name.
2. In the left pane, click **Organizations**.
3. Click the Organization for VM Apps (the default organization name is *`<vidm_hostname> Classic Tenant`*).
4. On the organization page, in the left pane, click **First User**.
5. Click **Add First User**.
6. In the **Create User**, configure the user name and password, configure the following roles, and click **Save**.

Setting	Description
Role	• **Assembler Administrator** • **Orchestrator Administrator** • **Organization Owner** • **Service Broker Administrator**

The first user account is created with full administrative access to the Organization for VM Apps. You can now use this account to access the organization if your VMware Identity Manager-based accounts become unavailable during the identity provider migration.

Add A New Identity Provider for the Organization for VM Apps

The Organization for VM Apps can have a single OIDC connection that is used by the legacy VMware Identity Manager configuration. Configure a new LDAP or SAML connection as the identity provider in the Organization portal to replace VMware Identity Manager-based authentication with direct identity provider integration.

This topic covers the integration with an external LDAP identity provider. You can also configure an integration with a SAML identity provider.

1. Log in to the VCF Automation Organization UI with the first user and `vidm Classic Tenant` as the organization name.
2. On the main navigation bar, click **Infrastructure**.
3. In the left pane, navigate to **Administration** > **Identity Providers**.
4. On the **Identity Providers** page, click the **LDAP** tab and click **Configure**.
5. In the **Edit LDAP Options**, select **Custom LDAP service** and click **Save**.
6. In the **LDAP** tab, click the **Custom LDAP** tab and click **Edit**.
7. In the **Edit Custom LDAP**, configure the LDAP connection.
   a) On the **Connection** tab, configure the settings.

Setting	Description
Server	The host name or IP address of the LDAP server.
Port	The port number on which the LDAP server is listening. For LDAP, the default port number is 389. For LDAPS, the default port number is 636.
Base Distinguished Name	VCF Automation connects to the base distinguished name (DN) in the LDAP directory. To connect to root level, enter only the domain components, for example, `DC=example,DC=com`. To connect to a node in the domain tree structure, enter the distinguished name for that node, for example, `OU=ServiceDirector,DC=example,DC=com`. Connecting to a node limits the scope of the directory available to VCF Automation.
Connector Type	The type of your LDAP server. Can be Active Directory or OpenLDAP.
Use SSL	If your server is LDAPS, select this check box.
Authentication Method	Simple authentication consists of sending the user's DN and password to the LDAP server. If you are using LDAP, the LDAP password is sent over the network in plain text. If you want to use Kerberos, you must configure the LDAP connection by using the vCloud API.
User name	Enter the full LDAP distinguished name (DN) of a service account with domain admin rights. VCF Automation uses this account to query the LDAP directory and retrieve user information. If the anonymous read support is enabled on your LDAP server, you can leave these text boxes blank.
Password	The password for the service account that connects to the LDAP server. If the anonymous read support is enabled on your LDAP server, you can leave these text boxes blank.

b) On the **User Attributes** tab, examine the default values for the user attributes, and, if your LDAP directory uses different schema, modify the values.
c) On the **Group Attributes** tab, examine the default values for the group attributes, and, if your LDAP directory uses different schema, modify the values.
d) On the **Button Label** tab, configure a custom name for the button label and click **Save**.

   You can enter up to 24 symbols. You can use special characters and accented letters. If you want to revert to the default text, delete the custom label. The default button label is localized, and depending on your browser language settings, the text might appear in a different language. Custom labels always appear as you enter them.
e) If you selected the **Use SSL** check box, and if the certificate of the LDAPS server is not yet trusted, in the **Trust Certificate** dialog box, confirm if you trust the certificate presented by the server endpoint.

8. To test the LDAP connection settings and the LDAP attribute mappings:
   a) Click **Test**.
   b) Enter the password of the LDAP server user that you configured and click **Test**.

      If connected successfully, a green check mark is displayed.

      The retrieved user and group attribute values are displayed in a table. The values that are successfully mapped to LDAP attributes are marked with green check marks. The values that are not mapped to LDAP attributes are blank and marked with red exclamation marks.
   c) To exit, click **Cancel**.
9. To synchronize VCF Automation with the configured LDAP server, click **Sync**.

The custom LDAP connection is registered as the identity provider for the Organization for VM Apps in VCF Automation. Users and groups from this LDAP directory can now be imported into the organization.

Migrate User Groups to the New Identity Provider for the Organization for VM Apps
After you configure a new LDAP identity provider for the Organization for VM Apps, import the corresponding groups by using the Organization Portal to replace the existing VMware Identity Manager-based groups.

- Every VMware Identity Manager-based group in the Organization for VM Apps has a corresponding group configured in the new identity provider.
- Each imported group must have at least the **Organization Member** role assigned.

After you configure the new external identity provider for the Organization for VM Apps, the groups in the Organization for VM Apps still point to the VMware Identity Manager OIDC identity provider and are not automatically migrated to the new identity provider connection. You must create new corresponding groups in the new identity provider for every existing VMware Identity Manager-based group in the OIDC identity provider. You then import the newly created groups and assign the corresponding roles in the Organization for VM Apps.

1. Log in to the VCF Automation Organization Portal with the first user and `vidm Classic Tenant` as the organization name.
2. On the main navigation bar, click **Administer**.
3. In the left pane, click **Access Control**.
4. Click the **Groups** tab and click **Import Groups**.
5. In the **Import Groups** dialog box, configure the settings and click **Save**.

Setting	Description
Source	Select the LDAP identity provider configured in the prerequisite task.
Search	Enter the name of the group you are importing and select it from the results.
Assign Role	Select the organization role or roles to assign. Assign the same roles that the corresponding vIDM-based group holds.

6. Repeat the steps for each remaining vIDM-based group.

Existing VMware Identity Manager-based groups are not automatically removed when you import their new identity provider equivalents. After you verify that all groups are successfully migrated and users can authenticate through the new identity provider, remove the VMware Identity Manager-based groups from the Organization for VM Apps.

Migrate User Accounts to the New Identity Provider and Import them in the Organization for VM Apps
After you import groups from the new identity provider, update the existing user accounts in the Organization for VM Apps to point to the new identity provider instead of the VMware Identity Manager OIDC connection. Use the bulk update feature to migrate all identity provider-backed user accounts in a single operation.

When Aria Automation is upgraded to VCF Automation 9.1, user accounts in the Organization for VM Apps authenticate through the VMware Identity Manager OIDC connection. After you configure a new identity provider, existing user accounts do not automatically switch to the new connection. You must update each user account to point to the new identity provider.

Two categories of user accounts exist in the Organization for VM Apps after an upgrade:

- Identity-provider-backed user accounts authenticate through VMware Identity Manager by using the OIDC connection. These accounts are updated in the VCF Automation Organization UI to point to the new identity provider by using the bulk update feature.
- Local VMware Identity Manager user accounts are created directly in VMware Identity Manager and have no corresponding accounts in the new external identity provider. These accounts cannot be migrated using the bulk update feature. They must be recreated as identity-provider-backed accounts.

Reconfigure Identity Provider Backed User Accounts to Point to the New Identity Provider for the Organization for VM Apps
By using the bulk update feature in the VCF Automation Organization UI, you reconfigure the existing identity-provider-backed accounts from VMware Identity Manager to point to the new identity provider for the Organization for VM Apps authentication.

1. Log in to the VCF Automation Organization UI with the first user and `vidm Classic Tenant` as the organization name.
2. On the main navigation bar, click **Administer**.
3. In the left pane, click **Access Control**.
4. Click the **Users** tab and click **Bulk Update**.
5. In the **Export Users** section, to download all user accounts as a `CSV` file, click **Export Users**.
6. Make a back-up of the `CSV` file.
7. Prepare the `CSV` file for migrating the user accounts.

   Note:

   Do not update or remove lines in the `CSV` file that reference VMware Identity Manager-based local user accounts during the bulk update procedure. Modifying those lines causes errors.

   a) For each identity-provider-backed user account, replace every reference to `OAUTH` with the name of your target identity provider source (in this example, use `LDAP`).
   b) For each identity-provider-backed user account, update the `email (readonly)` field in the format `username@domain` (for example, `user01@domain.local`).
   c) Save the `CSV` file.
8. On the **Bulk Update** page, in the **Export Users** section, click **Next**.
9. In the **Upload CSV File** section, click **Select CSV File**, select the `CSV` file you edited, click **OK**, and click **Next**.
10. In the **Update Users** section, click **Update**.

    The `Full Name` and `Effective Roles` fields may be empty for users that have not logged in. These fields are updated when the user logs in.

Verify that the updated user accounts appear with the correct identity provider source and email address.

Recreate Local VMware Identity Manager User Accounts to the New Identity Provider and Import them in the Organization for VM Apps

After recreating the VMware Identity Manager local user accounts in the new identity provider, you import the user accounts in the VCF Automation Organization UI.

- For each VMware Identity Manager local user account in the Organization for VM Apps, you must create a corresponding user account the new identity provider.

1. Log in to the VCF Automation Organization UI with the first user and `vidm Classic Tenant` as the organization name.
2. On the main navigation bar, click **Administer**.
3. In the left pane, click **Access Control**.
4. Click the **Users** tab and click **Import Users**.
5. In the **Search** bar, search for a user account, recreated in the new identity provider.
6. Select the account from the table.
7. From the **Assign Role** drop-down menu, select the corresponding roles that the original VMware Identity Manager local account held.
8. Turn off the **Inherit roles from group** toggle switch.
9. Click **Import**.

Remove the VMware Identity Manager Configuration for Organization for VM Apps

After you verify that all users and groups authenticate successfully through the new identity provider, remove the legacy VMware Identity Manager configuration from the Organization for VM Apps.

The legacy VMware Identity Manager configuration in the Organization for VM Apps consists of three components:

- The VMware Identity Manage-based group entries that reference the OIDC connection.
- The VMware Identity Manage-based user accounts registered in projects.
- The OIDC connection itself through which VMware Identity Manager provides authentication.

The three components must be removed in sequence. The OIDC connection cannot be removed while VMware Identity Manage-based groups or user accounts still reference it.

Do not start any procedure in this section until you verify that all users and groups authenticate successfully through the new LDAP identity provider. If you remove the OIDC connection before authentication is verified, users whose accounts were not migrated correctly lose access to the organization and cannot log in.

After completing all procedures in this section, the Organization for VM Apps uses only the new identity provider for authentication.

Remove VMware Identity Manager-Based Groups for the Organization for VM Apps

VMware Identity Manager-based groups in the Organization for VM Apps use the VMware Identity Manager OIDC connection and must be removed before you can delete the OIDC connection. Disable and delete each VMware Identity Manager-based group in the VCF Automation Organization UI.

A group cannot be deleted while it is in an active state. Disable each group first, then delete it.

1. Log in to the VCF Automation Organization UI with the first user and `vidm Classic Tenant` as the organization name.
2. On the main navigation bar, click **Administer**.
3. In the left pane, click **Access Control**.
4. Click the **Groups** tab.
5. Disable the VMware Identity Manager-based group.
   a) Click the vertical ellipsis next to the VMware Identity Manager-based group and, from the drop-down menu, select **Disable**.
   b) In the **Disable Group** dialog box, click **Disable** to confirm.
6. Delete the VMware Identity Manager-based group.
   a) Click the vertical ellipsis next to the VMware Identity Manager-based group and, from the drop-down menu, select **Delete**.
   b) In the **Delete Group** dialog box, click **Delete** to confirm.
7. Repeat this procedure for the remaining VMware Identity Manager-based groups in the Organization for VM Apps.

Remove VMware Identity Manager-Based Accounts for the Organization for VM Apps
VMware Identity Manager-based local user accounts must be removed from all projects and from the Organization for VM Apps before you can delete the VMware Identity Manager OIDC connection. Remove each account from every project, transfer any user-owned objects to prevent orphaned resources, then disable and delete the account from the VCF Automation Organization UI.

Remove VMware Identity Manager-based user accounts from the organization projects.

1. Log in to the VCF Automation Organization UI with the first user and `vidm Classic Tenant` as the organization name.
2. On the main navigation bar, click **Manage and Govern**.
3. In the left pane, click **Projects**.
4. On the **Projects** page, click the project that has VMware Identity Manager-based accounts assigned to it.
5. Click the **Users** tab, select the VMware Identity Manager-based accounts, and click **Remove**.
6. Click **Save**.
7. Repeat these steps for all remaining projects with assigned VMware Identity Manager-based accounts.

Remove VMware Identity Manager-based accounts from the organization.

8. On the main navigation bar, click **Administer**.
9. In the left pane, click **Access Control**.
10. Click the **Users** tab.
11. Disable the VMware Identity Manager-based accounts.
    a) Click the vertical ellipsis next to the VMware Identity Manager-based account and, from the drop-down menu, select **Disable**.
    b) In the **Disable User** dialog box, click **Disable** to confirm.
12. Delete the VMware Identity Manager-based account.
    a) Click the vertical ellipsis next to the VMware Identity Manager-based account and, from the drop-down menu, select **Delete**.
    b) To transfer any objects that are owned by this account to your administrator account and prevent them from becoming orphaned, select the **Transfer the user's objects to me** checkbox.
    c) In the **Delete User** dialog box, click **Delete** to confirm.
13. Repeat these steps to delete the remaining VMware Identity Manager-based accounts.

Remove the VMware Identity Manager OIDC Connection
The VMware Identity Manager OIDC connection is the final component of the legacy VMware Identity Manager configuration in the Organization for VM Apps. To complete the authentication migration to the new identity provider, set the OIDC connection to inactive, then delete it

1. Log in to the VCF Automation Organization UI with the first user and `vidm Classic Tenant` as the organization name.
2. On the main navigation bar, click **Administer**.
3. In the left pane, navigate to **Connections** > **Identity Providers**.
4. Click the **OIDC** tab and click **Edit**.
5. In the **Configure OpenID Connect** dialog, set the `Status` toggle to **Inactive**, then click **Save**.
6. Click **Delete**.

The VMware Identity Manager OIDC connection is removed. The Organization for VM Apps now uses the new LDAP identity provider for authentication. The authentication migration from VMware Identity Manager to direct identity provider integration is complete.

Upgrading to VCF Operations for Networks 9.1

You use VCF Operations to manage the lifecycle of VCF Operations for networks. You can import and upgrade VMware Aria Operations for Networks 6.14.0 Patch 6, 6.14.1 or later. You directly upgrade VCF Operations for networks 9.0.x to 9.1.

> **Note:**
>
> You can only import and upgrade one instance of VMware Aria Operations for Networks to VCF Operations for networks.

Import an Existing VMware Aria Operations for Networks Instance in VCF Operations

If you have a VMware Aria Operations for Networks instance, to upgrade to VCF Operations for networks 9.1, you first import VMware Aria Operations for Networks to VCF Operations and then upgrade the imported component.

- Verify that you have configured a software depot. See Binary Management for VMware Cloud Foundation.
- Verify that the subject alternative name field for all certificates contains the FQDNs and IPs for all platform and collector nodes.
  - If your Aria Operations for Networks instance is managed by VMware Aria Suite Lifecycle, see Replace certificate for VMware Aria Suite Lifecycle products.

  - If your Aria Operations for Networks instance is not managed by VMware Aria Suite Lifecycle, see Broadcom knowledge base article 324471.
- Verify the VCF Operations for Networks platform appliances are updated with valid certificates that contain IPs in SAN fields of certificates. See Broadcom knowledge base article 440459.
- Verify that you have taken a snapshot of your environment before performing an upgrade or scale-out operation. The snapshot must be taken manually from the relevant vCenter instance. For more information, see Broadcom knowledge base article 314428.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Build**.
3. In the left pane, click **Lifecycle**.
4. Click **VCF Management** and click the **Components** tab.
5. From the **Add Component** drop-down menu, select **VCF Operations for networks**.

   The **Add VCF Operations for Networks** wizard opens.
6. On the **Deployment Type** page, select **Import Aria Operations for Networks 6.14.x or VCF Operations for networks 9.0.x appliance**, configure the settings, and click **Next**.

Setting	Description
VCF Instance	Select the VCF Instance to host the imported VCF Operations for networks component.
Platform IP address	IP address for the VCF Operations for networks component.
Admin user password	The password for the **admin** account.

7. On the **Summary** page, review the details and click **Finish**.
8. To synchronize the VCF Operations inventory and continue with the upgrade, on the **Components** tab, click **Sync**.

Infrastructure data, such as DNS and NTP settings, is not available in fleet lifecycle until VMware Aria Operations for Networks is upgraded to VCF Operations for networks 9.1.

Perform Upgrade to VCF Operations for Networks 9.1

Use VCF Operations to upgrade an imported VMware Aria Operations for Networks Instance or a VCF Operations for networks 9.0.x instance to 9.1.

- Verify that all certificates meet FQDN requirements. The certificates must have FQDNs and IPs for all nodes VCF Operations for networks nodes. To replace non-compliant certificates, see Replace a Certificate with a Configured CA-Signed Certificate.
- Verify that the Subject Alternative Name of the VCF Operations for networks certificate contains the FQDNs and IP addresses of all platform nodes. See Broadcom knowledge base article 440459.

1. Log in to the VCF Operations.
2. From the navigation bar at the top, click **Build**.
3. In the left pane, click **Lifecycle**.
4. Click **VCF Management** and click the **Upgrade** tab.
5. Next to the **VCF Operations for networks** component, click **Run Prechecks**.

   Upgrade prechecks start. To view the progress of the workflow, click **Precheck details**.

   After the process completes, remediate any failures and run the prechecks again.
6. In the **Upgrade** tab, next to the **VCF Operations for networks** instance, click **Upgrade**.

   Upgrade starts. To view the progress of the workflow, click **Upgrade details**.

Upgrade to Log Management 9.1

Use VCF Operations to transfer VCF Operations for logs 9.0.x to the 9.1 log management component. As the log management component is now part of VCF management services, the upgrade operation deploys log management 9.1 and transfers the configuration from the 9.0.x VCF Operations for logs instance.

- Prepare a new FQDN for log management 9.1 that resolves to an IP address that is in the same network with the VCF management services.

No direct upgrade path for VMware Aria Operations for Logs 8.x to log management 9.1 exists. If your environment has VMware Aria Operations for Logs 8.x, you deploy the 9.1 log management component and migrate the configuration from your existing 8.x instance. For 8.x to 9.1 scenarios, follow the deployment guidance here: Deploy Log Management.

The log management component cannot deployed on a custom NSX overlay network. The architectural changes introduced in VCF 9.1 require that the log management component is deployed on the management network that hosts VCF management services.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Build**.
3. In the left pane, click **Lifecycle**.
4. Click **VCF Management** and click the **Upgrade** tab.
5. Next to the **log management** component, click **Configure**.
6. In the **Configure Log Management Update** wizard, configure settings and click **Finish**.

Setting	Description
Log management FQDN	New and unique FQDN for the log management component. The FQDN must not point to an IP address in the IP range that is assigned to VCF management services. **Note:** Do not use capital letters in the FQDN.
Size	• Small: a single replica. • Medium: from 3 to 12 replicas. • Large: from 3 to 18 replicas. For maximum limits and sizing specifics, see VMware Configuration Maximums.
Number of replicas	Dependent on the deployment size. Small deployments require one replica. Medium and large deployments require at least three replicas.

7. In the **Upgrade** tab, next to the **VCF Operations for logs** instance, click **Upgrade**.

   Upgrade starts. To view the progress of the workflow, click **Upgrade details**.

8. After the upgrade completes, review the details of the procedure and download the detailed report.
   a) Click the **Tasks** tab.
   b) Click the log management deployment/upgrade task to see the details.
   c) Click the **Upgrade report** icon to download the report and review the details.

After your log management component is successfully upgraded, to fully transition to the new 9.1 component, you perform the necessary configuration steps.

Post-Upgrade Action	Description	Documentation
1. Configure agents	• If you are upgrading from 9.0 to 9.1 and have log collection configured in VCF Operations 9.0 UI, the agent and log sources reconfiguration is automatic. • If you are upgrading from 8.18 to 9.1 and have log collection configured in the VMware Aria Operations for Logs UI, you must manually reconfigure the agents and log sources. In these scenarios, legacy appliances continue collecting data until you manually reconfigure log sources and agents to send logs to the 9.1 component.	Configuring Agents
2. Configure log sources	Reconfigure required log sources to point to your new instance if they are not automatically transitioned.	Configuring Log Sources for Log Management
3. Configure log forwarding	Reconfigure your log forwarding based on your upgrade scenario: • If any log forwarder is sending logs to the existing VMware Aria Operations for Logs 8.18 or VCF Operations for logs 9.0 cluster, you must reconfigure the log forwarder to send logs to the new 9.1 log management component FQDN. • When importing the configuration from VMware Aria Operations for Logs 8.18 or VCF Operations for logs 9.0, forwarders are copied over but are not active. You must manually activate them.	Configuring Log Forwarding
4. Migrate custom content	Custom content (dashboards, queries, alerts) is not transferred automatically. Create a Content Pack for your custom content, use the conversion tool to generate a Management Pack, and install it into VCF Operations.	Converting Content Packs to Management Packs
5. Transfer historical log data	You can access historical log data in the following ways: • You can use the data transfer utility. • You can import archived data and can query it in the 9.1 log management component UI. • For 90 days after the upgrade, you can query logs from the old cluster by using the 9.1 log management component UI.	Log Data Transfer

After you complete the post-upgrade configuration, you can shut down and remove the legacy appliances.

Upgrade to Identity Broker 9.1

Use VCF Operations to upgrade an appliance mode VCF Identity Broker 9.0.x instance to identity broker 9.1.

- Ensure that your VCF Identity Broker in appliance mode is deployed on the VCF management network or on the same network and datastore as VCF management services. See Redeploy VCF Identity Broker 9.0.x to a Supported Network and Datastore for Upgrade to 9.1.
- As VCF Identity Broker in appliance mode transitions to VCF management services, ensure that your identity broker component can reach your identity provider.

If your VCF Operations is already on version 9.1, and if your VCF Identity Broker 9.0.x is deployed on an NSX overlay segment or on a network or datastore that are different from the network or datastore of VCF management services, the upgrade to identity broker 9.1 fails. To remediate the upgrade, see Broadcom knowledge base article 444734.

VCF Identity Broker 9.0.x embedded upgrades to 9.1 as part of the vCenter instance upgrade. See Upgrading Embedded Identity Broker.

1. Log in to VCF Operations.
2. From the navigation bar at the top, click **Build**.
3. In the left pane, click **Lifecycle**.
4. Click **VCF Management** and click the **Upgrade** tab.
5. Next to the **Identity broker** component, click **Run Prechecks**.

   Upgrade prechecks start. To view the progress of the workflow, click **Precheck details**.

   After the process completes, remediate any failures and run the prechecks again.
6. In the **Upgrade** tab, next to the **Identity broker** component, click **Upgrade**.

   Upgrade starts. To view the progress of the workflow, click **Upgrade details**.

After you complete the upgrade to 9.1 of your management components, you can remove the old VCF Identity Broker appliance.

Upgrade to VCF Operations Orchestrator 9.1

You can upgrade your deployment to the latest version of VCF Operations orchestrator by using a mounted ISO image.

Verify that you are upgrading from VMware Aria Automation Orchestrator 8.18.1 or later. For more information on the supported upgrade path, see the Product Interoperability Matrix.

1. Download and mount the ISO image.
   a) Download the ISO image from the official Broadcom download site.
   b) Connect the CD-ROM drive of the appliance virtual machine in vSphere. See the *vSphere Virtual Machine Administration* documentation.

      Note:

      After connecting the CD-ROM drive, navigate to your appliance VM settings page and verify that **Connect At Power On** is enabled.
   c) Mount the ISO image to the CD-ROM drive of the appliance virtual machine in vSphere. See the *vSphere Virtual Machine Administration* documentation.
2. Log in to the VCF Operations orchestrator appliance command line as **root**.
3. Run the `blkid` command, and note the device name for the appliance CD-ROM drive.
4. Mount the CD-ROM drive.

   ```
   mount /dev/xxx /mnt/cdrom
   ```

   Important:

   For clustered VCF Operations orchestrator deployments, you must perform steps 2 and 3 on all nodes in the cluster.

5. Back up your VCF Operations orchestrator deployment by taking a virtual machine snapshot. See Take a Snapshot of a Virtual Machine.

   CAUTION:

   VCF Operations orchestrator does not currently support memory snapshots. Before taking the snapshot of your deployment, verify that the **Include virtual machine's memory** option is deactivated.

6. To initiate the upgrade, run one of the following commands depending on your product version.

   - For upgrades from Automation Orchestrator 8.18.1, run the following command on one of the nodes in your deployment:

     ```
     vracli upgrade exec -y --repo cdrom://
     ```

   - For upgrades from VCF Operations orchestrator 9.0 or later, run the following command on one of the nodes in your deployment:

     ```
     vracli upgrade exec -y --profile lcm --repo cdrom://
     ```

     Note:

     For VCF Operations orchestrator deployments authenticated with vSphere, you must provide the credentials of the user who registered your deployment with the vCenter Single Sign-On (SSO) service. You can verify the identity of this user either in the `/data/vco/usr/lib/vco/app-server/conf/sso.properties` path in the virtual appliance, or in the output of the `vracli vro authentication` command. You can either enter the password manually or export your password as an environmental variable. This can be useful for scenarios where you are using an automated script to upgrade multiple VCF Operations orchestrator deployments. To export the SSO password, run the `export VRO_SSO_PASSWORD=`*`your_sso_password`* command.

   During the upgrade, you are automatically logged out of your terminal, because the VCF Operations orchestrator appliance reboots.

7. Log in to the VCF Operations orchestrator appliance command line as **root** and follow the upgrade progress by running the `vracli upgrade status --follow` command.

   CAUTION:
   The `vracli upgrade status --follow` command can occasionally display a false error message that indicates that the upgrade has failed. To troubleshoot this problem, see step 8.

8. Optional: If you receive an error message while running the `vracli upgrade status --follow` command, follow these steps:

   a) Verify that you receive the following error message:

   ```
   Running health check after upgrade for nodes and pods.
   Health check after upgrade for nodes and pods failed.
   ... Upgrade terminated due to critical error. Follow the upgrade guide to recover
    the system. ...
   ```

   b) Navigate to the `/var/log/vmware/prelude/upgrade-report-latest` and confirm that you receive the following error:

   ```
   Pod: vco-app-xxxx is not in Ready or Completed state. All pods must be in either of these states
   ```

   c) Run the `kubectl get pods -n prelude -w | grep -E 'vco|orchestration-ui'` command and verify that the status of all 3 `vco-app` pods and the `orchestration-up-app` pod is `RUNNING`.

   Note:
   It can take up to 5-10 minutes after receiving the error message for all pods to enter the `RUNNING` state.

   ```
   orchestration-ui-app-xxxx    1/1     Running   0          5h42m
   vco-app-xxxx              3/3     Running   0          5h47m
   ```

d) Run the `curl -k https://<your_orchestrator_FQDN>/vco/api/healthstatus` command and verify that the health check is returning a `RUNNING` status.

```
{"state":"RUNNING","health-status":{"state":"OK","time":1615296823325},"instance-
id":"your_orchestrator_FQDN"}
```

Important:

The preceding command must run in an environment different from the VCF Operations orchestrator command line. You can run the command from the command line of a different virtual machine. You can also view the health status information in a browser by navigating directly to https://*<your_orchestrator_FQDN>*/vco/api/healthstatus.

You have upgraded your VCF Operations orchestrator deployment. If you encounter a notification stating that the upgrade has failed, see False Upgrade Failure Notification.

Validate that the upgrade was successful by running the `vracli version` command in the command line of the appliance. By running this command, you can validate the product version and build number of VCF Operations orchestrator.

Upgrade Failure Notification During VMware Aria Automation Orchestrator Upgrade

The upgrade log indicates that the upgrade process has failed, but the individual nodes of the deployment are upgraded.

After the upgrade script finishes running, you receive the following message in your Automation Orchestrator Appliance indicating that the upgrade has failed:

```
Upgrade failed and left the system in non-working state. Check the error report below
 to correct the problem. Once addressed, you can continue the upgrade by running 'vracli
 upgrade exec --resume'
```

However, the upgrade log lists that the nodes of your Automation Orchestrator deployment are upgraded.

```
Hostname:               <your_orchestrator_node_FQDN>
Status:                 Upgraded
Cluster Member:         Yes
Version Before:         <build_before_upgrade>
Version After:          <build_after_upgrade>
Description:            The node is upgraded successfully.
```

To resolve this problem, verify that the nodes are running, and resume the upgrade.

1. Verify that your Automation Orchestrator nodes are running.

   ```
   kubectl get all pods
   ```

2. If your Automation Orchestrator nodes are running, resume the upgrade process.

   ```
   vracli upgrade exec --resume
   ```

Upgrading NSX Global Manager Nodes in a Federated Environment

Before you can upgrade any VCF instances and NSX Local Managers, you must upgrade standby and active NSX Global Managers in the NSX Federation.

If NSX Federation is configured between two VCF Instances, VCF does not manage the lifecycle of the NSX Global Manager nodes. For each VCF instance, you upgrade the Upgrade Coordinator for NSX Federation, before you upgrade the standby and active NSX Global Manager nodes. To upgrade the NSX Global Managers, you must manually download the NSX upgrade precheck and upgrade bundles.

The NSX Upgrade Coordinator is a self-contained web application that runs in NSX Manager. It orchestrates the upgrade of the Upgrade Coordinator itself and NSX Global Manager nodes in the cluster. The Upgrade Coordinator provides a centralized interface that enables administrators to track and manage the upgrade process end to end.

Attention:

In a federated environment, you must upgrade the standby Global Manager nodes first, before you proceed with the active Global Manager nodes upgrade.

1. Download the NSX Global Manager Upgrade Precheck and Upgrade Bundles.
   a) Log in to the Broadcom Support Portal and browse to **My Downloads** > **VMware Cloud Foundation** > **VMware Cloud Foundation 9**.
   b) Click your VCF version and on the **VMware NSX** row, click **View Group**.
   c) Locate the NSX *version* Upgrade Bundle and NSX *version* Precheck Bundle.

      The upgrade precheck and upgrade bundles have the following format:

      - `VMware-NSX-upgrade-bundle-`*`versionnumber.buildnumber`*`.mub`
      - `VMware-NSX-upgrade-bundle-`*`versionnumber.buildnumber`*`-pre-check.pub`
   d) Agree to the *Terms and Conditions* and *Compliance Reporting Terms and Conditions*.
   e) Click the download icons to download the upgrade precheck and upgrade bundles to the system where you access the NSX Global Manager UI.
2. In a web browser, log in to the NSX Global Manager for the domain at https://*nsx_gm_vip_fqdn*/.
3. From the navigation panel, select **System** > **Upgrade**.
4. Under **Global Manager**, click **Upgrade**.
5. On the **Check Upgrade Readiness** tab, complete the following tasks.
   a) Click **Select local file**, navigate to the location you downloaded the upgrade precheck bundle file, and click **Upload**.

      Alternatively, paste the URL where the `.pub` file is located.
   b) Click **Next**.
   c) Click **Run Pre-Checks** > **All Pre-Checks** to verify that all NSX components are ready for upgrade.

      The pre-check checks for component connectivity, version compatibility, and component status.

      Review and address the errors and warnings to avoid problems during the upgrade.
6. On the **Prepare for Upgrade** tab, complete the following tasks.
   a) Click **Select local file**, navigate to the location you downloaded the upgrade bundle file, and click **Upload**.

      Alternatively, paste the URL where the upgrade bundle `.mub` file is located.
   b) Click **Next**.
   c) Click **Begin Upgrade**.

      Upgrade Coordinator is upgraded.
7. Click **Next**.
8. On the **NSX Global Managers** tab, click **Start upgrade**.

   After the upgrade finishes and the management plane restarts, the NSX Manager user interface, API, and CLI are accessible.

Upgrade the Remaining Components in the Management Domain to 9.1

To upgrade your core components to version 9.1, the management domain components must be at version 5.2 or later. If your platform is at a version earlier than 5.2, you must upgrade the management domain and all workload domains to 5.2 or later, before you can upgrade to 9.1.

Prerequisites for Upgrading the VCF Core Components

Before you upgrade the VCF management domain, verify that the following prerequisites are satisfied.

Table 979: Update Prerequisites

Prerequisite	Additional Information
Allocate a temporary IP address for each vCenter appliance upgrade.	Required for each vCenter upgrade. Must be allocated from the management subnet. The IP address can be reused.
Verify that no vCenter appliance is joined to an Active Directory domain.	See Broadcom knowledge base article 373004.
Verify that there are no expired or expiring passwords.	Review the password management dashboard in SDDC Manager.
Verify that there are no expired or expiring certificates.	Review the **Certificates** tab in SDDC Manager for each workload domain.
Verify ESX host TPM module status.	If ESX hosts have TPM modules in use, verify that they are running the latest 2.0 firmware. If not in use, they must be disabled in the BIOS. See Broadcom knowledge base article 312159.
Verify that ESX hardware is compatible with the target version.	See ESX Requirements and Broadcom Compatibility Guide.
Manually update the vSAN HCL database to ensure that it is up to date.	See Broadcom knowledge base article 2145116.
Back up SDDC Manager and all vCenter and NSX Manager instances.	Take file-based backups or image-based backups of SDDC Manager and all vCenter and NSX Manager instances. Take a cold snapshot of SDDC Manager.
Verify that there are no failed workflows in your system and none of the SDDC Manager resources are in activating or error state.	**CAUTION:** If any of these conditions are true, contact Broadcom Support before starting the upgrade.
Download the upgrade binaries.	See Binary Management for VMware Cloud Foundation.
Review the *Release Notes* for known issues that are related to upgrades.	
In the vSphere Client, verify that there are no active alarms on hosts or vSphere clusters.	

Transitioning from vSphere Lifecycle Manager Baselines to vSphere Lifecycle Manager Images

vSphere Lifecycle Manager baselines are not supported with VMware Cloud Foundation 9.0 or later. Before you can upgrade to ESX 9.0 or later, you must transition any clusters that use vSphere Lifecycle Manager baselines to use vSphere Lifecycle Manager images.

You can transition to vSphere Lifecycle Manager images by using one of the following methods.

- A PowerShell script
  - Menu-driven interface
  - Command-line interface (CLI)
- The SDDC Manager API

After you upgrade SDDC Manager to version 9.0 or later, you must transition all clusters in the management domain and in all workload domains. If the transition is not possible or you have vSphere Supervisor (Workload Management) enabled, transition any remaining vSphere Lifecycle Manager baseline clusters after upgrading vCenter and before upgrading ESX.

Learn about the planning considerations, prerequisites, and tasks for transitioning ESX clusters and standalone hosts from vSphere Lifecycle Manager baselines to vSphere Lifecycle Manager images. This process is for transitioning ESX clusters and standalone hosts that are already managed by SDDC Manager. For ESX clusters and standalone hosts that are not managed by SDDC Manager, you can transition to vSphere Lifecycle Manager images using the vSphere Client. See Convert a Cluster or a Host That Uses Baselines Into a Cluster or a Host That Uses vSphere Lifecycle Manager Images.

Warning:

For clusters and standalone hosts that are managed by SDDC Manager, the transition must be performed by SDDC Manager supported mechanisms only.

Prerequisites for Transitioning to vSphere Lifecycle Manager Images

Before you can transition a vSphere cluster or standalone host from vSphere Lifecycle Manager baselines to vSphere Lifecycle Manager images, you must meet certain prerequisites.

vSphere Lifecycle Manager Method Transition Prerequisites

<table>
<tr><th>Category</th><th>Prerequisite</th></tr>
<tr><td>SDDC Manager version</td><td>Clusters: 5.2.2 or later
Standalone hosts: 9.1 or later</td></tr>
<tr><td>Network access for SDDC Manager and vCenter</td><td>HTTPS (TCP 443)</td></tr>
<tr><td>SDDC Manager access privileges</td><td>Administrator (ADMIN)</td></tr>
<tr><td>ESX hosts</td><td>Ensure all the ESX hosts that you are attempting to transition are healthy before beginning the workflow.</td></tr>
<tr><td>vSphere Lifecycle Manager image</td><td>To use the PowerShell script or SDDC Manager API, you must have at least one vLCM image available. While these images are originally authored in vCenter, the required format has changed:
• Version 8.0.3 and earlier: Images must be created a zero-host, vLCM-managed cluster.
• Version 9.0 and later: Images should be authored using the vCenter Image Library.
For information about creating a temporary cluster with a vLCM image, see Creating and Managing vSphere Lifecycle Manager Clusters. Choose the documentation section corresponding to the version in the following table.
Note:
<table><tr><th>vCenter Version</th><th>Documentation Section</th></tr><tr><td>8.0 Update 3</td><td>Create a Cluster That Uses a Single Image by Importing an Image from a Host</td></tr><tr><td>8.0 through 8.0 Update 2</td><td>Create a Cluster That Uses a Single Image by Composing a New Image</td></tr></table>For information about creating an Image Library vLCM image in vCenter 9, see Create an Image in Image Library Manually.
Note:
The ESX version in the image must match the current version of the ESX hosts in the cluster you are transitioning.
Note:
Use a descriptive name for the images that you create.</td></tr>
<tr><td>PowerShell version</td><td>7.2 or later
See Installing Powershell.</td></tr>
</table>

Category	Prerequisite
PowerShell script	Download the script from Broadcom knowledge base article 385617.
VCF PowerCLI version	9.1 or later. **Important:** VMware.PowerCLI conflicts with VCF.PowerCLI. VMware.PowerCLI must be removed before installing VCF.PowerCLI. Uninstall VMware.PowerCLI. Installing VCF.PowerCLI.
Accurate client time settings	Windows Time Settings MacOS Time Settings Linux time settings are different depending on your distribution

The processes for transitioning to vSphere Lifecycle Manager images support the following topologies.

VMware Cloud Foundation Architecture Support

Topology	vSAN Ready Nodes Minimum Workload Domain Version
Standard architecture Cluster	5.0
Standalone host	5.0 *
Consolidated architecture Cluster	5.1
vSAN Stretched Cluster	5.1
DR Protected Cluster	5.1
vSphere Supervisor (Workload Management) enabled Cluster	Split BOM 9.0 **
Heterogeneous Hardware Cluster	9.0 ***

* Standalone host transition support is only possible in combination with SDDC Manager 9.1 or later.

** The workload domain vCenter must first be upgraded to vCenter 9.0 or later, while ESX must remain on version 8.0 Update 1a or later.

*** Clusters with heterogeneous hardware cannot transition to vSphere Lifecycle Manager images until all hosts are running ESX 9.0 or later. Before you start this procedure on a heterogeneous hardware cluster, upgrade it to ESX 9.0 or later. See Broadcom knowledge base article 424294.

Important Considerations

- You cannot combine the process of transitioning to vSphere Lifecycle manager images with the process of upgrading ESX.
- vSphere clusters that use vSphere Lifecycle Manager baselines must be transitioned to use images before they can be upgraded to ESX 9.0 or later.
- During the transition process, you cannot perform any other operations on the cluster or their underlying hosts until the transition is complete. This restriction includes, but is not limited to ESX password rotation.

Maximum Compliance Checks and Transitions for Clusters

- A maximum of 25 simultaneous cluster compliance checks might be run per SDDC Manager. Extra compliance checks are discarded.
- A maximum of 25 simultaneous cluster transitions might be run per SDDC Manager. Extra transition requests are discarded.

Maximum Compliance Checks and Transitions for Standalone Hosts

- A maximum of 25 simultaneous standalone host compliance checks might be run per Workload Domain. Extra compliance checks are discarded.
- A maximum of 25 simultaneous standalone host transitions might be run per Workload Domain. Extra transition requests are discarded.

Image Seeding Support for vLCM Image Auto-Generation

- A minimum workload domain version of 5.2.1 is required to activate the option of image seeding (to complement the option of constructing and applying your own vLCM image).

Using Image Seeding

Baseline-managed standalone hosts or clusters with compatible ESX and vCenter sBOM support allow for image seeding. This process auto-generates a vLCM image-based on a single host (standalone or cluster member). In a cluster environment, the created image must be uniform across all hosts, with matching VIBs. Upon completion, the image is stored in the vCenter depot.

Transitioning vSphere Lifecycle Manager Baseline Clusters to vSphere Lifecycle Manager Image Clusters Using PowerShell

You can use the PowerShell script through a menu-driven interface or a command-line interface (CLI). The menu-driven interface provides a guided transition process and is best suited for smaller environments. The command-line interface is useful for larger environments and can be used to operate against a single cluster at a time or, at scale, using JSON files.

> **Attention:**
>
> After you start a task with PowerShell, entering `Ctrl-C` or exiting PowerShell will not halt many tasks (for example, image import, compliance checks, or transition tasks). Those tasks continue to run in the background in SDDC Manager or vCenter.

> **Notice:**
>
> If the script execution system does not trust the certificate of the SDDC Manager, execute the following lines before running the script:
>
> ```
> Set-PowerCLIConfiguration -InvalidCertificateAction Ignore -Scope User
> ```
>
> ```
> $Env:SkipCertificateCheck="enabled"
> ```

Logging

The script records all actions to time-stamped files in a `logs` sub-directory that is created under the directory containing the script. This functionality facilitates troubleshooting.

If you run the script and this directory does not exist, it is automatically created.

```
PS C:\Users\Admin> .\VcfBaselineClusterTransition.ps1 -Connect
LogFolder not found, creating C:\Users\Admin\logs
```

Transition to vSphere Lifecycle Manager Images Using the PowerShell Script Menu-Driven Interface

The menu-driven interface provides a guided experience for the vLCM baseline to vLCM image management transition process.

The PowerShell script imports vSphere Lifecycle Manager images from the vCenter Image Catalog in vCenter 9.0 or later and from vSphere clusters and standalone-hosts in vCenter 8.0.3 and earlier.

Notice:

If the vLCM image is vSphere cluster-based and contains no hosts, the script will delete the vSphere cluster after completing the import and safety checks.

Attention:

SDDC Manager enforces image uniqueness based on two criteria: **name** and **manifest**.

First, image names must be unique across all vCenters; for example, importing **esx-803b** from one vCenter blocks a later import of the same name from another. Second, image manifests (metadata) must be unique.

If two differently named images—such as **esx-803-m01** and **esx-803-w01**—share identical components (e.g., base image **ESX 8.0.3a** and **Vendor A Version 1.2.3**), SDDC Manager rejects the second image. This manifest validation occurs during the final stage, after the full image has been uploaded.

1. Launch the script.

   ```
   .\VcfBaselineClusterTransition.ps1
   ```

2. Connect to SDDC Manager and select a vCenter that manages the vSphere clusters you want to transition.
   a) From the main menu of the script, select `Connect to SDDC Manager and select vCenter`.

      If a credentials file is not found on disk, you are prompted to provide the SDDC Manager FQDN and administrative credentials. If an SDDC credential file is present, the script uses those credentials to connect to the SDDC Manager referenced in the file.
   b) Select the vCenter(s) to which you want to connect.

      You can connect to all vCenters, a single vCenter, or a comma-delimited list of multiple vCenters.

      ```
      The following vCenter(s) were detected in SDDC Manager "vcf01.example.com":

        Id vCenter                      Workload Domain
        -- -------                      ---------------
        1  <ALL VCENTERS>               N/A
        2  m01-vc01.example.com         m01
        3  w01-vc01.example.com         w01

      This script can only transition vLCM baseline-managed clusters for connected vCenter(s).

      Enter an id, a comma-delimited list of ids of vCenters to connect to, or type 'c' to cancel: 1

      [INFO] Preparing to connect to vCenter(s)...

      [INFO] Successfully connected to vCenter "m01-vc01.example.com".
      [INFO] Successfully connected to vCenter "w01-vc01.example.com".
      ```

3. Import a vSphere Lifecycle Manager image into SDDC Manager from any connected vCenter.

   Manually importing a vLCM image into SDDC Manager is only required if your clusters or standalone hosts do not support image seeding.

a) From the main menu of the script, `Import vLCM images from vCenter(s) into SDDC Manager.`
b) Select the ID of the image to import and wait for the task to complete.

```
[INFO] Scanning vCenter(s) for vLCM images...

  Id vCenter Image Name vCenter Name
  -- ------------------ ------------
  1  esx-803b-vendor-a  m01-vc01.example.com
  2  esx-803b-vendor-b  m01-vc01.example.com

Enter an id or comma-delimited list of ids of image(s) to import into SDDC Manager or press 'c' to cancel: 1

[INFO] Successfully created SDDC Manager image "esx-803b-vendor-a" into SDDC Manager "vcf01.example.com"
[INFO] Safety check passed - No ESX hosts detected in cluster "esx-803b-vendor-a" in vCenter "m01-vc01.example.com".
```

If the import task fails with the error message `An image named <image-name> already exists in the SDDC Manager`, rename the source image in vCenter or delete the conflicting image from SDDC Manager before retrying. To delete the conflicting image in SDDC Manager, return to the main menu and select `(Optional) Delete SDDC Manager Image`.

c) Repeat these steps for all the images you want to import.

4. Check vSphere Lifecycle Manager baseline clusters or standalone hosts for compliance with a vSphere Lifecycle Manager image.

   Be mindful of the executing logic:

   The menu-driven interface always processes clusters sequentially. Each compliance check of a cluster must finish before the next one begins. Parallel checks are only possible using the command-line interface with the `-Parallel` option.

   Standalone hosts, however, are processed in parallel, so long as the following conditions are met:

   - Each batch does not exceed 25 standalone hosts
   - Each batch contains identical images or uses image seeding
   - Each batch uses the same workload domain vCenter.

   Attention:

   The compliance check mis-identifies certain non-user managed VIBs and components as non-compliant and in need of removal.

   The underlying files are not missing, but rather the metadata describing the entities is not yet installed. The metadata installation occurs automatically post-vLCM image transition.

   If the compliance check identifies the removal of these entities as the **exclusive trigger for maintenance mode**, the hosts will not **transition into** maintenance mode during the operation, The transition instead can be performed live.

Solution	Type	VIB/Component Name
HA (High Availability)	VIB	vmware-fdm
NSX	Component	nsx-lcp-bundle
Spherelet (vSphere Supervisor)	Component	VMware-Spherelet
vSphere Replication	VIB	vmware-hbr-agent

   For more information, see Broadcom knowledge base article 311882.

a) From the main menu of the script, select `Check existing cluster(s)/standalone host(s)' vLCM image compliance.`
b) Enter a comma-delimited list of the IDs of the vSphere clusters and/or standalone hosts that you want to scan for compliance.
c) When image seeding is supported across all selected clusters, you have the following two options.
   - Yes: Automatically generate an image based on the current cluster state
   - No: Manually select a vLCM image from the SDDC Manager inventory.

   The script returns a high-level status and summary and saves a JSON file with the full findings for each vSphere cluster or standalone host.
d) Review the summary report to determine if a vSphere cluster or standalone host is ready to be transitioned.

   Resolve any issues for vSphere clusters with a status of `INCOMPATIBLE` or `UNKNOWN`.

Status	Description
`NON_COMPLIANT`	The vSphere cluster or standalone host is not currently compliant with the selected vSphere Lifecycle Manager image, but is ready to be transitioned.
`INCOMPATIBLE`	The vSphere cluster or standalone host cannot be transitioned to the selected vSphere Lifecycle Manager image without taking extra steps. Review the summary and JSON file and resolve any issues that are blocking the transition. Contact Broadcom Support if you are unable to resolve one or more issues.
`UNKNOWN`	Compatibility could not be determined. Make sure that all the hosts in the vSphere cluster, or the standalone host, or the is/are healthy and connected to vCenter. Contact Broadcom Support if you are unable to resolve the issue.
`COMPLIANT`	All hosts are confirmed to be compliant the vLCM image.

Important:

The compliance check mis-identifies certain non-user managed VIBs and components as non-compliant and in need of removal.

The underlying files are not missing, but rather the metadata describing the entities is not yet installed. The metadata installation occurs automatically post-vLCM image transition.

If the compliance check identifies the removal of these entities as the **exclusive trigger for maintenance mode**, the hosts will not **transition into** maintenance mode during the operation, The transition instead can be performed live.

Solution	Type	VIB/Component Name
HA (High Availability)	VIB	vmware-fdm
NSX	Component	nsx-lcp-bundle
Spherelet (vSphere Supervisor)	Component	VMware-Spherelet
vSphere Replication	VIB	vmware-hbr-agent

For more information, see Broadcom knowledge base article 311882.

Example of a compliance check of a vSphere cluster using a specified image.

```
[INFO] Scanning for vLCM baseline (VUM) managed clusters/standalone-hosts...

Displaying vLCM Baseline (VUM) Managed Clusters/Standalone Hosts in Connected vCenter(s):

  Id Resource Name               Resource Type   vCenter Name                 Compliance Status SDDC Manager Image Name
  -- -------------               -------------   ------------                 ----------------- ------------------------
  1  m01-cl01                    Cluster         m01-vc01.example.com         NOT_CHECKED_YET   N/A
  2  w01-cl01                    Cluster         w01-vc01.example.com         NOT_CHECKED_YET   N/A
  3  esxi-1.example.com          Standalone Host w01-vc01.example.com         NOT_CHECKED_YET   N/A
  4  esxi-2.example.com          Standalone Host w01-vc01.example.com         NOT_CHECKED_YET   N/A

[INFO] Select the vLCM baseline cluster(s)/standalone-host(s) to check for compliance with vLCM image management.

Enter an id, a comma-delimited list of ids, or 'c' to cancel: 1

[INFO] Checking for image seeding support...

All selected clusters and standalone-hosts are eligible for image seeding.  Would you like to auto-generate vLCM images
 for them using image seeding?
[Y] Yes  [N] No  [?] Help (default is "Y"): N

[INFO] vLCM images available in SDDC Manager "sddc-manager.vrack.vsphere.local":

Id    SDDC Manager Image Name   Base Image            Components            Addons                         Hardware Support
--    -----------------------   ----------            ----------            ------                         ----------------
1     esx-803-vendor-a          8.0.3-24280767        N/A                   {vendor-a addon for Vendor S.. N/A
2     esx-803-vendor-b          8.0.3-24280767        N/A                   {vendor-a addon for Vendor S.. N/A

Please enter the id of an image or press 'c' to cancel: 1

[INFO] Beginning compliance check for Cluster "m01-cl01" in workload domain "w01" and beginning compliance check against image "esx-803-vendor-a"...

[INFO] Check image compliance of cluster m01-cl01 completed successfully.

[INFO] Cluster "m01-cl01" in Workload Domain "m01" has status "NON_COMPLIANT".

[WARNING] Remediation involving maintenance mode or a reboot, or both, is required to transition "m01-cl01" to vLCM image management.

[INFO] Summary of compatibility check cluster "m01-cl01" in Workload Domain "m01" against image "esx-803-vendor-a"

  Component Name Component Vendor Version Change Impact                    Severity Message
  -------------- ---------------- -------------- ------                    -------- --------
  N/A            N/A              No             MAINTENANCE_MODE_REQUIRED WARNING  The following VIBs on the host are missing from the
                                                                                     image and will be removed from the host during
                                                                                     remediation: vmware-fdm(8.0.3-24792044).
  ---------------------------------------------------------------------------------------------------------------------------------------
  N/A            N/A              No             N/A                       WARNING  These hosts have standalone vibs installed which may
                                                                                     get removed on remediating:
                                                                                     m01-esx01.example.com, m01-esx01.example.com,
                                                                                     m01-esx03.example.com, m01-esx04.example.com,
                                                                                     Review their compliance details carefully
                                                                                     before proceeding.
  ---------------------------------------------------------------------------------------------------------------------------------------
  N/A            N/A              No             MAINTENANCE_MODE_REQUIRED INFO     None
  ---------------------------------------------------------------------------------------------------------------------------------------
  NSX LCP Bundle VMware           Yes            MAINTENANCE_MODE_REQUIRED INFO     Components NSX LCP Bundle are removed in the image.
                                                                                     They will be removed from the host during remediation.
  ---------------------------------------------------------------------------------------------------------------------------------------
  N/A            N/A              No             MAINTENANCE_MODE_REQUIRED INFO     Quick Boot is supported on the host.
  ---------------------------------------------------------------------------------------------------------------------------------------
  N/A            N/A              No             MAINTENANCE_MODE_REQUIRED INFO     The host will be placed in maintenance mode before
                                                                                     remediation.
  ---------------------------------------------------------------------------------------------------------------------------------------

[INFO] Full compliance findings stored for cluster "m01-cl01" in Workload Domain "m01" have been saved to \Users\Administrator\logs\cluster-compatibility-for-m01-cl01-on-m01-with-esx-803-vendor-a.json
```

Example of a compliance check for a vSphere cluster using image seeding.

```
[INFO] Scanning for vLCM baseline (VUM) managed clusters/standalone-hosts...

Displaying vLCM Baseline (VUM) Managed Clusters/Standalone Hosts in Connected vCenter(s):

  Id Resource Name               Resource Type   vCenter Name                 Compliance Status SDDC Manager Image Name
  -- -------------               -------------   ------------                 ----------------- ------------------------
  1  m01-cl01                    Cluster         m01-vc01.example.com         NON_COMPLIANT     esx-803-vendor-a
  2  w01-cl01                    Cluster         w01-vc01.example.com         NOT_CHECKED_YET   N/A
  3  esxi-1.example.com          Standalone Host w01-vc01.example.com         NOT_CHECKED_YET   N/A
  4  esxi-2.example.com          Standalone Host w01-vc01.example.com         NOT_CHECKED_YET   N/A

[INFO] Select the vLCM baseline cluster(s)/standalone-host(s) to check for compliance with vLCM image management.

Enter an id, a comma-delimited list of ids, or 'c' to cancel: 2

[INFO] Checking for image seeding support...

All selected clusters and standalone-hosts are eligible for image seeding.  Would you like to auto-generate vLCM images
 for them using image seeding?
[Y] Yes  [N] No  [?] Help (default is "Y"): Y

[INFO] Image seeding workflow selected.

[INFO] Auto-generating vLCM image for Cluster "w01-cl01" in workload domain "w01" and beginning compliance check ...

[INFO] Auto-generating vLCM image for "w01-cl01" in workload domain "w01" and beginning compliance check ...

[INFO] Check image compliance of cluster w01-cl01 completed successfully.

[INFO] Cluster "w01-cl01" in Workload Domain "w01" has status "NON_COMPLIANT".

[WARNING] Remediation involving maintenance mode or a reboot, or both, is required to transition "w01-cl01" to vLCM image management.

[INFO] Summary of compatibility check cluster "w01-cl01" in Workload Domain "w01" against image "Autogenerated-Image"

  Component Name Component Vendor Version Change Impact                    Severity Message
  -------------- ---------------- -------------- ------                    -------- --------
  N/A            N/A              No             MAINTENANCE_MODE_REQUIRED WARNING  The following VIBs on the host are missing from the
                                                                                     image and will be removed from the host during
                                                                                     remediation: vmware-fdm(8.0.3-24792044).
  ---------------------------------------------------------------------------------------------------------------------------------------
  N/A            N/A              No             N/A                       WARNING  These hosts have standalone vibs installed which may
                                                                                     get removed on remediating:
                                                                                     w01-esx01.example.com, w01-esx01.example.com,
                                                                                     w01-esx03.example.com, w01-esx04.example.com,
                                                                                     Review their compliance details carefully
                                                                                     before proceeding.
  ---------------------------------------------------------------------------------------------------------------------------------------
  N/A            N/A              No             MAINTENANCE_MODE_REQUIRED INFO     None
  ---------------------------------------------------------------------------------------------------------------------------------------
  NSX LCP Bundle VMware           Yes            MAINTENANCE_MODE_REQUIRED INFO     Components NSX LCP Bundle are removed in the image.
                                                                                     They will be removed from the host during remediation.
  ---------------------------------------------------------------------------------------------------------------------------------------
  N/A            N/A              No             MAINTENANCE_MODE_REQUIRED INFO     Quick Boot is supported on the host.
  ---------------------------------------------------------------------------------------------------------------------------------------
  N/A            N/A              No             MAINTENANCE_MODE_REQUIRED INFO     The host will be placed in maintenance mode before
                                                                                     remediation.
  ---------------------------------------------------------------------------------------------------------------------------------------

[INFO] Full compliance findings stored for cluster "w01-cl01" in Workload Domain "w01" have been saved to \Users\Administrator\logs\cluster-compatibility-for-w01-cl01-on-w01-with-Autogenerated-Image.json
```

---

5. Transition `NON_COMPLIANT` (ready to transition) vSphere clusters or standalone hosts to vSphere Lifecycle Manager images.

   CAUTION:

   After you start a compliance or transition task, it cannot be paused or stopped. The task continues to run on the SDDC Manager even if the PowerShell window is closed.

   a) From the main menu, `Transition vLCM baseline (VUM) cluster/standalone host to vLCM image management.`
   b) Enter a comma-delimited list of the IDs of the vSphere clusters or standalone hosts that you want to transition to vSphere Lifecycle Manager images.
   c) Review the compliance findings from those vSphere clusters and/or standalone hosts, and if you wish to proceed, enter `Y` to begin the workflow.
   d) After you select your clusters and/or standalone hosts, you might choose to temporarily override remediation policies for the transition operation. See Configuring Remediation Settings for further details.

      Example of a cluster transition with an override remediation policy

```
[INFO] Scanning for vLCM baseline (VUM) managed clusters/standalone hosts...

Displaying vLCM Baseline (VUM) Managed Clusters/Standalone Hosts in Connected vCenter(s):

  Id Resource Name               Resource Type   vCenter Name                 Compliance Status SDDC Manager Image Name
  -- -------------               -------------   -------------                ----------------- -----------------------
  1  m01-cl01                    Cluster         m01-vc01.example.com         NON_COMPLIANT     esx-803-vendor-a
  2  w01-cl01                    Cluster         w01-vc01.example.com         NON_COMPLIANT     <Autogenerated-Image>
  3  esxi-1.example.com          Standalone Host w01-vc01.example.com         NOT_CHECKED_YET   N/A
  4  esxi-2.example.com          Standalone Host w01-vc01.example.com         NOT_CHECKED_YET   N/A


[INFO] Select the vLCM baseline clusters/standalone hosts to transition to vLCM image management.

Enter an id, a comma-delimited list of ids in order, or 'c' to cancel: 1

Confirm you have reviewed the image compliance findings stored in "C:\Users\Administrator\logs" before proceeding.

Would you like to continue?
[Y] Yes  [N] No  [?] Help (default is "N"): Y

[INFO] Clustert "m01-cl01" transition to vLCM image management accepted.

Would you like to use the default host remediation options?
[Y] Yes  [N] No  [?] Help (default is "Y"): N

[INFO] For information more information on host remediation, please visit: https://techdocs.broadcom.com/us/en/vmware-cis/vsphere/vsphere-sdks-tools/8-0/automation-sdks-programming-guide/managing-the-lifecycle-of-hosts-and-clusters/conf
iguring-remediation-settings.html

Enter your preferred remediation action (FAIL or RETRY) or press 'c' to cancel: RETRY

Enter the number of times to retry the failed operation (between 3 and 100) or press 'c' to cancel: 3

Enter the seconds to wait between retry operations (between 300 and 3600) or press 'c' to cancel: 300

Choose one preRemediationPowerAction from the following list: POWER_OFF_VMS, SUSPEND_VMS, DO_NOT_CHANGE_VMS_POWER_STATE, SUSPEND_VMS_TO_MEMORY or press 'c' to cancel: DO_NOT_CHANGE_VMS_POWER_STATE

Do you want to enable Quickboot?  Enter yes, no, or 'c' to cancel: yes

[INFO] Beginning vLCM image transition for Standalone Host "m01-cl01" in Workload Domain "m01-cl01"

...
```

   e) Monitor the status of the transition task.

      If the task fails, try the following steps:

      - Return to the main menu .
      - Select Option 6, `(Optional) Retry incomplete transition tasks..`
      - Wait some time then select Option 9, `(Optional) Show cluster transition status..`
      - If the process succeeds, no further action is required, if it fails, contact Broadcom Support.

6. Disconnect from SDDC Manager and vCenter.

   Close the PowerShell window or enter `Q` to quit.

Transition to vSphere Lifecycle Manager Images Using the PowerShell Script Command-Line Interface

The PowerShell script command-line interface (CLI) is useful for larger environments. The CLI can be used to transition a single resource command-line arguments or multiple resources using JSON input files.

See PowerShell Script CLI Options for Transitioning to vSphere Lifecycle Manager Images for information about all the CLI options.

Attention:

For bulk operations using a JSON file, use a JSON validator to ensure that your file is properly formed before using it as an input.

Attention:

Appending `-Silence` to any command suppresses screen output and any progress indicators. Pertinent data is still sent to the log file and task operation status might still be queried through relevant check commands. This process can be useful in scripting operations.

1. Run the script with `-Version` and compare it to the version listed on Broadcom knowledge base article 385617 to determine if you are running the latest copy of the script.

```
PS C:\Users\Admin>.\VcfBaselineClusterTransition.ps1 -Version
```

2. Connect to SDDC Manager and its vCenter instances.

   You can provide your SDDC Manager credentials interactively or by including the path to a JSON with the credentials.

Command	Description
`.\VcfBaselineClusterTransition.ps1 -Connect`	Enter the following information for SDDC Manager: • FQDN • ADMIN user name • ADMIN password Optionally, you can choose to save these credentials in a JSON file.
`.\VcfBaselineClusterTransition.ps1 -Connect -JsonInput` ***`path_to_file`***	For the *`path_to_file`*.json, enter the location of the file containing the SDDC Manager credentials. For example: `C:\Users\Admin\SddcManagerCredentials.json`

   The script connects to SDDC Manager and all vCenter instances that it manages.

   Note:
   If you see the message `Failed to connect to vCenter`, verify that you can access the vCenter from PowerShell and that it returns a 200 status code.

```
Invoke-WebRequest -Method HEAD -SkipCertificateCheck -Uri https://vCenter_FQDN | Select-Object
 StatusCode
```

   If it does not, verify you can login to that vCenter from a browser. If you cannot, contact Broadcom Support.

3. Identify vSphere clusters and standalone hosts across all workload domains that are managed using vSphere Lifecycle Manager baselines.

Command	Description
`.\VcfBaselineClusterTransition.ps1 -ShowBaselineResources`	Displays a list of baseline resources (clusters and standalone hosts).
`.\VcfBaselineClusterTransition.ps1 -ShowBaselineResources -JsonOutput` ***`path_to_file`***	Displays a list of baseline-managed resources (clusters and standalone hosts) and then saves the output to a file.

```
PS C:\Users\Administrator> .\VcfBaselineClusterTransition.ps1 -ShowBaselineResources

[INFO] Scanning for vLCM baseline (VUM) managed clusters/standalone-hosts...

[INFO] vLCM Baseline (VUM) Managed Clusters/Standalone Hosts:

 Resource Name              Resource Type   vCenter Name               Compliance Status SDDC Manager Image Name
 -------------              -------------   ------------               ----------------- ------------------------
 m01-cl01                   Cluster         m01-vc01.example.com       NOT_CHECKED_YET   N/A
 w01-cl01                   Cluster         w01-vc01.example.com       NOT_CHECKED_YET   N/A
 esxi-1.example.com         Standalone Host w01-vc01.example.com       NOT_CHECKED_YET   N/A
 esxi-2.example.com         Standalone Host w01-vc01.example.com       NOT_CHECKED_YET   N/A
```

4. Display and/or save all the vSphere Lifecycle Manager images in connected vCenter(s).

Command	Description
`.\VcfBaselineClusterTransition.ps1 -ShowImagesInVcenter`	Displays vCenter images on screen.

Command	Description
`.\VcfBaselineClusterTransition.ps1 -ShowImagesInVcenter -JsonOutput path_to_file`	Displays vCenter images on screen and saves the output to a file.
`.\VcfBaselineClusterTransition.ps1 -ShowImagesInVcenter -JsonOutput path_to_file -Silence`	Only saves vCenter image output to a file, and does not display output to the screen.

```
PS C:\Users\Admin> .\VcfBaselineClusterTransition.ps1 -ShowImagesInVcenter

[INFO] Scanning vCenter(s) for vLCM images...

[INFO] vLCM images in attached vCenter(s):

  vCenter Image Name vCenter Name
  ------------------ ------------
  esx-803b-vendor-a  m01-vc01.example.com
  esx-803b-vendor-b  m01-vc01.example.com
```

5. Import vSphere Lifecycle Manager images into SDDC Manager from connected vCenter instances.

Command	Description
`.\VcfBaselineClusterTransition.ps1 -ImportImagesFromVcenter -VcenterImageName image_name- VcenterName vCenter_FQDN`	Import a single image from vCenter and display progress on screen.
`.\VcfBaselineClusterTransition.ps1 -ImportImagesFromVcenter -VcenterImageName image_name -VcenterName vCenterFQDN -Silence`	Import a single image from vCenter and do not display progress on screen.
`\VcfBaselineClusterTransition.ps1 -ImportImagesFromVcenter -JsonInput Path_To_File -VcenterName vCenter_FQDN`	Import multiple images from vCenter from a JSON input file serially and display progress on screen. Example JSON file for bulk import: `[ { "VcenterImageName": "image_name", "VcenterName": "vCenter_FQDN" }, { "VcenterImageName": "another_image_name", "VcenterName": "vCenter_FQDN" } ]`
`.\VcfBaselineClusterTransition.ps1 -ImportImagesFromVcenter -JsonInput path_to_file -VcenterName vCenter_FQDN -Parallel`	Import multiple images from vCenter from a JSON input file in parallel.

Note:

If the script is called without the flag `-Parallel`, it attempts to delete empty vSphere clusters after a successful import. The script does not delete vSphere clusters with hosts or images from the vCenter image catalog.

To check the task status, run the following command:

```
.\VcfBaselineClusterTransition.ps1 -CheckTaskStatus -TaskType SddcManagerImageUpload
```

Note:
If the import task fails with the error message `An image named <image-name> already exists in the SDDC Manager`, you can rename the source image in vCenter or delete the conflicting image from SDDC Manager before retrying. To delete the conflicting image in SDDC Manager, run the following command:

```
.\VcfBaselineClusterTransition.ps1 -DeleteImageFromSddcManager -SddcManagerImageName image_name
```

6. Review the images available in SDDC Manager.

Command	Description
`./VcfBaselineClusterTransition.ps1 -ShowImagesInSddcManager`	Displays a list of all vSphere Lifecycle Manager images in SDDC Manager.
`Get-Content logs/Image_Name.json`	Displays detailed information about a specific vSphere Lifecycle Manager image.

```
PS C:\Users\Admin> .\VcfBaselineClusterTransition.ps1 -ShowImagesInSddcManager

[INFO] vLCM images available in SDDC Manager "sfo-vcf01.sfo.rainpole.io":

Id    SDDC Manager Image Name          Base Image                Components                Addons                          Hardware Support
--    -----------------------          ----------                ----------                ------                          ----------------
1     esx-803b-vendor-a                8.0.3-24280767            N/A                       {Vendor-b addon for Vendor Se.. N/A
2     esx-803b-vendor-b                8.0.3-24280767            N/A                       {Vendor-a addon for Vendor Se.. N/A


PS C:\Users\Admin> Get-Content .\logs\esx-803b-vendor-a.json
{
    "add_on": {
        "name": "VEND-ESXi",
        "version": "803.123456-A02"
    },
    "alternative_images": null,
    "base_image": {
        "version": "8.0.3-0.35.24280767"
    },
    "components": null,
    "hardware_support": null,
    "removed_components": null,
    "solutions": null
}
```

7. **Attention:**

   The compliance check mis-identifies certain non-user managed VIBs and components as non-compliant and in need of removal.

   The underlying files are not missing, but rather the metadata describing the entities is not yet installed. The metadata installation occurs automatically post-vLCM image transition.

   If the compliance check identifies the removal of these entities as the **exclusive trigger for maintenance mode**, the hosts will not **transition into** maintenance mode during the operation, The transition instead can be performed live.

Solution	Type	VIB/Component Name
HA (High Availability)	VIB	vmware-fdm
NSX	Component	nsx-lcp-bundle
Spherelet (vSphere Supervisor)	Component	VMware-Spherelet
vSphere Replication	VIB	vmware-hbr-agent

For more information, see Broadcom knowledge base article 311882.

Check vSphere Lifecycle Manager baseline-managed vSphere clusters for compliance with a vSphere Lifecycle Manager image.

Command	Description
.\VcfBaselineClusterTransition.ps1 -ComplianceCheck -ResourceType [Standalone Host|Cluster] -ResourceType *Resource_Name* -WorkloadDomainName *workload_domain* -SddcManagerImageName *image_name*	Executes a compliance check on a baseline-managed resource (cluster or host) against the vSphere Lifecycle Manager image, displaying the results after the process is complete.
.\VcfBaselineClusterTransition.ps1 -ComplianceCheck -JsonInput *path_to_file*	Performs sequential compliance checks on multiple vSphere clusters and/or standalone hosts against vSphere Lifecycle Manager images using a JSON input file. **Important:** Note: • If your "ResourceName" is a cluster, the "ResourceType" must be defined as a "Cluster". • If your "ResourceName" is a standalone host, the "ResourceType" must be defined as a "Standalone Host". • If "SddcManagerImageName" is not specified, host seeding is assumed. Example JSON input file: `[{ "ResourceName": "standalone_host_name", "ResourceType": "Standalone Host", "WorkloadDomainName": "workload_domain", "SddcManagerImageName" :"image_name", }, { "ResourceName": "vsphere_cluster_name", "ResourceType": "Cluster", "WorkloadDomainName": "workload_domain", "SddcManagerImageName" : "image_name" }]`
.\VcfBaselineClusterTransition.ps1 -ComplianceCheck -JsonInput *path_to_file* -Parallel	Performs parallel compliance checks for multiple baseline-managed clusters and/or standalone hosts against vSphere Lifecycle Manager images using a JSON input file, without displaying results upon completion. When performing compliance checks using the parallel or silent flags, use the following command to verify the status of the operation. `.\VcfBaselineClusterTransition.ps1 -CheckTaskStatus -TaskType ComplianceCheck`

```
PS C:\Users\Administrator> ./VcfBaselineClusterTransition -ComplianceCheck -JsonInput .\BaselineResources.json -Parallel

[INFO] No SDDC Manager image named specified for "esx-1.example.com" in workload domain "w01", but image seeding is supported.
[INFO] No SDDC Manager image named specified for "esx-2.example.com" in workload domain "w01", but image seeding is supported.
[INFO] Auto-generating vLCM image for Standalone Host "esx-1.example.com" in workload domain "w01" and beginning compliance check...
[INFO] Auto-generating vLCM image for Standalone Host "esx-2.example.com" in workload domain "w01" and beginning compliance check...
```

8. For compliance checks performed in parallel or with the silence flag, review the compliance results after the compliance check is complete.

Command	Description
`.\VcfBaselineClusterTransition.ps1 -ReviewComplianceResults -ShowAllResources`	Displays a summary of the compliance findings for all vSphere clusters and standalone hosts.
`.\VcfBaselineClusterTransition.ps1 -ReviewComplianceResults -ShowAllResources -ShowExtendedResults`	Displays a summary and detailed compliance findings for all vSphere and standalone hosts.
`.\VcfBaselineClusterTransition.ps1 -ReviewComplianceResults -ResourceName resource_name -ResourceType [Cluster|Standalone Host] -WorkloadDomainName workload_domain`	Displays a summary of the compliance findings for a specific cluster or standalone host.
`.\VcfBaselineClusterTransition.ps1 -ReviewComplianceResults -ResourceName resource_name -ResourceType [Cluster|Standalone Host] -WorkloadDomainName workload_domain -ShowExtendedResults`	Displays a summary and detailed compliance findings for a specific cluster or standalone host.

9. Review the compliance findings to determine if a vSphere cluster is ready to be transitioned.

   Resolve any issues for vSphere clusters or standalone hosts with a status of `INCOMPATIBLE` or `UNKNOWN`.

NON_COMPLIANT	The vSphere cluster or standalone host is not currently compliant with the selected vSphere Lifecycle Manager image, but is ready to be transitioned.
INCOMPATIBLE	The vSphere cluster or standalone host cannot be transitioned to the selected vSphere Lifecycle Manager image without taking extra steps. Review the summary and JSON file and resolve any issues that are blocking the transition. Contact Broadcom Support if you are unable to resolve one or more issues.
UNKNOWN	Compatibility could not be determined. Make sure that all the hosts in the vSphere cluster, or the standalone host, or the is/are healthy and connected to vCenter. Contact Broadcom Support if you are unable to resolve the issue.
COMPLIANT	All hosts are confirmed to be compliant the vLCM image.

10. Transition `NON_COMPLIANT` (ready to transition) clusters to vSphere Lifecycle Manager images.

Command	Description
`.\VcfBaselineClusterTransition.ps1 -CreateHostRemediationOptionsFile -JsonOutput path_to_file`	(Optional) Create an override host remediation option setting for the transition process.
`./VcfBaselineClusterTransition.ps1 -TransitionResource -ResourceName resource_name -ResourceType [Cluster|Standalone Host] -WorkloadDomainName workload_Name`	Transitions a single cluster or standalone host to use a Lifecycle Manager image and displays progress.
`./VcfBaselineClusterTransition.ps1 -TransitionResource -ResourceName resource_name -ResourceType [Cluster|Standalone Host] -WorkloadDomainName workload_domain -HostRemediationOptionsFile path_to_file`	Transition a cluster or standalone host to a Lifecycle Manager image using a custom remediation file. Progress is displayed during the process.
`./VcfBaselineClusterTransition.ps1 -TransitionResources -JsonInput path_to_file`	Transitions multiple vSphere clusters sequentially to use a vSphere Lifecycle Manager image and displays progress. Example JSON input file: `[`

Command	Description

	<pre>{
	"ResourceName": "*vSphere_cluster_name*",
	"ResourceType": "*Cluster*",
	"WorkloadDomainName": "*workload_domain*",
	},
	{
	"ResourceName": "*standalone host name*",
	"ResourceType": "*Standalone Host*",
	"WorkloadDomainName": "*workload_domain*"
	}
	]</pre>

./VcfBaselineClusterTransition.ps1 -TransitionResources -JsonInput path_to_file -Parallel	Transitions multiple vSphere clusters and/or standalone hosts in parallel to use a vSphere Lifecycle Manager image and does not display progress.

You can run compliance checks in parallel, but there is a concurrent task limit of 25 compliance and transition tasks. If you exceed this limit, the task is rejected, rather than queued.

Important:

The system transitions the cluster or standalone hosts using the last image that you checked against. If you run the compliance check multiple times against multiple images, run `./VcfBaselineClusterTransition.ps1 -ShowBaselineResources` to review the final association before proceeding.

CAUTION:

After you start a compliance or transition task, it cannot be paused or stopped. The task continues to run on the SDDC Manager even if the PowerShell window is closed.

a) Run one of the transition commands listed in the preceding table.

Example of a standalone host transition with a host remediation file.

```
PS C:\Users\Administrator> .\VcfBaselineClusterTransition.ps1 -TransitionResource -ResourceName esx-1.example.com
-ResourceType "Standalone Host" -WorkloadDomainName "m01" -HostRemediationOptionsFile HostRemediationFile.json

[INFO] No errors detected in HostRemediationOptionsConfig file .

[INFO] Preparing for cluster(s)/standalone-host(s) transition(s)...

[INFO] Beginning vLCM image transition for Standalone Host "esx-1.example.com" in Workload Domain "m01".
[INFO] Standalone Host "esx-1.example.com" in Workload Domain "m01" has been successfully converted to vLCM image
management.

[INFO] Full details of the transition written to C:\Users\Administrator\logs\VcfBaselineClusterTransition-02-03-20
26.log.
```

b) For parallel and silenced operations, check the status of the transition.

```
.\VcfBaselineClusterTransition.ps1 -CheckTransitions
```

11. If the transition task fails, try the following steps.
    a) Identify which vSphere cluster and/or standalone host has failed to transition to vLCM image management.

    ```
    .\VcfBaselineClusterTransition.ps1 -CheckTransitions
    ```

    b) Retrieve instructions on how to retry the transition.

    ```
    .\VcfBaselineClusterTransition.ps1 -CheckTransitions -ResourceName resource_name -
    ResourceType [Cluster|Standalone Host] -WorkloadDomain workload_domain
    ```

    c) Resume the failed task using the instructions.

    ```
    .\VcfBaselineClusterTransition.ps1 -RetryTransition -TaskId task_id
    ```

d) Run the check transitions command to see if the vSphere cluster and/or standalone host transition is `In Progress`.

```
.\VcfBaselineClusterTransition.ps1 -CheckTransitions
```

e) Periodically re-run `-CheckTransitions` to see if the transition completes successfully or it fails. If it fails again, contact Broadcom Support.

12. Disconnect from the SDDC Manager and all vCenter instances.

```
./VcfBaselineClusterTransition.ps1 -Disconnect
```

PowerShell Script CLI Options for Transitioning to vSphere Lifecycle Manager Images

This list covers the CLI options for the PowerShell script that transitions clusters or standalone hosts from baselines to vSphere Lifecycle Manager images.

You can run the script with the `-Help` argument to see these options.

```
PS C:\Users\Admin>.\VcfBaselineClusterTransition.ps1 -Help
```

For examples of how to use these options, see Transition to vSphere Lifecycle Manager Images Using the PowerShell Script Command-Line Interface.

Parameter	Argument	Importance
-CheckTaskStatus	-TaskType SddcManagerImageUpload or -TaskType ComplianceCheck	Required
	-Silence	Optional
-CheckTransitions	-ResourceName *resource_name* -ResourceType [`Cluster|Standalone Host`] -WorkloadDomainName *workload_domain*	Optional
	-Silence	Optional
-Connect	-JsonInput *path_to_file*	Optional
-ComplianceCheck	-ResourceName *resource_name* -ResourceType [`Cluster|Standalone Host`] -SddcManagerImageName *image_name* -WorkloadDomainName *workload_domain*	You must provide either all of the individual arguments or use the -JsonInput argument.
	-JsonInput *path_to_file*	
	-Parallel	Optional
	-Silence	Optional
-DeleteImageFromSddcManager	-SddcManagerImageName *image_name*	Required
	-Silence	Optional
-DeleteTemporaryCluster	-ResourceName *resource_name* -ResourceType [`Cluster|Standalone Host`] -VcenterName *vCenter_FQDN*	Required
	-Silence	Optional
-Disconnect	-Silence	Optional
-Help		N/A
-ImportImagesFromVcenter	-VcenterImageName *image_name* -VcenterName *vCenter_FQDN*	You must provide either all of the individual arguments or use the -JsonInput argument.
	-JsonInput *path_to_file*	
	-Parallel	Optional
	-Silence	Optional

Parameter	Argument	Importance
-ReviewComplianceResults	-ResourceName *resource_name* -ResourceType [Cluster|Standalone Host] -WorkloadDomainName *workload_domain*	Optional
	-ShowExtendedResults	Optional
	-ShowAllResources	Optional
	-Silence	Optional
-RetryTransition	-TaskId <Task Id>	Required
-ShowBaselineResources	-JsonOutput *path_to_file*	Optional
	-Silence	Optional
-ShowImagesInSddcManager	-Silence	Optional
-ShowImagesInVcenter	-JsonOutput *path_to_file*	Optional
	-Silence	Optional
-TransitionResource	-ResourceName *resource_name* -ResourceType [Cluster|Standalone Host] -WorkloadDomainName *workload_domain*	You must provide either all of the individual arguments or use the -JsonInput argument.
	-JsonInput *path_to_file*	
	-Parallel	Optional
	-Silence	Optional
-Version		N/A

Transition to vSphere Lifecycle Manager Images Using the SDDC Manager API

If you do not want to use the PowerShell script to transition vSphere clusters to vSphere Lifecycle manager images, you can use the SDDC Manager API instead.

The following procedure describes how to use the SDDC Manager Developer Center to transition a single vSphere cluster or a set of standalone hosts to vSphere Lifecycle Manager Images.

CAUTION:

The API workflow does not contain the same guardrails and error handling as the PowerShell script. Using the SDDC Manager API is only recommended if the PowerShell script does not meet your business requirements.

Cluster API Workflow

1. Retrieve the names and IDs of vSphere clusters that are managed using vSphere Lifecycle Manager baselines.
   a) Log in to the SDDC Manager UI.
   b) Click **Developer Center** > **API Explorer**.
   c) Expand **Clusters** and click **GET /v1/clusters**.
   d) Enter the value `false` for the `isImageBased` parameter and click **Execute**.
   e) Click **Download** to save the results.

      You can rename the saved file using a more meaningful name.
   f) Review the file to identify the vSphere cluster ID associated with each vSphere cluster name.

      Example:

```
{
    "elements": [
        {
            "id": "7b0fadd9-c6dd-46d6-ab12-4a2cce93bb59",
            "domain": {
                "id": "108355fd-e758-4b19-ac93-1c5efc6fd7df"
            },
            "name": "m01-cl01",
            "status": "ACTIVE",
            "primaryDatastoreName": "m01-m01-cl01-vsan01",
            "primaryDatastoreType": "VSAN",
            "hosts": [
                {
                    "id": "a392202a-1c3a-491e-96ea-b3175a4d10b8"
                },
                {
                    "id": "f3f180cc-69cd-43a9-8c27-ed9018d41061"
                },
                {
                    "id": "fff2e3aa-2322-4a25-9b85-2bfd9bcb4228"
                }
            ],
            "isStretched": false,
            "isDefault": true,
            "failuresToTolerate": 1,
            "isImageBased": false,
            "capacity": {
…
```


2. Import vSphere Lifecycle Manager images from vCenter into SDDC Manager using the SDDC Manager UI.
   a) In the navigation pane, click **Lifecycle Management** > **Image Management** .
   b) Click **Import Image**.
   c) Select **Import from a vCenter**.
   d) From the **vCenter** drop-down menu, select the vCenter containing your vSphere Lifecycle Manager image.

      Click `Refresh` to see the current list of images or vSphere clusters.
   e) Select the image (vCenter 9.0 and later) or vSphere cluster (vCenter 8.0.3 and below).

      For vCenter 8.0.3 and below, enter a name for your image.
   f) Click **Import**.

      You can monitor the status in the **Tasks** panel.
   g) Repeat the process for each unique image you want to import.
3. Check a vSphere Lifecycle Manager baseline-managed vSphere cluster for compliance with a vSphere Lifecycle Manager image.

   You can run a vSphere cluster compliance check multiple times, but only the latest check (and associated image) is used for transitioning the vSphere cluster.

Attention:

The compliance check mis-identifies certain non-user managed VIBs and components as non-compliant and in need of removal.

The underlying files are not missing, but rather the metadata describing the entities is not yet installed. The metadata installation occurs automatically post-vLCM image transition.

If the compliance check identifies the removal of these entities as the **exclusive trigger for maintenance mode**, the hosts will not **transition into** maintenance mode during the operation, The transition instead can be performed live.

Solution	Type	VIB/Component Name
HA (High Availability)	VIB	vmware-fdm
NSX	Component	nsx-lcp-bundle
Spherelet (vSphere Supervisor)	Component	VMware-Spherelet
vSphere Replication	VIB	vmware-hbr-agent

For more information, see Broadcom knowledge base article 311882.

a) Click **Lifecycle Management** > **Image Management**.
b) Identify the name of the vSphere Lifecycle Manager image that you want to use for the compliance check.
c) Click **Developer Center** > **API Explorer**.
d) Expand **Personalities** and click **GET /v1/personalities**.
e) Enter the name of the image for the `personalityName` parameter and click **Execute**.
f) Retrieve the `personalityId` from the response.
g) Expand **Clusters** and click **PATCH /v1/clusters/{id}**.
h) Enter the vSphere cluster ID for the `id` parameter.
i) Enter the following information for the `body` parameter.

Use the personality ID that you retrieved for the `clusterImageId` value.

```
{
"clusterImageComplianceCheckSpec": { "clusterImageId": "personalityId" }
}
```

j) Click `Execute`.

4. Review the compliance findings to determine if a cluster is ready to be transitioned.

Resolve any issues for clusters with a status of `INCOMPATIBLE` or `UNKNOWN`.

Status	Description
`NON_COMPLIANT`	The vSphere cluster is not currently compliant with the selected vSphere Lifecycle Manager image, but is ready to be transitioned.
`INCOMPATIBLE`	The vSphere cluster cannot be transitioned to the selected vSphere Lifecycle Manager image without taking additional steps. Review the summary and JSON file and resolve any issues that are blocking the transition. Contact Broadcom Support if you are unable to resolve the issue(s).
`UNKNOWN`	Compatibility could not be determined. Make sure that all the hosts in the vSphere cluster are available. Contact Broadcom Support if you are unable to resolve the issue.
`COMPLIANT`	All hosts in the vSphere cluster confirm to the vLCM image.

a) Expand **Clusters** and click **GET /v1/clusters/{id}/image-compliance**
b) Enter the vSphere cluster ID for the `id` parameter.
c) Click `Execute`.
d) Click `Download` to save the results.
e) Review the results using the tool of your choice.

5. Transition `NON_COMPLIANT` (ready to transition) clusters to vSphere Lifecycle Manager images.
   a) Expand **Clusters** and click **PATCH /v1/clusters/{id}**.
   b) Enter the vSphere cluster ID for the `id` parameter.
   c) Enter the following information for the `body` parameter.

```
{ "clusterTransitionSpec": { } }
```

Optionally, you can specify more options:

```
{
"clusterTransitionSpec":
{
 "remediationOptionsSpec":
 {
  "disableDpm": false,
  "disableHac": false,
  "enableQuickboot": true,
  "evacuateOfflineVms": true,
  "preRemediationPowerAction": "DO_NOT_CHANGE_VMS_POWER_STATE",         "remediationFailureAction":
  {
   "action": "RETRY",
   "retryCount": 3,
   "retryDelay": 1200
  }
 }
}
```

6. Monitor the task in the **Tasks** panel.

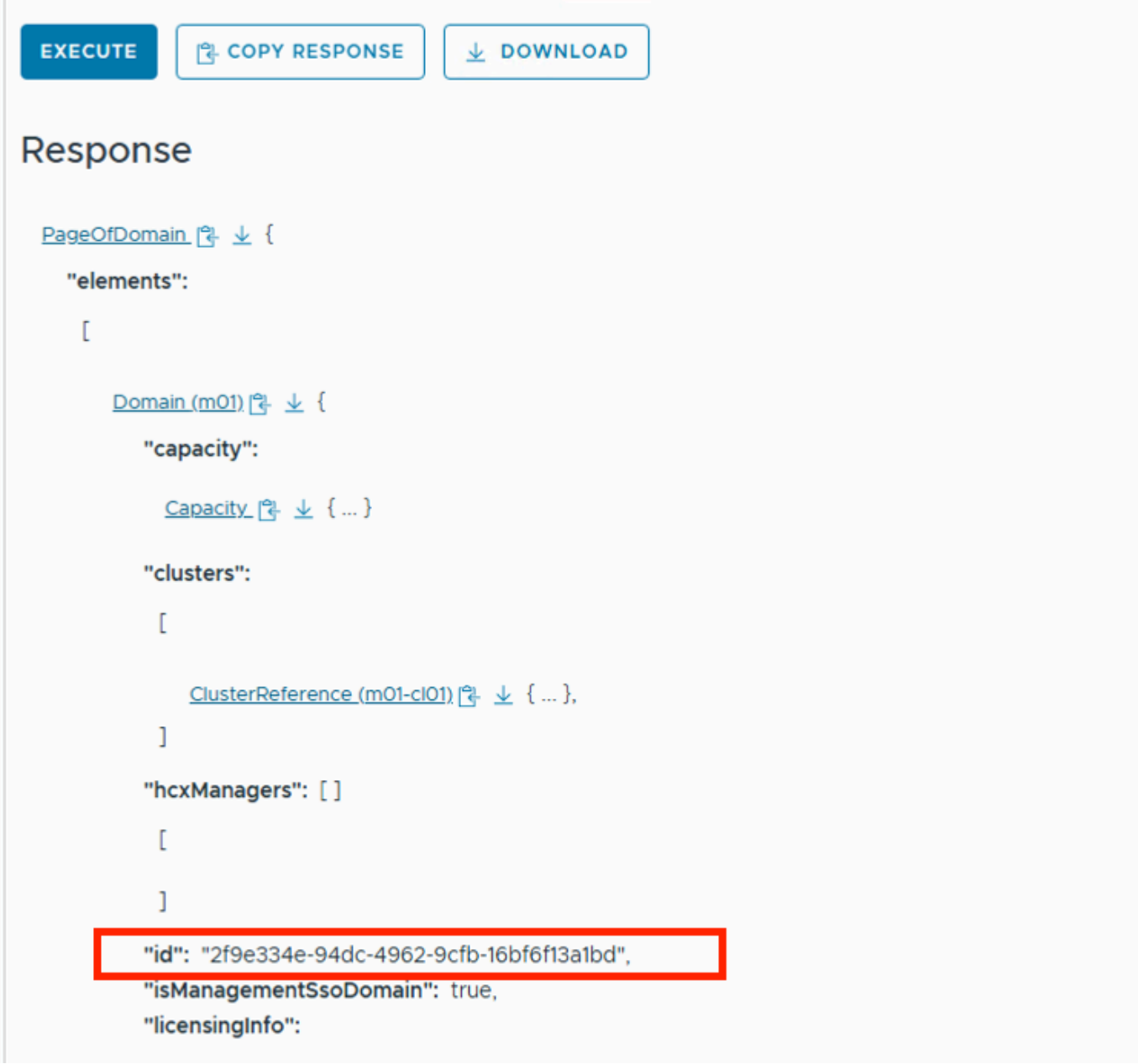
   If the task completes successfully, you can repeat these steps for any additional vSphere clusters you want to transition.

   If the task fails, resolve the issues and try again.

Standalone Hosts API Workflow

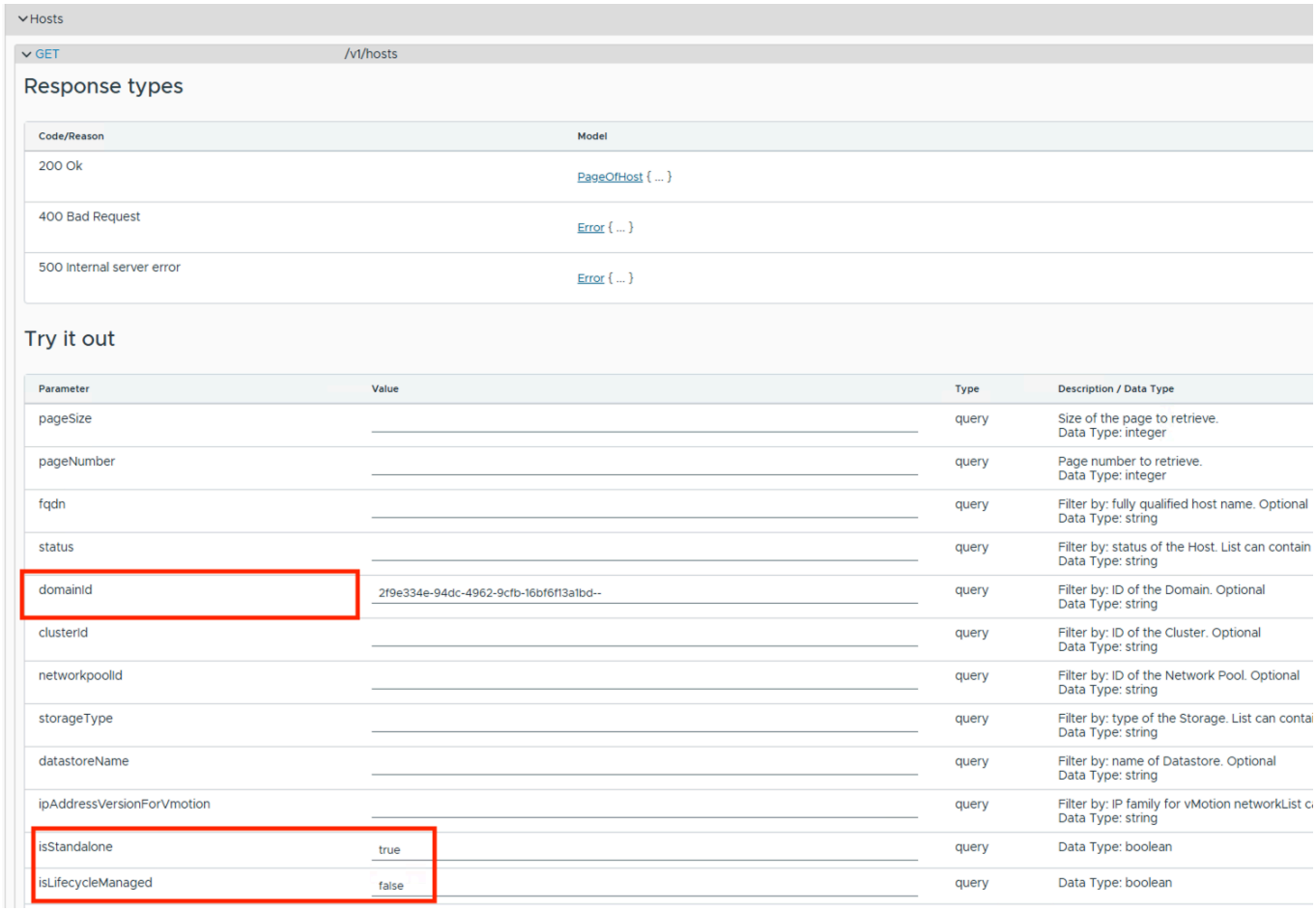
1. Access Developer Center.
2. a) Log in to the SDDC Manager UI.
   b) **Developer Center** > **API Explorer**.
3. Lookup Domain ID of Workload Domain containing Standalone Hosts.
   a) Expand **Domains**.
   b) Expand **Domains** and click **GET /v1/domains**.
   c) Enter the name of the workload domain, in the **name** field, and click **Execute**.
   d) Under **Response** > **PageOfDomain** > **elements**Click on the name of the workload domain.
   e) Copy down the **ID** at the first sub-level of your domain.

   Example Response



4. Retrieving the list of vLCM baseline-managed standalone hosts in a workload domain.
   a) Expand **Hosts** and click **GET /v1/hosts**.
   b) Enter **ID** from step 3e into the `domainId` field.
   c) Enter `true` in the **isStandalone** field.
   d) Enter `false` in the **isStandalone** field.
   e) Click **Execute**.

   Querying standalone hosts



f) Copy and paste the top level of the `elements` stanza.

Example Response



g) Save this list of host IDs for a future step.

5. (Optional) Get ID of vLCM Images.

   This step is only required if you wish to specify a vLCM image rather than using image seeding.

   a) Expand **Personalities** and click **GET /v1/personalities**.
   b) Click **Execute**.
   c) Under **Response** > **PageOfDomain** > **elements** Click on **Personality** to expand it.
   d) Locate the *`personalityId`* matching the *`personalityName`* of the image that you wish to use.
   e) Save that Id for the next step.

6. Run Compliance Check
   a) Expand **Domains** and click **GET /v1/domains/{id}**.
   b) Enter the domain ID from step 3e into the `id (required)` field.
   c) Copy and past the following code snippet into the `body` section. Substitute the IDs of each standalone host from step 4f you wish to transition into the ID field for each resource. Add or subtract resource stanzas as necessary to accommodate your hosts, up to the maximum payload limit of 25 standalone hosts. To use image seeding, choose option 1. To use a specified image, choose option 2 with the `personalityID` from step 5e.

      Option 1: Code for Image Seeding

```
{
```

```
  "imageComplianceCheckSpec": {
    "resources": [
      {
        "id": "<HOST ID 1>",
        "type": "HOST"
      },
      {
        "id": "<HOST ID> 2",
        "type": "HOST"
      }
    ]
  }
}
```

Option 2: Code for Specified Image:

```
{
  "imageComplianceCheckSpec": {
    "imageId": "<peronality ID>",
    "resources": [
      {
        "id": "<HOST ID 1>",
        "type": "HOST"
      },
      {
        "id": "<HOST ID 2>",
        "type": "HOST"
      }
    ]
  }
}
```

a) Click **Execute**.
b) In the `Response`, record the value of the *`id`*.

7. Look up the query ID for your compliance findings.
   a) Within **Domains** click **POST /v1/domains/{id}/image-compliance/queries**
   b) Enter the *`id`* from step 3e in the *`domainid`* field.
   c) In the `body`, copy and paste the following code snippet (substituting the IDs of each standalone hosts from step 6c).

```
{
  "resources": [
    {
      "id": "<Host ID 1",
      "type": "HOST"
    },
    {
      "id": "<HOST ID 2>",
      "type": "HOST"
    }
  ]
}
```

d) Click **Execute**.
e) In the `Response`, expand `queryInfo`, and record the value of the *`queryId`*.

8. Review compliance findings.
   a) Within **Domains** click **GET /v1/domains/{id}/image-compliance/queries**.
   b) Under `domainId (required)` enter your workload domain from step 3e.
   c) Under `queryid (required)` enter your query id from step 7e.
   d) Click **Execute**.
   e) Review in the browser (or **Download** and review offline) the compliance findings for your standalone hosts.
   f) Review the compliance findings to determine if the vSphere standalone hosts are ready to be transitioned.

      Resolve any issues for vSphere clusters or standalone hosts with a status of `INCOMPATIBLE` or `UNKNOWN`.

`NON_COMPLIANT`	The vSphere standalone host is not currently compliant with the selected vSphere Lifecycle Manager image, but is ready to be transitioned.
`INCOMPATIBLE`	The vSphere standalone host cannot be transitioned to the selected vSphere Lifecycle Manager image without taking extra steps. Review the summary and JSON file and resolve any issues that are blocking the transition. Contact Broadcom Support if you are unable to resolve one or more issues.
`UNKNOWN`	Compatibility could not be determined. Make sure the standalone host is healthy and connected to vCenter. Contact Broadcom Support if you are unable to resolve the issue.
`COMPLIANT`	The hosts is confirmed to be compliant the vLCM image.

9. Run vLCM baseline to vLCM images transition.
   a) Within **Domains** click **PATCH /v1/domains/{id}**.
   b) Enter the *`id`* from step 3e in the *`domainid`* field.
   c) In the `body` field, copy and paste the following code snippet (substituting the IDs of each standalone hosts from step 6c).

```
{
  "transitionSpec": {
    "resources": [
      {
        "id": "<Host ID 1>",
        "type": "HOST"
      },
      {
        "id": "<Host ID 2>",
        "type": "HOST"
      }
    ]
  }
}
```

10.

Troubleshooting API Error Messages

Review the following common error messages that you might see when transitioning vSphere clusters from vSphere Lifecycle Manager baselines to vSphere Lifecycle Manager images. Refer to the remediation information for the next steps.

The underlying cause of a failed step might be within an asynchronous task or the original API call.

Error Message	Possible cause	Remediation
Personality *<Image Name>* with same specification already exists	A vSphere Lifecycle Manager image with the exact same parameters already exists in SDDC Manager. You cannot import duplicate images, even if they have different names.	No action required. Use the existing image in SDDC Manager to transition the vSphere cluster(s).
Failed to import the image. Personality with the specified name/id already exists.	A vSphere Lifecycle Manager image with the same name already exists in SDDC Manager.	For vCenter 8.0.3 and below, repeat the workflow with a new name. For vCenter 9.0 and above, rename the source image and repeat the workflow.
Highest ESX version *<Version>* in cluster: <Cluster Name> does not match version: *<Version>* in cluster image: *<Image Name>*	The vSphere Lifecycle Manager base image does not match the ESX version running on the hosts in the vSphere cluster.	Update the vSphere Lifecycle Manager image to match the ESX version on the hosts.

Plan a Domain Upgrade

To upgrade the core components in your domain, plan a domain upgrade by setting a target VCF version and VCF component versions, and defining the upgrade sequence for domain components.

Starting with VCF 9.1, vCenter and NSX Manager are upgraded in a common step in the upgrade process. You can select between the following upgrade scenarios.

- Optimized maintenance window upgrade
  - Reduces the required time for the maintenance window for the upgrade. The vCenter reduced downtime upgrade preparation stage is performed first and does not require downtime. After this stage is completed, within the same maintenance window, the NSX Manager nodes are upgraded followed by the vCenter instance switchover stage.
- Sequential upgrade
  - Upgrade operations for each component are performed in separate stages in different maintenance windows. The upgrade starts with the NSX Manager nodes upgrade followed by the vCenter instance upgrade. You can select between a vCenter in place upgrade or vCenter reduced downtime upgrade.

    Note:

    If you are upgrading from version 9.0 or later, both upgrade options are available. If you are upgrading from version 5.2.x, the only available option is vCenter reduced downtime upgrade.

Depending on your choice, you upgrade the remaining core components in one of the following sequences.

Optimized Maintenance Window Upgrade to 9.1	Sequential Upgrade to 9.1
1. Prepare the vCenter instance for upgrade 2. Upgrade the NSX Manager 3. Switchover of the vCenter instance 4. Upgrade the ESX hosts 5. Upgrade the NSX Edge Cluster and finalize NSX Upgrade	1. Upgrade the NSX Manager 2. Upgrade the vCenter instance – Option 1: vCenter in place upgrade – Option 2: Preparation and switchover 3. Upgrade the ESX hosts 4. Upgrade the NSX Edge Cluster and finalize NSX Upgrade

In both upgrade scenarios, you must complete the upgrade plan steps in a sequential order.

Note:

To create an optimized maintenance window upgrade, your vCenter target version and SDDC Manager current version must be the same.

1. Log in to VCF Operations at `https://<vcf_operations_fqdn>` with a user assigned the **Administrator** role.
2. Select **Build** > **Lifecycle**, expand **VCF Instances**, and select your domain.
3. On the **Upgrades** tab, click **Plan Domain Upgrade.**

   The **Plan Domain Upgrade** dialog box appears.
4. On the **Target Version** page, from the drop-down menu, select a VCF target version and click **Next**.
5. Optional: On the **Customize Target Version** page, click **Customize Upgrade**, select a custom target version for one or more components, and click **Next**.
6. On the **vCenter and NSX Upgrade Options** page, select an upgrade scenario and click **Next**.
7. On the **Plan Overview** page, review the upgrade plan and click **Submit Plan**.

Perform Upgrade Prechecks

To ensure that your environment is ready for an upgrade, before applying an upgrade, you can perform prechecks in VCF Operations.

You can choose which pre-checks to run and either plan pre-check for all components or choose one or more components. You can run general prechecks or prechecks against a specific VCF target version. Alternatively, after you configure a component for upgrade, you can run a component precheck from the dedicated component card.

- After you start a precheck of a component, you cannot cancel the precheck and it is not recommended to start or schedule an upgrade.
- After a precheck completes, only the latest results are saved. Previous results are not visible.
- Even when one or more precheck fails, you can attempt to upgrade the component. The recommendation is to review the errors and warnings, before attempting the upgrade of the component.

If you silence a vSAN Skyline Health alert in the vSphere Client, SDDC Manager skips the related precheck and indicates which precheck it skipped. Click **RESTORE PRECHECK** to include the silenced precheck. Silenced prechecks do not send warnings or block upgrades.

1. Log in to VCF Operations at `https://<vcf_operations_fqdn>` with a user assigned the **Administrator** role.
2. Select **Build** > **Lifecycle**, expand **VCF Instances**, and select your domain.
3. On the **Upgrades** tab, under **Prechecks**, click **Run Prechecks**.
4. Select precheck option for the components in the domain and click **Run Prechecks**.

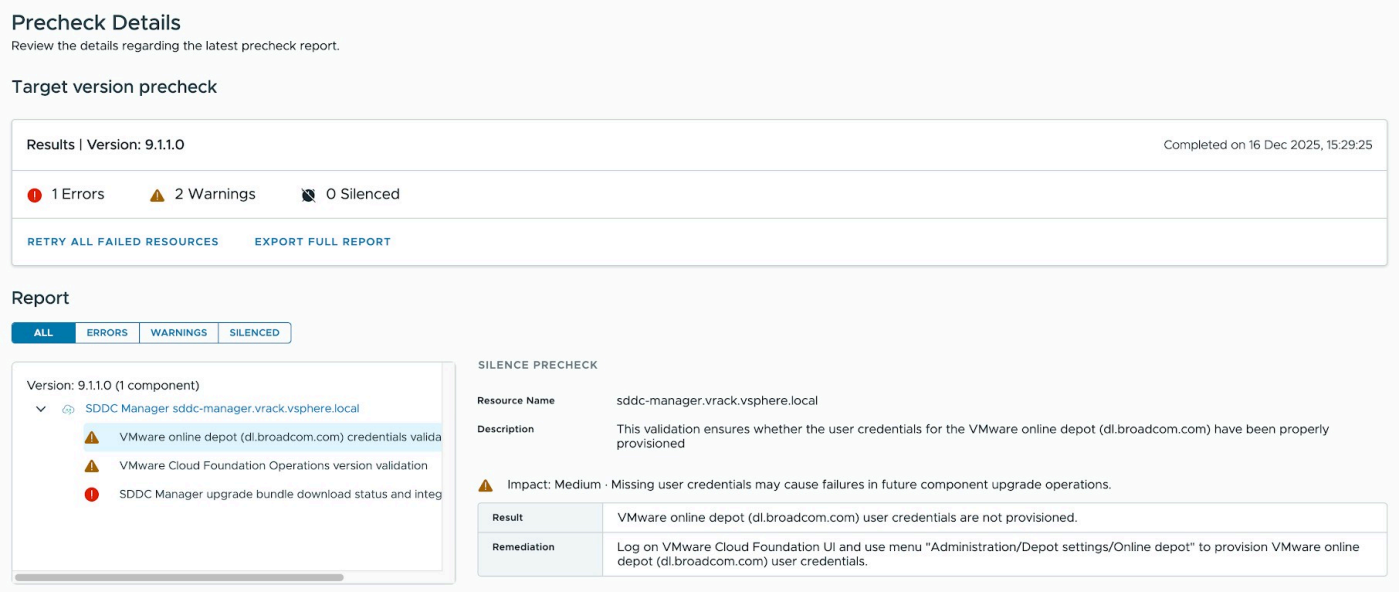
   You can select **Target version** and run prechecks for the selected version. Alternatively, if you are planning an upgrade to future release of VCF, you can select **Upgrade readiness**. All components in the domain are selected by default.

Target version precheck	Upgrade readiness
• Select a target version. • Select precheck scope. – You can select all or specific components from the list.	• Select precheck scope. – You can select all or specific components from the list.

   If there are pending upgrade binaries available, then the **Target version** drop-down contains the available VCF versions to upgrade to. If there is an available VCF upgrade version, there will be extra checks: bundle-level prechecks for hosts, vCenter, and so on. The version-specific prechecks run only prechecks on components that have available upgrade binaries downloaded.
5. After the prechecks complete, under **Prechecks**, click **View Details**.
6. To browse through the results, click through **All**, **Errors**, **Warnings**, and **Silenced** to filter the results.

   Only **Errors**, **Warnings**, **Silenced** results are displayed for **Target version** and **Upgrade readiness** prechecks. The threshold value is 500. To change the value, see Broadcom knowledge base article 413530.

To see all prechecks and precheck details, select **Export Full Report**. The exported report is in `CSV` file format and includes all precheck results.



7. To see details for a task, click the expander arrow.

   If a precheck task failed, you can fix the issue, and click **Retry** to run the task again. You can also click **Retry All Failed Resources** to retry all failed tasks.

   - The **Result** field describes the issue.
   - The **Remediation** field provides information how to resolve an issue.
   - The **Impact** field describes the severity and specfic impact that might happen if the issue is not resolved.

8. Optional: If the domain contains a host that includes pinned VMs, the precheck fails at the Enter Maintenance Mode step. If the host can enter maintenance mode through the vCenter UI, you can suppress this check for NSX and ESX in VCF by following the steps below.

   1. Log in to SDDC Manager by using a Secure Shell (SSH) client with the user name vcf and password.
   2. Open the `/opt/vmware/vcf/lcm/lcm-app/conf/application-prod.properties` file.
   3. Add the following line to the end of the file:

      ```
      lcm.nsxt.suppress.dry.run.emm.check=true
      lcm.esx.suppress.dry.run.emm.check.failures=true
      ```

   4. Restart the lcm service by typing the following command in the console window.

      ```
      systemctl restart lcm
      ```

   5. After the lcm service is restarted, run the precheck again.

Upgrading vCenter and NSX Manager

Starting with VCF 9.1, you upgrade your NSX Manager cluster and NSX Edge clusters separately. Depending on the upgrade option you select in the domain upgrade plan, you proceed with updating your vCenter instance and NSX Manager instances in one of the following ways: Perform an Optimized Upgrade of vCenter and NSX Manager or Perform a Sequential Upgrade of vCenter and NSX Manager. The NSX Upgrade Coordinator is upgraded as part of these workflows.

Checklist for NSX Upgrade by Using VCF Operations

Use the following checklist to track your work through the automated upgrade process.

Automated Upgrade Task	Instructions
Evaluate the operational impact of the upgrade.	See Operational Impact of the NSX Upgrade.
Complete the pre-upgrade tasks.	See Pre-Upgrade Tasks for Upgrading NSX.
Verify that the NSX environment is in a healthy state.	See Verify the Current State of NSX.
Make the required changes in your system configuration if your deployment contains features that are not supported in the target release.	See Manage Unsupported Features.

Shared NSX Instances

For NSX instances that are shared across multiple domains, you must upgrade the NSX, vCenter, and ESX components on all domains, where the NSX instance is shared. It is not recommended to cancel the automatically created upgrade plan on a dependent domain, as this action cancels the upgrade plan on all domains that share the NSX instance. If the upgrade plan is cancelled, you must create a new upgrade plan.

For example, if you create a domain upgrade plan or shared NSX component upgrade plan for domain A, you must add and upgrade the vCenter instance and all ESX hosts on this domain. Before you can finalize the upgrade, you must visit domain B, add the vCenter and ESX components to the component upgrade plan, and upgrade all three components. After you upgrade the vCenter, ESX, and NSX components on domain B and all other dependent domains, you can finalize the upgrade on domain A and the dependent domains.

If an NSX instance is shared between a management domain and workload domain(s), when all other components are upgraded, you must start the **VMware NSX Edge Clusters Upgrade and NSX Upgrade Finalize** upgrade step from the management domain only.

> **Note:**
>
> When you select the **Optimized Upgrade** sequence, the NSX Manager upgrade cannot proceed until every domain that shares the NSX instance completes the vCenter Reduced Downtime Upgrade (RDU) switchover.

Upgrading Embedded Identity Broker

For environments with embedded VCF Identity Broker 9.0.x, as the component is embedded in the management vCenter instance, the upgrade to embedded identity broker 9.1 is automated during the upgrade of the vCenter instance.

After the embedded identity broker is at version 9.1, it can be migrated to an instance that is part of VCF management services for high availability. See Migration of Identity Broker Embedded to Identity Broker Instance.

Checklist for Upgrading NSX

Use the following checklist to track your work through the NSX upgrade.

Upgrade Task	Instructions
Evaluate the operational impact of the upgrade.	See Operational Impact of the NSX Upgrade.
Complete the pre-upgrade tasks.	See Pre-Upgrade Tasks for Upgrading NSX.
Verify that the NSX environment is in a healthy state.	See Verify the Current State of NSX.
Make the required changes in your system configuration if your deployment contains features that are not supported in the target release.	See Manage Unsupported Features.
Verify the upgrade.	See Verify the NSX Upgrade.

Operational Impact of the NSX Upgrade

This documentation page discusses the operational state of NSX components during an upgrade.

Management Plane Upgrade

During Upgrade	After Upgrade
• If you make configuration changes during the Management plane upgrade, there might be a delay before the changes take effect. • The NSX Manager UI is unavailable for a short period. • When upgrading from NSX 4.x, API service is momentarily unavailable. When upgrading from VCF 9.x, API downtime is negligible.	• You cannot configure new features introduced by the upgrade until the **VMware NSX Edge Clusters Upgrade and NSX Upgrade Finalize** upgrade step is completed. • Perform configuration tasks only after the upgrade process is complete.

NSX Edge Cluster Upgrade

During Upgrade	After Upgrade
• During the edge upgrade, you might experience the following traffic interruptions: – North-south datapath is affected if the edge is part of the datapath. – East-west traffic between tier-1 routers using edge firewall, NAT, or load balancing. – Temporary Layer 2 and Layer 3 interruption. • Configuration changes are not blocked on NSX Manager but might be delayed. • The hardware versions of edge node VMs change or remain the same as follows: – For edge nodes auto-deployed through the NSX Manager UI or the API, the VM hardware is automatically updated to the maximum version supported by the underlying ESX host. – For edge nodes manually deployed through vSphere Client, the VM hardware remains unchanged at the same version. For more details, see Broadcom knowledge base article 319091.	• Configuration changes are allowed. • You cannot configure new features introduced by the upgrade until the **VMware NSX Edge Clusters Upgrade and NSX Upgrade Finalize** upgrade step is completed. • Run post checks to make sure that the upgraded edge cluster and NSX do not have any problems.

In-Place Upgrades

In-place upgrades are integrated with the live patch process in VMware Cloud Foundation lifecycle management. When upgrading to an NSX hot patch or express patch, VMware Cloud Foundation initiates a live patch when possible, and initiates the in-place upgrade.

In-place upgrades are not supported in the following scenarios:

- More than 1000 vNICs are configured on the ESX host and the vNICs on the VMs connect to a single VDS. If the host has multiple VDS for NSX, this vNIC limit is per VDS.
- Layer 7 firewall rules or Identity Firewall rules are enabled.
- A VProbe-based packet capture is in progress.
- The `nsx-cfgagent` service is not running on the host.
- IDS/IPS or distributed malware prevention is enabled for your NSX environment.

Pre-Upgrade Tasks for Upgrading NSX

Before you upgrade NSX, perform the pre-upgrade tasks to ensure that the upgrade is successful.

1. Ensure that your transport node profiles include the appropriate transport zones. Otherwise, NSX Manager might not display the list of transport node profiles.
2. Run the Certificate Analyzer, Results and Recovery (CARR) script. The goal of running the CARR script is to ensure that NSX certificates, such as transport node (TN) certificates, do not expire within 90 days. For more information, see Broadcom knowledge base article 369034.
3. Ensure that you back up the NSX Manager before you start the upgrade process. See Backing Up and Restoring NSX Manager or Global Manager.
4. Ensure that your host OS is supported for NSX Manager. For more information, see the Broadcom Compatibility Guide.
5. Turn off automatic backups before you start the upgrade process. For more information, see Configure Backups.
6. Terminate any active SSH sessions or local shell scripts that you might have invoked on the NSX Manager or the NSX edge nodes, before you begin the upgrade process.
7. Ensure that the appropriate communication ports are open from the host transport nodes and edge nodes to the NSX Manager nodes. For more information on ports, see VMware Ports and Protocols.
8. Delete all expired user accounts before you begin the upgrade. Upgrade for NSX on vSphere fails when your exception list for vSphere lockdown mode includes expired user accounts. For more information on accounts with access privileges in lockdown mode, see *Specify Lockdown Mode Exception Users in the VMware Host Client* in the vSphere documentation.
9. VCF 9.x does not support the multi NSX feature. Ensure that you switch off the multi NSX features for all the compute managers that are registered with NSX Manager. Alternatively, you can map the NSX Manager instances to individual vCenter instances before upgrading.
10. Starting with VCF 9.0, a service account is created on vCenter for NSX to communicate with a registered compute manager. If you are upgrading to VCF 9.0 and any registered compute manager lacks a service account, a pre-check error is raised. To proceed with the upgrade, you must enable the **Create Service Account** setting on each registered compute manager.

Verify the Current State of NSX

Before you begin the upgrade process, it is important to test the NSX working state. Otherwise, you cannot determine if the upgrade caused post-upgrade problems or if the problem existed before the upgrade.

> **Note:**
>
> Do not assume that everything is working before you start to upgrade the NSX infrastructure.

1. Identify and record the administrative user IDs and passwords.
2. Verify that you can log in to the NSX Manager web user interface.
3. Inspect the **Dashboard**, system overview, host transport nodes, edge transport nodes, NSX edge cluster, HA status of the edge, and all logical entities. Ensure that all status indicators are green, deployed, and do not show any warnings.
4. Validate North-South connectivity by pinging out from a VM.
5. Validate that there is an East-West connectivity between any two VMs in your environment.
6. Record BGP states on the edge devices.
   - `get logical-routers`
   - `vrf <vrf>`
   - `get bgp`
   - `get bgp neighbor`

Manage Unsupported Features

Certain features in the source version of NSX might not be supported in the target version. If your deployment contains an unsupported feature, make the required changes to your system configuration before starting the upgrade. The upgrade precheck step of the upgrade process returns an error when it detects an unsupported feature.

Multi-NSX

Starting with VCF 9.0, the multi-NSX feature is not supported. Before the upgrade, turn off the multi-NSX feature for all registered compute managers, else the upgrade precheck step returns an error.

Alternatively, you can map the NSX Manager instances to individual vCenter instances before upgrading.

Compute Managers Without Service Accounts

Starting with VCF 9.0, a service account is created on vCenter for NSX to communicate with a registered compute manager. If you are upgrading to VCF 9.0 or later and a registered compute manager lacks an enabled service account, a pre-check error is thrown.

To proceed with the upgrade, you must enable the service account for those compute managers. Note that for compute managers that already have a service account feature enabled, the same service account is used post upgrade for those compute managers. The old privileges are deleted for such a service account and the predefined privileges are added.

Manager API

Starting with VCF 9.0, the Manager APIs for logical networking and security are removed. Before the upgrade, use the Policy promotion tool in the source NSX deployment to migrate any Manager objects in your environment to the equivalent Policy objects.

In addition, perform the following steps for specific use cases.

Retrieve Manager IDs of logical objects

After the upgrade, Manager IDs will no longer be available. To ensure the operation of any API that requires the Manager ID of a logical object as an input, perform these steps before starting the upgrade:

1. Create the object using the equivalent Policy API.
2. Retrieve the Unique ID value of the Policy object. The Unique ID is equivalent to the Manager ID of the logical object.
3. Use the retrieved Unique ID as input to the API requiring the Manager ID.

For example, the API for transport node profile creation takes the Manager IDs of IP pool and logical switch objects as inputs. After creating the IP pool and segment objects using Policy APIs, you can input the Unique ID values of these objects to the API for transport node profile creation.

Perform intermediate upgrade for mixed-mode configurations

A configuration that contains a combination of policy and manager objects is known as a mixed-mode configuration. Some examples of mixed-mode configurations are:

- NAT rules created through Manager APIs attached to routers created through Policy APIs
- Groups created through Policy APIs used in DFW rules created through Manager APIs

If your source deployment uses mixed-mode configurations, you must use the following workflow to upgrade to VCF 9.0 or later.

1. Upgrade the deployment to NSX 4.2.1 or later.
2. In NSX 4.2.1 or later, use the Promotion feature to promote the mixed-mode objects to policy objects.
3. Upgrade to VCF 9.0 or later.

Convert mixed-mode IP allocations

The upgrade precheck process in the NSX Manager upgrade coordinator does not detect mixed-mode IP allocations or warn you of the presence of this unsupported condition.

Before the upgrade, verify whether mixed-mode IP allocations exist in your environment. If present, use the Promotion feature in the source NSX deployment to migrate the mixed-mode IP allocations to Policy equivalents.

Replace member types related to grouping

Certain member types are deprecated or no longer supported. If your source deployment uses any of these member types, replace them with their supported equivalents which are functionally the same.

Deprecated or Unsupported Member Type	Supported Equivalent
IPSET (deprecated)	Group
Logical Switch (not supported)	Segment
Logical Port (not supported)	Segment Port

Remove any remaining Manager objects after the upgrade

After the upgrade, NSX Manager will raise an alarm if it detects any pure Manager logical objects that have not been migrated to Policy equivalents.

Configure NSX Edge Failover Pre-checks

NSX Edge failover pre-checks ensure that routing services can continue without downtime across clusters during an upgrade. As each edge node prepares to shut down for the upgrade, pre-checks validate the availability of peer nodes in that edge cluster and alert the user if a peer is not ready for failover.

By default, edge failover pre-checks are activated during host and NSX Edge upgrades. You can deactivate them using the NSX API. You can also selectively deactivate failover pre-check warnings for specific edges, edge clusters, and hosts due to known issues in your environment.

In addition, you can use the API to run failover pre-checks before an upgrade, as part of your pre-upgrade planning process.

Running Failover Pre-checks Before an Upgrade

To plan for an upgrade and identify areas of failover readiness that require attention, you can run the failover pre-checks on demand. Use the following API call and request payload.

```
POST https://<nsx-manager>/api/v1/upgrade/pre-upgrade-checks/host/planned-maintenance

{
    "vcenter-uuid": "<compute manager UUID>",
    "entity-id": "<host moref ID>"
}
```

The pre-checks report any failover readiness issues in the response body.

Failover Pre-check Workflows During an Upgrade

The failover pre-checks run during both the host upgrade and edge upgrade steps of the upgrade process.

- When the host upgrade step begins, the pre-checks consider all the peers of edges on the candidate host to be upgraded. If a peer is found not ready for failover, the upgrade pauses on the candidate host until the user makes the peer edge available for failover. After this host completes its upgrade, the failover pre-checks run sequentially on the next candidate host.
- When the edge upgrade step begins, the pre-checks validate the peers of each candidate node to be upgraded. If a peer itself has recently been upgraded and is still starting up, the upgrade process implicitly waits for the peer to become available before retrying the pre-check.

Deactivate Edge Failover Pre-checks

To deactivate some or all edge failover pre-checks, use the API call `PATCH https://<NSX Manager>/policy/api/v1/system-config` with a request payload consisting of key/value pairs as described in the following table.

By default, the `EdgeFailoverReadiness_DisablePreChecks` key is set to `false`, which activates all edge failover pre-checks.

Key	Value	Description
EdgeFailoverReadiness_DisablePreChecks	true	Deactivates all edge failover pre-checks.
	false (default)	Activates all edge failover pre-checks, with the option to selectively turn off certain pre-checks.
EdgeFailoverReadiness_DisablePreChecksForVmotionableEdges	false (default)	Activates failover pre-checks for all edges on a given host, including edges eligible for vMotion migration.
	true	Deactivates failover pre-checks for edges eligible for vMotion migration.
EdgeFailoverReadiness_DisablePreCheckForEdgeClusterID	Comma-separated list of edge cluster UUIDs	Deactivates failover pre-checks for the specified edge clusters.
EdgeFailoverReadiness_DisablePreCheckForEdgeNodeID	Comma-separated list of edge node UUIDs	Deactivates failover pre-checks for the specified edge nodes.
EdgeFailoverReadiness_DisablePreCheckForHostNodeID	Comma-separated list of hosts, represented in either of these formats: • Host UUID • Compute manager UUID : Host MO-REF ID	Deactivates failover pre-checks for the specified hosts.

For example, the following request activates failover pre-checks in general, but turns off pre-checks for these entities:

- Edge node with UUID `5b21679b-87bd-49c9-80fb-a9964bab34e9`
- Edge node with UUID `5b21679b-87bd-49c9-80fb-a9964bab3413`
- Host with MO-REF `host-11` on vCenter with UUID `b07c9a01-16eb-43fb-968a-44c57ca5e270`
- Host with UUID `4ac05527-94fb-4f12-9ed6-b11fbf48a1dd`

```
PATCH https://<NSX Manager>/policy/api/v1/system-config

{
      "keyValuePairs" : [{
            "key" : "EdgeFailoverReadiness_DisablePreChecks",
            "value" : false
        },
        {
            "key" : "EdgeFailoverReadiness_DisablePreCheckForEdgeNodeID",
            "value" : "5b21679b-87bd-49c9-80fb-a9964bab34e9, 5b21679b-87bd-49c9-80fb-a9964bab3413"
        },
        {
            "key" : "EdgeFailoverReadiness_DisablePreCheckForHostNodeID",
            "value" : "b07c9a01-16eb-43fb-968a-44c57ca5e270:host-11, 4ac05527-94fb-4f12-9ed6-b11fbf48a1dd"
        }]
}
```

Activate Edge Failover Pre-checks

To reactivate the edge failover pre-checks after they have been deactivated, use the following API call and request payload:

```
PATCH https://<NSX Manager>/policy/api/v1/system-config

{
      "keyValuePairs" : [{
            "key" : "EdgeFailoverReadiness_DisablePreChecks",
```

```
            "value" : false
        }]
}
```

Perform an Optimized Upgrade of vCenter and NSX Manager

If you choose the optimized option during the domain upgrade plan, you must upgrade NSX Manager and vCenter in the following order: prepare the vCenter instance for migration, upgrade NSX, and switchover to the new vCenter instance.

- Verify that you planned a domain upgrade and selected the optimized maintenance window upgrade option. See Plan a Domain Upgrade.
- Verify that you upgraded SDDC Manager. See Upgrade to SDDC Manager 9.1.
- Verify that you downloaded the required upgrade binaries. See Download Binaries to Software Depot.
- Verify that you performed all upgrade prechecks. See Perform Upgrade Prechecks.
- If your workload domain contains vSphere Supervisor clusters, the supported target release depends on the version of Kubernetes (K8s) currently running in the cluster. Older versions of K8s might require a specific upgrade sequence. For the supported upgrade sequence for Supervisor clusters running older Kubernetes versions, see Broadcom knowledge base article 92227.
- Pre-version 9 licenses for NSX security features are no longer supported. To avoid service interruption after the upgrade, assign a valid vDefend license or your NSX security features enter evaluation mode for 90 days.

The optimized upgrade workflow reduces the maintenance window by moving vCenter instance upgrade preparation outside the scheduled downtime and before the NSX Manager upgrade, consolidates the NSX and vCenter upgrades into a single maintenance window, and ensures that the vCenter instance switchover occurs after the NSX Manager upgrade is complete.

Note:

You can upgrade NSX in workload domains before or after upgrading NSX in the management domain.

After a new vCenter instance preparation stage completes and the NSX Manager nodes are upgraded, you complete the upgrade and perform the vCenter switchover stage, where the current vCenter data and configuration is copied to the new instance.

Note:

You must complete the upgrade tasks in a sequential order.

Optimized upgrade of vCenter and NSX Manager is not available for straddle states. For example, if SDDC Manager is at version 9.1, while vCenter target version is below 9.1.

1. Log in to VCF Operations at `https://<vcf_operations_fqdn>` with a user assigned the **Administrator** role.
2. Select **Build** > **Lifecycle**, expand **VCF Instances**, and select your domain.
3. On the **Upgrades** tab, under **VMware vCenter Preparation Stage**, click **Configure**.
4. Configure the preparation stage.
   a) Review the upgrade mechanism information and click **Next**.

      A vCenter Reduced Downtime Upgrade (RDU) uses a migration-based approach. In this approach, a new vCenter appliance is deployed and the current vCenter data and configuration is copied to it.

      During the preparation phase of a reduced downtime upgrade, the source vCenter instance and all resources remain online. The only downtime occurs when the source vCenter instance is stopped, the configuration is switched over to the target vCenter , and the services are started. The downtime is expected to take approximately 5 minutes under ideal network, CPU, memory, and storage provisioning.

      Once the target vCenter instance is operational, SDDC Manager deletes the source vCenter appliance as part of the upgrade. If the upgrade fails before the switchover completes, vCenter RDU performs an automatic rollback and the configuration reverts to the source vCenter instance. You can take a file-based backup before starting the upgrade so that you can restore it if the automatic rollback fails.

The RDU behavior in VCF is different than the behavior in vSphere Foundation, see Prepare Upgrade and Switch Over to the Target vCenter appliance.

b) Select a backup option and click **Next**.

Before starting the upgrade, best practice is to have a file-based backup of the vCenter appliance. See On-Demand Backup of vCenter. After taking a backup, do not make any changes to the vCenter inventory or settings until the upgrade completes successfully.

c) Provide a temporary network to be used only during the upgrade and click **Next**.

Setting	Value
Automatic	SDDC Manager delegates temporary network configuration to the vCenter RDU process, which generates a temporary link-local address for the parallel appliance. No DHCP or available IP addresses on the existing subnet are required.
Static	Enter an IP address, subnet mask, and gateway. The IP address must be in the management subnet.

d) Select a deployment size and storage size.

Based on your selection, the resources that are required for the vCenter deployment change.

e) Review the upgrade configuration and click **Finish**.

If you are upgrading a management domain, after you confirm the domain configuration, the management domain upgrade is scheduled or in progress. Therefore, upgrade operations for workload domains are unavailable.

5. Optional: Under **VMware vCenter Preparation Stage**, click **Run Prechecks** to run the upgrade prechecks.

   You can click on **View Details** to monitor the progress of the preheck. You can review the report and evaluate all errors and warnings.

6. Under **VMware vCenter Preparation Stage**, click **Schedule** and select scheduling options for the preparation and switchover stages of the vCenter upgrade.

   a) Schedule an option for the preparation stage.

   You can start the vCenter preparation stage immediately or specify another time.

   b) Schedule an option for the switchover stage.

   Specify the switchover behavior after the target instance is created with the selected configuration and after NSX Manager is upgraded.

   - Automatic: Proceed with an automatic switchover after the preparation stage is complete.
   - Manual: You must manually click **Switchover Now** to start the switchover.

   c) Click **Schedule**.

   Your new vCenter instance is deployed (preparation stage) and the current vCenter data and configuration is copied to it

7. Under **VMware NSX Manager Nodes Upgrade**, click **Configure**.

   To complete the upgrade of an NSX instance, you must upgrade the vCenter and ESX components of this domain. After you upgrade the vCenter, ESX, and NSX Manager components, you can finalize the upgrade.

8. Optional: Under **VMware NSX Manager Nodes Upgrade**, click **Run Prechecks** to run the upgrade prechecks.

   The NSX precheck runs on all workload domains in your environment that share the NSX Manager cluster.

9. Under **VMware NSX Manager Nodes Upgrade**, click **Upgrade Now**.

   Alternatively, you can click **Schedule** and specify another time for the upgrade.

   When all NSX Manager nodes are upgraded successfully, a message with a green background and check mark is displayed.

10. If NSX is shared across multiple domains, confirm that the vCenter RDU switchover is complete on every domain that shares that NSX instance.
11. Under **VMware vCenter Switchover Stage**, click **Switchover Now**.

The migration to the new vCenter instance starts and you can monitor the progress. See Monitoring VMware Cloud Foundation Upgrades. If you cancel the switchover, the configuration reverts to the source vCenter instance.

- Allow some time (approximately 15 minutes) for post-upgrade auto-reconciliation of configuration drift fixes before scheduling another upgrade. If you schedule a second upgrade to run before the remediation task completes, the upgrade might be cancelled or auto-reconciliation might fail.
- Use the vSphere Client to change the vSphere DRS Automation Level setting back to the original value (before you took a file-based backup) for each vSphere cluster that is managed by the vCenter. See Broadcom knowledge base article 87631 for information about using VMware PowerCLI to change the vSphere DRS Automation Level.
- Transition any clusters that use vSphere Lifecycle Manager baselines to use vSphere Lifecycle Manager images before upgrading ESX hosts. See Transitioning from vSphere Lifecycle Manager Baselines to vSphere Lifecycle Manager Images.

Perform a Sequential Upgrade of vCenter and NSX Manager

If you choose the sequential option during the domain upgrade plan, you must first upgrade NSX Manager followed by a vCenter upgrade.

- Verify that you planned a domain upgrade and selected the sequential upgrade option. See Plan a Domain Upgrade.
- Verify that you upgraded SDDC Manager. See Upgrade to SDDC Manager 9.1.
- Verify that you downloaded the required upgrade binaries. See Download Binaries to Software Depot.
- Verify that you performed all upgrade prechecks. See Perform Upgrade Prechecks.
- If your workload domain contains vSphere Supervisor clusters, the supported target release depends on the version of Kubernetes (K8s) currently running in the cluster. Older versions of K8s might require a specific upgrade sequence. See Broadcom knowledge base article 92227.
- Pre-version 9 licenses for NSX security features are no longer supported. To avoid service interruption after the upgrade, assign a valid vDefend license or your NSX security features enter evaluation mode for 90 days.

The sequential upgrade workflow allows upgrade operations for each component to be performed in separate stages in different maintenance windows. The upgrade starts with the NSX Manager nodes upgrade and is followed by the vCenter instance upgrade. You can select between a **vCenter In Place Upgrade** or **vCenter Reduced Downtime Upgrade**. If you choose the latter mechanism, you can schedule a time for the vCenter preparation and switchover stages. By decoupling the preparation stage from the final switchover, the workflow allows for shorter maintenance windows.

> **Note:**
>
> The upgrade binary for vCenter is used to upgrade the vCenter instances managed by SDDC Manager. Upgrade vCenter in the management domain before upgrading vCenter instances in workload domains. You can upgrade NSX in workload domains before or after upgrading NSX in the management domain.

> **Note:**
>
> You must complete the upgrade tasks in a sequential order.

1. Log in to VCF Operations at `https://<vcf_operations_fqdn>` with a user assigned the **Administrator** role.
2. Select **Build** > **Lifecycle**, expand **VCF Instances**, and select your domain.
3. Optional: On the **Upgrades** tab, under **VMware NSX Manager Nodes Upgrade**, click **Run Prechecks** to run the upgrade prechecks.

   The NSX precheck runs on all workload domains in your environment that share the NSX Manager cluster.
4. Under **VMware NSX Manager Nodes Upgrade**, click **Upgrade Now** or click **Schedule** to upgrade later.

   To complete the upgrade of an NSX instance, you must upgrade the vCenter and ESX components of this domain. After you upgrade the vCenter, ESX, and NSX Manager components, you can finalize the upgrade.

Note:

NSX 9.x features are not available until all ESX hosts in all vSphere clusters that share an NSX Manager are upgraded to 9.x and the NSX finalize procedure completes.

5. Under **VMware vCenter Upgrade**, click **Configure**.

Note:

If you are upgrading from version 9.0 or later, both upgrade options are available. If you are upgrading from version VCF 5.2.x or vSphere 8.x, the only available option is **vCenter Reduced Downtime Upgrade**.

vCenter Reduced Downtime Upgrade

A **vCenter Reduced Downtime Upgrade** (RDU) uses a migration-based approach. In this approach, a new vCenter appliance is deployed and the current vCenter data and configuration is copied to it. During the preparation phase of a reduced downtime upgrade, the source vCenter instance and all resources remain online. The only downtime occurs when the source vCenter instance is stopped, the configuration is switched over to the target vCenter, and the services are started.

1. Select the **vCenter Reduced Downtime Upgrade** mechanism and click **Next**.
2. Select a backup option and click **Next**.
   Best practice is to have a file-based backup of the vCenter appliance. See On-Demand Backup of vCenter. After taking a backup, do not change the vCenter inventory or settings until the upgrade completes successfully.
3. Provide a temporary network to be used only during the upgrade and click **Next**.

Setting	Value
Automatic	SDDC Manager delegates temporary network configuration to the vCenter RDU process, which generates a temporary link-local address for the parallel appliance. No DHCP or available IP addresses on the existing subnet are required.
Static	Enter an IP address, subnet mask, and gateway. The IP address must be in the management subnet.

4. Select a deployment size and storage size and click **Next**.
   Based on your selection, the resources that are required for the vCenter deployment change.
5. Review the upgrade configuration and click **Finish**.
6. (Optional) Under **VMware vCenter Upgrade**, click **Run Prechecks** to run the upgrade prechecks.
7. Under **VMware vCenter Upgrade**, click **Schedule**.
   a. Schedule a time for the vCenter preparation stage.
      You can start the preparation immediately or specify another time.
   b. Specify the switchover behavior after the target instance is created with the selected configuration.
      - Automatic: Proceed with an automatic switchover after the preparation stage is complete.
      - Manual: You must manually click **Switchover Now** to start the switchover.
      - Scheduled: Specify a time for the switchover.
8. If you selected the **Manual** switchover option, under **VMware vCenter Upgrade**, click **Switchover Now**.
   If you cancel the upgrade, the configuration reverts to the source vCenter instance.

vCenter Regular Upgrade

vCenter Regular Upgrade requires downtime of the vCenter instance during the entire upgrade process.

1. Select the **vCenter Regular Upgrade** mechanism and click **Next**.
2. Select a backup option and click **Next**.
   Best practice is to have a file-based backup of the vCenter appliance. See On-Demand Backup of vCenter. After taking a backup, do not change the vCenter inventory or settings until the upgrade completes successfully.
3. Review the upgrade configuration and click **Finish**.
4. (Optional) Under **VMware vCenter Upgrade**, click **Run Prechecks** to run the upgrade prechecks.
5. Under **VMware vCenter Upgrade**, click **Upgrade Now** or **Schedule**.

You can start the upgrade immediately or specify another time.

The upgrade of the vCenter instance starts and you can monitor the progress. See Monitoring VMware Cloud Foundation Upgrades.

- Allow some time (approximately 15 minutes) for post-upgrade auto-reconciliation of configuration drift fixes before scheduling another upgrade. If you schedule a second upgrade to run before the remediation task completes, the upgrade might be cancelled or auto-reconciliation might fail.
- Use the vSphere Client to change the vSphere DRS Automation Level setting back to the original value (before you took a file-based backup) for each vSphere cluster that is managed by the vCenter.See Broadcom knowledge base article 87631 for information about using VMware PowerCLI to change the vSphere DRS Automation Level.
- Transition any clusters that use vSphere Lifecycle Manager baselines to use vSphere Lifecycle Manager images before upgrading ESX. SeeTransitioning from vSphere Lifecycle Manager Baselines to vSphere Lifecycle Manager Images.

Upgrade ESX to 9.1

Prior to VCF 9.0, the product supported both vSphere Lifecycle Manager baseline clusters and vSphere Lifecycle Manager image clusters. With VMware Cloud Foundation 9.0 and later, only vSphere Lifecycle Manager image clusters are supported.

- Verify that all vSphere clusters in a domain are managed by vSphere Lifecycle Manager images. If any clusters are still using legacy vSphere Lifecycle Manager baselines, transition them to use vSphere Lifecycle Manager images. See Transitioning from vSphere Lifecycle Manager Baselines to vSphere Lifecycle Manager Images. For upgrade of heterogeneous ESX clusters, see Broadcom knowledge base article 424294.
- Verify that the domain does not have any ESX hosts or vSphere clusters in an error state. It is recommended that all hosts are connected and no health issues are present. You can resolve the error states or remove the ESX hosts and vSphere clusters with errors.
- Verify that you upgraded NSX Manager and vCenter before you can upgrade ESX hosts with a vSphere Lifecycle Manager image.
- Verify that you created a vSphere Lifecycle Manager image using the vSphere Client. See Working with Image Library.
  - Verify that the ESX version used is specified in the VCF BOM.
  - You can create a composite image for a cluster that has hosts from different vendors or from the same vendor but different generations, family, or model. A composite image contains one default image and one or more alternative images. All images in a composite image contain the same base image and solutions, but might include different vendor or firmware and driver add-ons. When you create alternate images, you also specify the rules that are used to filter the hosts to which the alternative image is assigned. See Edit a vSphere Lifecycle Manager Image on a Cluster or Host Level.
  - If you plan to use the **Import from a file** option to import a vSphere Lifecycle Manager image, you must first export the files from the vSphere Client.

Before VCF 9.0,the NSX VIBs on ESX hosts were upgraded as part of the NSX Manager upgrade. With VCF 9.0 and later, ESX host clusters are upgraded as part of the **VMware ESX Upgrade** step. As a result of this change, ESX hosts no longer go into maintenance mode during an NSX Manager upgrade.

If you are using external (non-vSAN) storage, the following procedure upgrades the ESX hosts attached to the external storage. However, updating and patching the storage software and drivers is a manual task and falls outside of SDDC Manager lifecycle management. To ensure supportability after an ESX upgrade, consult the vSphere HCL and your storage vendor.

You can configure multiple cluster upgrade tasks with different schedules without having to wait for the first cluster upgrade to complete. For example, you could schedule an upgrade for `cluster-1` and `cluster-2` to run immediately and then configure another upgrade task for `cluster-3` and schedule it to run overnight.

Note:

- If you are upgrading a stretched cluster with a physical vSAN Witness Host, upgrade the witness host manually, before upgrading the ESX hosts in the stretched cluster.

- Virtual vSAN Witness Hosts are upgraded as part of the cluster upgrade workflow.

1. Log in to VCF Operations at `https://<vcf_operations_fqdn>` with a user assigned the **Administrator** role.
2. Select **Build** > **Lifecycle**, expand **VCF Instances**, and select your VCF Instance.
3. Import a vSphere Lifecycle Manager image.
   a) Select **Image Management**.
   b) Click **Import Image**.
   c) Select the import method, provide the required information, and click **Import**.

Option	Description
Import from a vCenter	Select the vCenter where you created the image from the drop-down menu. Select the upgrade image.
Import from a file	Enter a name for the image and then click **Select File** to import each of the required files. Select the JSON, ZIP, and ISO files from your local computer. The ISO file is optional.

   You can view status in the **Tasks** panel.
4. Expand **VCF Instances** and select your domain.
5. On the **Upgrades** tab, under **VMware ESX Upgrade**, click **Configure**.
6. Configure the ESX cluster upgrade.
   a) Read the introductory information and click **Next**.
   b) Select the clusters to upgrade and click **Next**.
      - To upgrade specific clusters, select **Custom Selection** and select the clusters to upgrade.
        – The default setting is to upgrade all clusters, including single host clusters.
      - To select or deselect specific hosts in a cluster, select **Custom Selection**, select a cluster from the list, and click **Customize**. In the **Select Hosts** dialog box, select **Custom Selection**, select the hosts to upgrade, and click **Select**.
        – The default setting is to upgrade all hosts in the selected cluster. You can skip one or more hosts for an upgrade, for example, if there are issues or errors with the hosts.
   c) Assign an image to upgrade the selected hosts in the selected clusters.

      Depending on your configuration, you can upgrade ESX clusters in batches. For example, if you use different vSphere Lifecycle Manager images for your hosts, you can upgrade all ESX hosts using the same image in batch 1, then the hosts using a different image in batch 2 and so on.

      1. Select clusters from the list.
      2. Click **Assign Image** .
      3. Select an imported image to assign to the selected clusters, and click **Assign Image**.
      4. Click **Next**.
   d) Select the upgrade options and click **Next**.
      - Select **Enable sequential cluster upgrade** to upgrade the clusters in a subsequent order. If you do not select the check box, by default, the clusters are upgraded in parallel to reduce upgrade time. If you selected more than ten clusters to be upgraded, the first ten are upgraded in parallel and the remaining clusters are upgraded sequentially.
      - Select **Enable Quick Boot** to reduce the upgrade time by skipping the physical reboot of the host.
      - Under **VM power state**, select the desired virtual machine behavior during the host upgrade.
      - Select **Migrate Powered Off and Suspended VMs** to migrate the suspended and powered off virtual machines from the hosts that must enter maintenance mode to other hosts in the cluster.
      - Select **Retry policy** to retry the host upgrade. Specify a desired delay time between the host upgrade attempts and a number of allowed upgrade attempts.

- Select **Enable Live Patch** when the cluster image includes a Live Patch.
  - With the **Auto** option, hosts in the cluster are upgraded using Live Patch when possible. If Live Patch is not available, ESX hosts in the cluster are placed into maintenance mode and are rebooted.
  - With the **Enforce** option, only Live Patch is used for remediation without maintenance mode. ESX hosts are not rebooted and there is no need to migrate the virtual machines running on the hosts in the cluster. If a host in the cluster requires Maintenance Mode or Reboot, remediation is blocked.

Note:

Live Patch is not applicable for major upgrades. For major upgrades, for example, when upgrading your ESX hosts from version 9.0 to 9.1, either select the **Auto** option or deselect the **Enable Live Patch** check box.

e) Select standalone hosts to upgrade and click **Next**.

By default, all standalone hosts are selected for upgrade. To upgrade specific standalone host, click **Custom Selection** and select hosts from the list.

f) If you selected standalone hosts for upgrade, assign an image to the selected hosts.

1. Select hosts from the list.
2. Click **Assign Image**.
3. Select an imported image to assign to the selected hosts, and click **Assign Image**.
4. Click **Next**.

g) If you selected standalone hosts for upgrade, select host upgrade options.

- Select **Enable Quick Boot** to reduce the upgrade time by skipping the physical reboot of the host.
- Under **VM power state**, select the desired virtual machine behavior during the host upgrade.
- Select **Retry policy** to retry the host upgrade. Specify a desired delay time between the host upgrade attempts and a number of allowed upgrade attempts.

h) Review the upgrade configuration, and click **Finish**.

You can click **Export** to download the cluster upgrade information in a `CSV` file format. Standalone hosts are not included in the configuration export.

7. If you did not select all available ESX hosts for upgrade, you can select **Configure New Batch**.

   To create a new batch of ESX hosts for upgrade, you must perform step 5 again and specify the upgrade options for the new batch.

8. Optional: Under **VMware ESX Upgrade**, click **Run Prechecks** to run the upgrade prechecks.

   You can click on **View Details** to monitor the progress of the precheck. You can review the report and evaluate all errors and warnings.

9. Select **Upgrade Now**.

   Perform this step on all ESX host batches that are configured for upgrade.

   Alternatively, you can click **Schedule** and specify another time for the upgrade.

The upgrade of the ESX hosts starts. If you are upgrading multiple clusters or standalone hosts in parallel and one or more of the cluster or standalone host upgrades fails, the upgrade continues on the remaining hosts. In addition, you can configure a new upgrade batch for one or more failed clusters or standalone hosts without having to wait for the original upgrade batch to finish.

- You can monitor the progress. See Monitoring VMware Cloud Foundation Upgrades.

After the upgrade is complete, allow some time (approximately 15 minutes) for post-upgrade auto-reconciliation of configuration drift fixes before scheduling another upgrade. If you schedule a second upgrade to run before the remediation task completes, the upgrade might be canceled or auto-reconciliation might fail.

Upgrade the NSX Edge Cluster and Finalize NSX Upgrade

To complete the NSX upgrade and your domain core components, you must upgrade the NSX Edge cluster and finalize the NSX upgrade. You cannot configure new NSX features introduced by the upgrade until this upgrade step is completed.

1. Log in to VCF Operations at `https://<vcf_operations_fqdn>` with a user assigned the **Administrator** role.
2. Select **Build** > **Lifecycle**, expand **VCF Instances**, and select your domain.
3. On the **Upgrades** tab, under **VMware NSX Edge Clusters Upgrade and NSX Upgrade Finalize**, click **Configure**.
4. Configure the NSX Edge Cluster upgrade and finalize the NSX upgrade.
   a) Review the validations and click **Next**.

      Note:

      If you do not have NSX Edge clusters in your environment, skip the next two substeps.
   b) On the NSX Edge Clusters page, select the NSX Edge clusters you want to upgrade and click **Next**.

      By default, all NSX Edge clusters are upgraded. To select specific NSX Edge clusters, turn on the **Select specific NSX Edge clusters** toggle button and select the NSX Edges that you want to upgrade.
   c) On the Upgrade options page, select whether to upgrade the NSX Edge clusters sequentially.

      If you do not select the check box, the NSX Edge clusters are upgrade in parallel.
   d) On the Review page, review your settings and click **Finish**.
5. Optional: Under **VMware NSX Edge Clusters Upgrade and NSX Upgrade Finalize**, click **Run Prechecks** to run the upgrade prechecks.

   You can click on **View Details** to monitor the progress of the preheck. You can review the report and evaluate all errors and warnings.
6. Click **Upgrade Now**.

   Alternatively, you can click **Schedule** and specify another time for the process to complete.

- Monitor the upgrade progress. See Monitoring VMware Cloud Foundation Upgrades.
- Verify the NSX upgrade. See Verify the NSX Upgrade.

After the upgrade is complete, allow some time (approximately 15 minutes) for post-upgrade auto-reconciliation of configuration drift fixes before scheduling another upgrade. If you schedule a second upgrade to run before the remediation task completes, the upgrade might be canceled or auto-reconciliation might fail.

Verify the NSX Upgrade

After you upgrade NSX, you can verify whether the NSX components have been updated to the target version.

1. From your browser, log in as a local admin user to an NSX Manager at `https://nsx-manager-ip-address/login.jsp?local=true`.
2. Select **System** and under **Lifecycle Management**, select **Upgrade**.
3. Verify that the overall upgrade version, component version, and initial and target product version are accurate.
   a) Optional: Inspect the Dashboard, fabric hosts, edge cluster, transport nodes, and all logical entities. Ensure that all status indicators are green, normal, deployed, and do not show any warnings.
   b) Optional: Verify the status of several components.
      - Fabric nodes installation
      - Transport node Local Control Plane (LCP) and Management plane agent connectivity
      - Routers connectivity
      - NAT rules
      - DFW rules
      - DHCP lease
      - BGP details

- Flows in the IPFIX collector
- TOR connectivity to enable the network traffic

The status of the upgrade appears as Successful.

Note:

If you upgraded from NSX 4.1.x or earlier, new system-generated transport zones appear in addition to the transport zones originally configured in your source deployment.

- For clusters prepared for networking and security, new security transport zones are created. The system generates these security transport zones by default.
- For clusters prepared for NSX on DVPGs, new VLAN transport zones are created. The system generates these VLAN transport zones by default.

4. Modify the default admin password expiration.

   If the password expires, you cannot log in and manage components. Also, any task or API call that requires the password will fail. By default, passwords expire after 90 days. For more information, see Broadcom knowledge base article 338942.

   a) Reset the expiration period.

      The expiration period can range from 1 to 9999 days.

      ```
      nsxcli set user admin password-expiration <1 - 9999>
      ```

   b) Optional: You can disable password expiry so the password never expires.

      ```
      nsxcli clear user audit password-expiration
      ```

5. If you have VIDM enabled, access the local account at https://*nsx-manager-ip-address*/login.jsp?local=true.
6. Verify CPU and Memory values for edge VMs.

   After the upgrade, log in to the vSphere Client to verify if your existing edge VMs are configured with the following CPU and Memory values. If they are not, edit the VM settings to match these values.

NSX Edge Form Factor	Memory	vCPU
NSX Edge Small VM	4 GB	2
NSX Edge Medium VM	8 GB	4
NSX Edge Large VM	32 GB	8

7. (For upgrades from NSX 4.1.x or earlier) Create a certificate for the NSX cluster VIP, either self-signed or signed by a third-party Certificate Authority. The certificate must include the Subject Alternative Name and Subject Alternative IP for all NSX Manager nodes in the cluster.

   This step ensures a successful repository sync for any NSX Manager node that is redeployed in the future. For more information, See Broadcom knowledge base article 397340.

Upgrade vSphere Distributed Switch Versions

To take advantage of features that are available only in the later versions, you can upgrade the vSphere Distributed Switch in the vSphere Client.

- Verify that the ESX and vCenter upgrades are completed.

1. On the vSphere Client Home page, click **Networking** and navigate to the vSphere Distributed Switch.
2. Right-click the vSphere Distributed Switch and select **Upgrade** > **Upgrade Distributed Switch**.

   The **Upgrade Distributed Switch** wizard appears.

3. On the **Configure upgrade** page, select the vSphere Distributed Switch version that you want to upgrade the switch to and click **Next**.
4. On the **Check compatibility** page, verify that hosts assigned to the vSphere Distributed Switch are compatible with the selected version, and click **Next**.
5. On the **Ready to complete** page, review the upgrade and click **Finish**.

The vSphere Distributed Switch is successfully upgraded.

Upgrade vSAN On-Disk Format Versions

Upgrade the vSAN on-disk format version using the vSphere Client to take advantage of features that are available only in the later versions.

- ESX and vCenter upgrades are completed.
- Verify that the disks are in a healthy state. Navigate to the Disk Management page to verify the object status.
- Verify that your ESX hosts are not in maintenance mode. When upgrading the disk format, do not place the ESX hosts in maintenance mode.
- Verify that there are no component rebuilding tasks currently in progress in the vSAN cluster. For information about vSAN resynchronization, see Monitor the Resynchronization Tasks in vSAN Cluster.

The upgrade might cause temporary resynchronization traffic and use additional space by moving data or rebuilding object components to a new data structure.

1. In the vSphere Client, navigate to the vSAN cluster.
2. Click the **Configure** tab.
3. Under **vSAN**, select **Disk Management**.
4. Click **Pre-check Upgrade**. The upgrade pre-check analyzes the cluster to uncover any issues that might prevent a successful upgrade. Some of the items checked are host status, disk status, network status, and object status. Upgrade issues are displayed in the **Disk pre-check status text** box.
5. Click **Upgrade**.
6. Click **Upgrade** on the Upgrade dialog box to perform the upgrade of the on-disk format.

vSAN successfully upgrades the on-disk format. The On-disk Format Version column displays the disk format version of storage devices in the cluster.

Upgrade vSAN File Service

When you upgrade the file service, the upgrade is performed on a rolling basis.

Ensure that the following components are upgraded:

- ESX Hosts
- vCenter
- vSAN disk format

During the upgrade, the file server containers running on the VMs which are undergoing upgrade fails over to other virtual machines. The file shares remain accessible during the upgrade. During the upgrade, you might experience some interruptions while accessing the file shares.

1. In the vSphere Client, navigate to the cluster.
2. Click the **Configure** tab.
3. Under vSAN, click **Services**.
4. Under **File Service**, click **Edit**.
5. Click **Check upgrade** from the list.
6. In the Upgrade File Service dialog box, select one of the following deployment options and then click **Upgrade**.

Option	Action
Automatic approach	This is the default option. This option lets the system search and download the OVF. After the upgrade begins, you cannot cancel the task. **Note:** vSAN requires Internet connectivity for this option.
Manual approach	This option allows you to browse and select an OVF that is already available on your local system. After the upgrade begins, you cannot cancel the task. **Note:** If you select this option, you should upload all the following files: • `VMware-vSAN-File-Services-Appliance-`*`versionnumber.buildnumber`*`_OVF10` • `VMware-vSAN-File-Services-Appliance-`*`versionnumber.buildnumber`*`_OVF10` • `VMware-vSAN-File-Services-Appliance-`*`versionnumber.buildnumber`*`-system.vmdk` • `VMware-vSAN-File-Services-Appliance-`*`versionnumber.buildnumber`*`-cloud-components.vmdk` • `VMware-vSAN-File-Services-Appliance-`*`versionnumber.buildnumber`*`-log.vmdk` • `VMware-vSAN-File-Services-Appliance-`*`versionnumber.buildnumber`*`_OVF10`

Apply VCF Configuration Upgrades

VCF configuration upgrades identify and resolve any discrepancies between the intended/prescribed configuration and the actual configuration, ensuring that the deployment aligns with the recommended configuration. This process includes reconciling the configuration for 2nd party software components that are listed in the VCF Bill of Materials (BOM).

Configuration upgrades might be required after you apply software upgrades. After a configuration upgrade becomes available, you can apply it immediately or can wait until after you have applied all software upgrades. Configuration Upgrades must be performed during a maintenance window.

Configuration Upgrades can be applied to multiple domains in parallel. However, if a Configuration Upgrade is in progress, another configuration upgrade on the same domain should not be attempted.

1. Log in to VCF Operations at `https://`*`<vcf_operations_fqdn>`* with a user assigned the **Administrator** role.
2. Select **Build** > **Lifecycle**, expand **VCF Instances**, and select your domain.
3. On your domain page, click the **Upgrades** tab.
4. Optional: Under **Prechecks**, click **Run Prechecks**.

   Resolve any issues before proceeding with the upgrade.

5. Under **Configuration Upgrades**, click **Apply All**.
   - FEATURE: Configuration change required for a new feature.
   - FIX: Configuration change associated with a fix for a defect.

   Note:

   Pending Configuration Upgrades do not block future BOM upgrades.
6. Optional: Check the progress of a configuration upgrade by clicking the task in the Tasks panel.

   After the configuration upgrades are successfully applied, they will no longer appear in the table.

Monitoring VMware Cloud Foundation Upgrades

You can monitor in-progress upgrades for VCF components.

1. In the In-Progress Upgrades section, click **View Status** to view the high-level upgrade progress and the number of components to be upgraded.

   Details of the component being upgraded is displayed.
2. Click the arrow to see a list of tasks being performed to upgrade the component.

   After each task is completed successfully, it shows a green check mark.

   When all tasks to upgrade a component have been completed, the upgrade status for the component is displayed as `Upgraded`.

   If a component upgrade fails, the status is displayed as `Failed`. The reason for the failure and remediation steps are displayed. Resolve the issues and retry the upgrade.

Accessing the VMware Cloud Foundation Upgrade History

The Upgrade History page displays all upgrades that are applied to a domain.

1. Log in to VCF Operations at `https://<vcf_operations_fqdn>` with a user assigned the **Administrator** role.
2. Select **Build** > **Lifecycle**, expand **VCF Instances**, select your domain, and click the **Upgrade History** tab.

   All upgrades applied to this domain are displayed. If an upgrade bundle was applied more than once, click **View Past Attempts** to see more information.

Accessing the VMware Cloud Foundation Upgrade Log Files

You can check the log files for failed upgrades to help troubleshoot and resolve issues.

1. SSH in to the SDDC Manager appliance with the `vcf` user name and enter the password.
2. To access upgrade logs, navigate to the `/var/log/vmware/vcf/lcm` directory.
   - `lcm-debug` log file contains debug level logging information.
   - `lcm.log` contains information level logging.

   Note:

   After SDDC Manager is upgraded to 9.x, all logging information will be sent to the `lcm-debug` log file and `lcm.log` will no longer be upgraded.

Upgrading to vSphere Foundation 9.1.x

To upgrade your vSphere Foundation 5.2 or 9.0.x environment to version 9.1.x, you must upgrade your components in a specific subsequent order.

Forward-Only Upgrade Requirement

vSphere Foundation enforces strict upgrade path validation. You can only upgrade your vSphere Foundation environment to a target version that is released after the release date of your current version. The target version can be a major, minor, or a maintenance release. In general, you can upgrade directly to a patch release only if you can also upgrade to its parent maintenance release. For example:

- Supported upgrade to vSphere Foundation 9.1.0 (11 MAY 2026): vCenter 8.0 Update 3a (18 JUN 2024) and ESX 8.0 Update 3 (25 JUN 2024).
- Not supported upgrade to vSphere Foundation 9.1.0 (11 MAY 2026): vCenter 8.0 Update 3j (27 MAY 2026) and ESX 8.0 Update 3j (27 MAY 2026).
- Supported upgrade to vSphere Foundation 9.1.1 (03 SEP 2026): vCenter 8.0 Update 3j (27 MAY 2026) and ESX 8.0 Update 3j (27 MAY 2026).

VCF components might have further limitations, see VMware Interoperability Matrix.

VCF Upgrade Planner

You can now use the new VCF Upgrade Planning Tool to navigate you during the process of upgrading your existing infrastructure to vSphere Foundation 9.1.

Environments Without Aria Operations for Logs 8.x or VCF Operations for Logs 9.0.x

For environments that do not have Aria Operations for Logs or VCF Operations for Logs and are not subject to upgrade to log management 9.1.x, you can also follow Upgrading vSphere 8 and Optionally vSAN and Aria Operations 8 to 9.1.x. In this scenario, you do not deploy VCF management services as part of the upgrade to 9.1.x. You can deploy them later as a Day-N operation as they are required to deploy log management 9.1.x. You cannot use the capabilities of the 9.1.x components that are not yet deployed. For more information on the 9.1 components and capabilities, see Components in VCF and vSphere Foundation.

Upgrade Order for vSphere Foundation 5.2.x or 9.0.x to Version 9.1.x

#	Component	Upgrade to Version 9.1.x Procedure
1	Aria Operations / VCF Operations & Cloud Proxy	Upgrading to VCF Operations 9.1
2	Aria Operations Orchestrator	Upgrade to VCF Operations Orchestrator 9.1
3	vCenter	1. Upgrade vCenter. See Reduced Downtime Upgrade of the vCenter Appliance. 2. Deploy VCF Management Services and License Server for vSphere Foundation 3. If you are upgrading from 9.0.x, follow Transfer Licenses After You Upgrade VCF Operations from Version 9.0.x
4	Aria Operations for Logs / VCF Operations for logs	• If you are upgrading from Aria Operations for Logs 8.x, see Deploy Log Management • If you are upgrading from VCF Operations for logs 9.0, see Upgrade to Log Management 9.1
5	vSphere Kubernetes Service	Upgrade vSphere Kubernetes Service to 3.6.0 or later
6	Supervisor	Upgrade a Supervisor to 9.1
7	ESX	Upgrade the ESX Hosts
8	vSphere Distributed Switch	Upgrade a vSphere Distributed Switch to a Later Version
9	VMware Tools	Upgrade VMware Tools to 13.1
10	Virtual machines compatibility	Upgrade Virtual Machines
11	vSAN On-Disk Format Versions	Upgrade vSAN On-Disk Format Versions
12	vSAN File Service	Upgrade vSAN File Service

Deploy VCF Management Services and License Server for vSphere Foundation

To upgrade your VMware vSphere Foundation to version 9.1, after upgrading VCF Operations and vCenter, you deploy VCF management services and license server.

- Download the required binaries. See Downloading Binaries to VCF Installer.

- Verify that you deployed VCF Installer version 9.1.1 or later. See Deploy VCF Installer.
- Verify that you configured the required FQDNs and IP addresses for deployment. See VCF Components FQDNs and IP addresses.
- The VCF management services FQDN names are checked for validity, the FQDN validation relies on RFC1034 and RFC1123, and only top level domains of the internet are validated. The `.local` domain is not supported as it does not fall into the list of top-level domains in the Domain Name System (DNS) of the Internet.
- If your environment has additional management components and your VCF Operations has a configured proxy server with SSL termination, you must remove the proxy server configuration before you deploy VCF management services.

This process installs the following VCF management services components:

- VCF services runtime
- Fleet lifecycle
- SDDC lifecycle
- Software depot
- Telemetry

Additionally, as a part of this procedure, you deploy a license server. For environments, where a license server already exists, this workflow does not deploy a second license server.

VCF Installer deploys VCF management services for vSphere Foundation with a small component size. To change the component size, you can use a `JSON` specification file. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation.

If VCF Installer does not have access to the Internet, you can download the VCF management services and license server binaries by using the VCF Download Tool to an offline depot or to VCF Installer. See Download Installation Binaries to an Offline Depot or Manually Transfer Binaries to VCF Installer.

For information about VCF management services, see Components in VCF and vSphere Foundation.

1. In a web browser, log in to the VCF Installer appliance at `https://installer_appliance_FQDN`.

   Enter the `admin@local` user and password that you provided when you deployed the appliance and click **Log In**.

2. Click **Deployment Wizard** > **VMware vSphere Foundation**.
3. Review the introductory information and click **Next**.
4. Select **Deploy VCF Management Services**.
5. Select your network configuration.

   You can continue with the recommended values or click **Customize** to provide custom network settings.

Management network configuration	• VCF management network: For broader VCF management components, such as VCF Operations. You can share this network with the VM management network or use a dedicated network. You can use one of the networks to segment management traffic for security and compliance requirements. For more information, see Network Deployment Options for VCF Components.
IP Scheme	If Dual-stack is enabled for VCF Installer, you can enable the **Dual stack networking** toggle and use IPv4 and/or IPv6 Internet protocols for your management domain components. • vMotion network IP scheme (IPv4 or IPv6) • Storage network IP scheme (IPv4 or IPv6) Dual-stack networking allows communication with components that support both network protocols, ensuring compatibility and continuous connectivity. While not all components support IPv6 networking, components that are IPv4 only can still communicate with the other components. **Note:** The dual-stack networking option is not available, if VCF Installer does not have IPv4 and IPv6 configured. For more information, see Configure the Network on the ESX Hosts and Deploy VCF Installer.

6. Review the prerequisites for your deployment.

   Depending on your selections in the wizard, the prerequisites for the deployment are listed.

   You can click **PRE-FILL GENERATED FQDNs IN WIZARD** and specify the required FQDNs for the deployment. You can use the provided template or specify custom FQDNs. These values are later populated in the VCF Installer deployment workflows.

   You can download a JSON template that is based on your choices in the VCF Installer so far. You can populate it and deploy VCF by using a JSON specification file. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation for more information.

7. Enter the general configuration details for your deployment.

   With VCF 9.0 and later, passwords are auto-generated. During the deployment, you can review and save the passwords.

   Additional information:

Option	Description
Version	Choose which VMware vSphere Foundation version to install.
Customer Experience Improvement Program (CEIP)	Select an option to activate or deactivate CEIP across all deployed components. VMware's Customer Experience Improvement Program (CEIP) collects technical information from participating customers to provide better support. For more information, see Customer Experience Improvement Program.

Option	Description
Autogenerate passwords	Select this option to let the deployment generate secure passwords for all deployed components. You can retrieve the passwords after the deployment begins. If you do not select this option, you must provide each password manually.

8. Enter the existing VCF Operations details.

   The primary node FQDN and the admin account password for the existing VCF Operations instance are required to authenticate and register the VCF management services.

9. Enter the existing vCenter details.

10. Configure the network.

    - VCF Management Network - distributed port group separation for all management components
      If you chose to share this network with the VM management network, this option is not displayed.
    - VCF management services IP pool.
      - For target versions 9.1.0.0 through 9.1.0.300, you enter an IP range (in the format `start_IP - end_IP`) with a minimum of 10 IP addresses.
      - For target versions 9.1.0.400 and later:
        - You can now provide a comma-separated list of IP addresses. The IP addresses can be contiguous or non-contiguous.
        - You can now exclude IP addresses from the provided IP range or IP list. Excluded IP addresses cannot be used for deploying additional VCF components.

    If you use dual-stack networking, you must also provide an IPv6 address range or IPv6 address list.

11. Enter the details for your VCF management services details.

Each Fully Qualified Domain Name (FQDN) must point to a unique IP address that is outside of the IP range of the VCF services runtime.

Component	Field	Description
VCF Management Services	Fleet components FQDN	Enter the FQDN of the fleet-level components, such as fleet lifecycle, software depot, etc. For more information, see Components in VCF and vSphere Foundation. **Note:** Do not use capital letters in the FQDN. Do not use a `.local` FQDN.
	Instance components FQDN	Enter the FQDN of the instance-level components, such as SDDC lifecycle, and telemetry. For more information, see Components in VCF and vSphere Foundation. **Note:** Do not use capital letters in the FQDN. Do not use a `.local` FQDN.
	VCF services runtime FQDN	Enter the FQDN for VCF services runtime. VCF services runtime is the common architectural runtime deployed to deliver the shared cluster of management services. The hostname from this FQDN is prefixed to the names of its node VMs and related objects. **Note:** Do not use capital letters in the FQDN. Do not use a `.local` FQDN.

Component	Field	Description
	Internal cluster CIDR IPv4	Select the VCF services runtime internal IPv4 network. • 198.18.0.0/15 • 240.0.0.0/15 • 250.0.0.0/15 Different VCF services runtime instances on the same network can work with the same internal network. For example, you can have the two instances of VCF services runtime for your VCF management services and VCF Automation using the same internal network.
	Internal cluster CIDR IPv6	Select the VCF services runtime internal IPv6 network. • fd00::/111 (default) • fd00::0/111 • fc00::0004:0/111 • fc00::4:0/111 • fc00::/111 • fc00::0/111 This option is only available if you have dual-stack networking enabled in your environment.
License Server	License Server FQDN	Enter the FQDN of the license server. A license server is added to a VCF Operations instance to store the licenses you want to assign from this VCF Operations instance. You must add at least one license server to each VCF Operations instance that you use for license management. You can deploy license server as IPv4 only. In vSphere Foundation 9.1.1 and later, after the deployment, you can change the networking configuration to IPv6. See Configure IPv6 on a License Server.

12. Review the validation information.

    The VCF Installer validates the information you provided in the deployment wizard and reports any errors or warnings.

Warnings	Review any warnings and either address them or click **Acknowledge** to proceed with deployment.
Errors	Review all failed validations and resolve the issue(s) using the remediation information provided in the error message. You can navigate back to the relevant pages in the deployment wizard to make updates and then re-run the validations. You can also download the JSON specification file, make the necessary changes, and then upload the modified JSON specification file from the VCF Installer homepage.

13. After addressing all validation errors and warnings, click **Deploy**.
14. Monitor the deployment progress.

    If there are errors, review the failed tasks in the Tasks panel, address the issue(s), and retry the deployment.

After the VCF management services deployment completes, you can proceed with the upgrade of log management.

Upgrading vSphere 8 and Optionally vSAN and Aria Operations 8 to 9.1.x

As a first phase to upgrade your vSphere 8.x and optionally vSAN and Aria Operations 8.x virtual infrastructure to 9.1.x, follow the order of upgrade and deployment procedures.

Follow this scenario only if your environment has just vSphere and optionally vSAN and Aria Operations components and no SDDC Manager, NSX, or other Aria components. This scenario takes you to a stable environment that can later be extended to full vSphere Foundation or VCF 9.1.x environments.

VCF Operations and the license server components are required to license all 9.1.x environments. If you do not have Aria Operations to upgrade to 9.1.x, you deploy VCF Operations 9.1.x.

Note:

This scenario is a first phase for upgrading to a vSphere Foundation or VCF 9.1.x environments. You cannot use the capabilities of the 9.1.x components that are not yet deployed. For more information on the 9.1 components and capabilities, see Components in VCF and vSphere Foundation.

Supported 8.x Components for this Scenario	Not Supported Components for this Scenario
• vCenter 8 Update 3 and later • ESX 8 Update 3 and later • (Optional) vSAN 8 Update 3 and later • (Optional) Aria Operations 8.18.x	• SDDC Manager • NSX • Aria Operations for Logs

#	Component	Getting to Version 9.1.x Procedure
1	Aria Operations	Upgrade Aria Operations 8.18 and later or deploy VCF Operations 9.1
2	License server	Deploy a License Server
3	vCenter	Upgrade the vCenter Instance
4	ESX	Upgrade the ESX Hosts
5	vSAN On-Disk Format Versions	Upgrade vSAN On-Disk Format Versions
6	vSAN File Service	Upgrade vSAN File Service

VCF Upgrade Planner

You can now use the new VCF Upgrade Planning Tool to navigate you during the process of upgrading your existing infrastructure to 9.1.x.

Converging to Full VCF 9.1.x

After you have performed this scenario, you can use your existing 9.1 virtual infrastructure as building blocks to converge it to a full VCF 9.1.x deployment. See Converge a vCenter Instance, ESX Hosts, VMware Aria Operations, and Optionally vSAN and NSX.